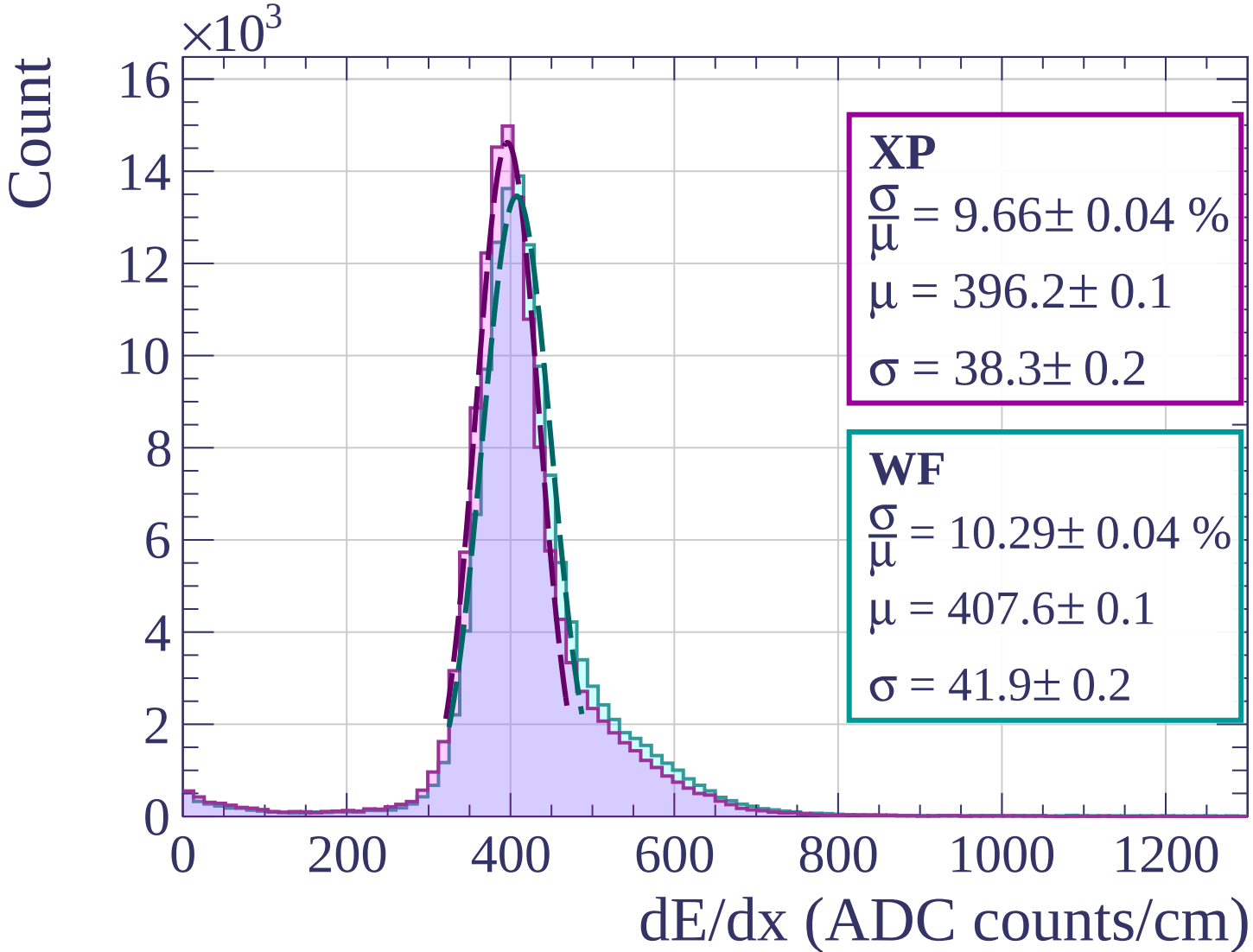
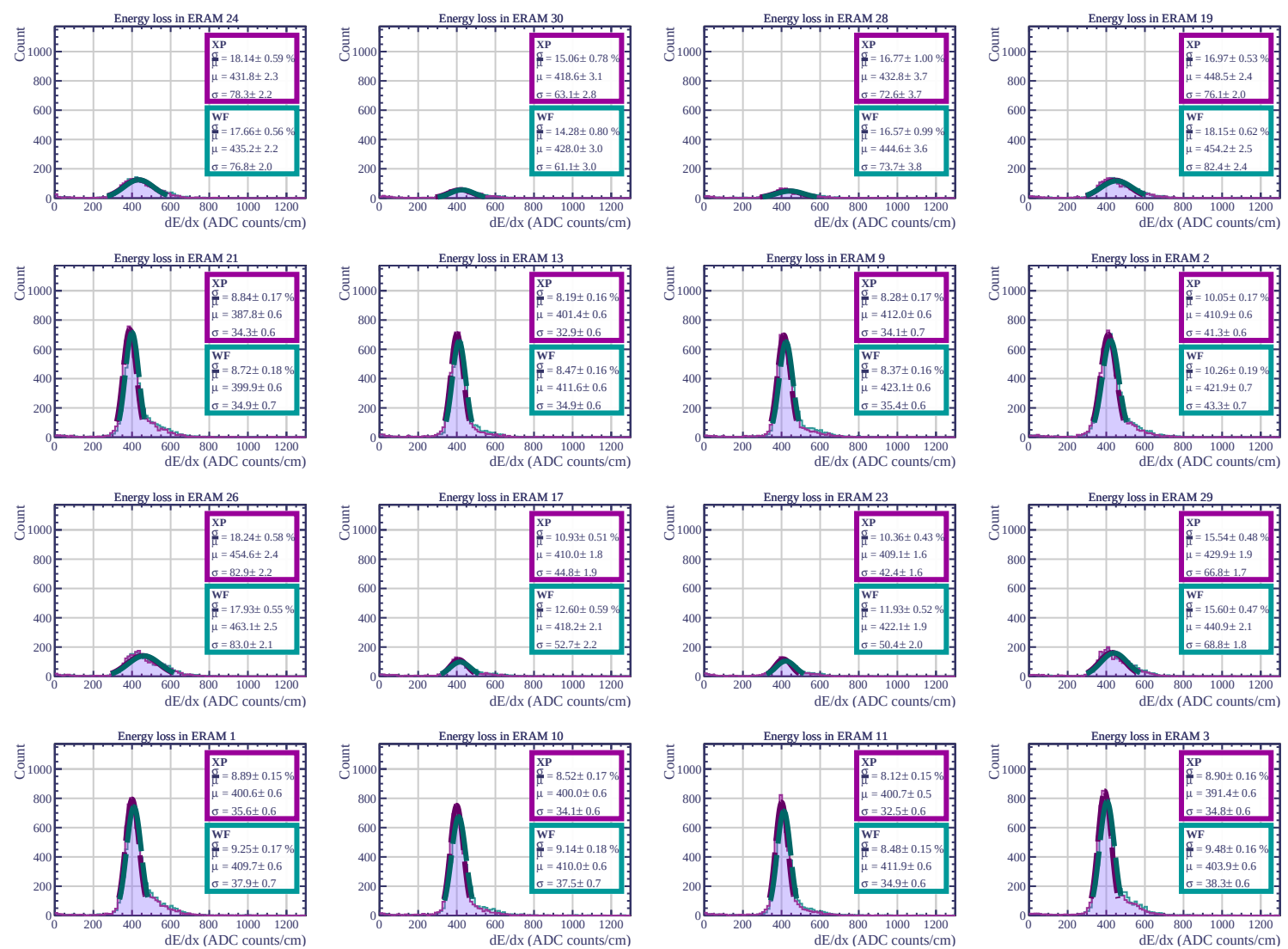
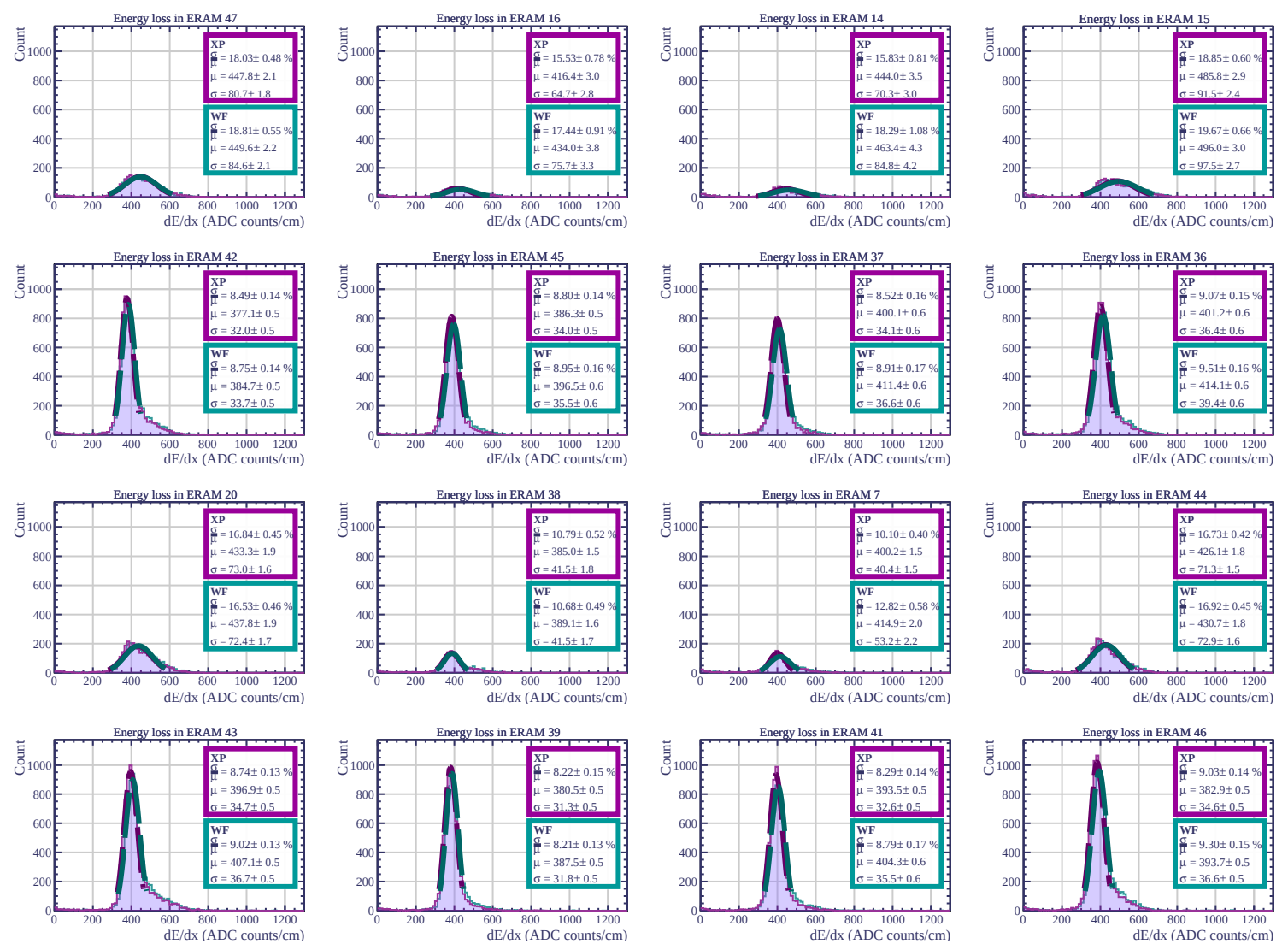
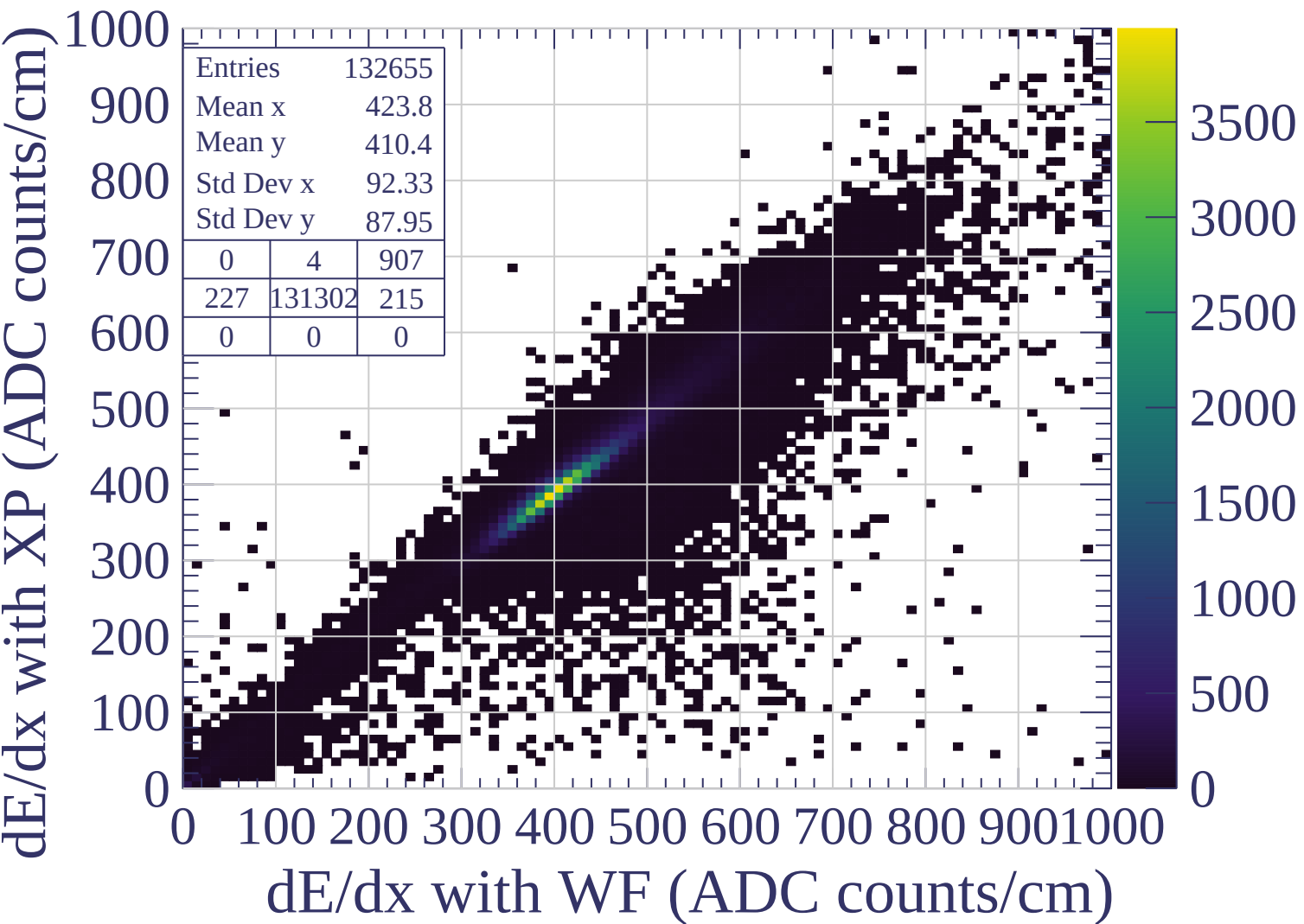
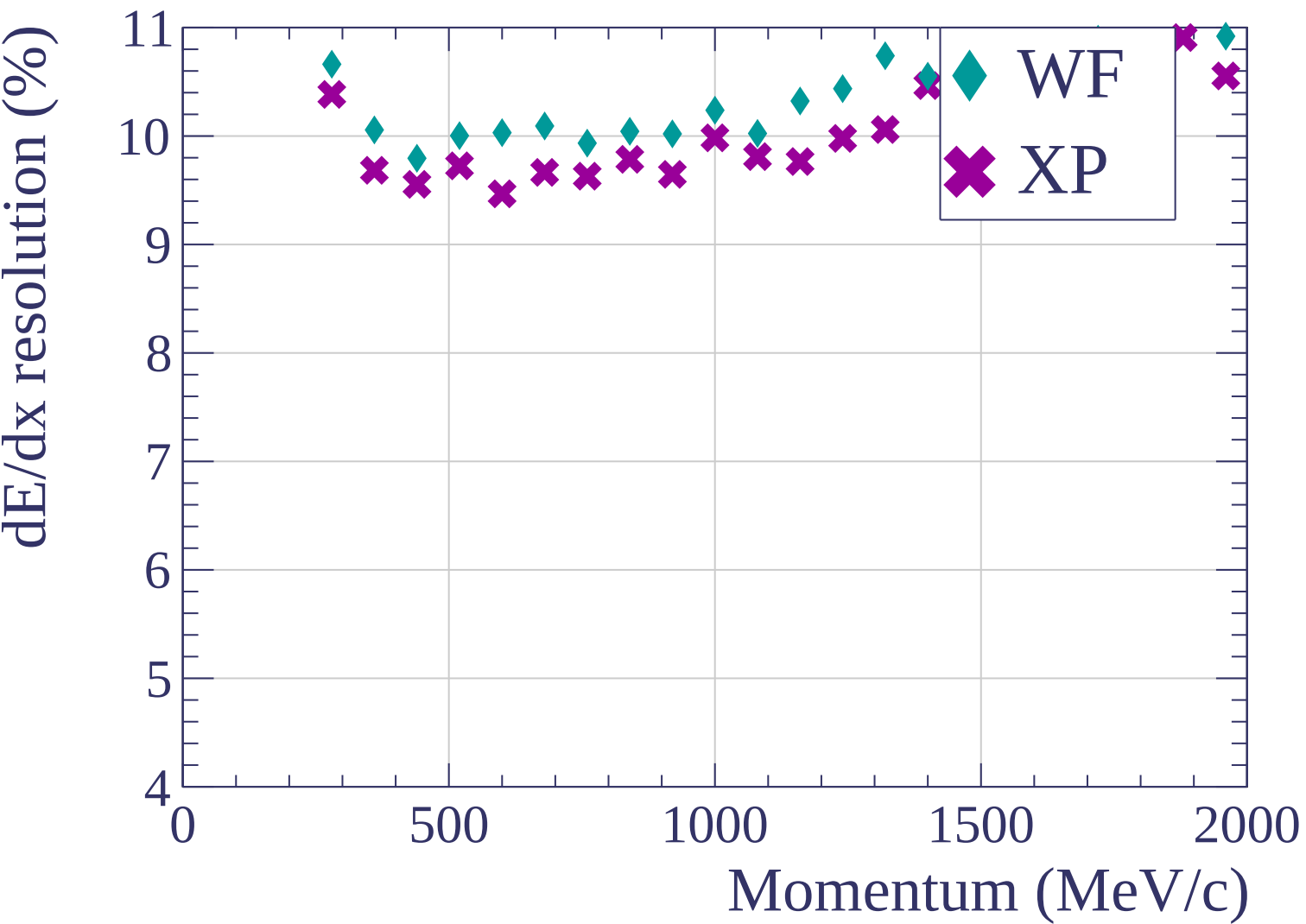


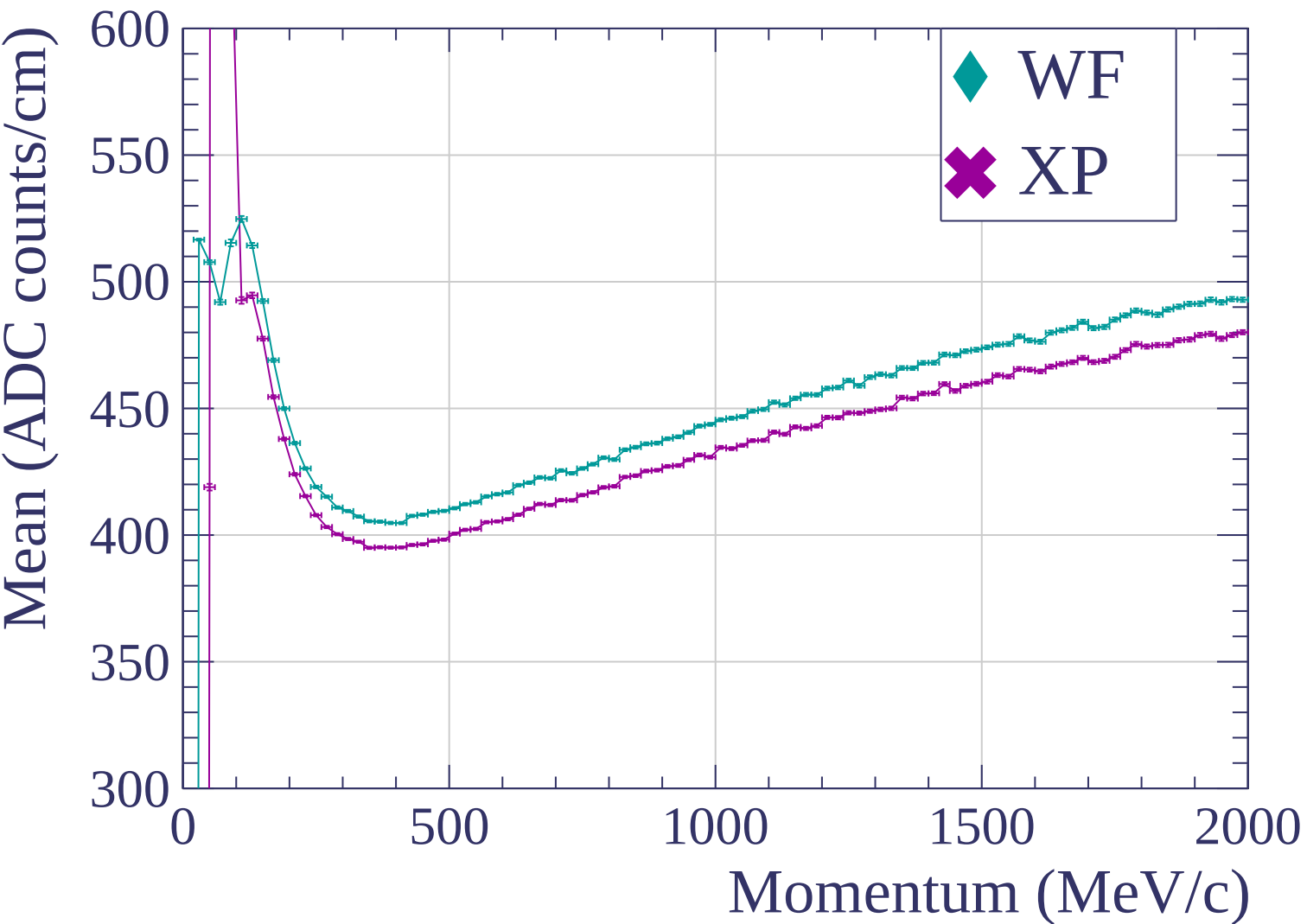


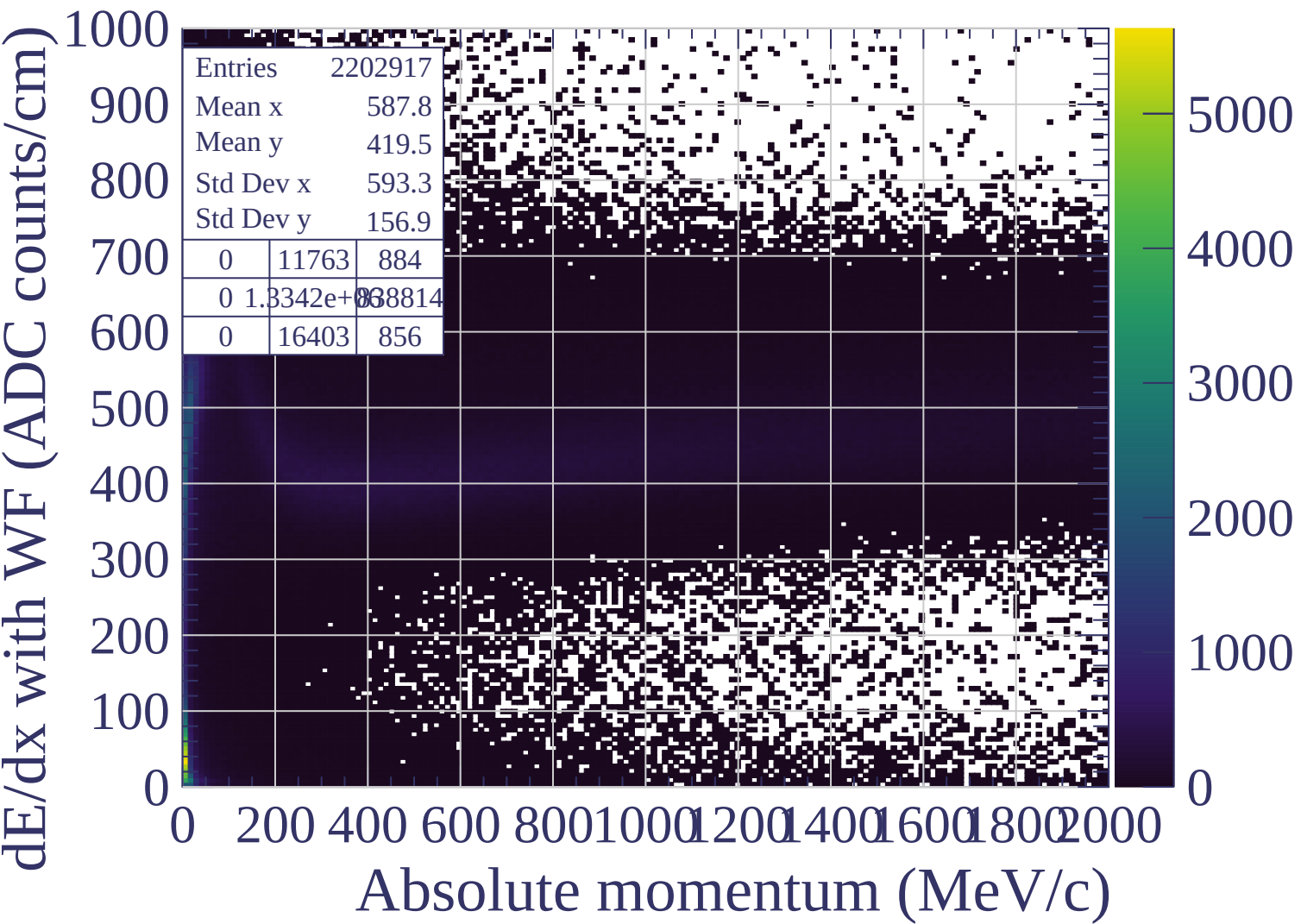




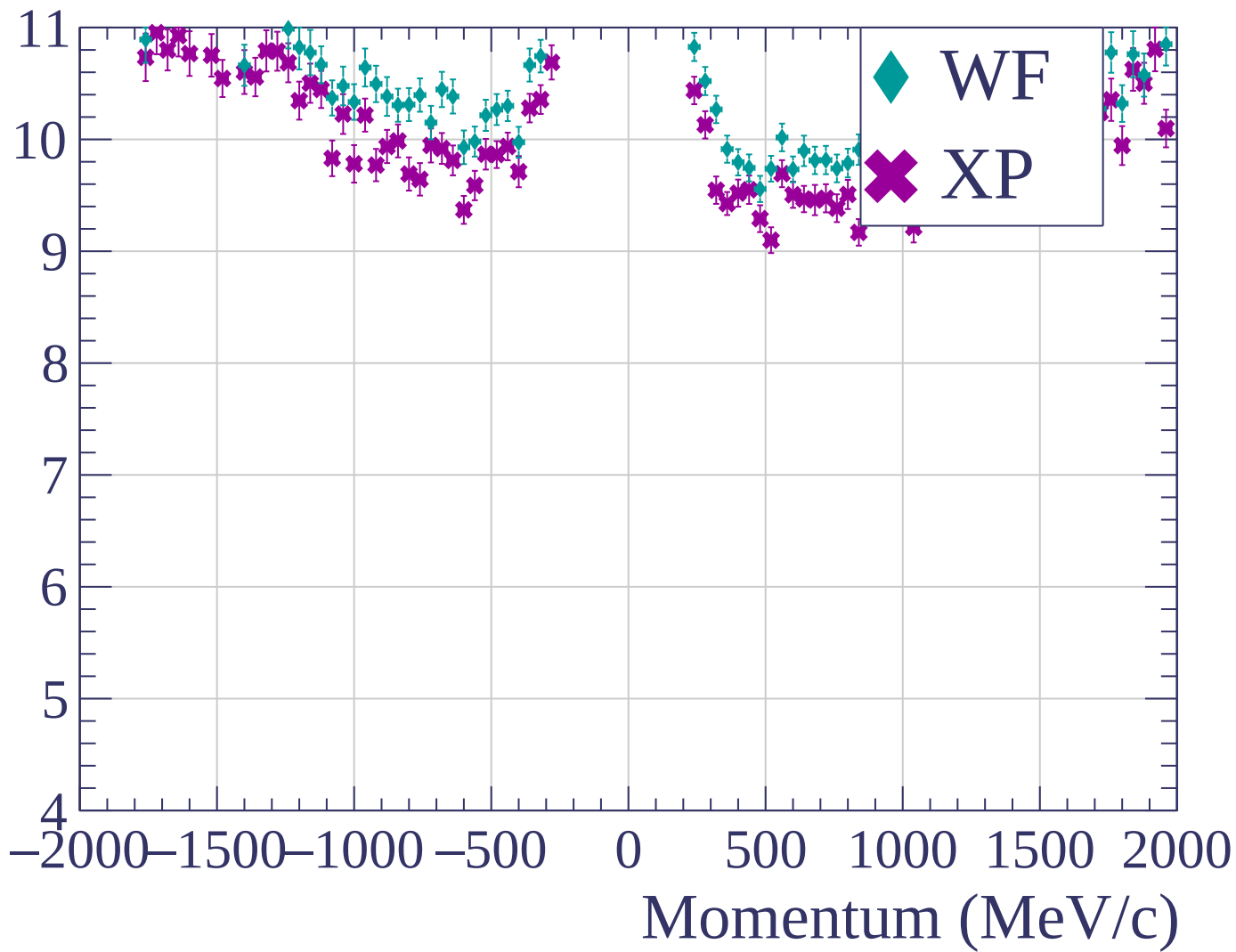


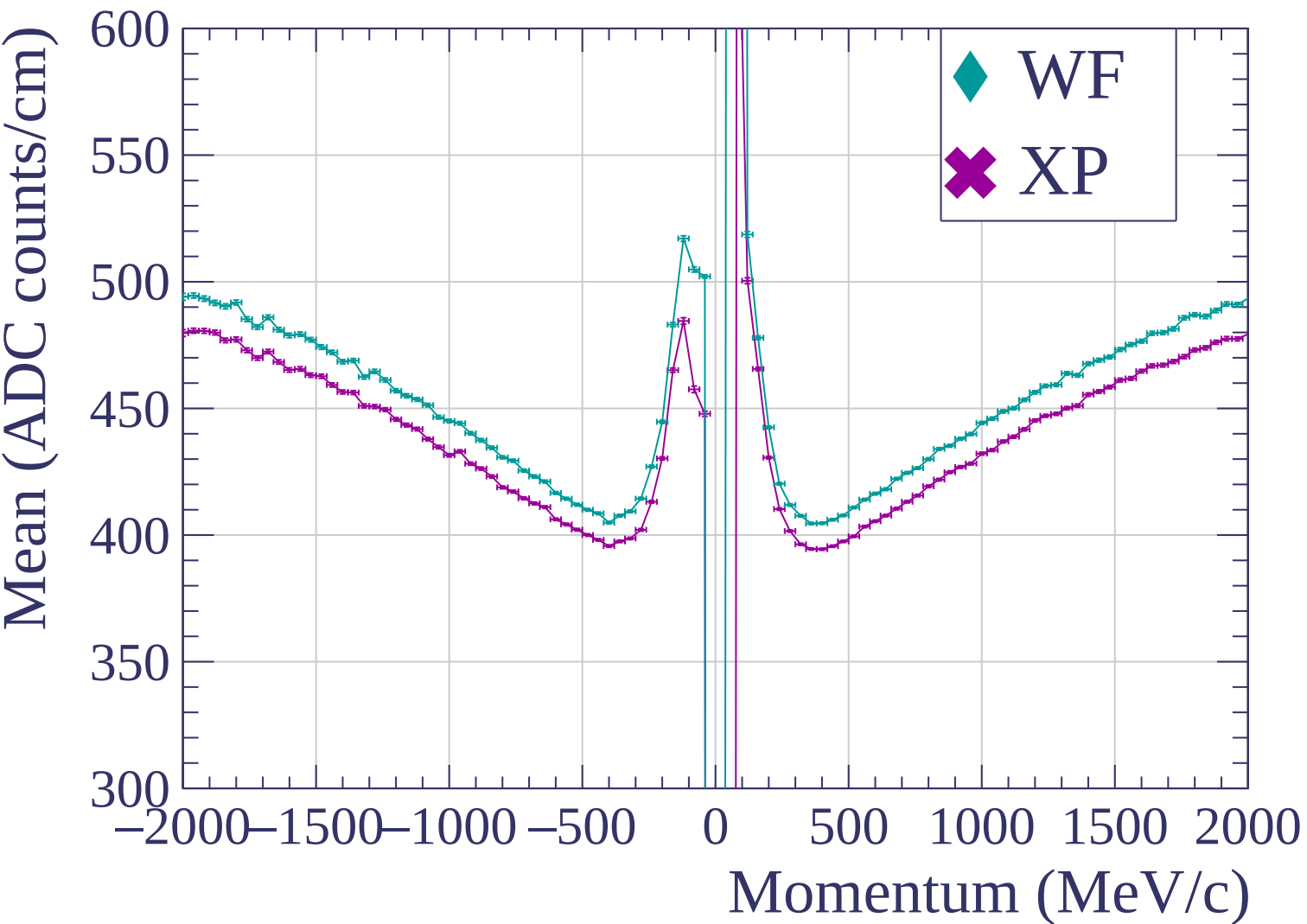


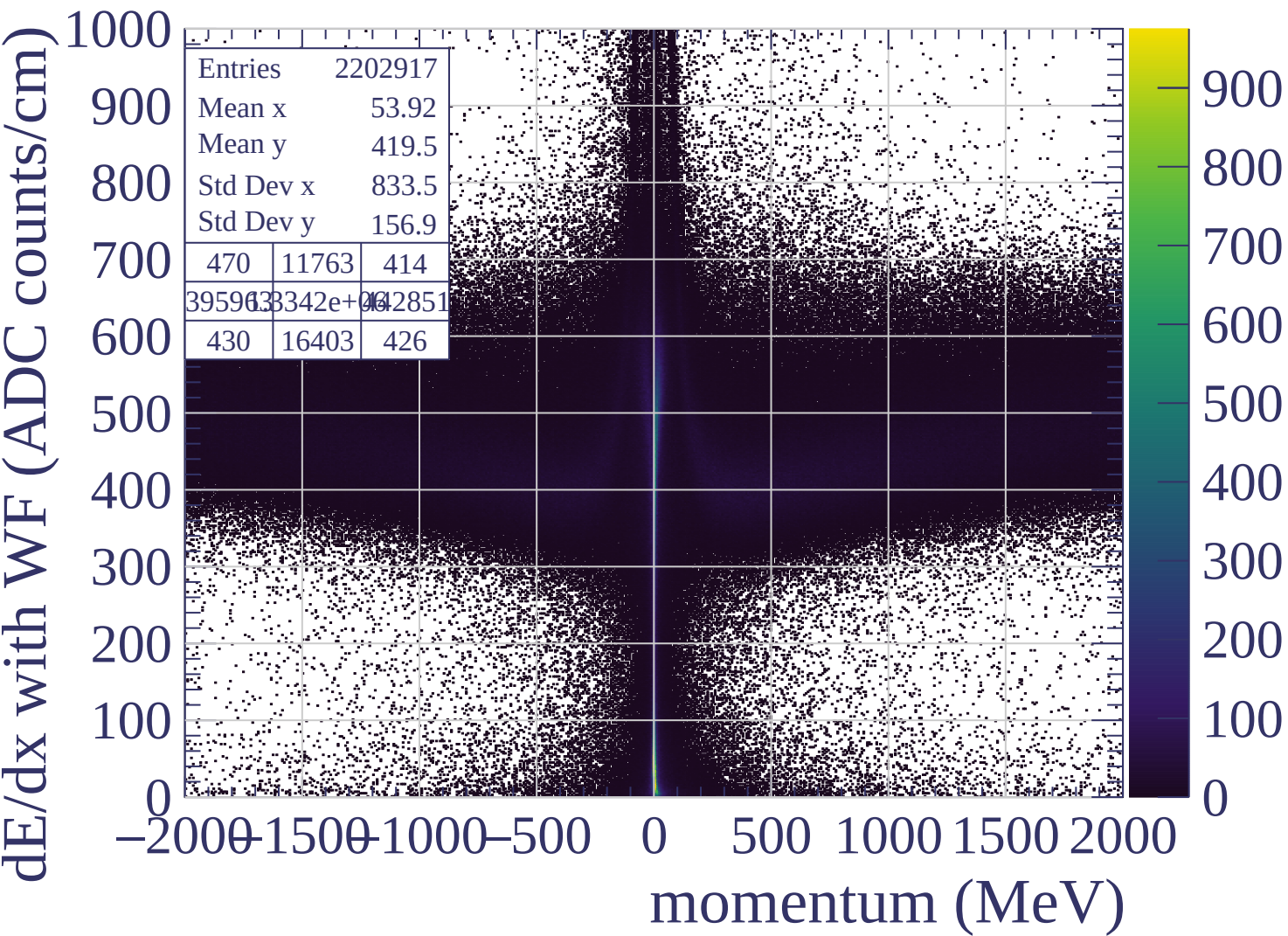


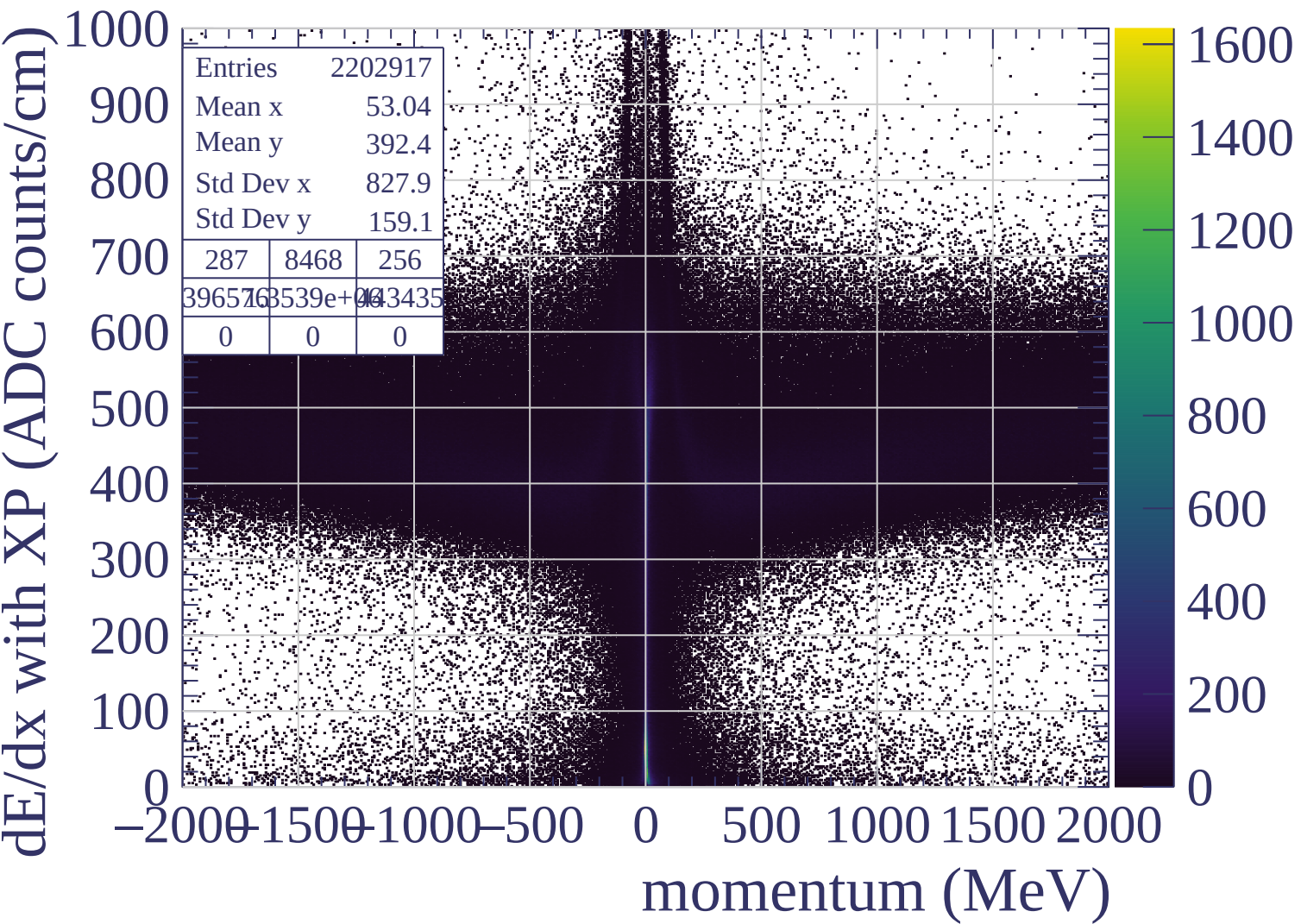


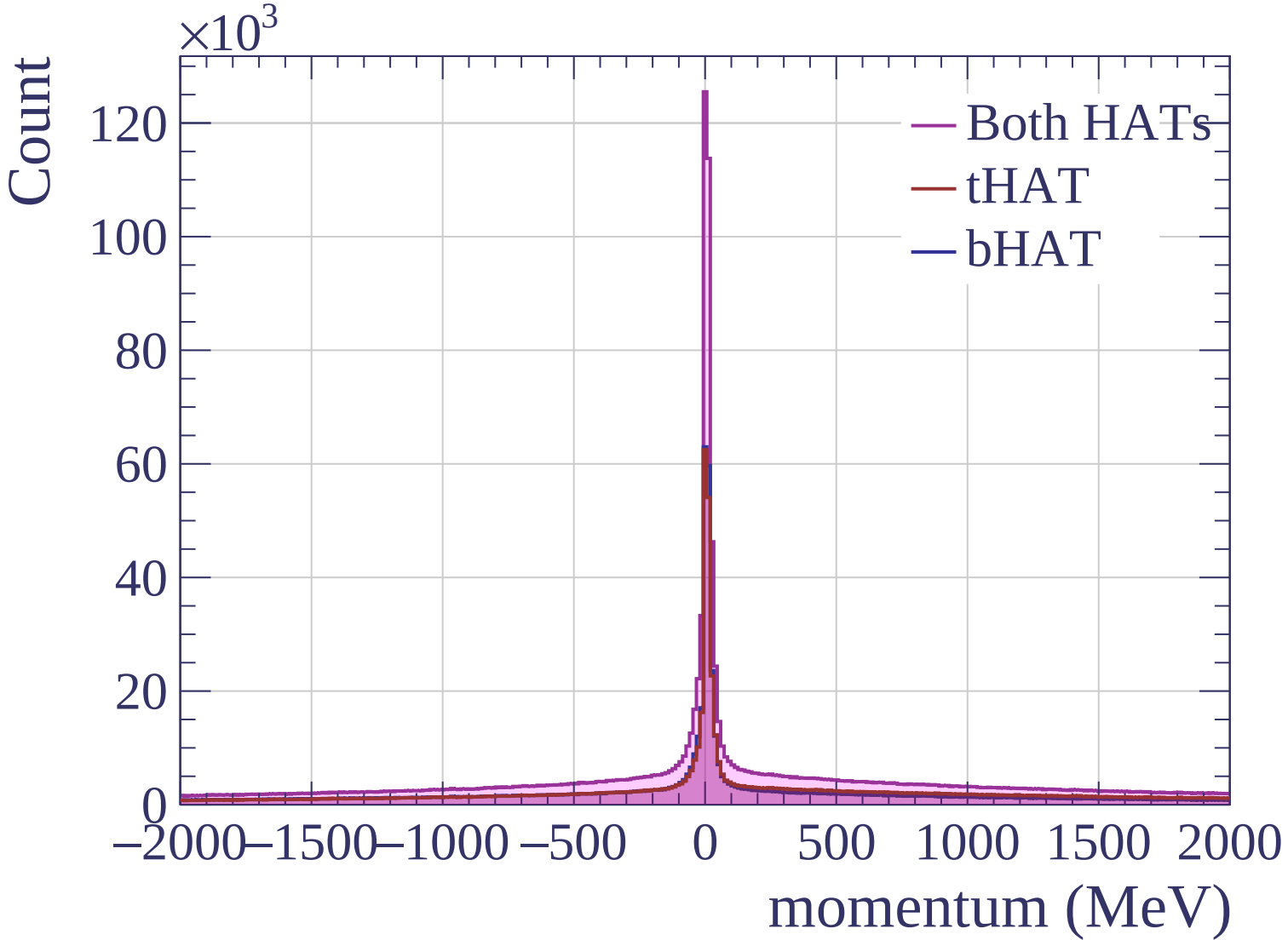
dE/dx resolution (%)

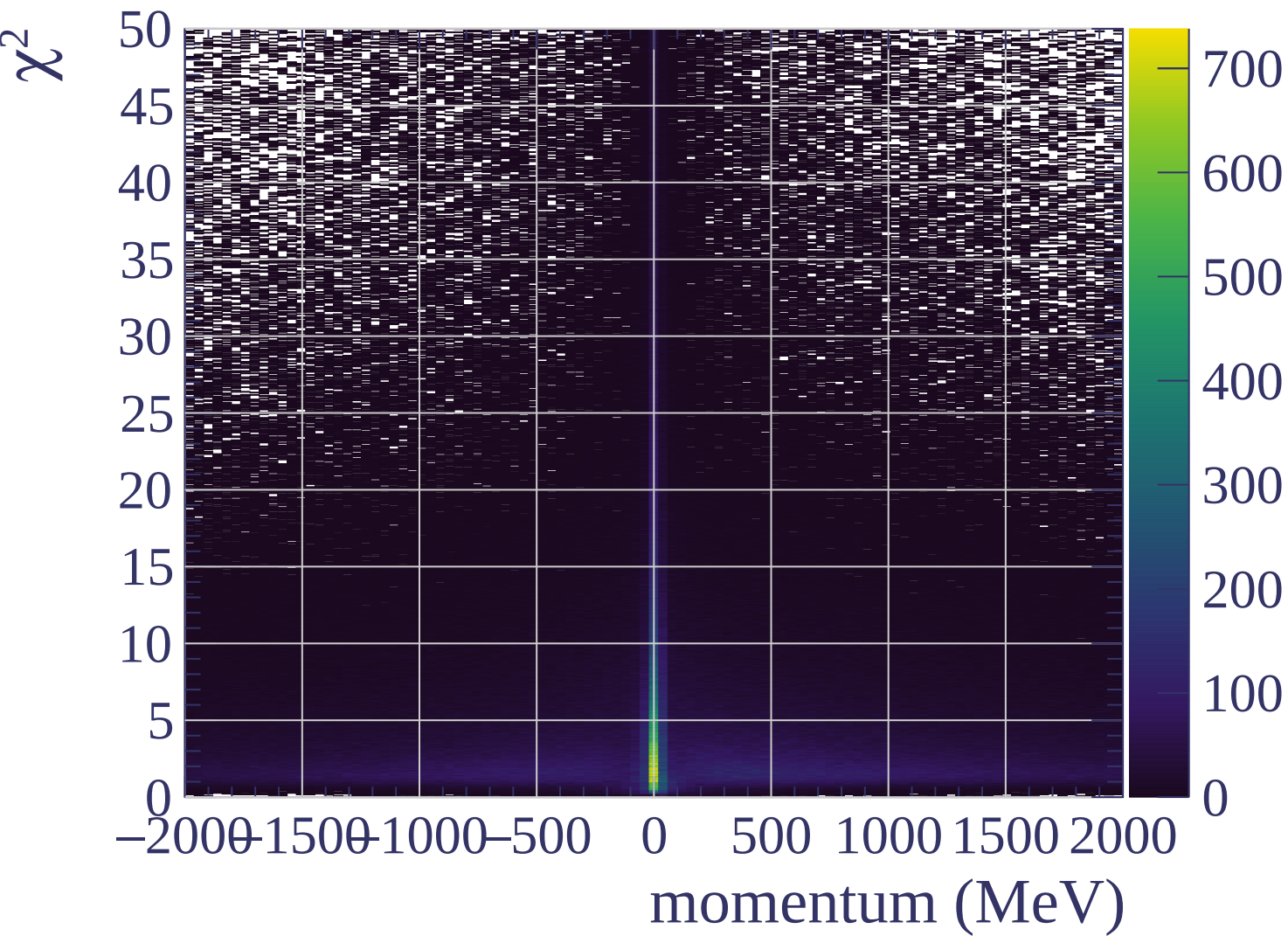


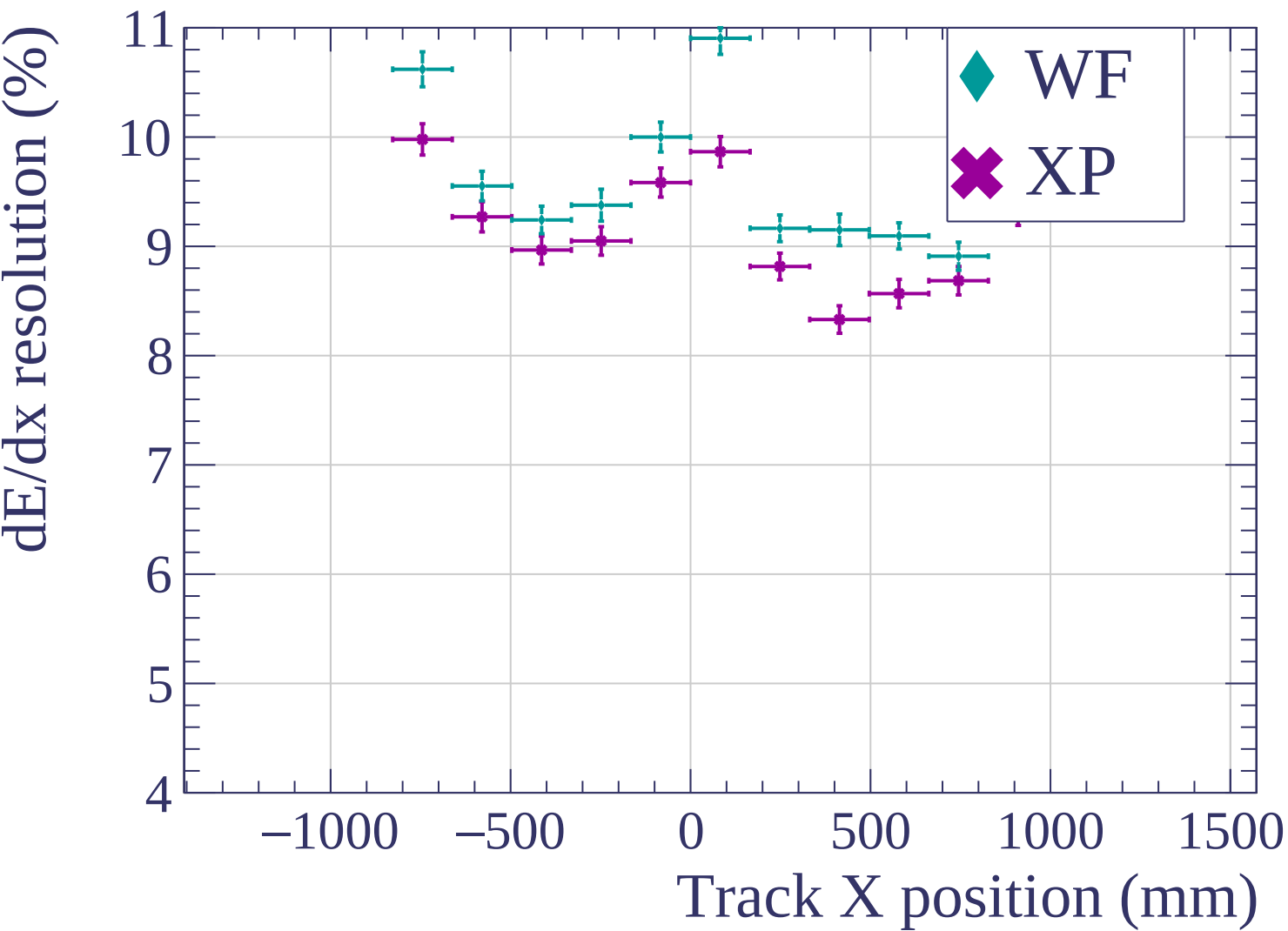


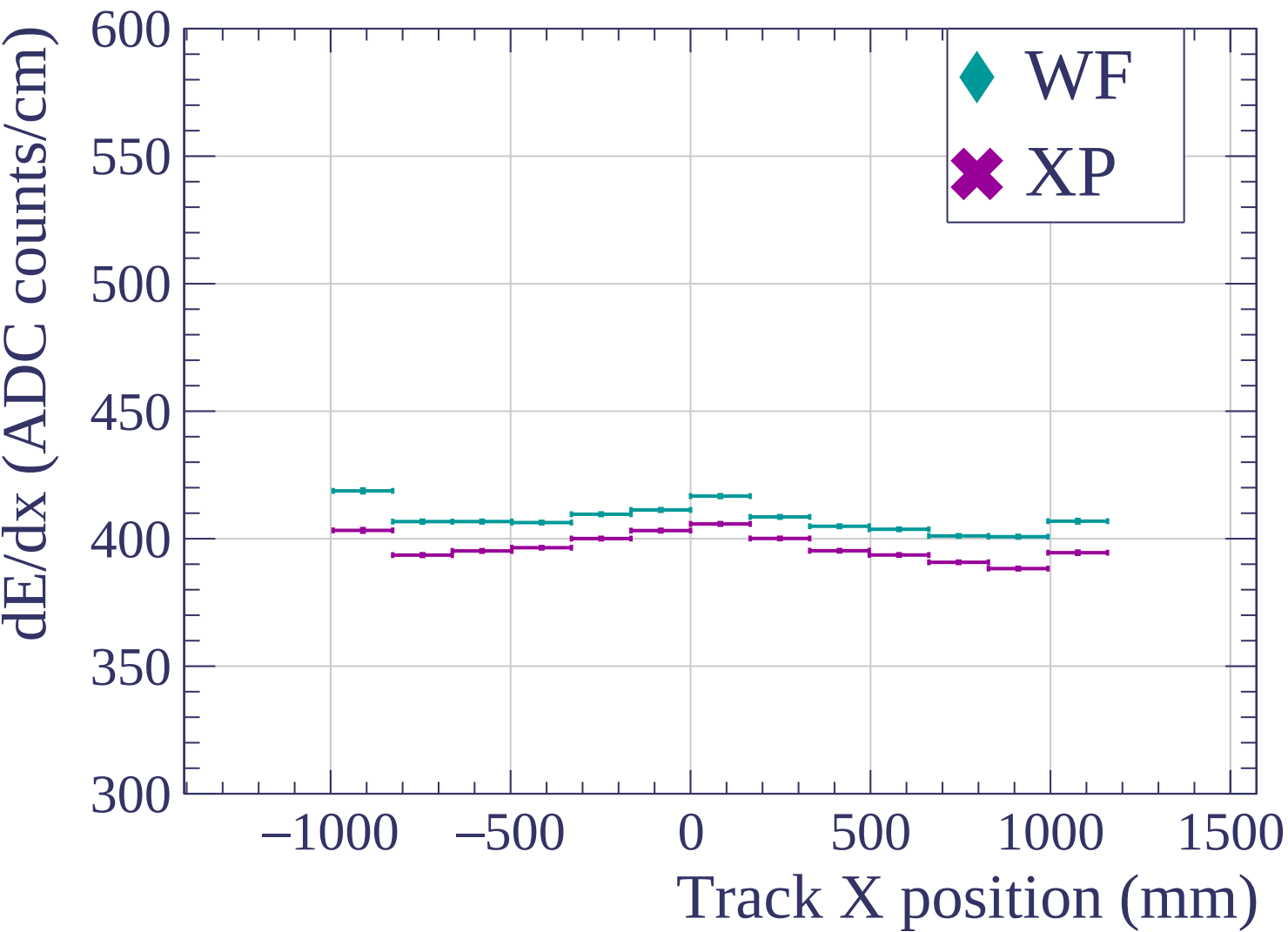


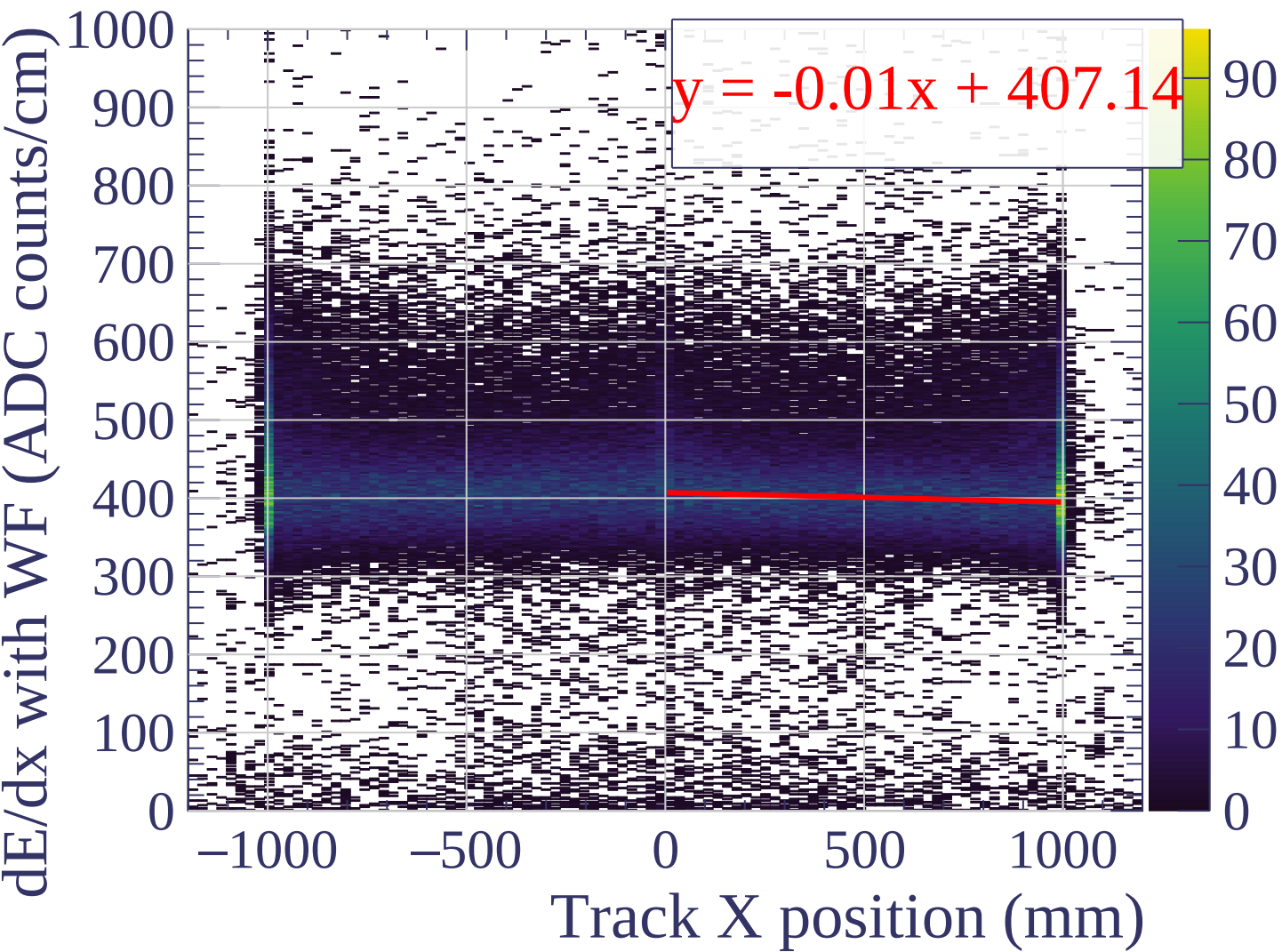


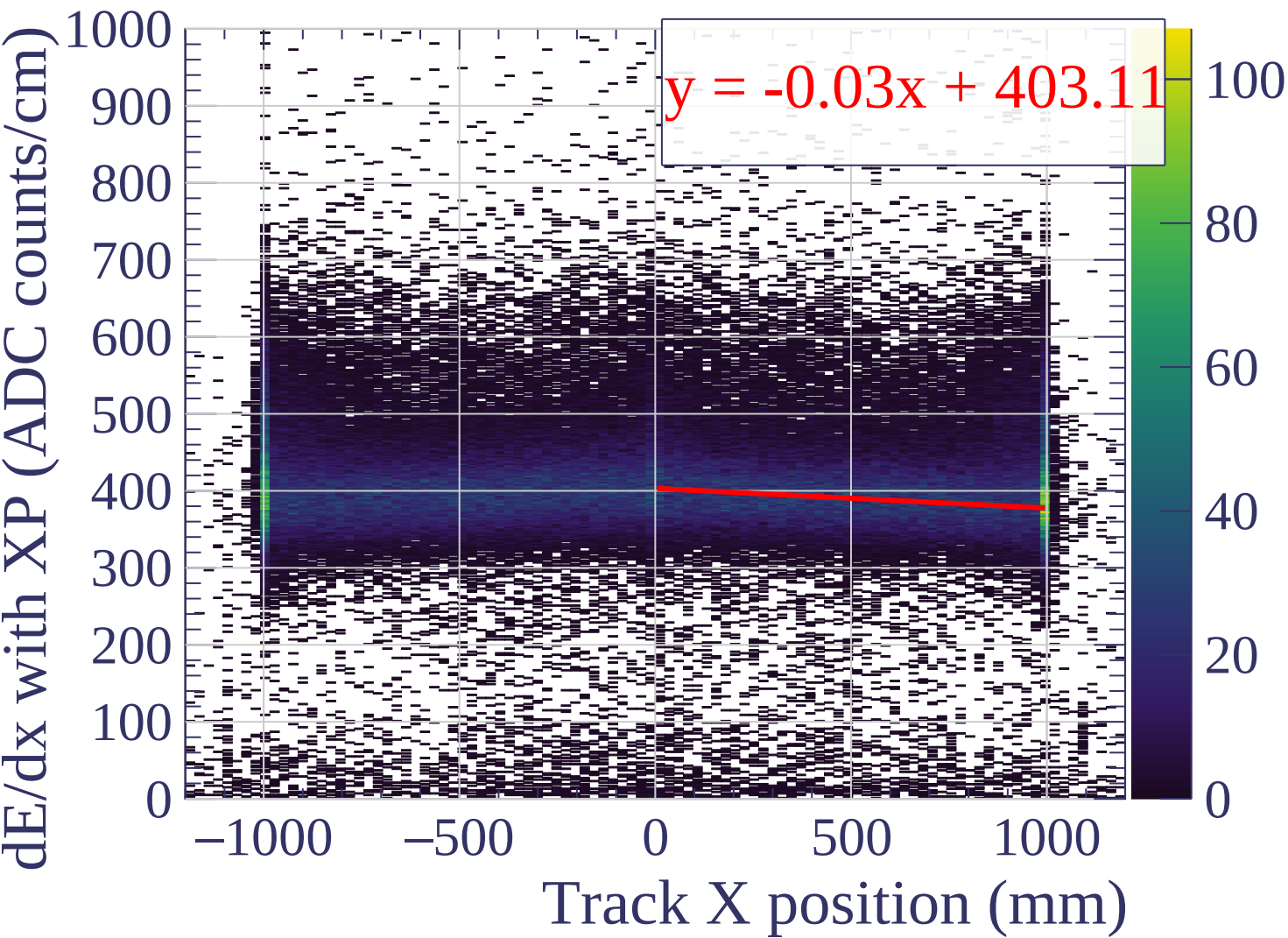


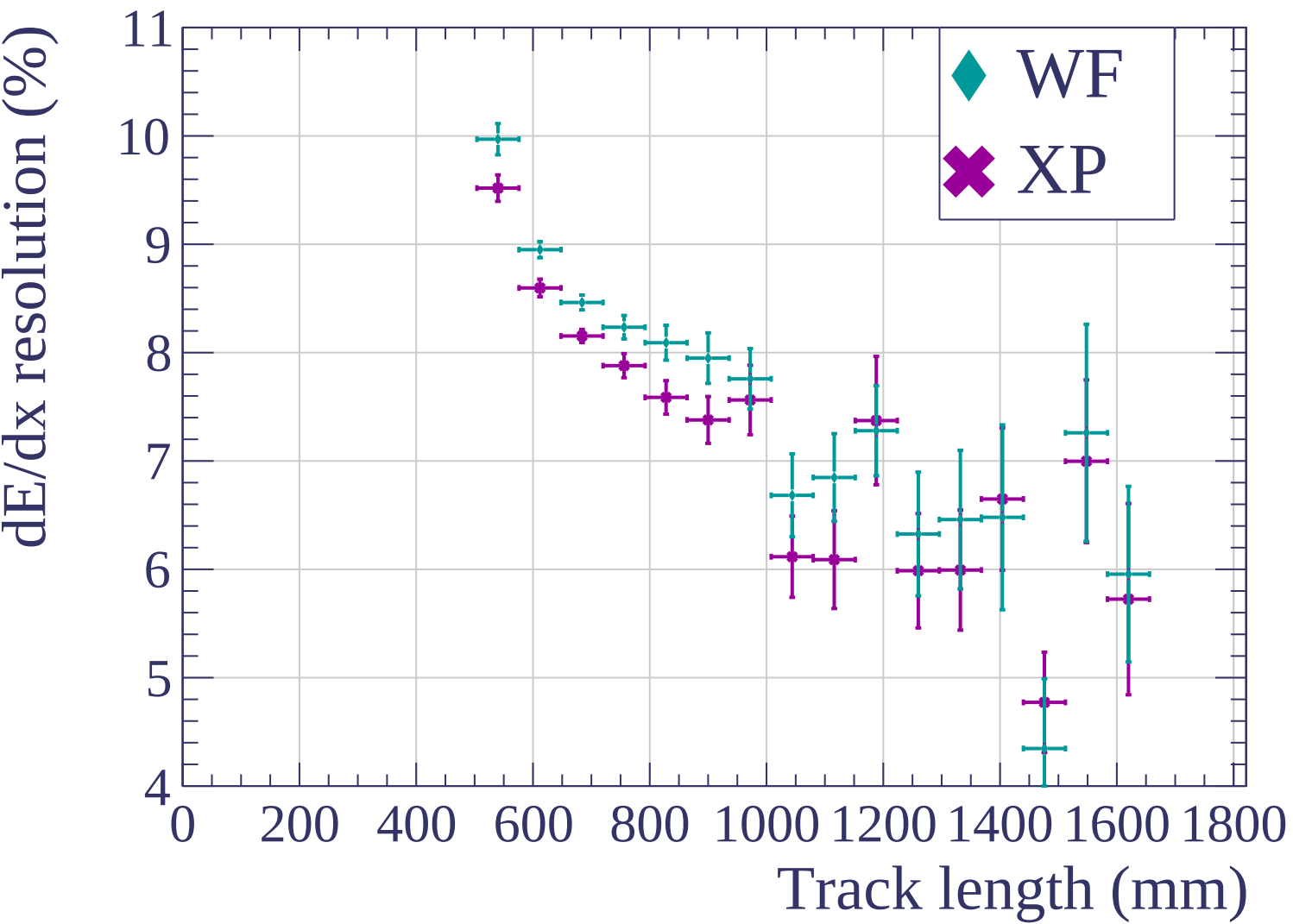


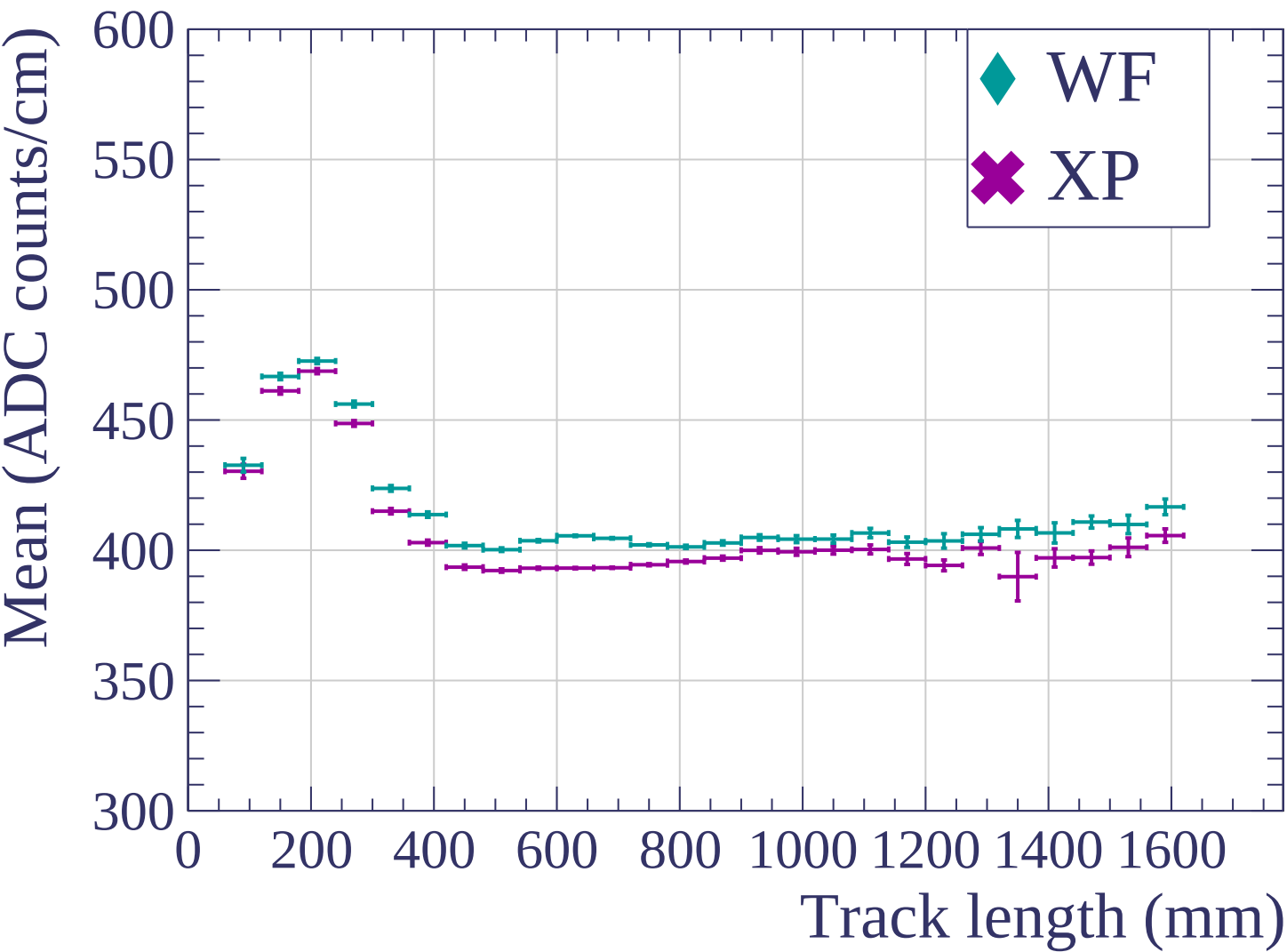


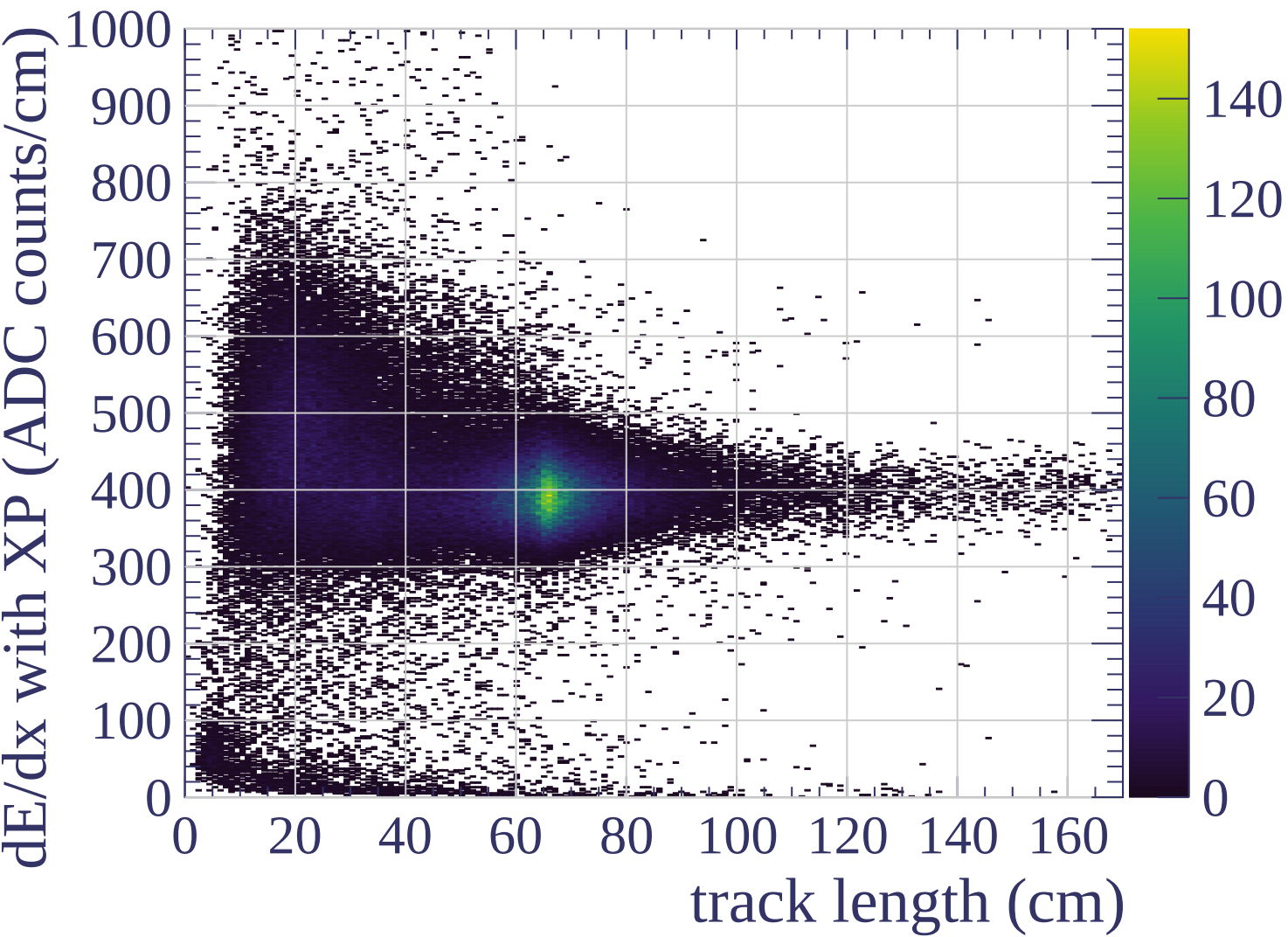


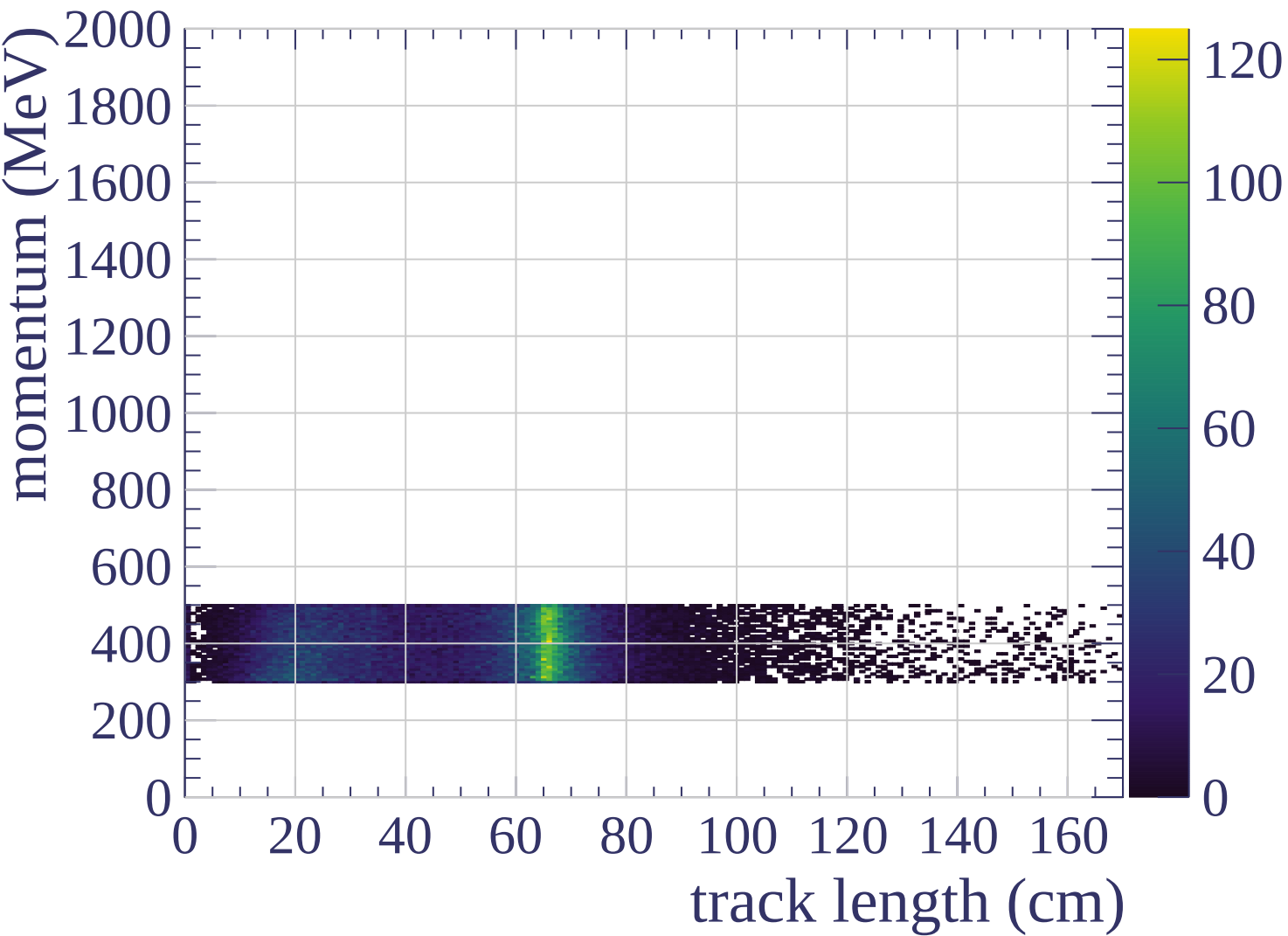


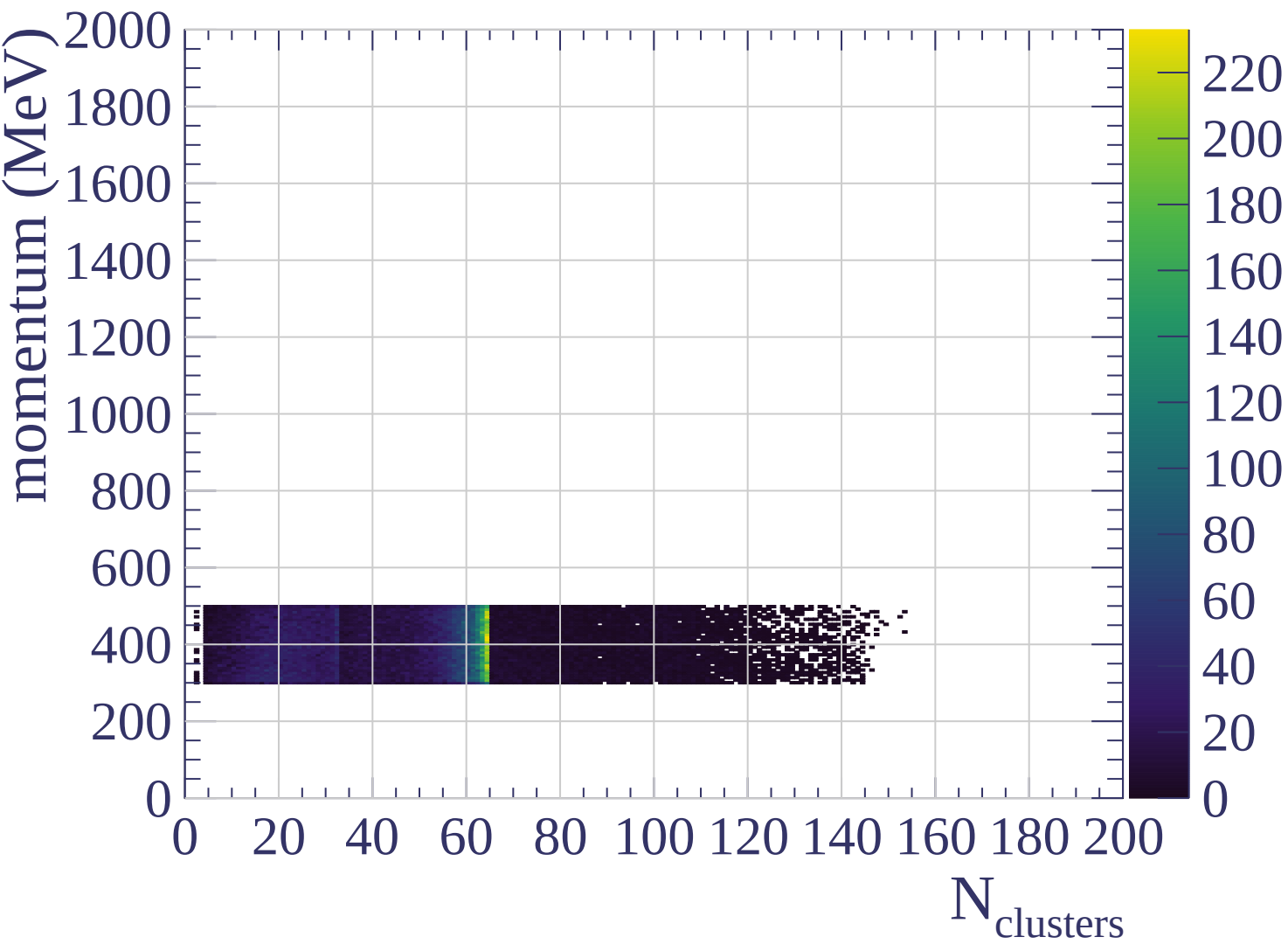




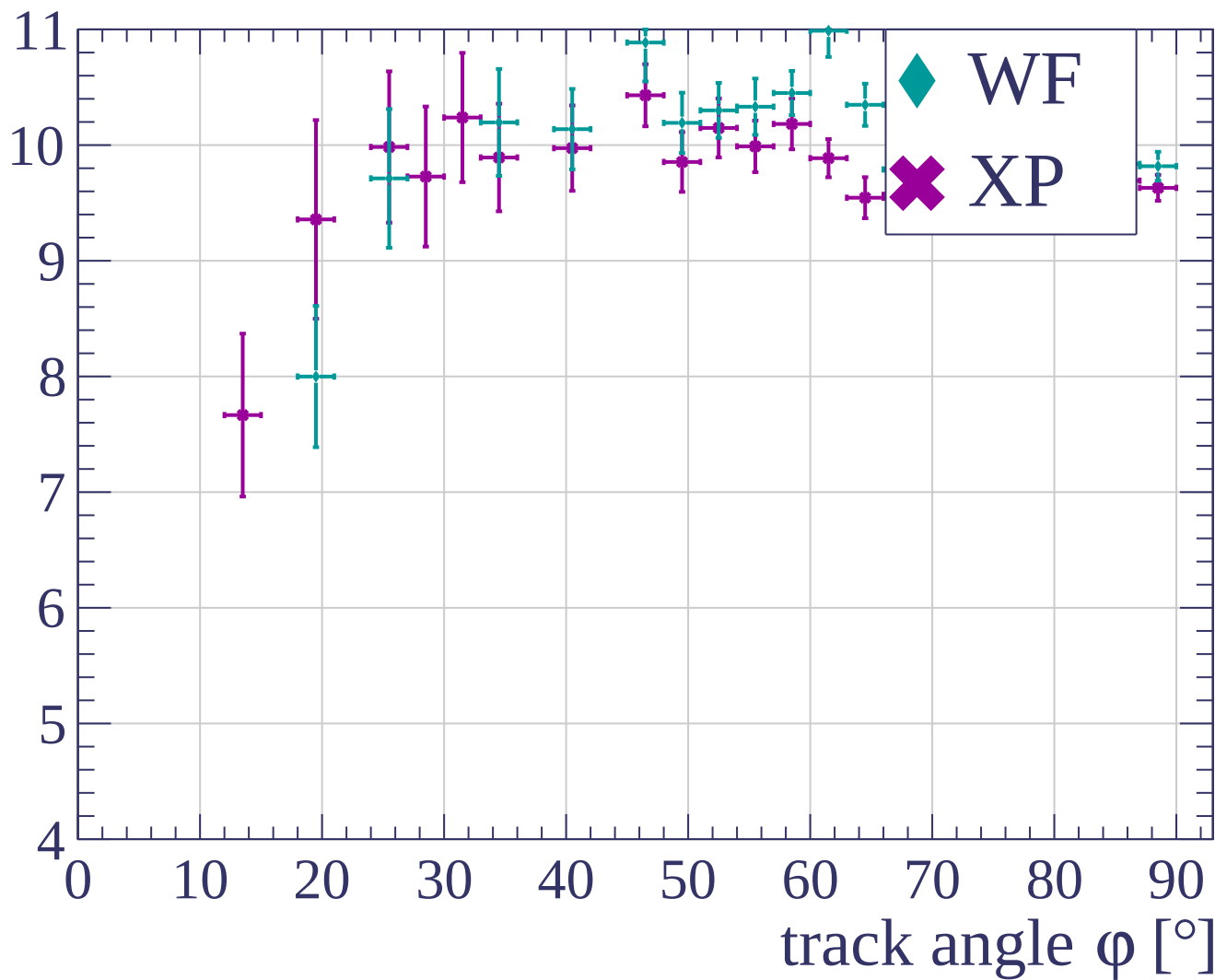


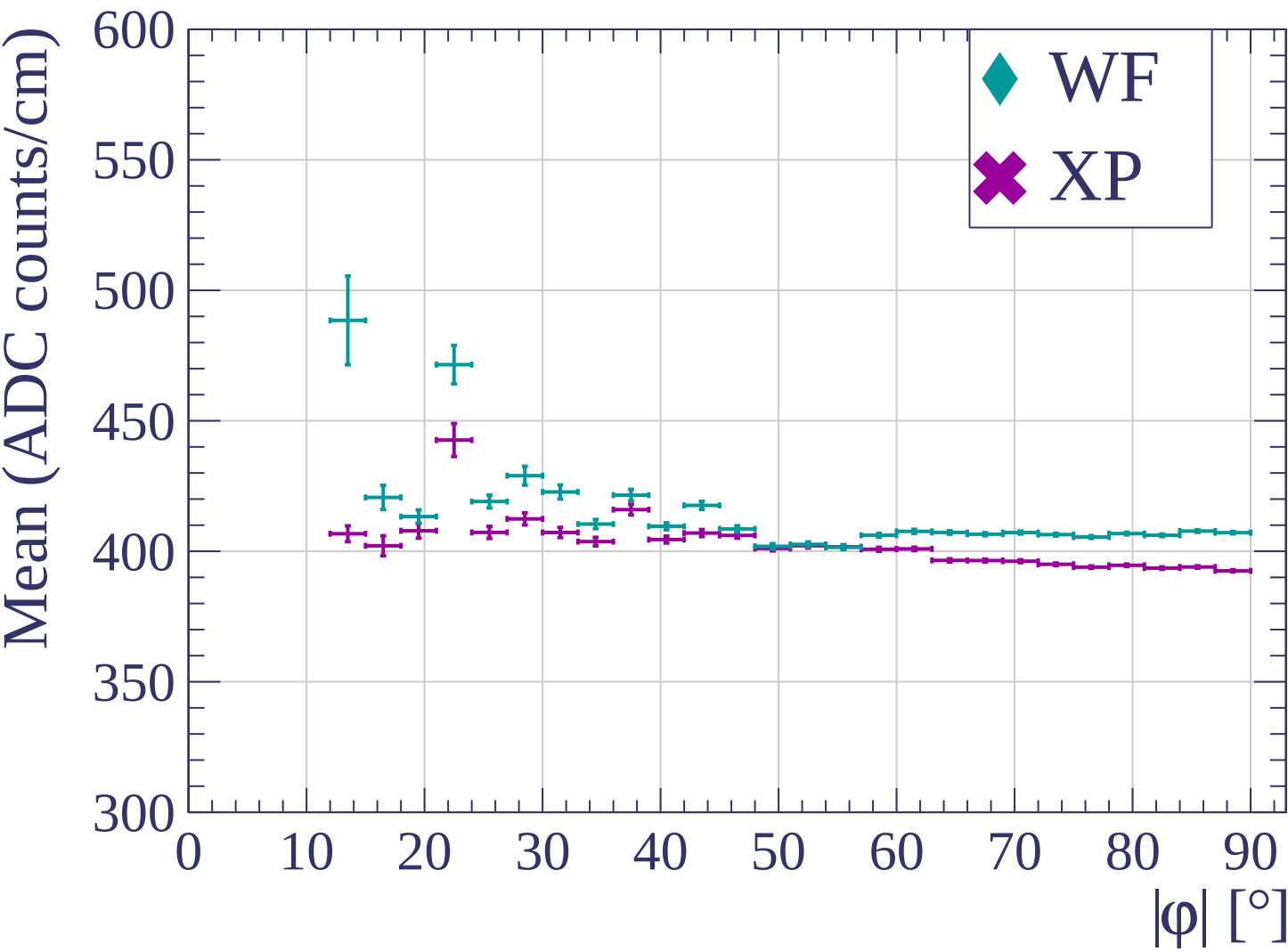




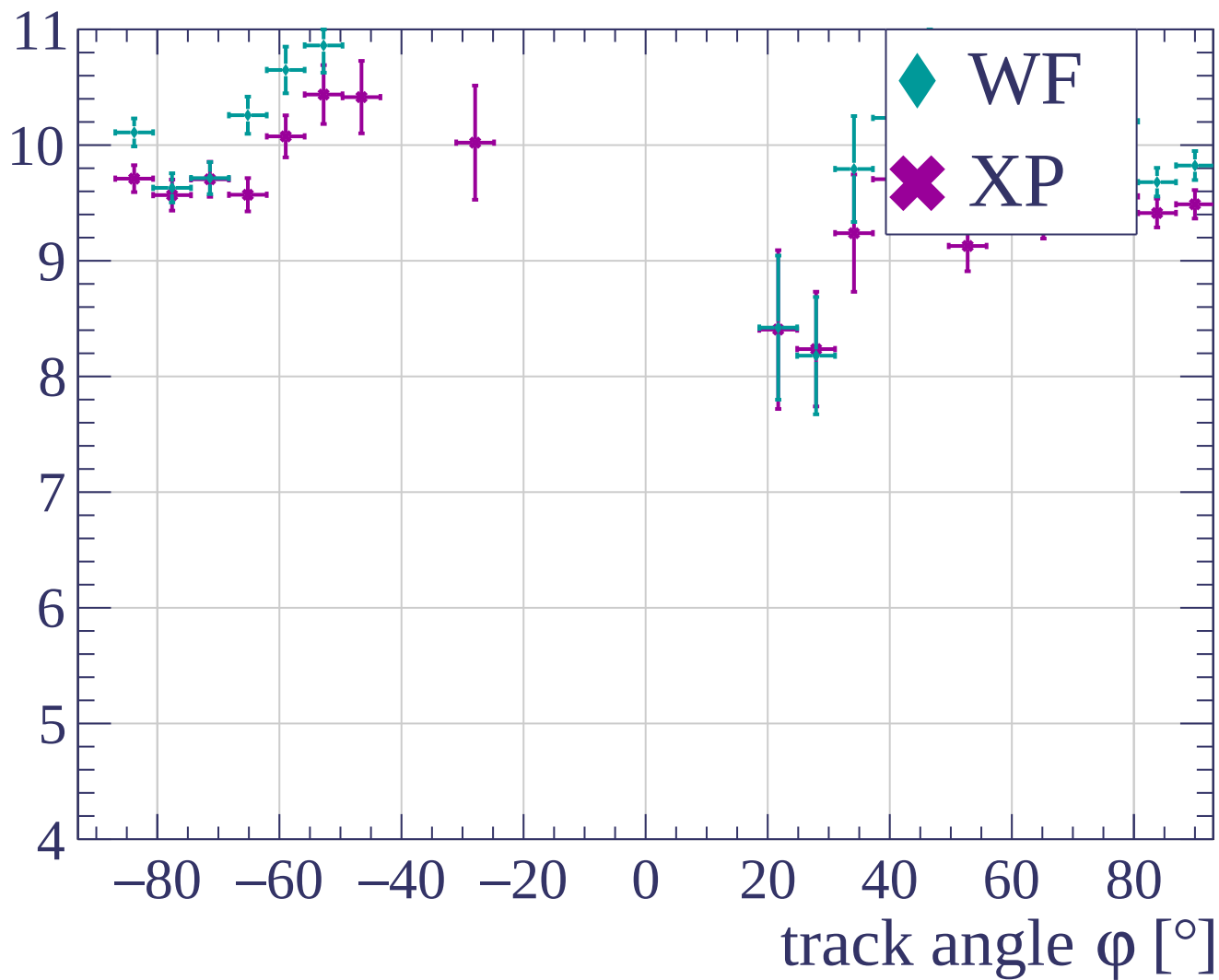


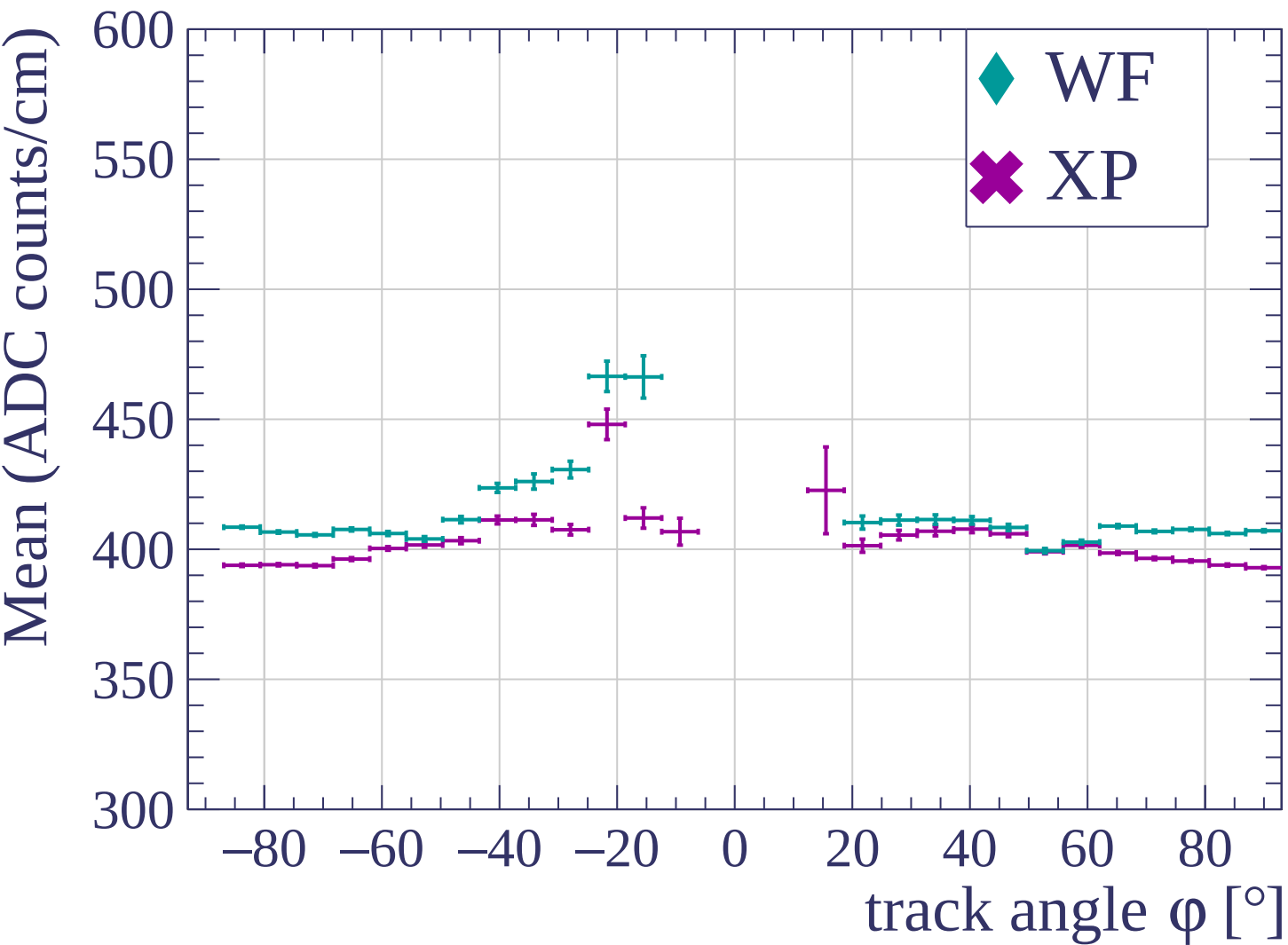
dE/dx resolution (%)

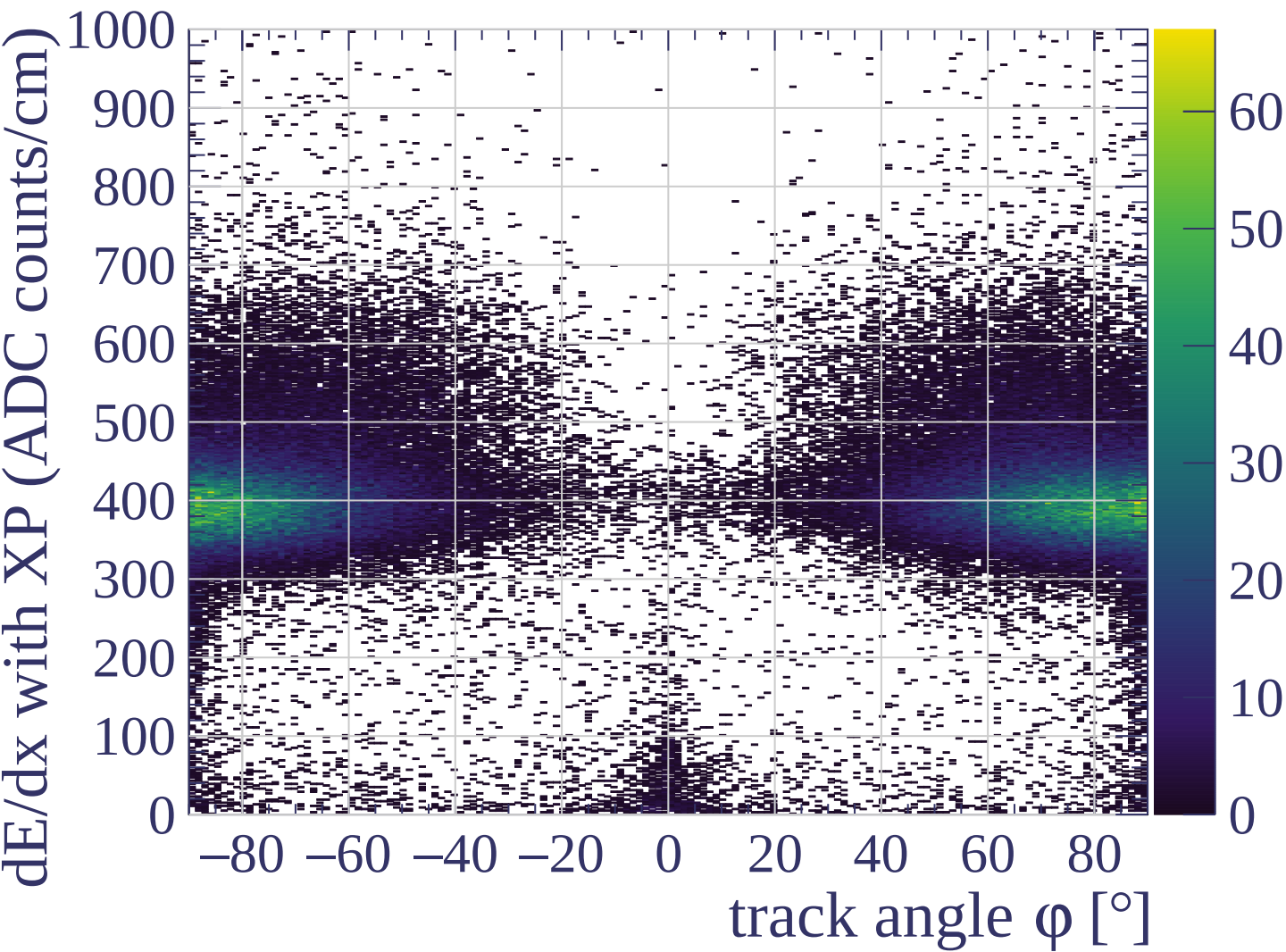


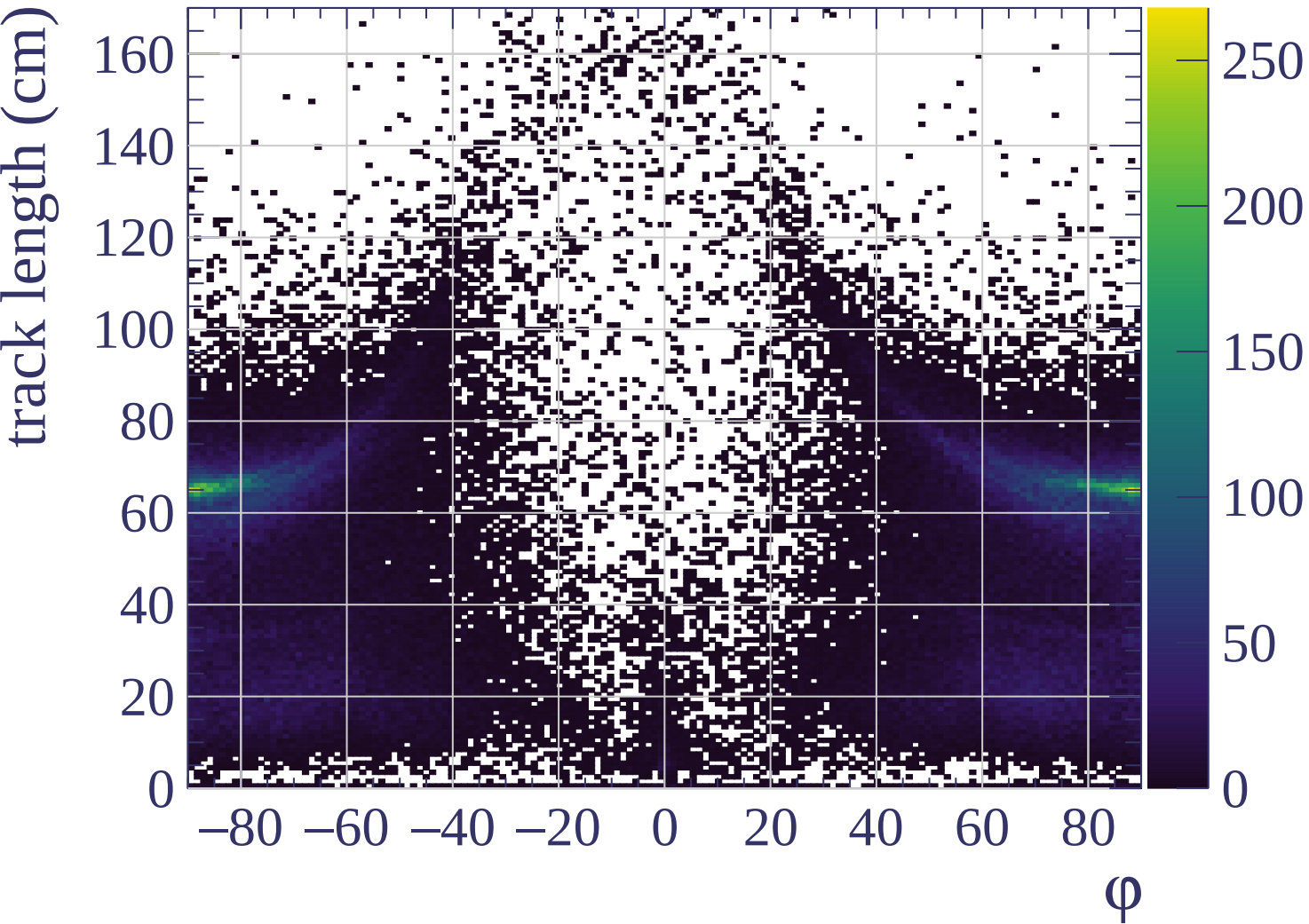


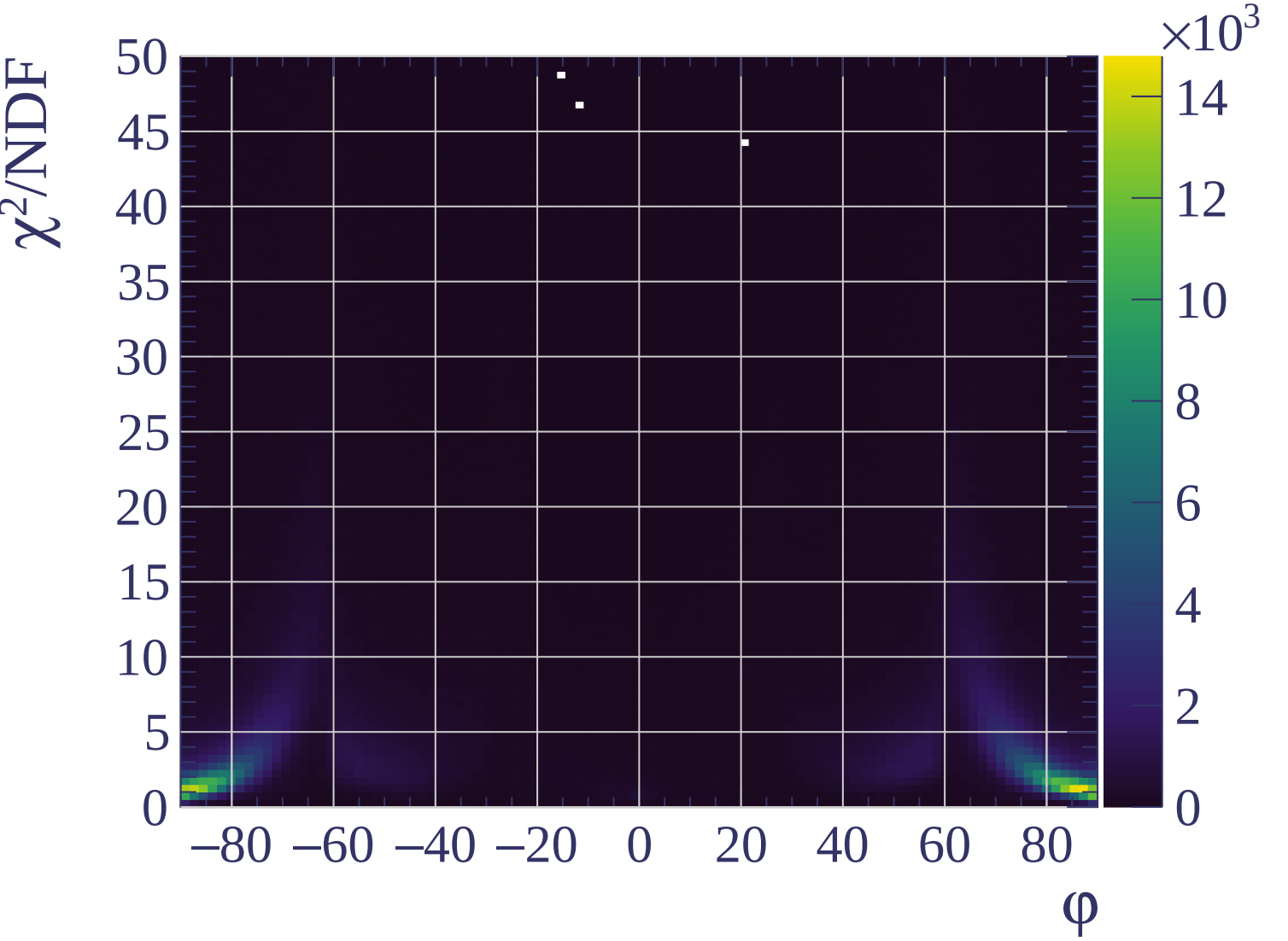
dE/dx resolution (%)

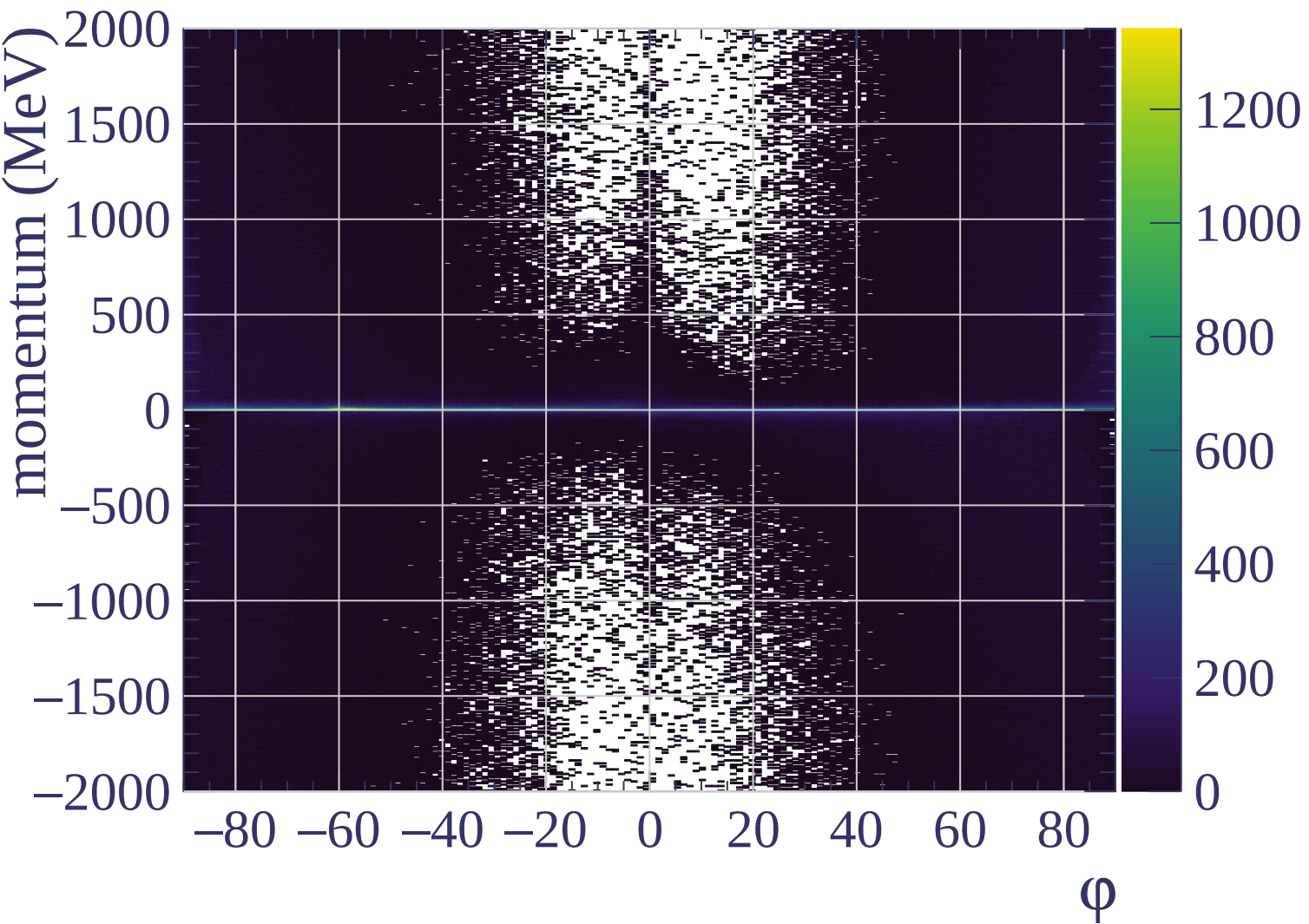


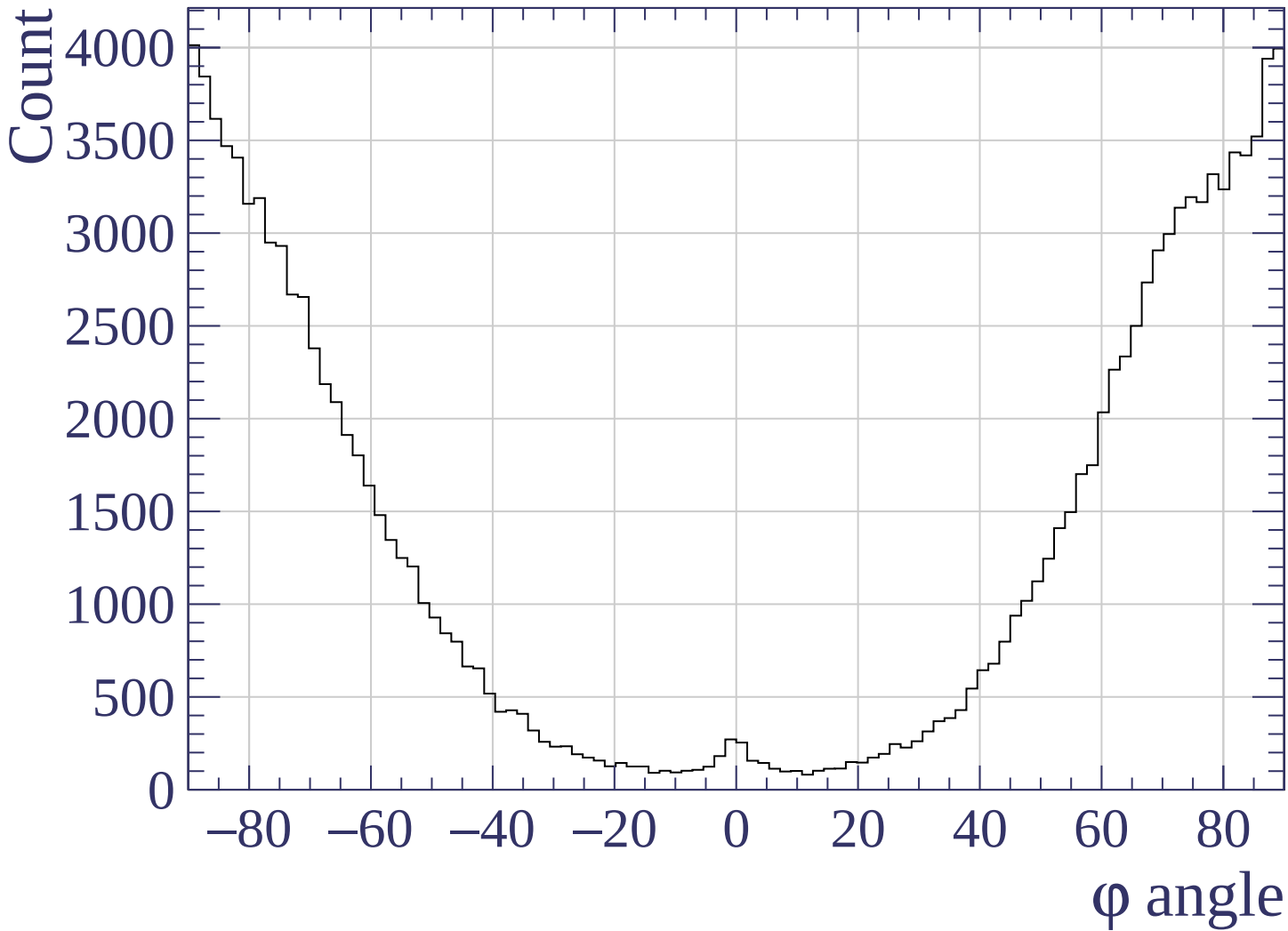


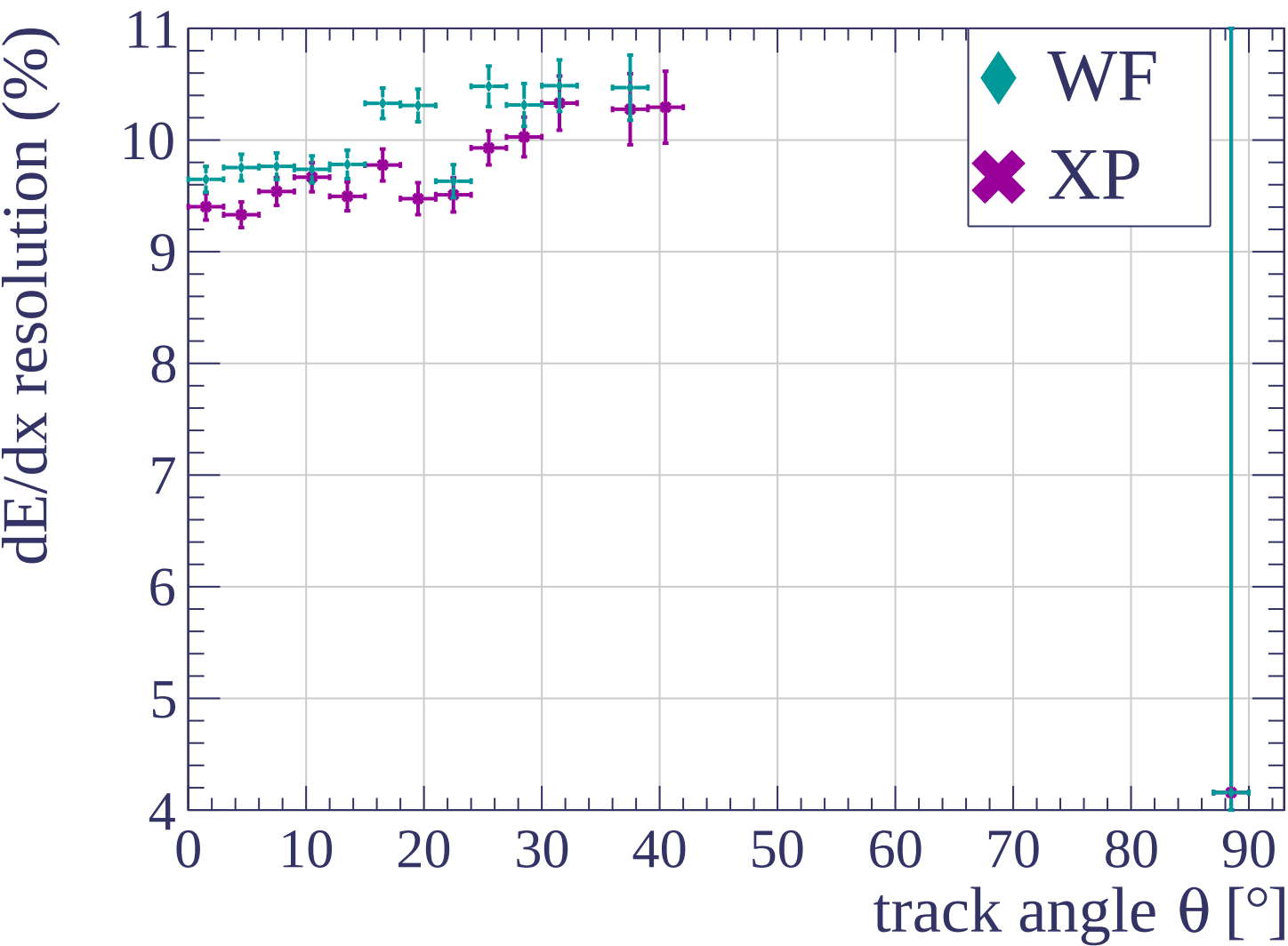


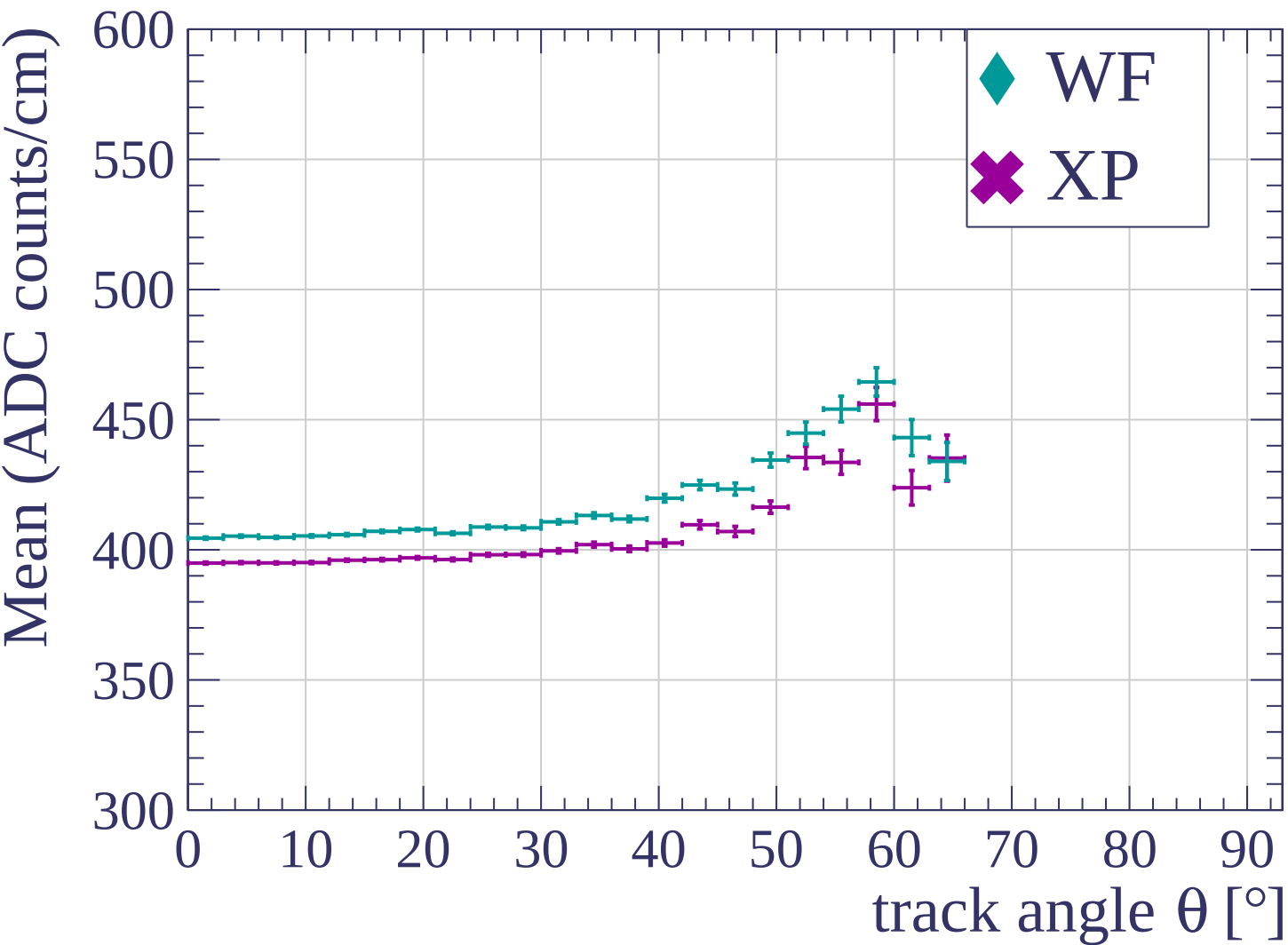


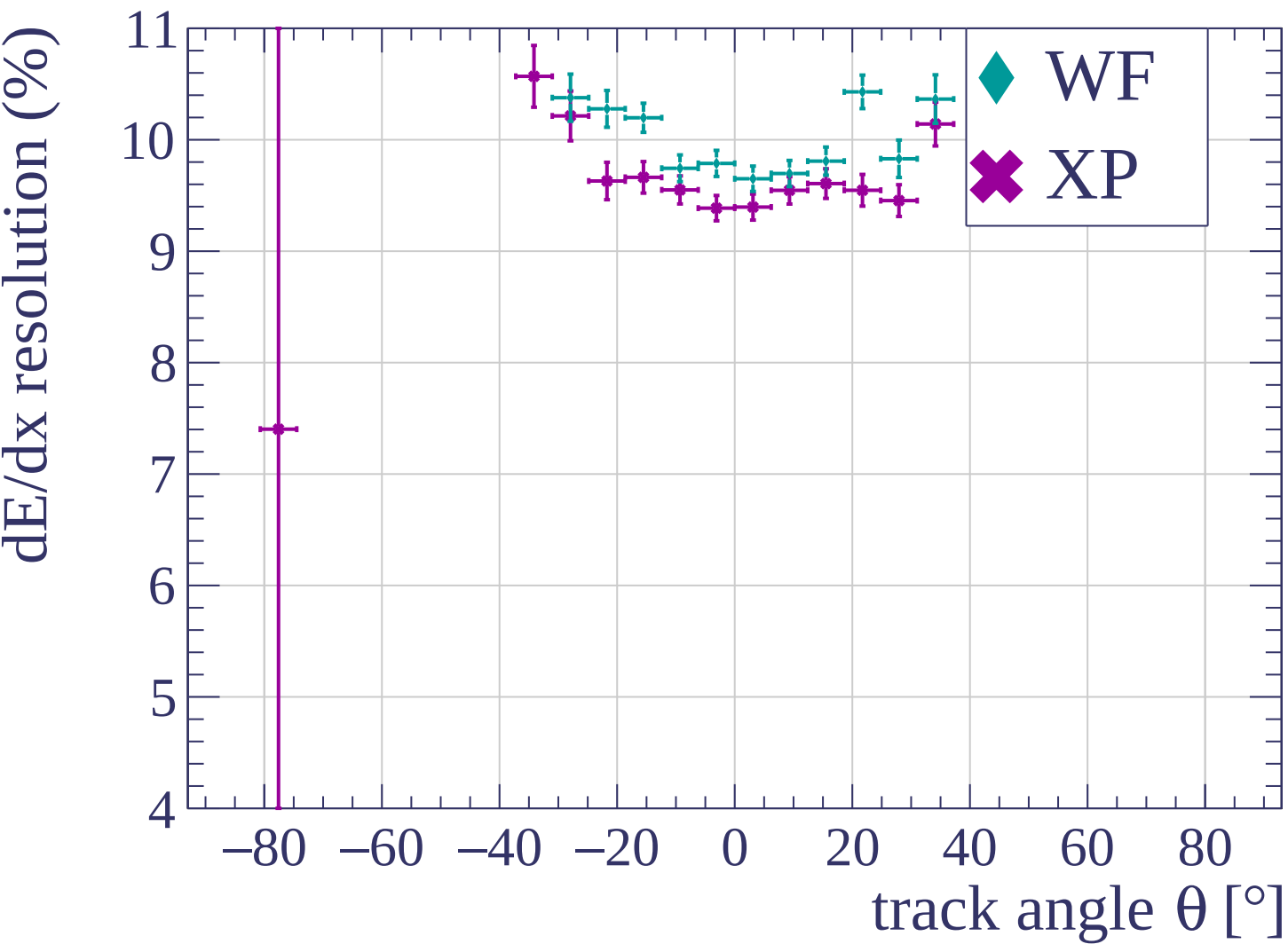


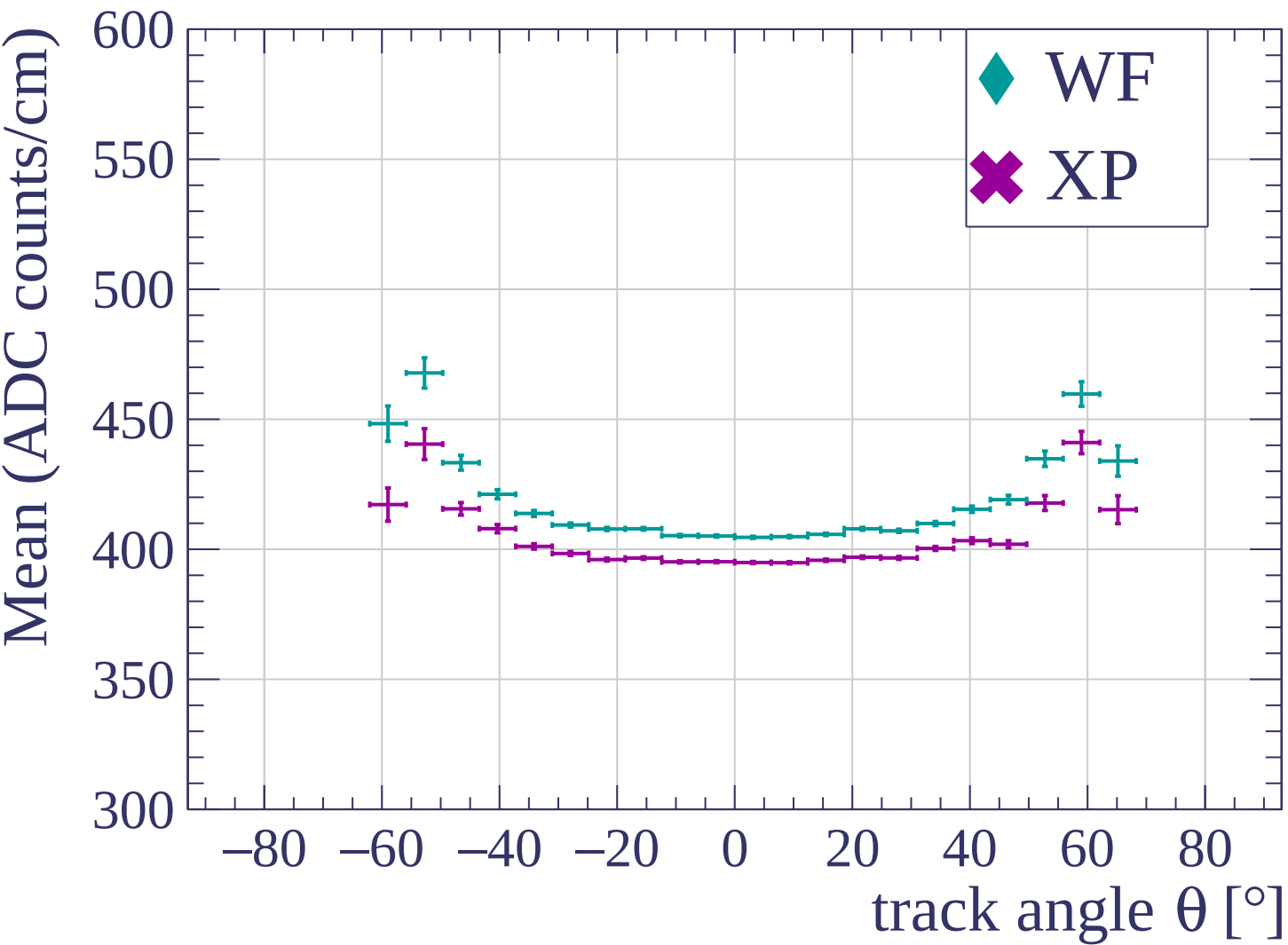


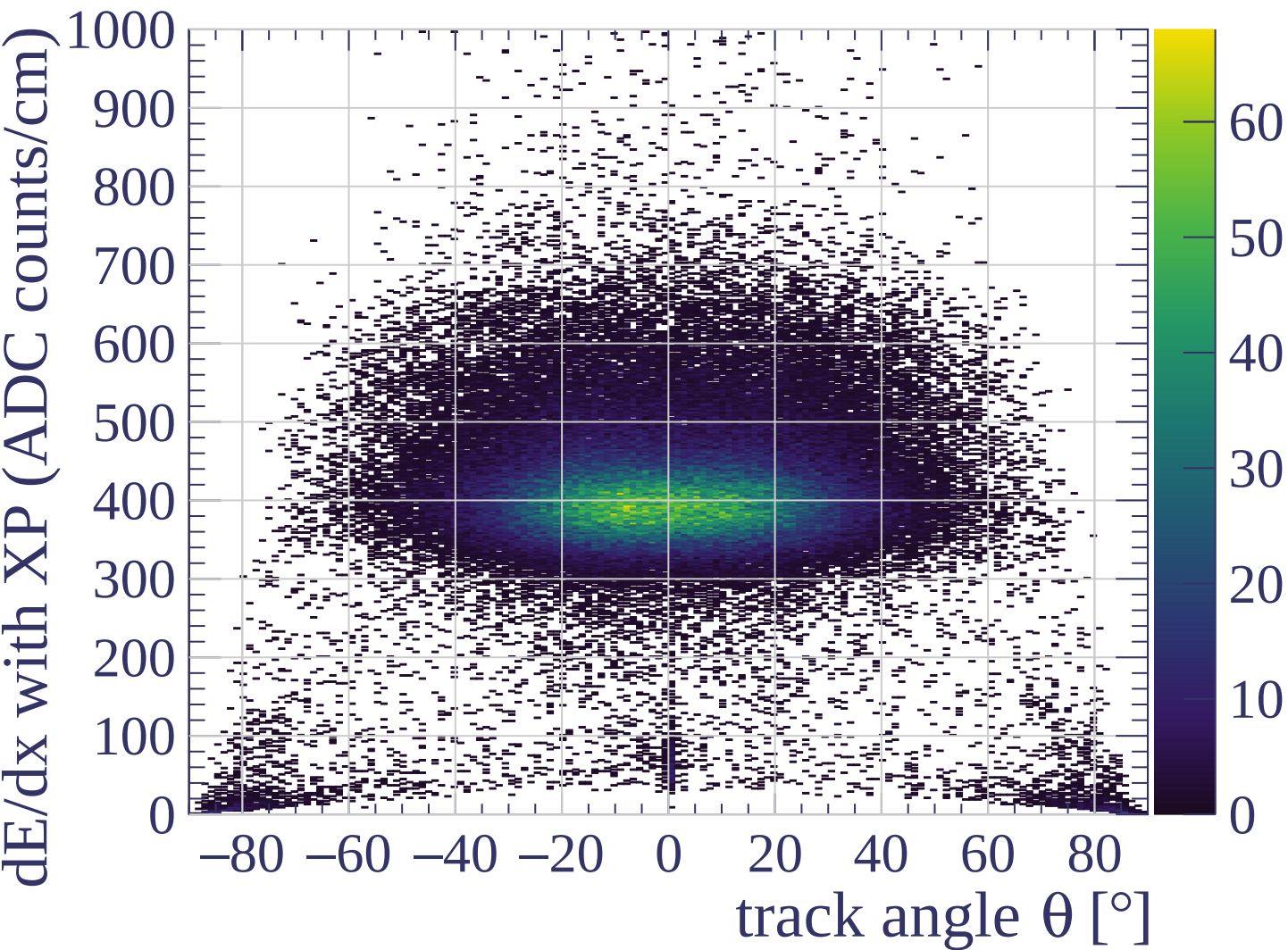


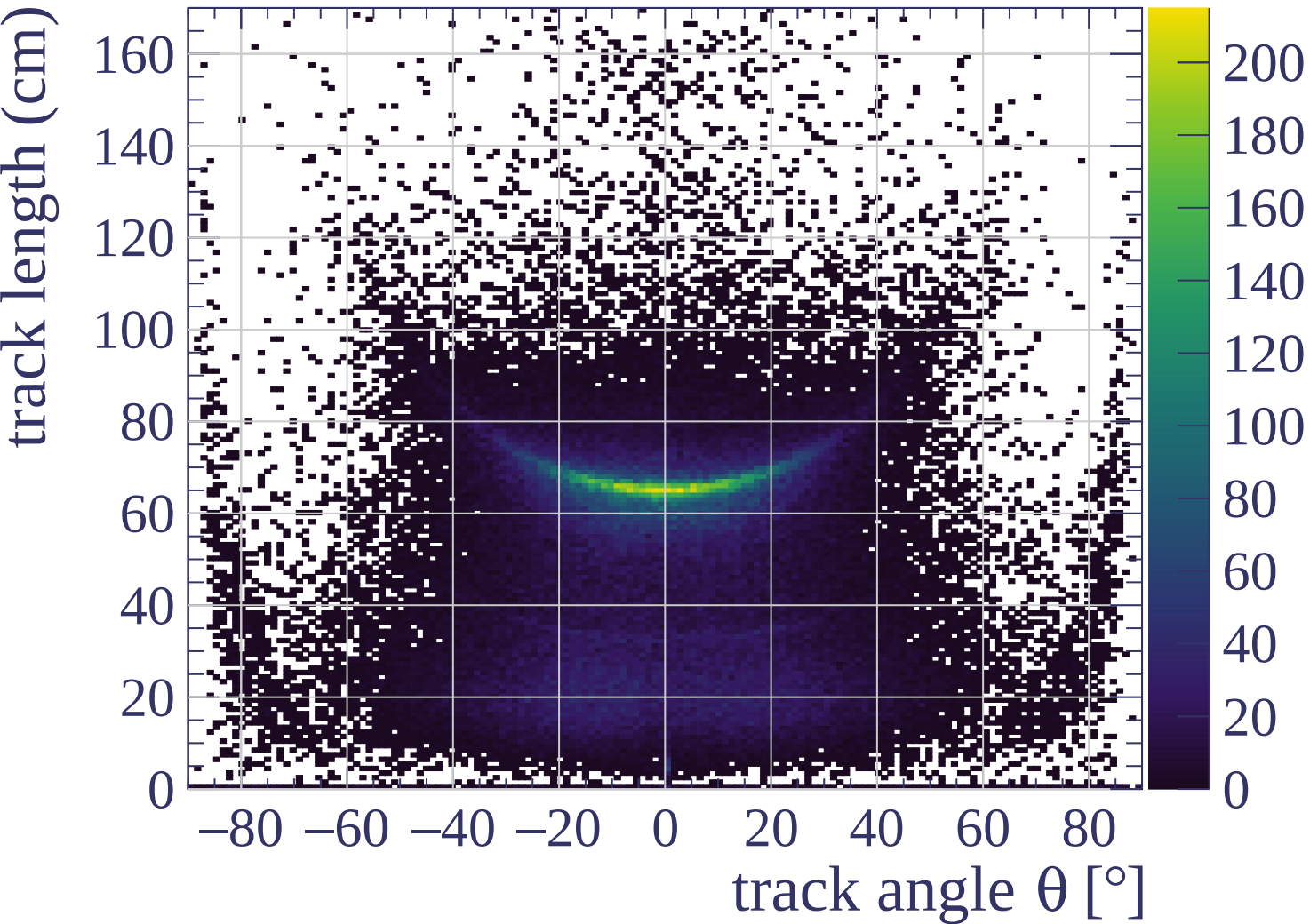


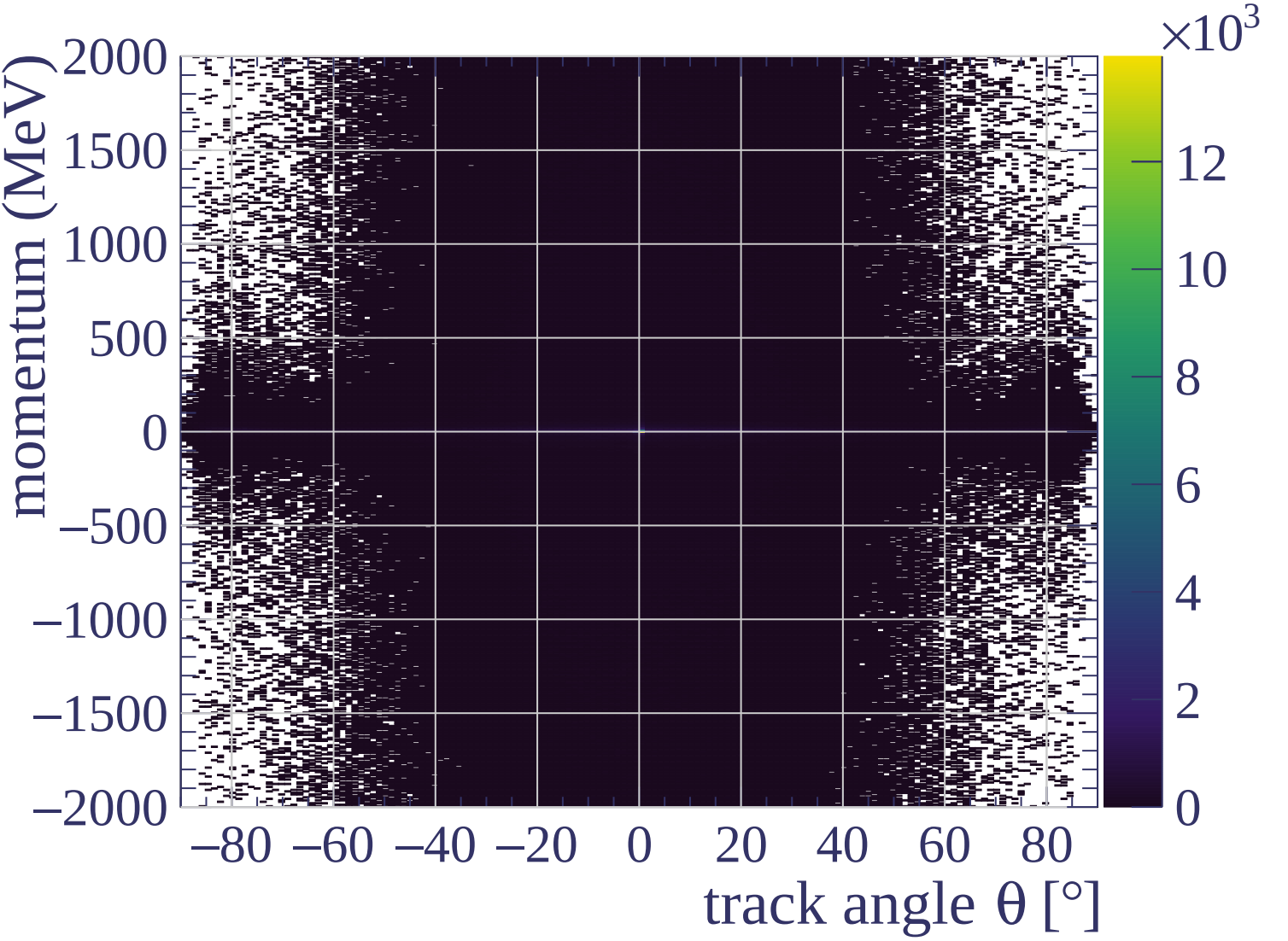








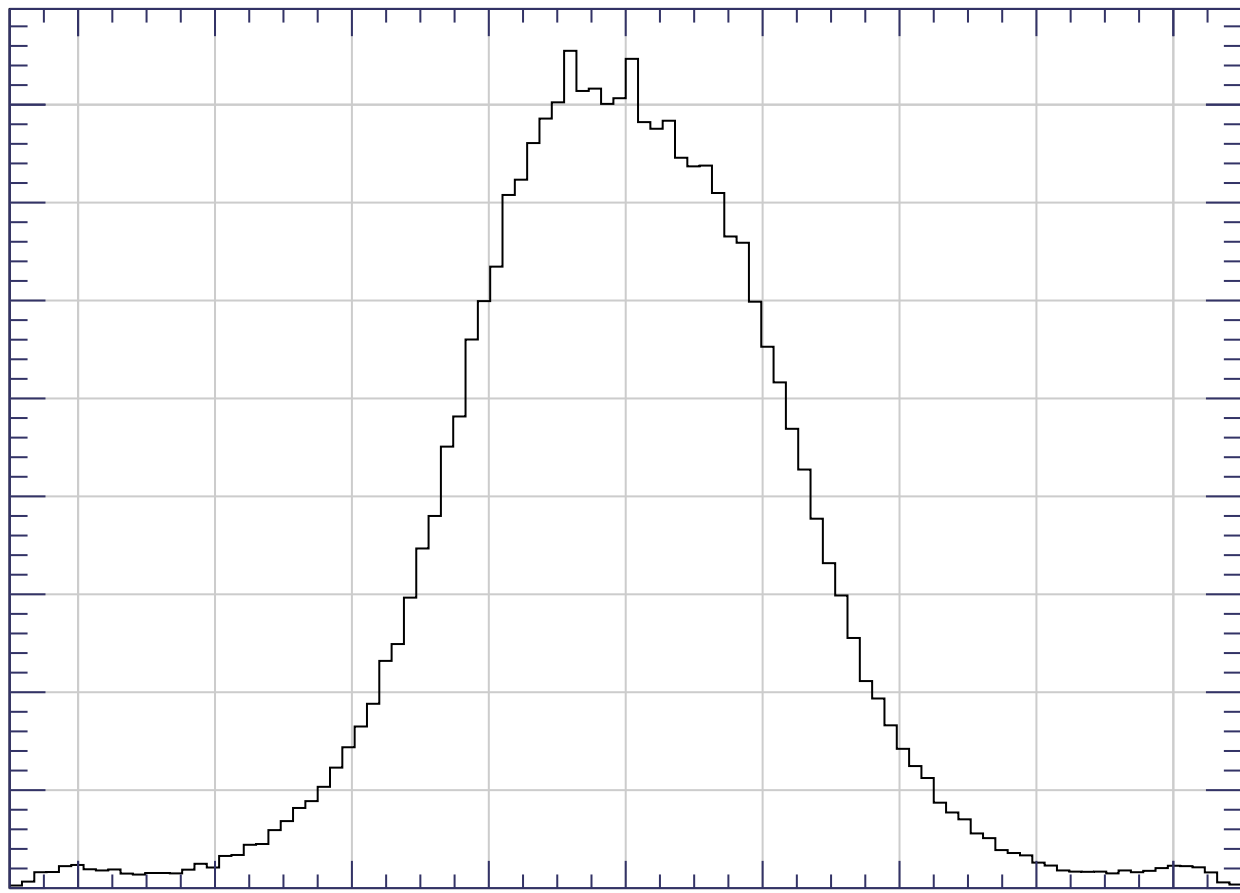




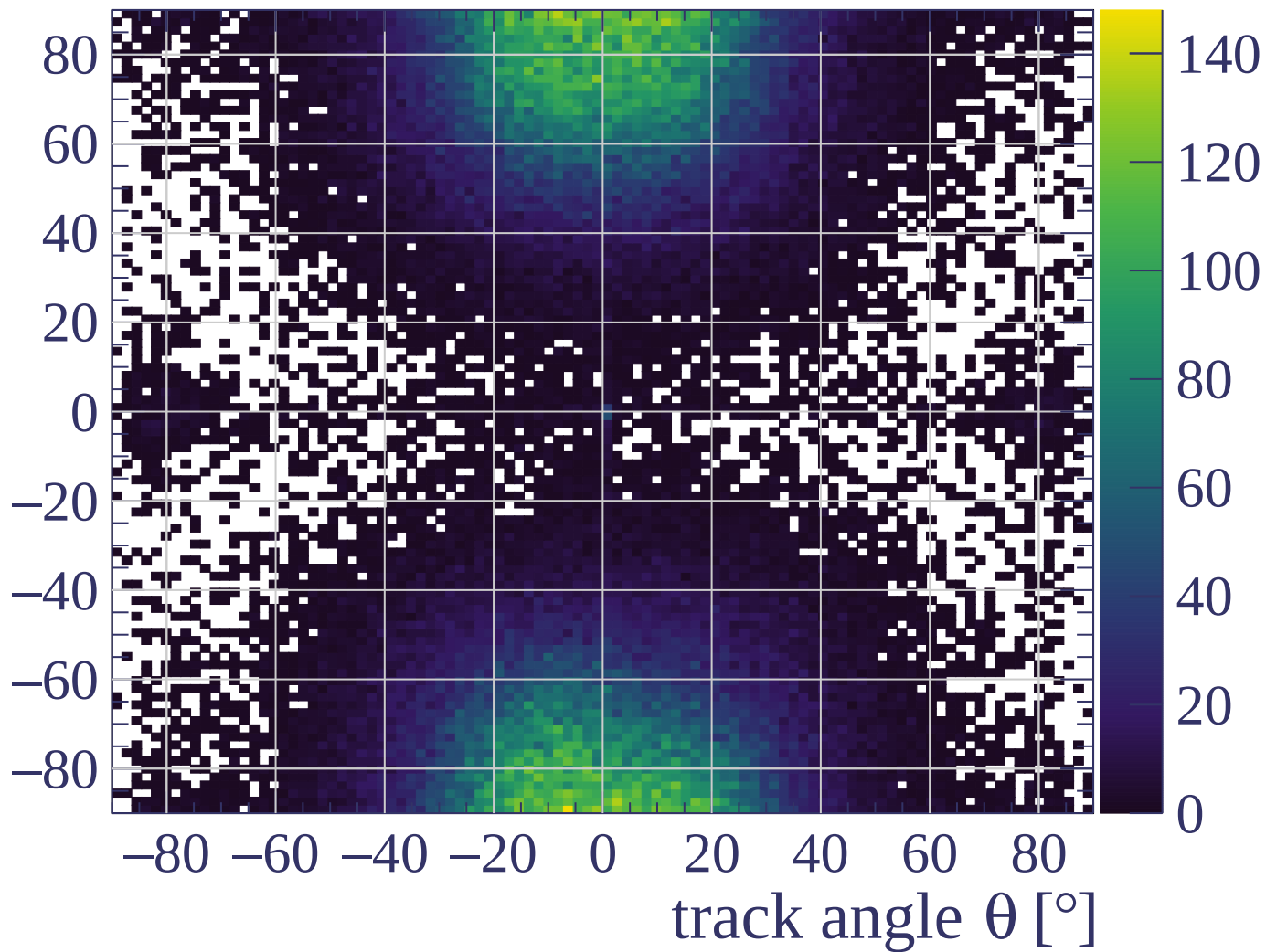
Count

4000
3500
3000
2500
2000
1500
1000
500
0

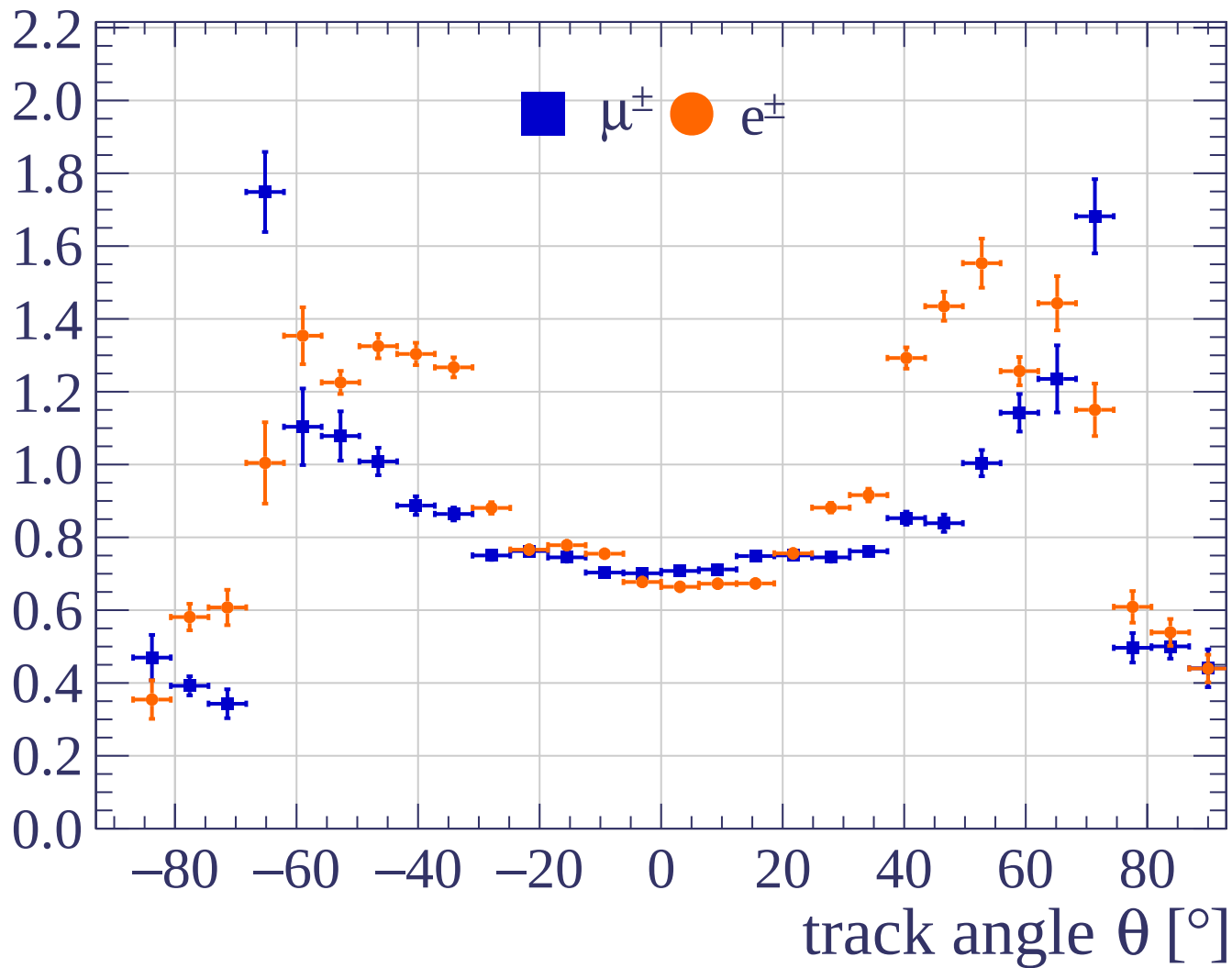
-80 -60 -40 -20 0 20 40 60 80
track angle θ [°]



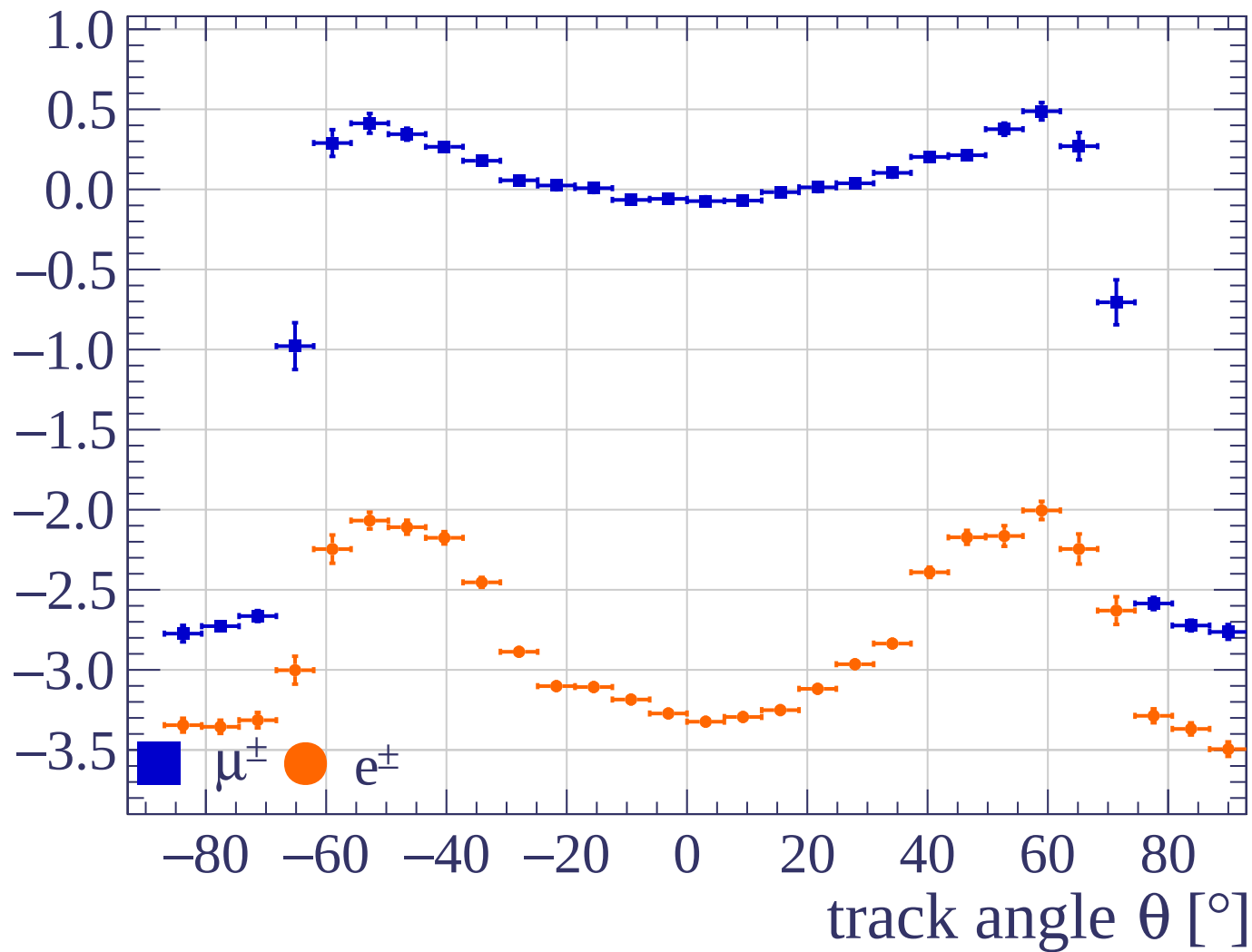
track angle ϕ [°]



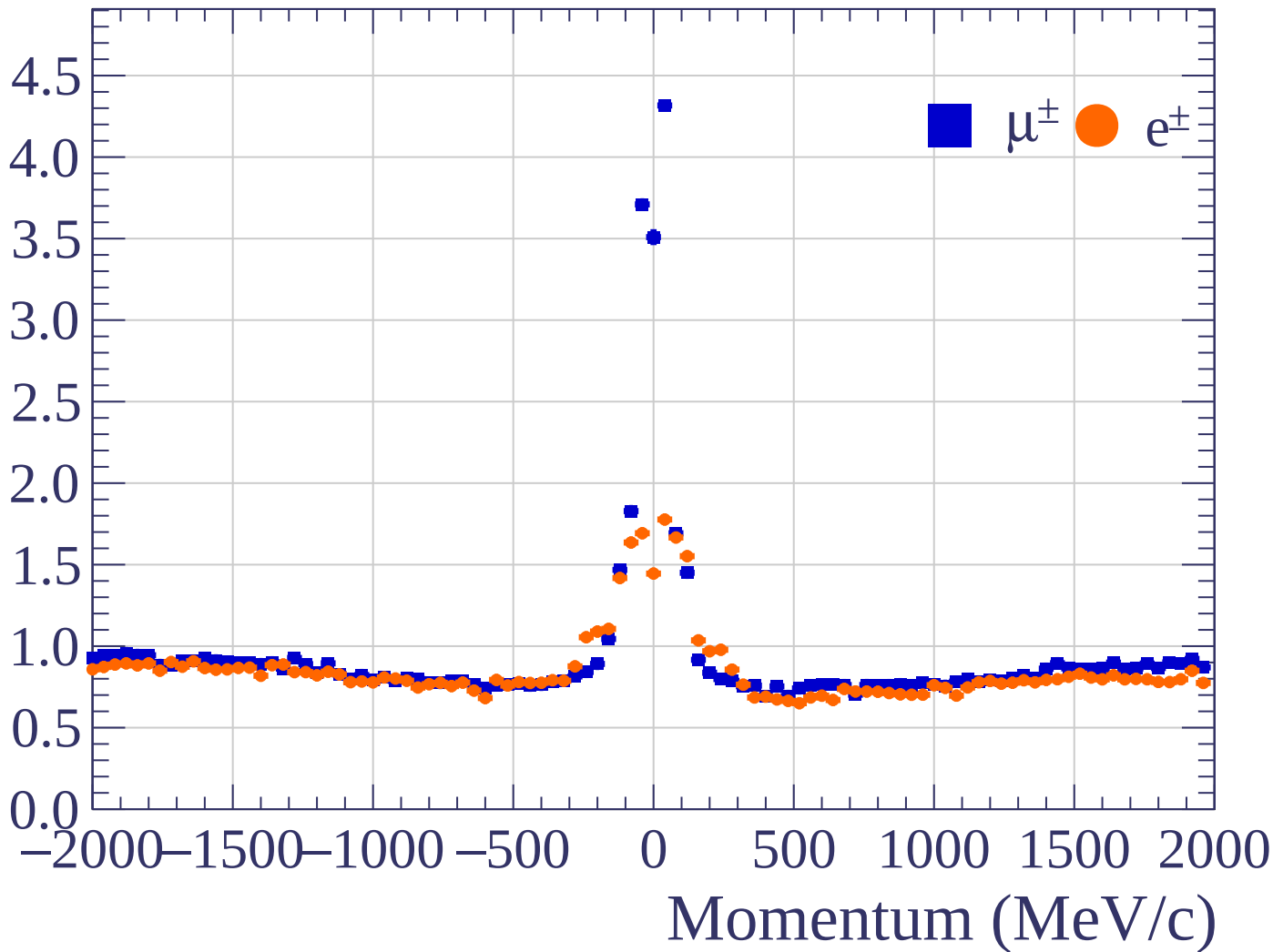
Pulls standard deviation



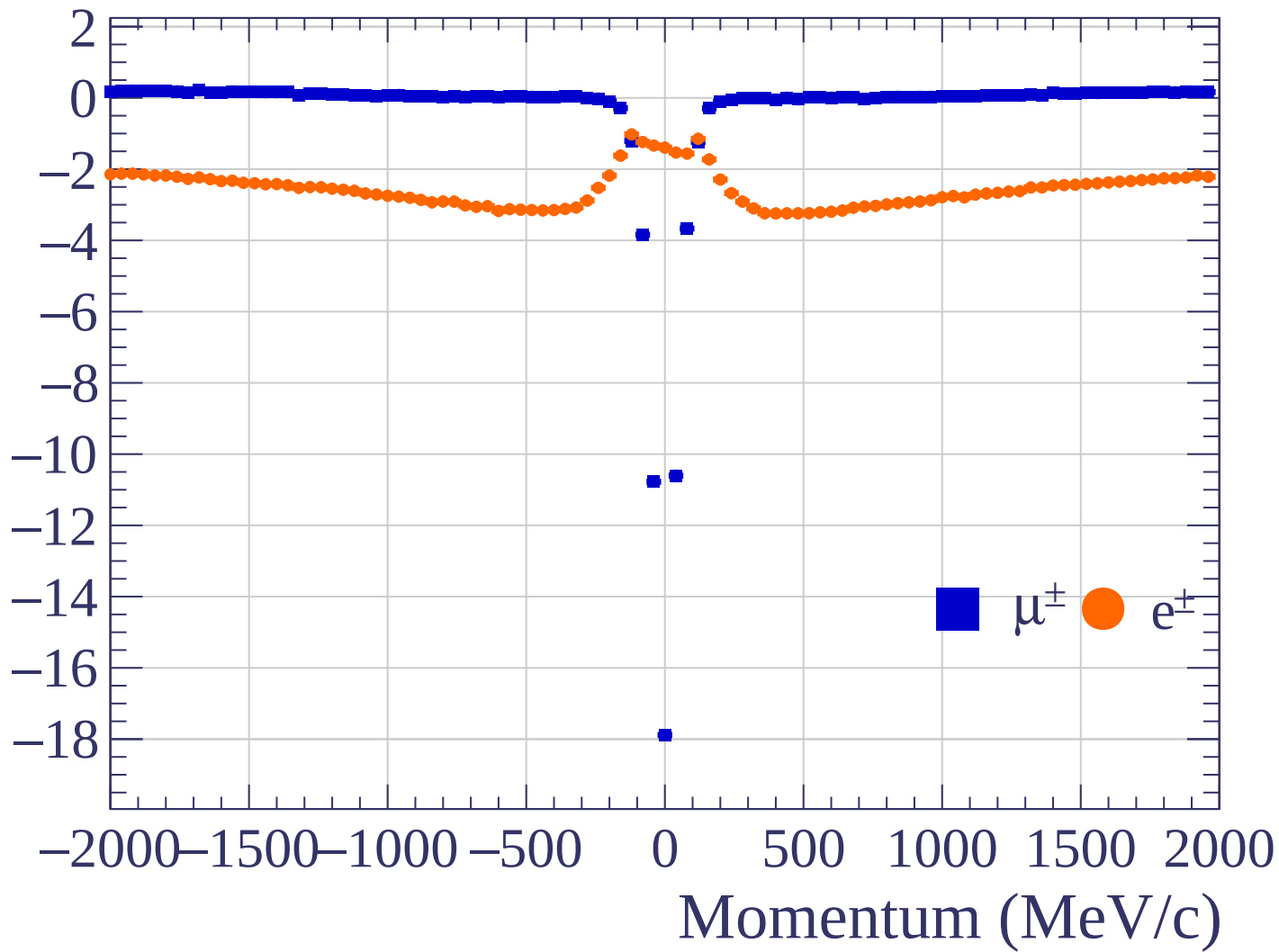
Pulls mean

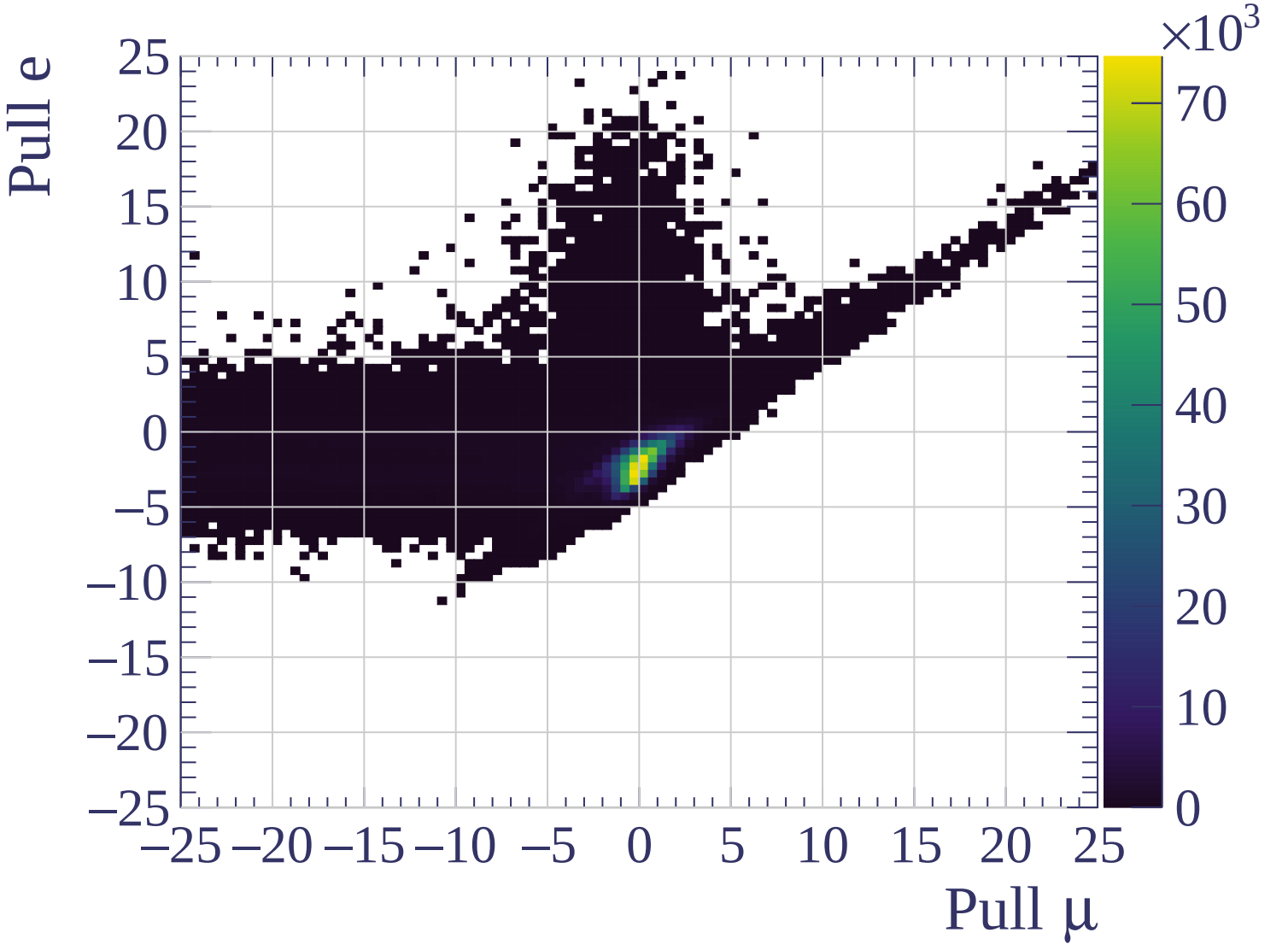


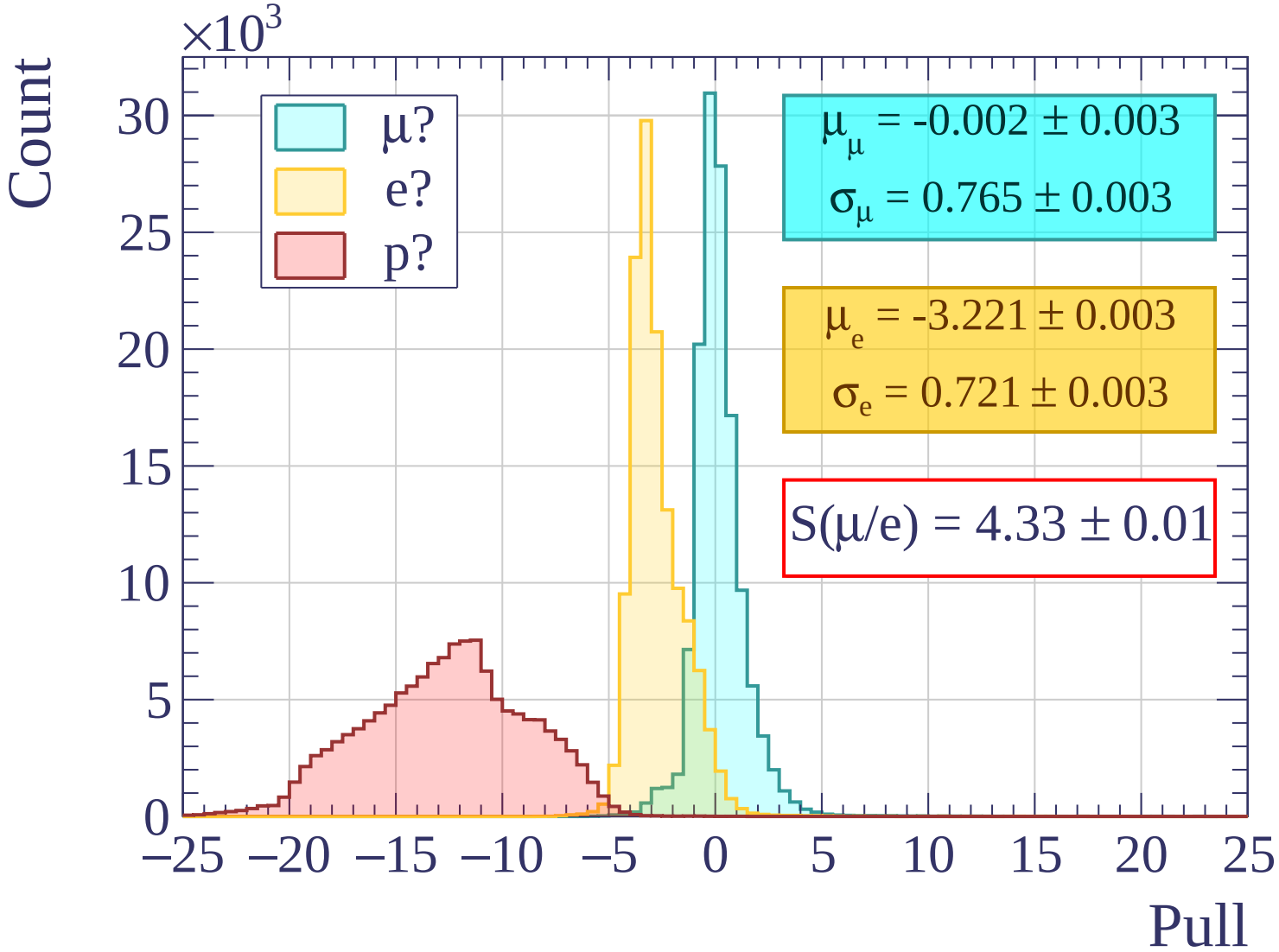
Pulls standard deviation

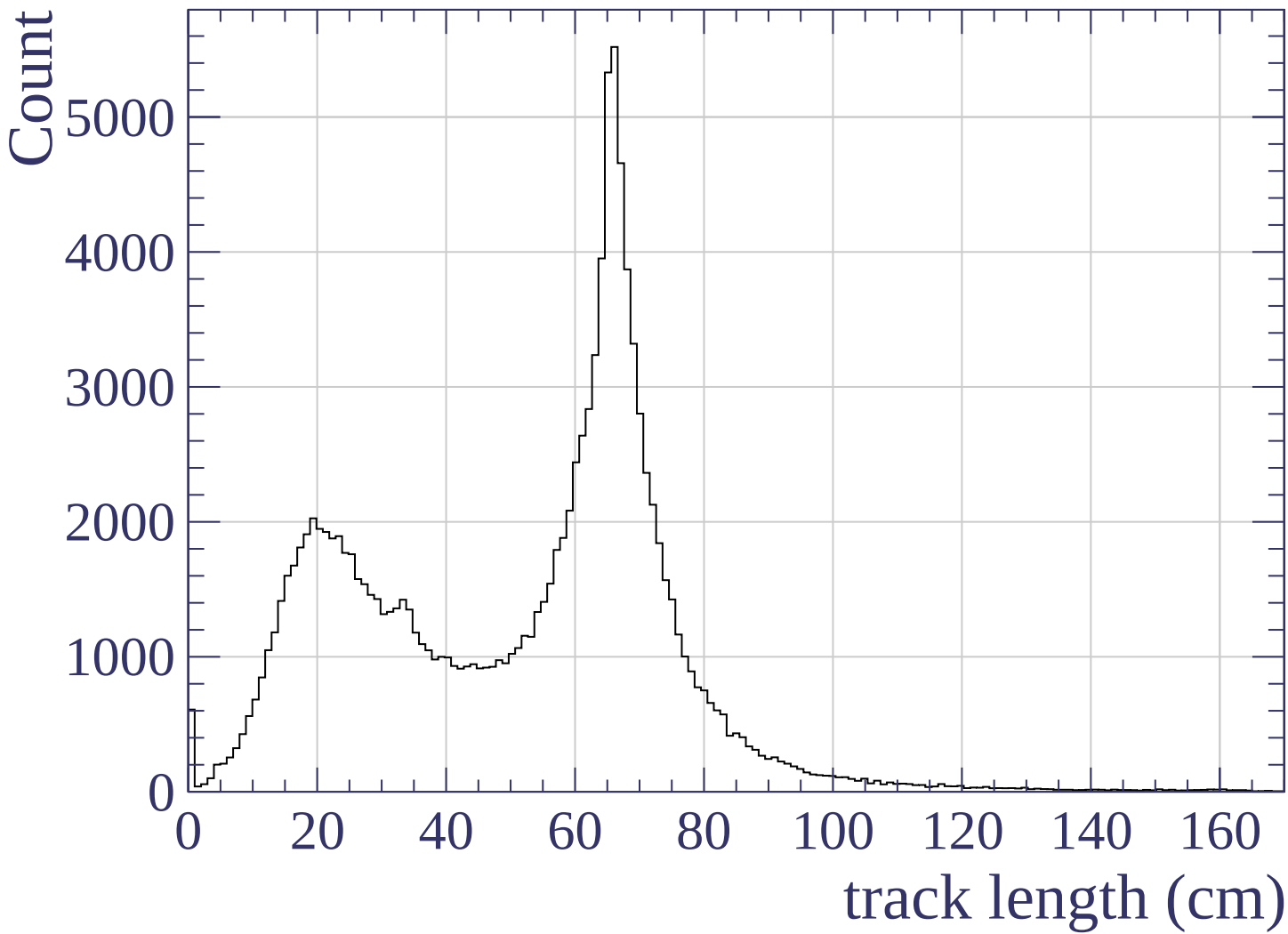


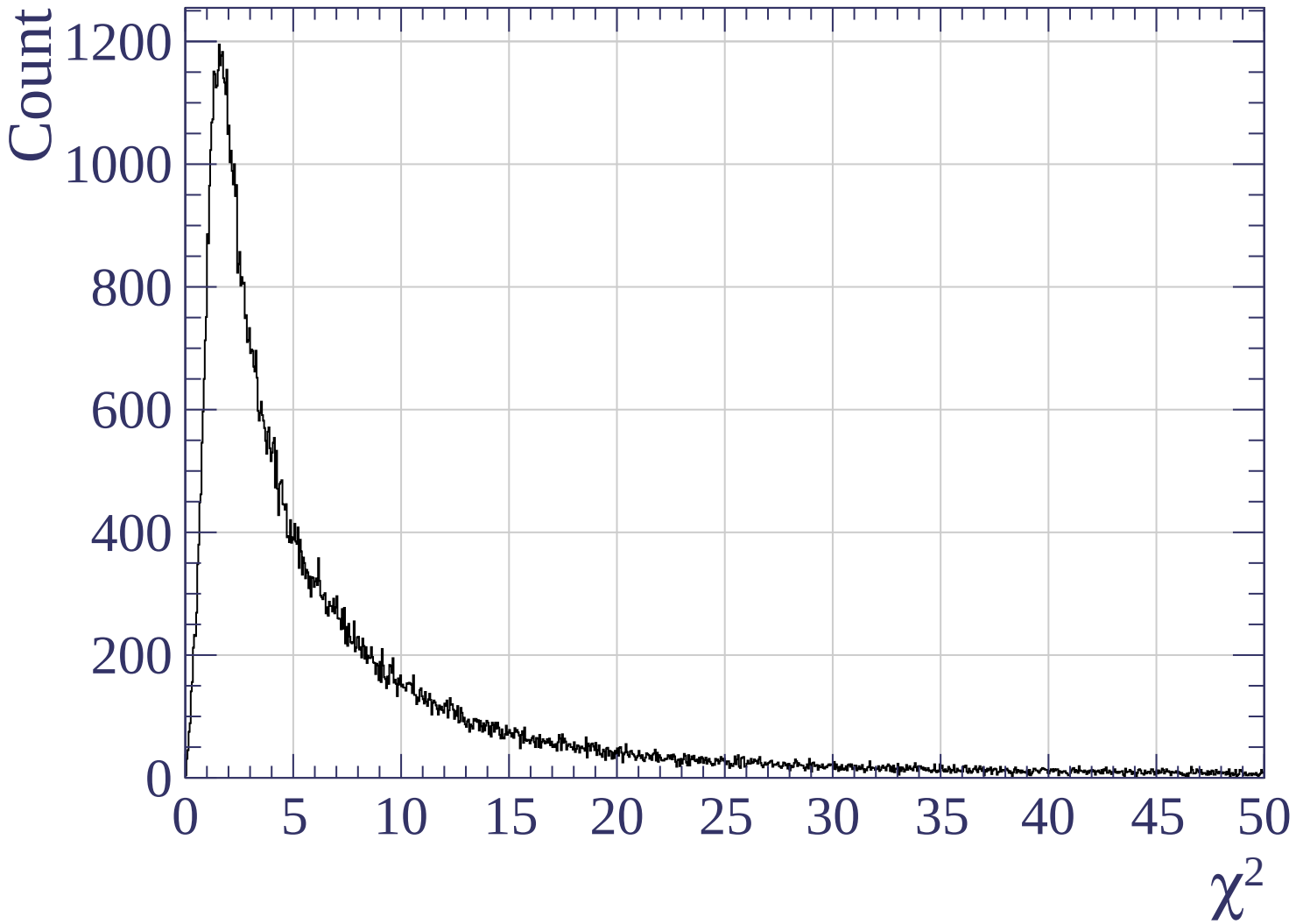
Pulls mean

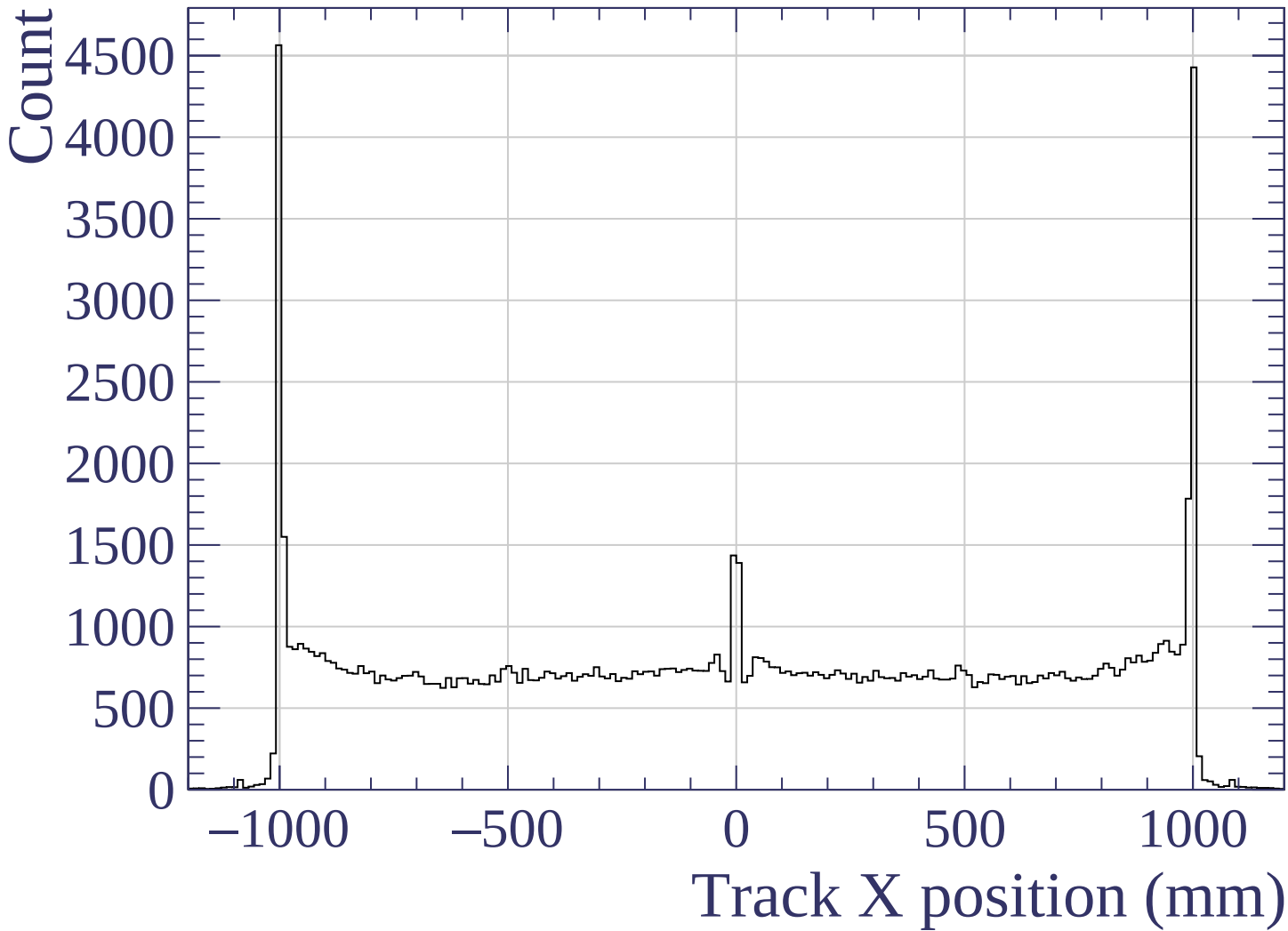


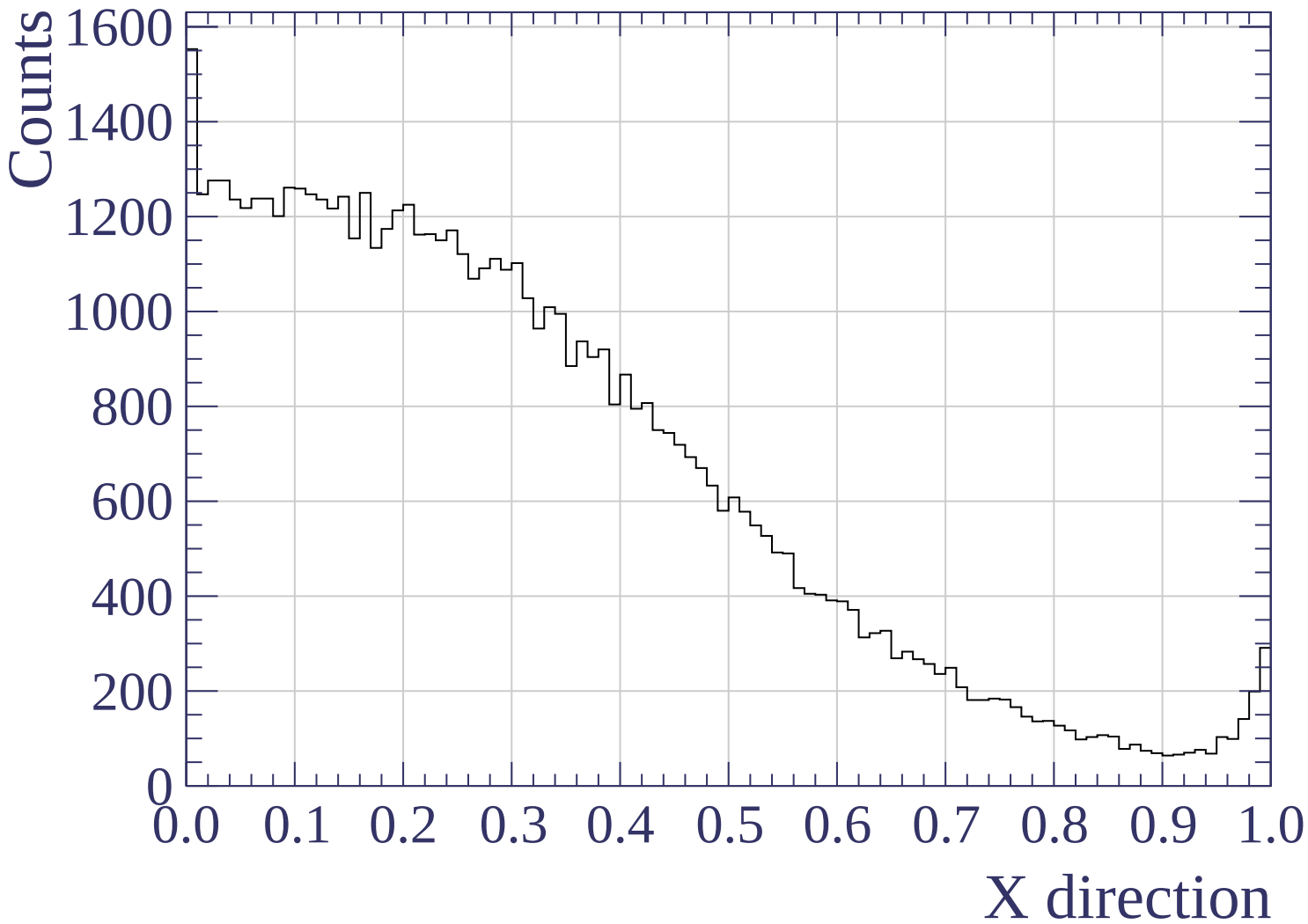


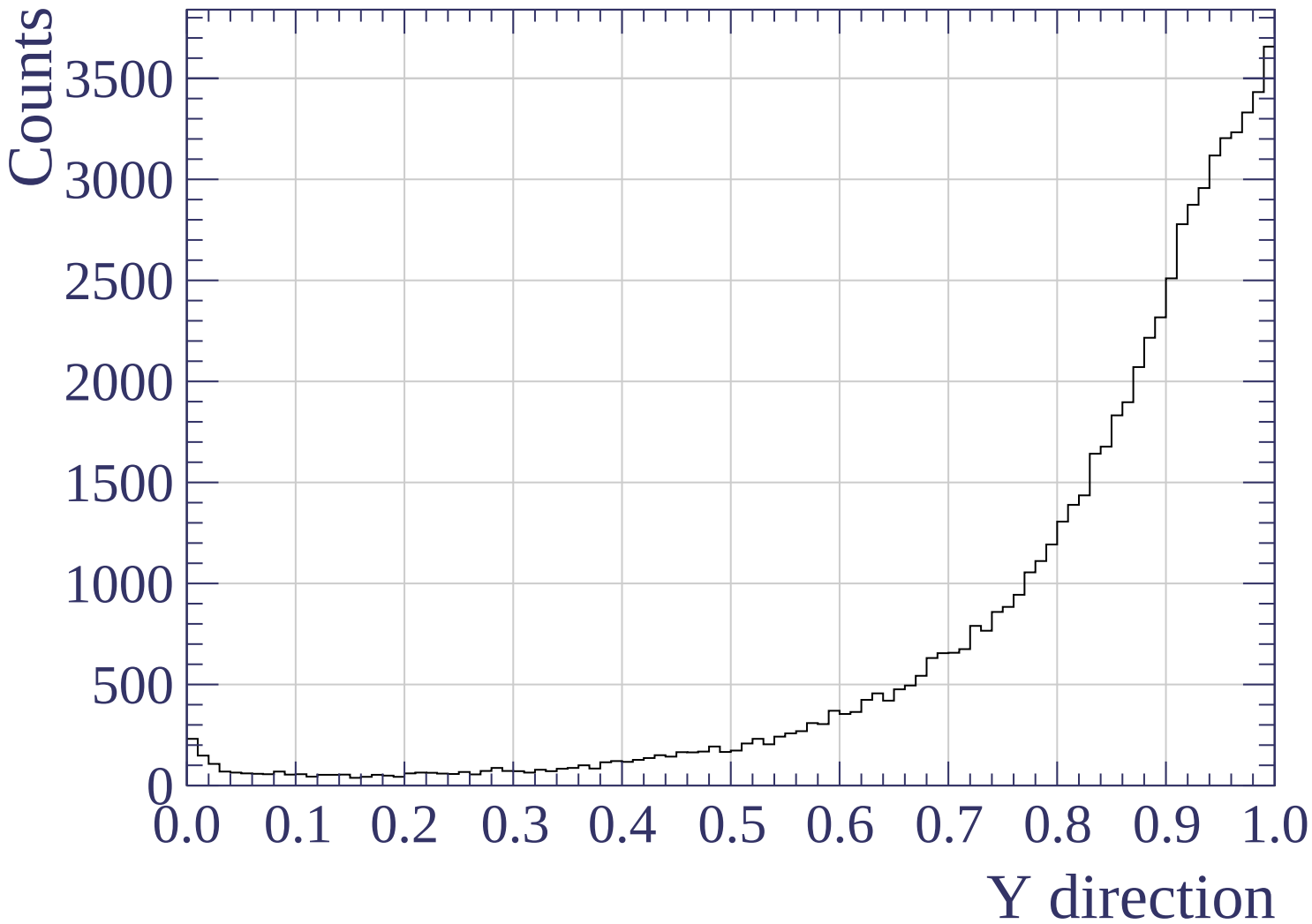


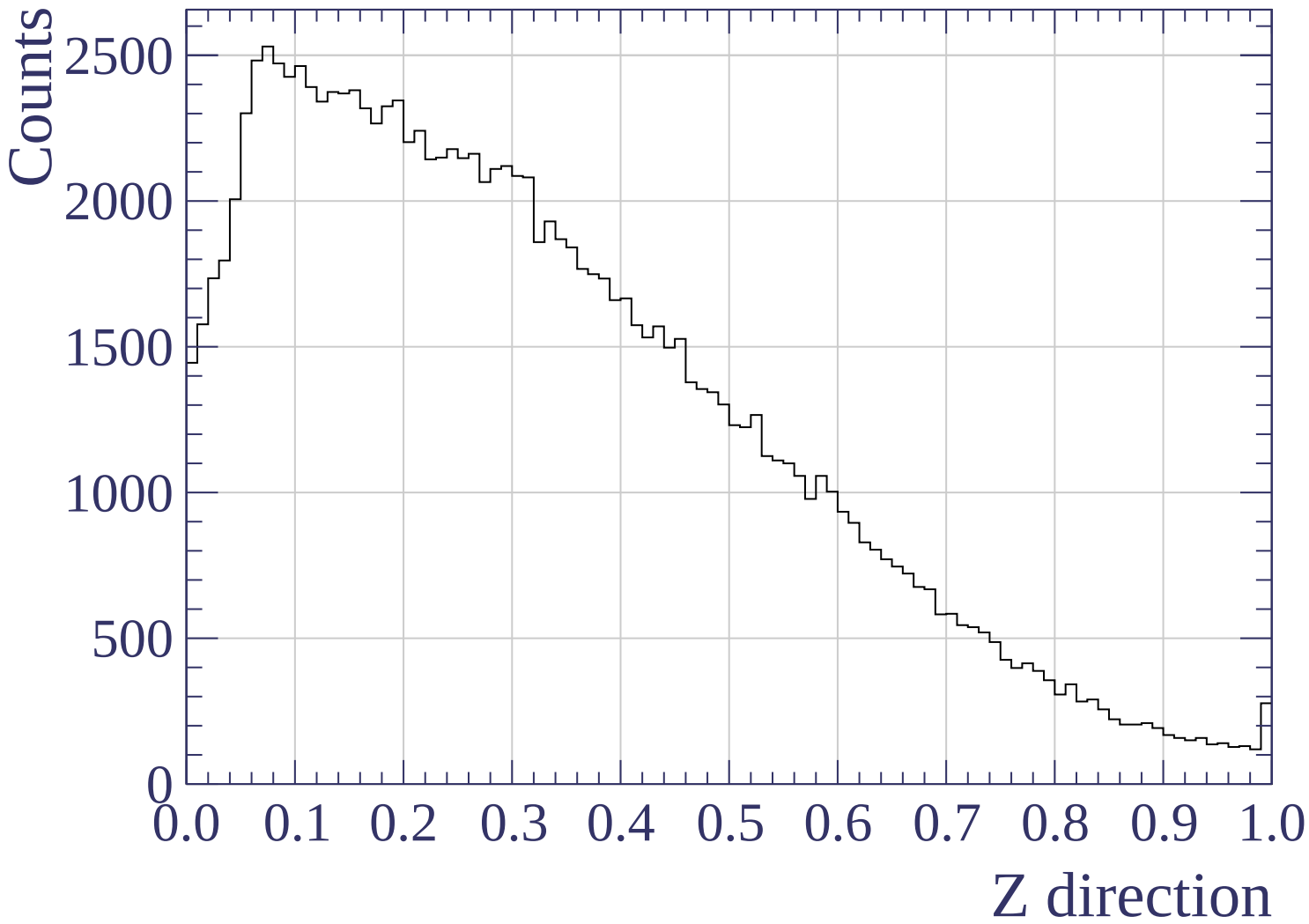


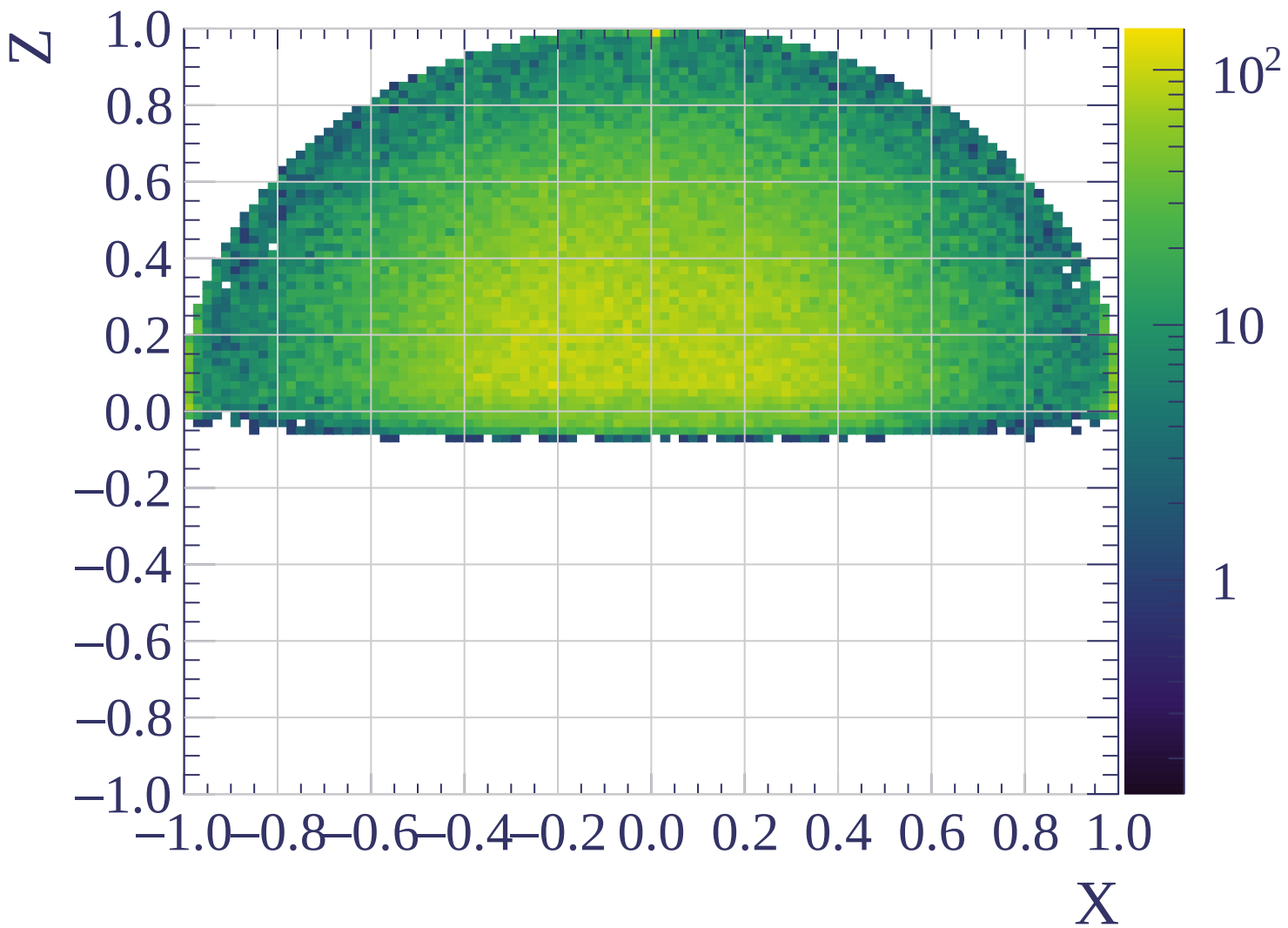


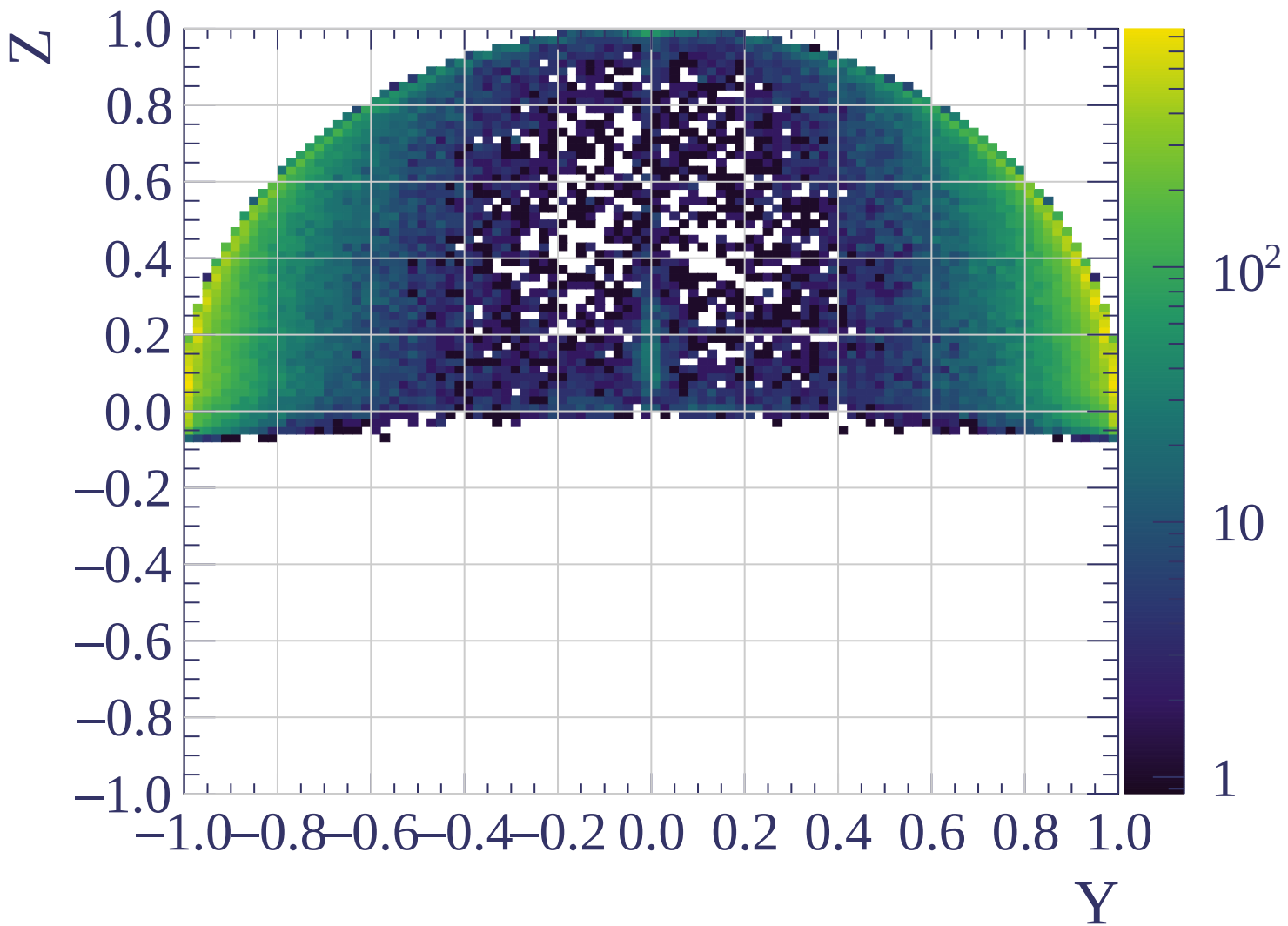


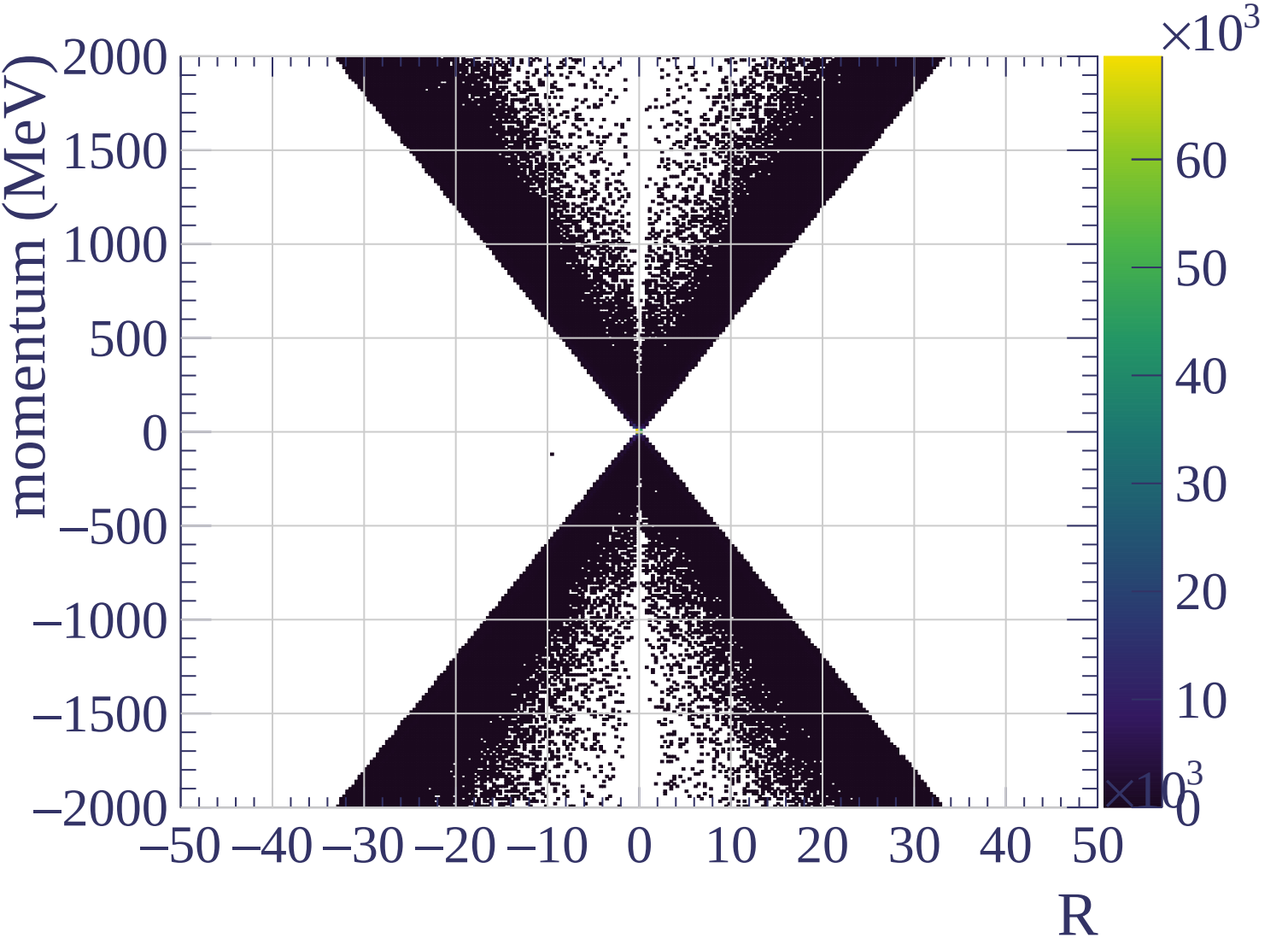




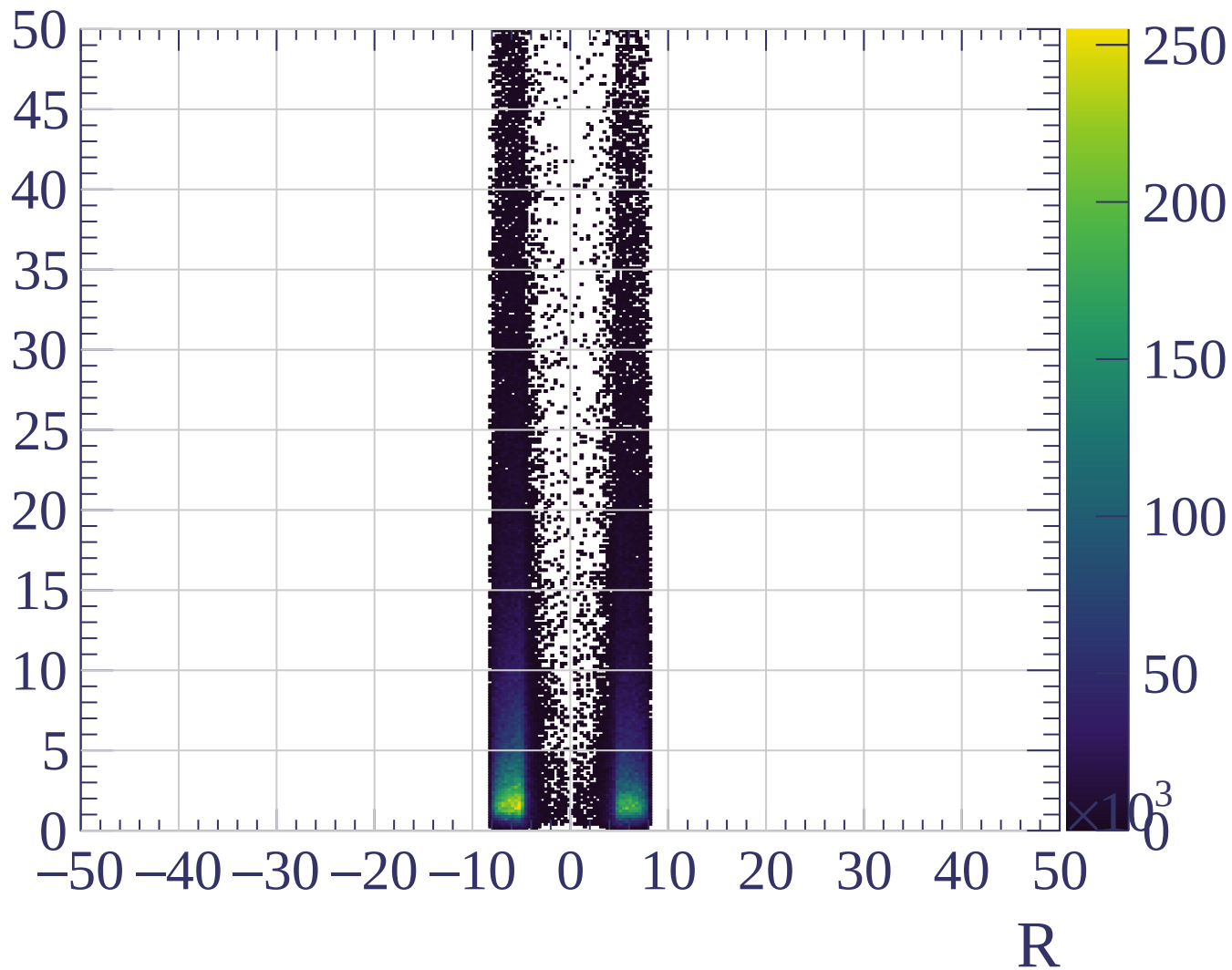








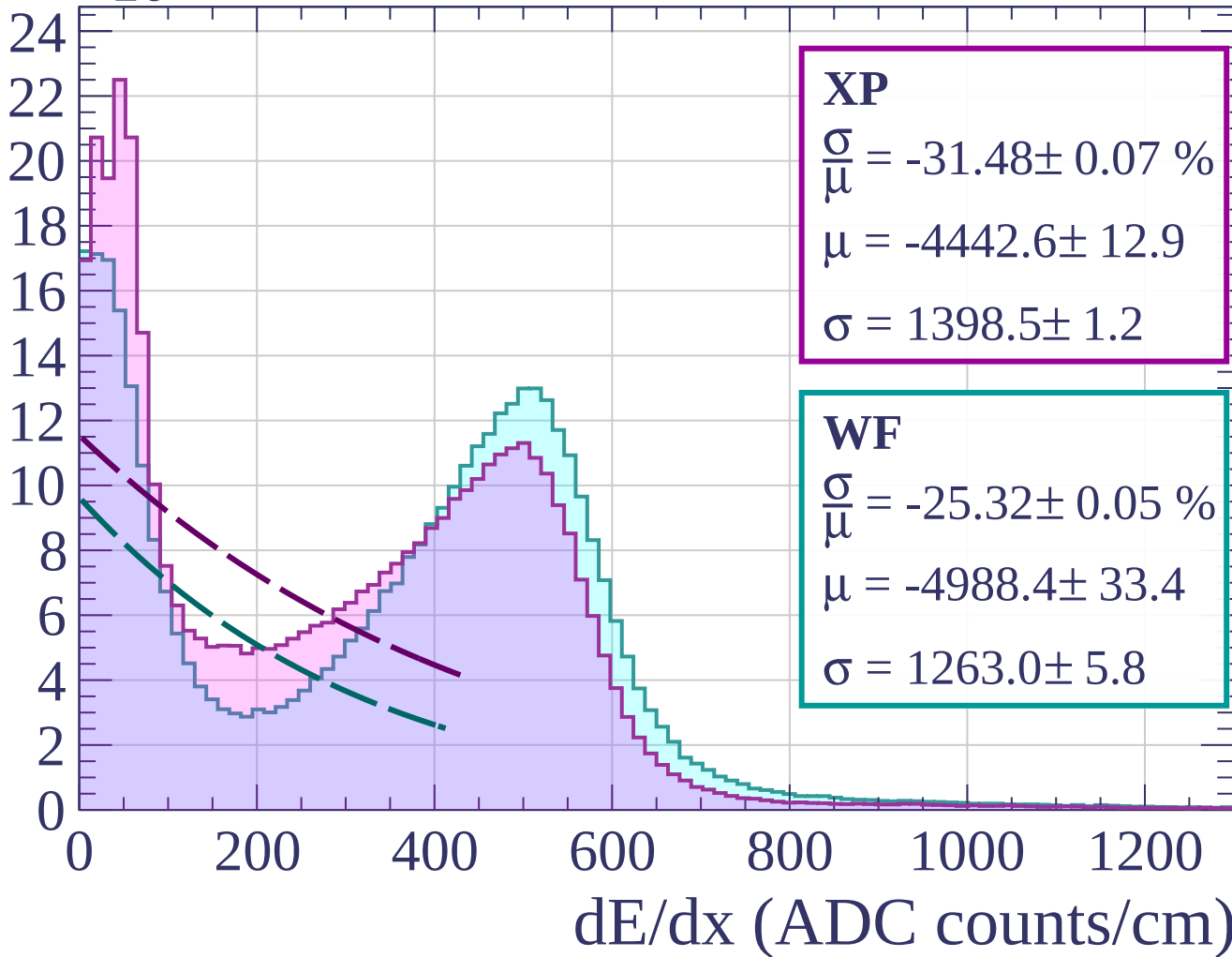
χ^2/NDF



Count

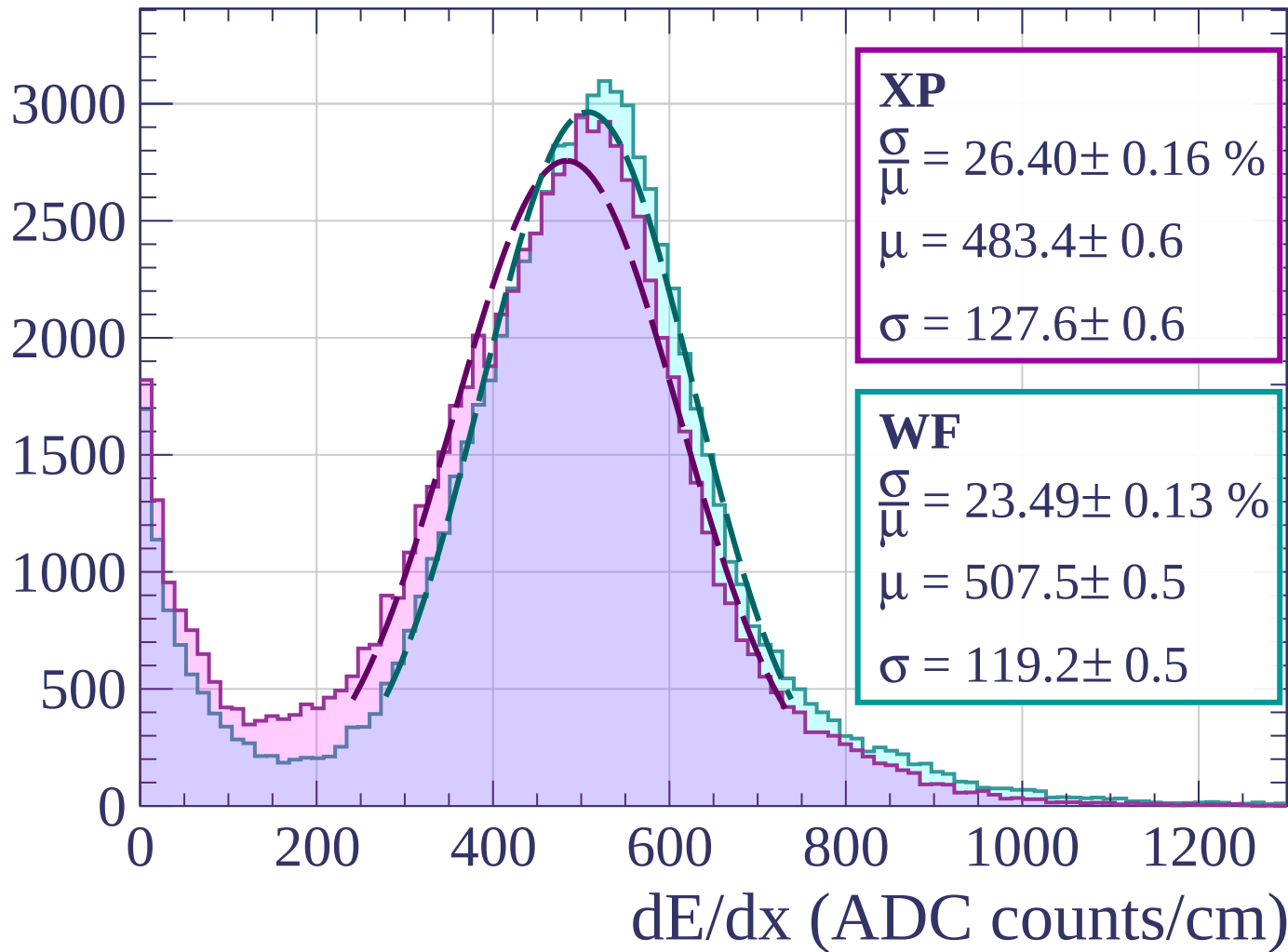
Energy loss | $0 < p < 80$

$\times 10^3$

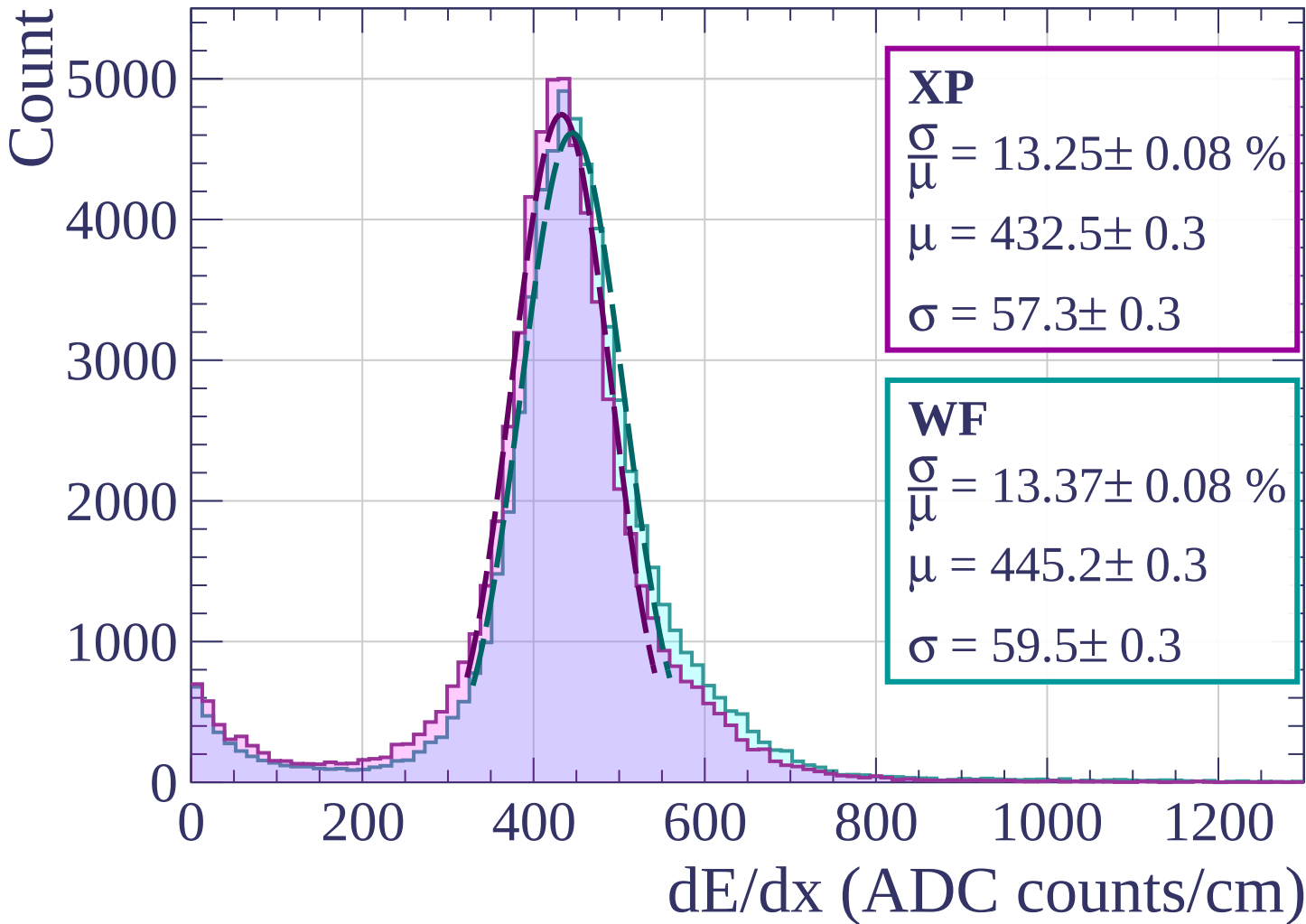


Energy loss | $80 < p < 160$

Count



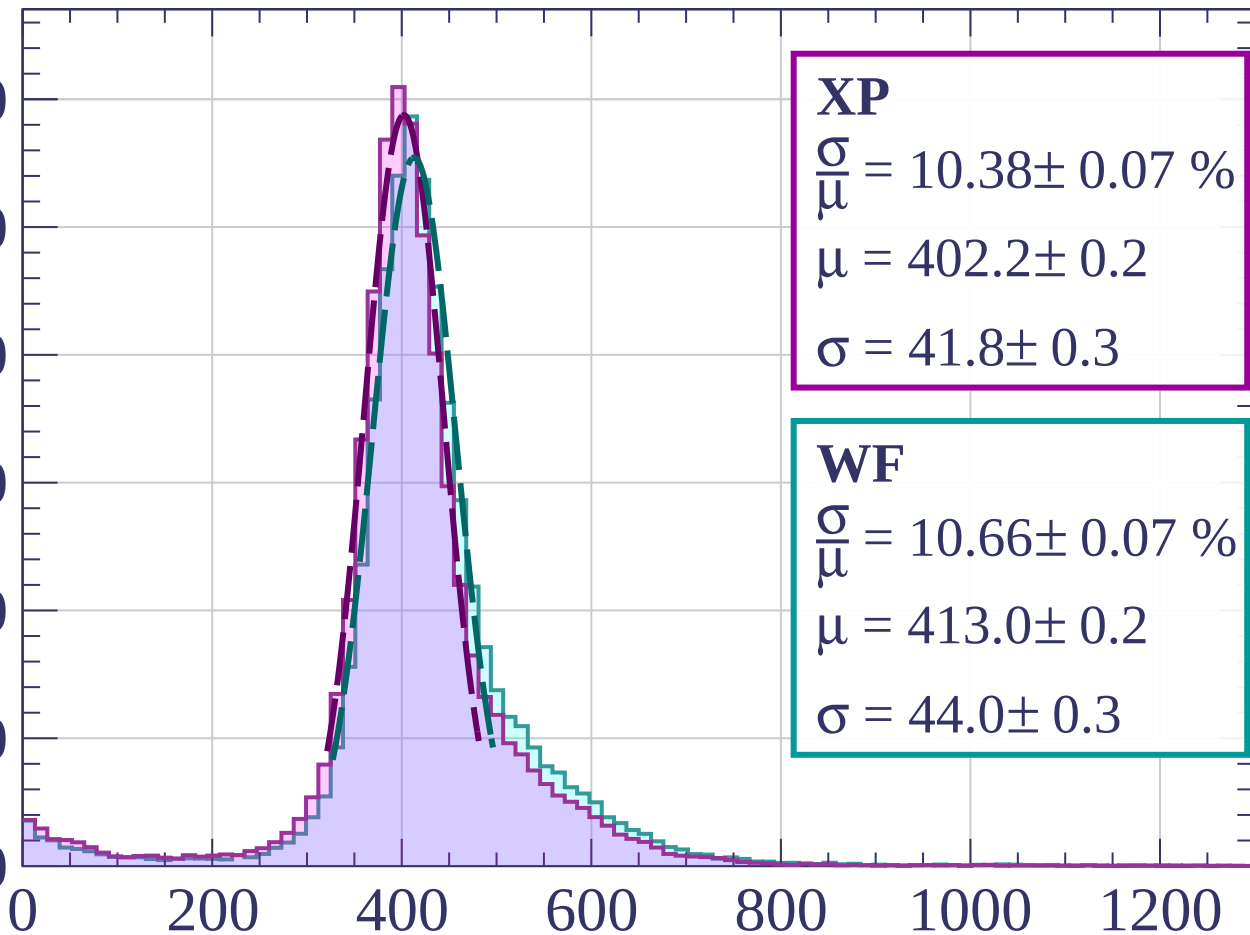
Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

Count

6000
5000
4000
3000
2000
1000
0



XP

$$\frac{\sigma}{\mu} = 10.38 \pm 0.07 \%$$

$$\mu = 402.2 \pm 0.2$$

$$\sigma = 41.8 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 10.66 \pm 0.07 \%$$

$$\mu = 413.0 \pm 0.2$$

$$\sigma = 44.0 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count

6000

5000

4000

3000

2000

1000

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.68 \pm 0.06 \%$$

$$\mu = 395.5 \pm 0.2$$

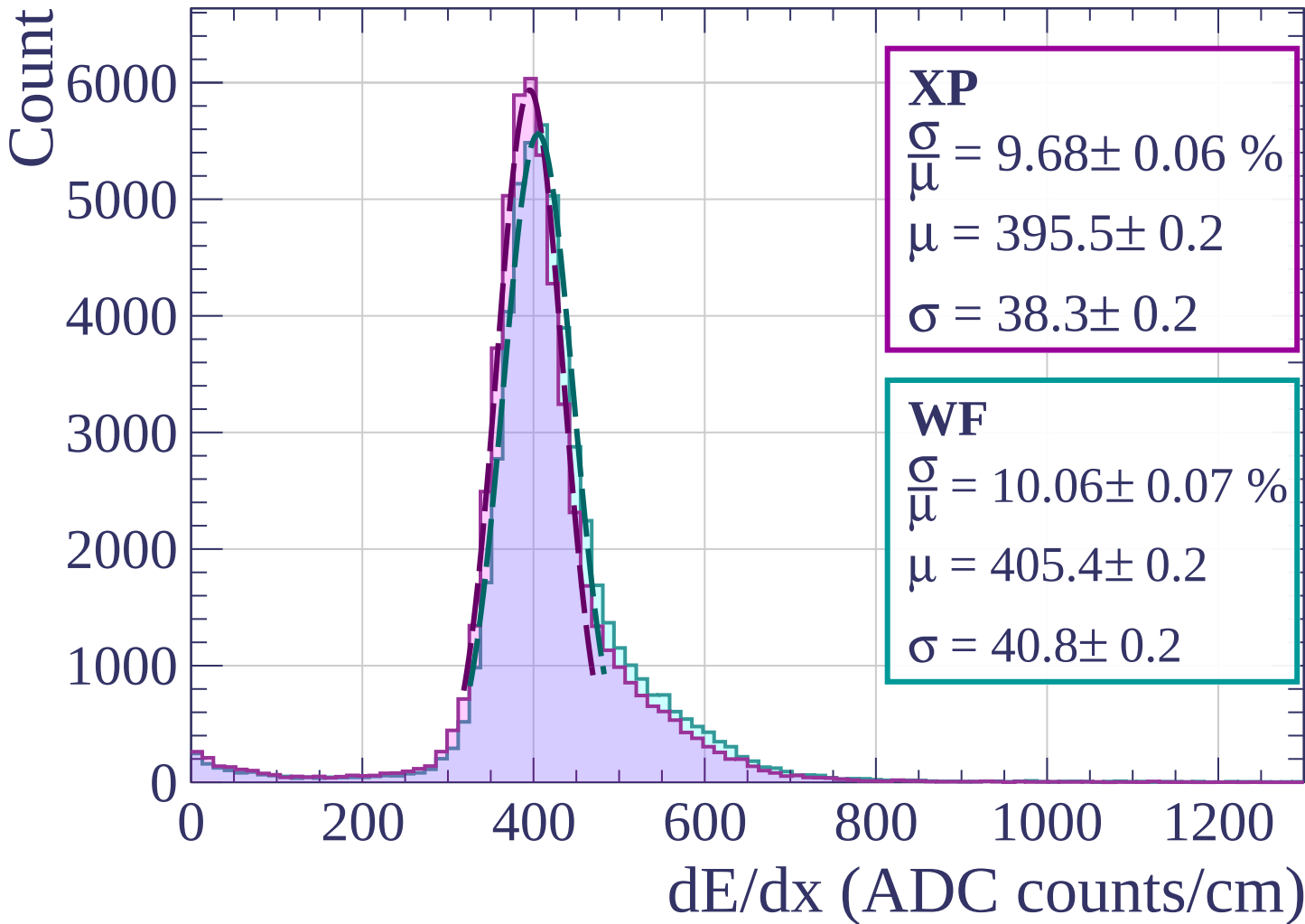
$$\sigma = 38.3 \pm 0.2$$

WF

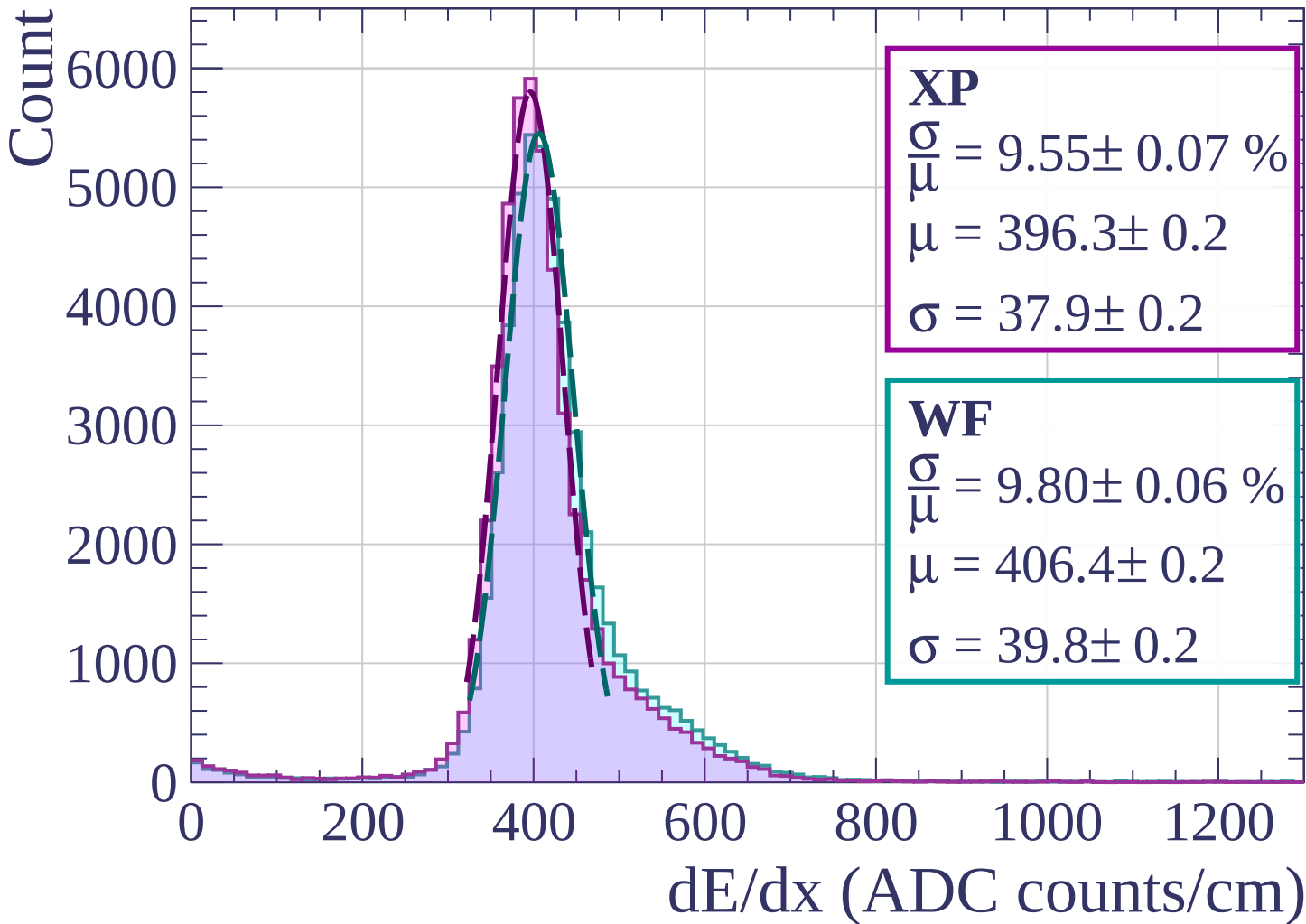
$$\frac{\sigma}{\mu} = 10.06 \pm 0.07 \%$$

$$\mu = 405.4 \pm 0.2$$

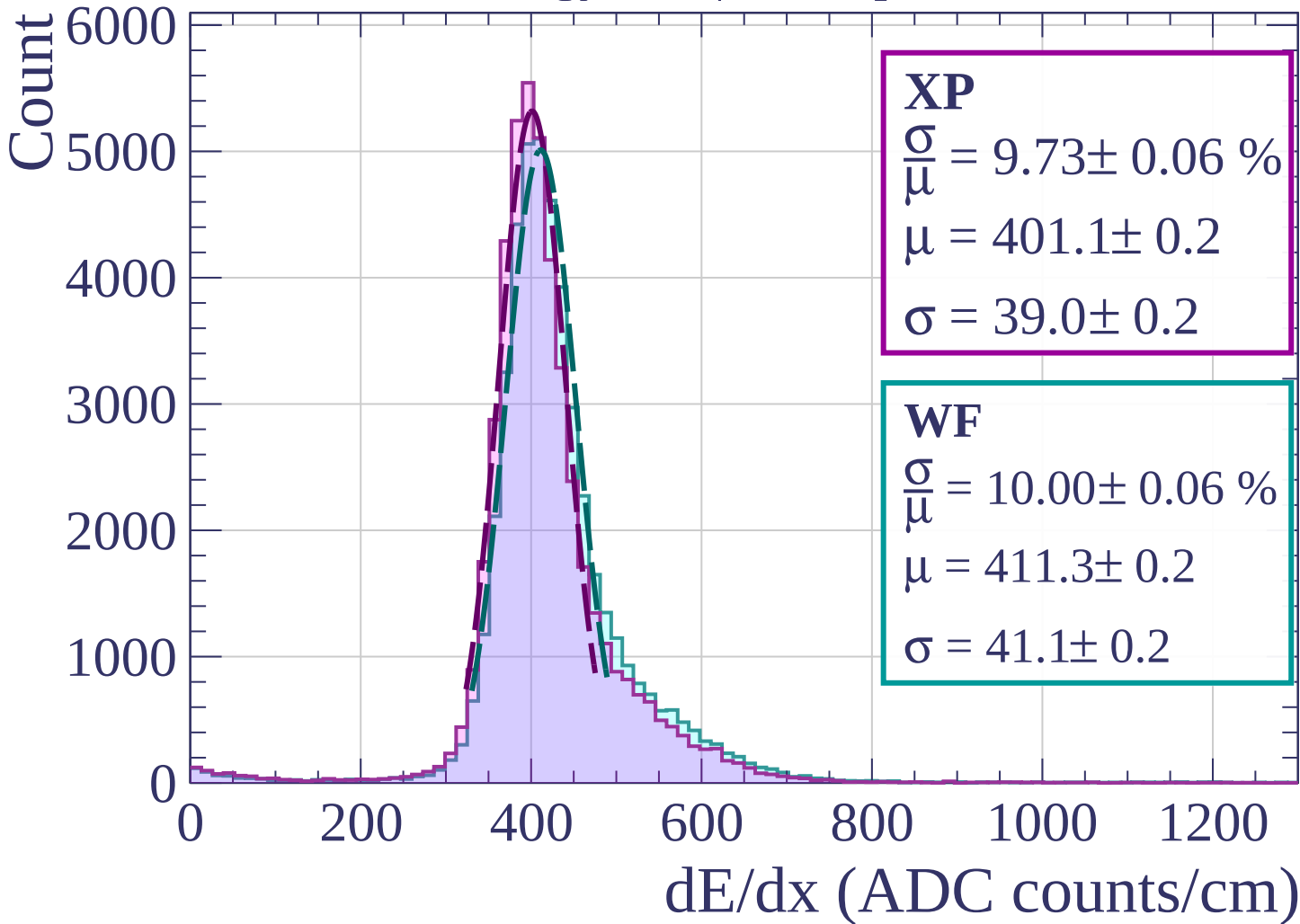
$$\sigma = 40.8 \pm 0.2$$



Energy loss | $400 < p < 480$



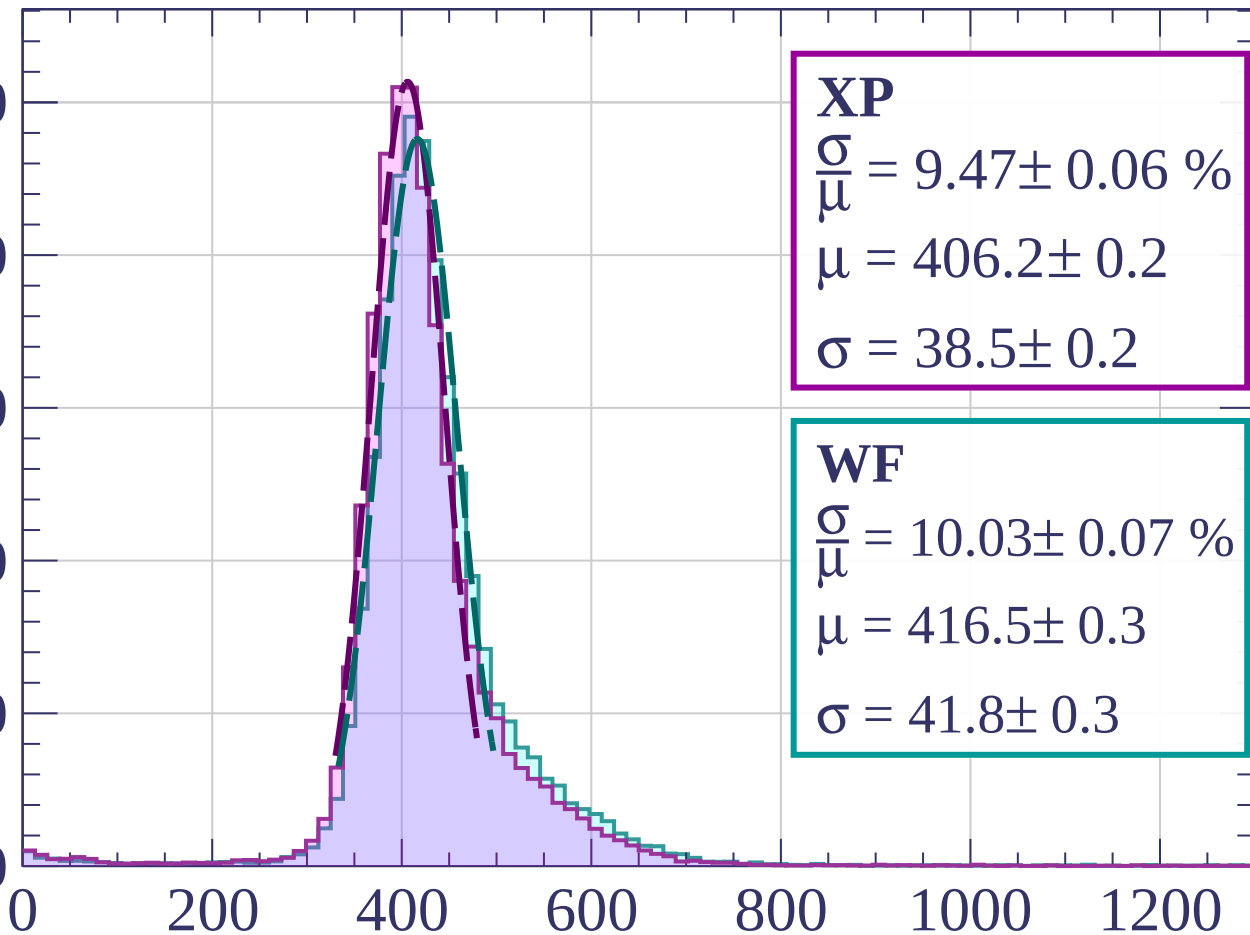
Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count

5000
4000
3000
2000
1000
0



XP

$$\frac{\sigma}{\mu} = 9.47 \pm 0.06 \%$$

$$\mu = 406.2 \pm 0.2$$

$$\sigma = 38.5 \pm 0.2$$

WF

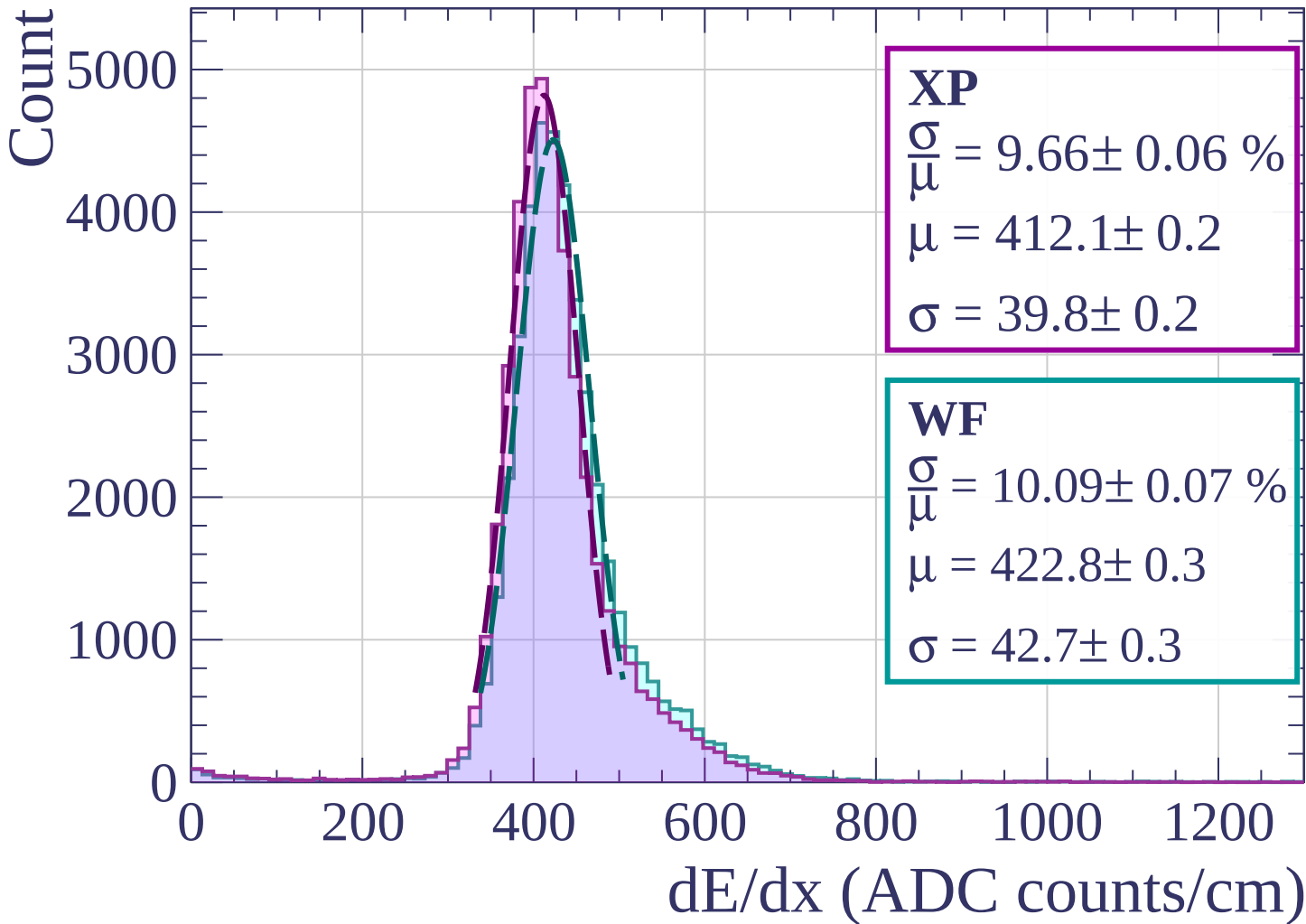
$$\frac{\sigma}{\mu} = 10.03 \pm 0.07 \%$$

$$\mu = 416.5 \pm 0.3$$

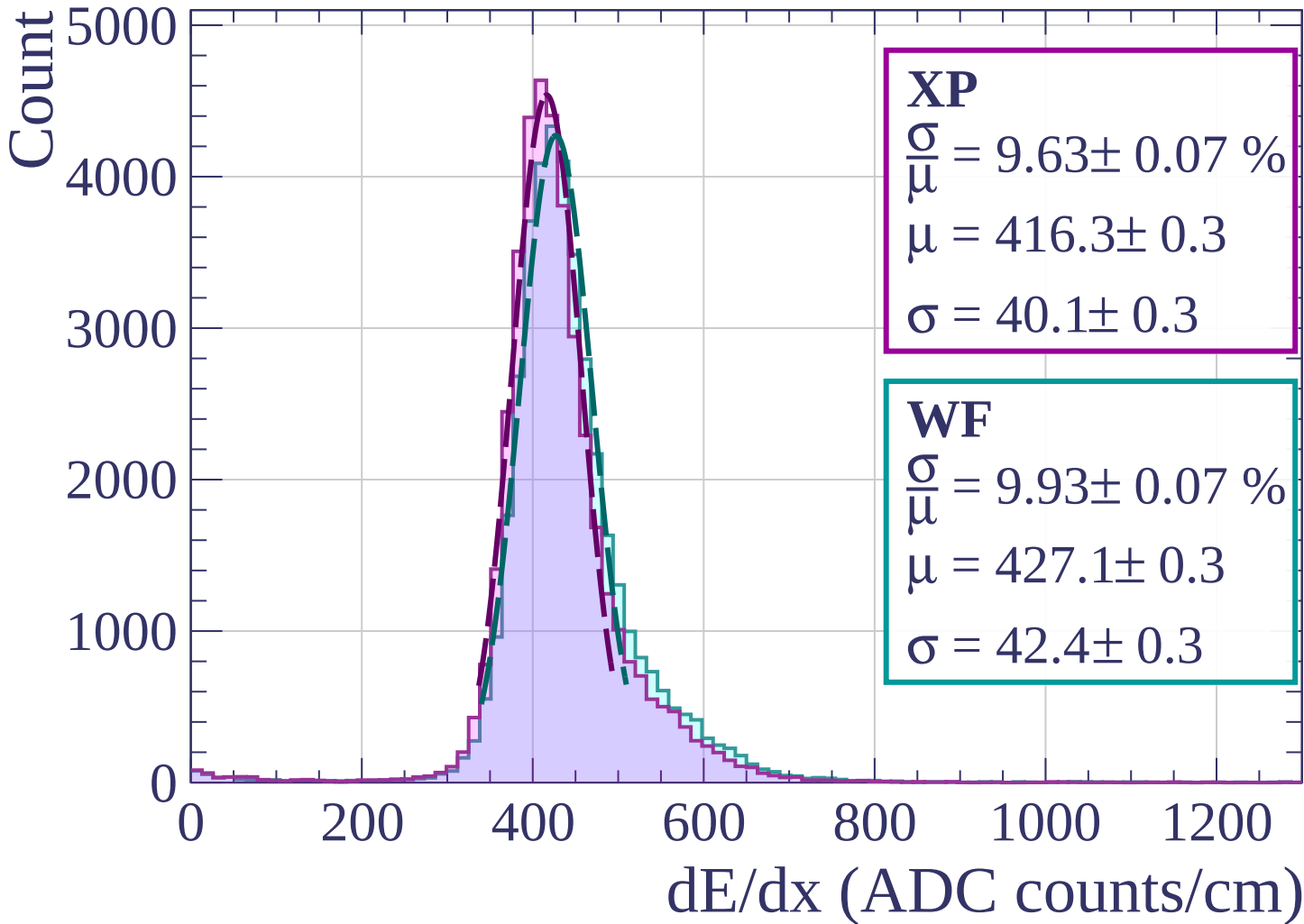
$$\sigma = 41.8 \pm 0.3$$

dE/dx (ADC counts/cm)

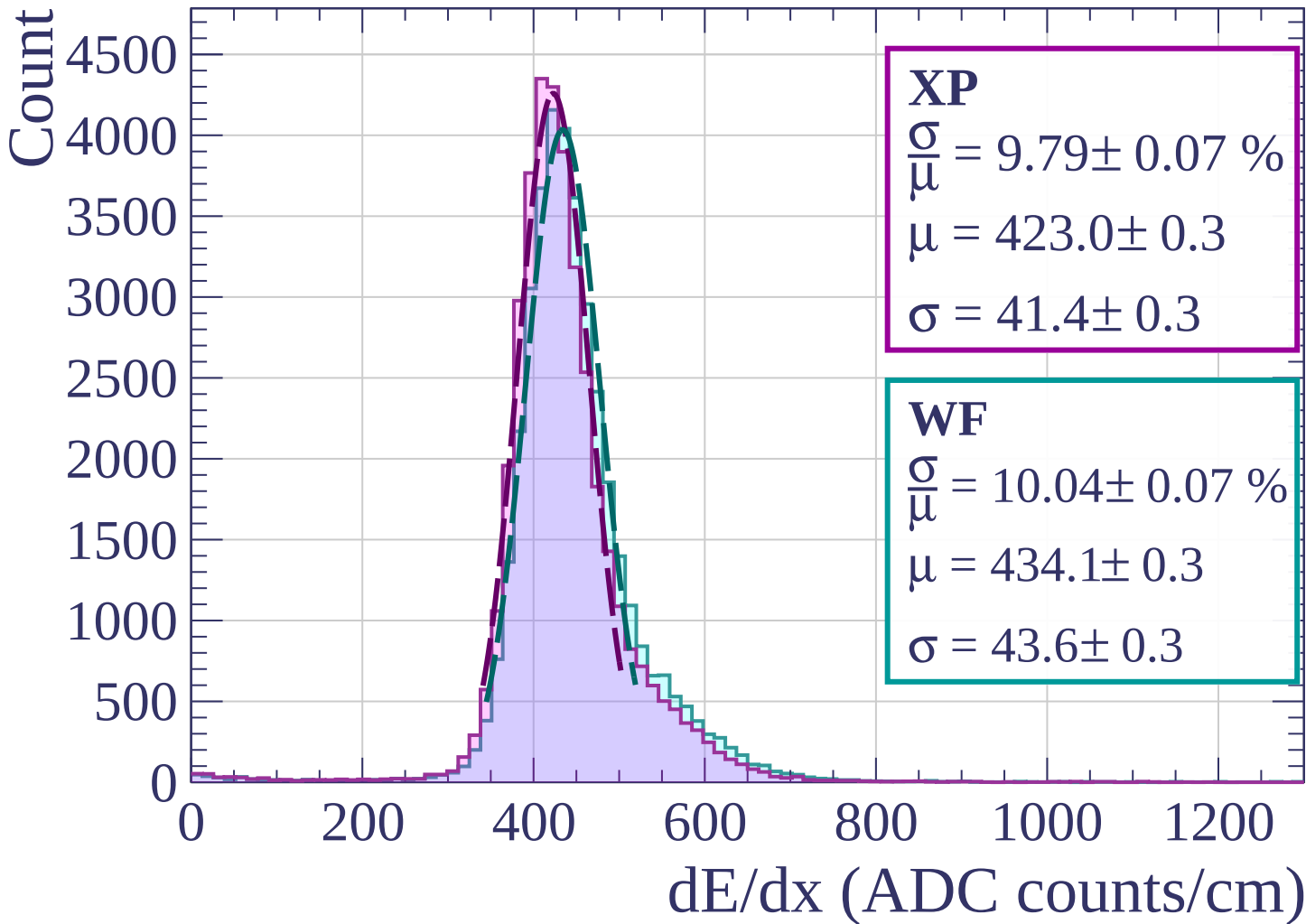
Energy loss | $640 < p < 720$



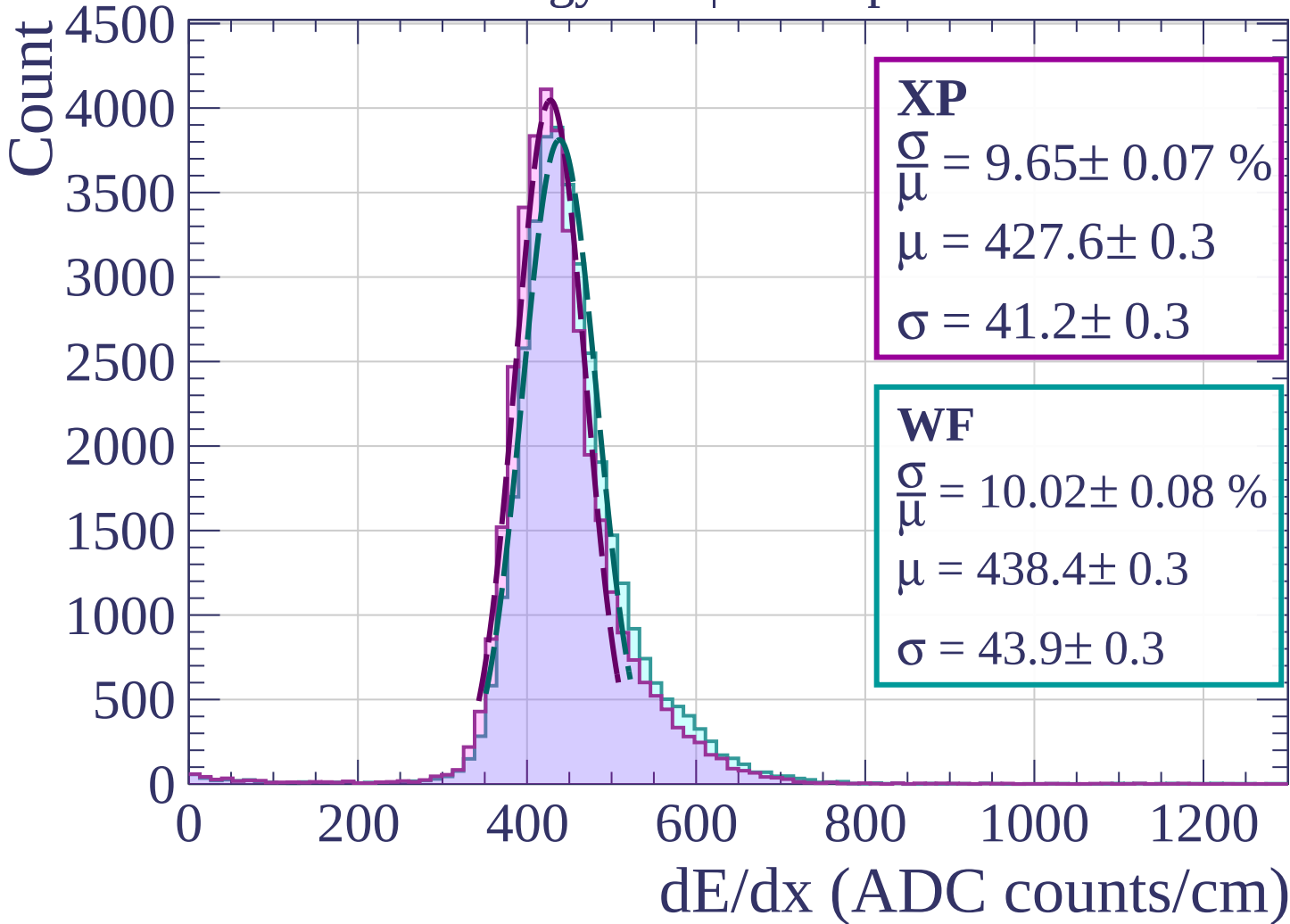
Energy loss | $720 < p < 800$



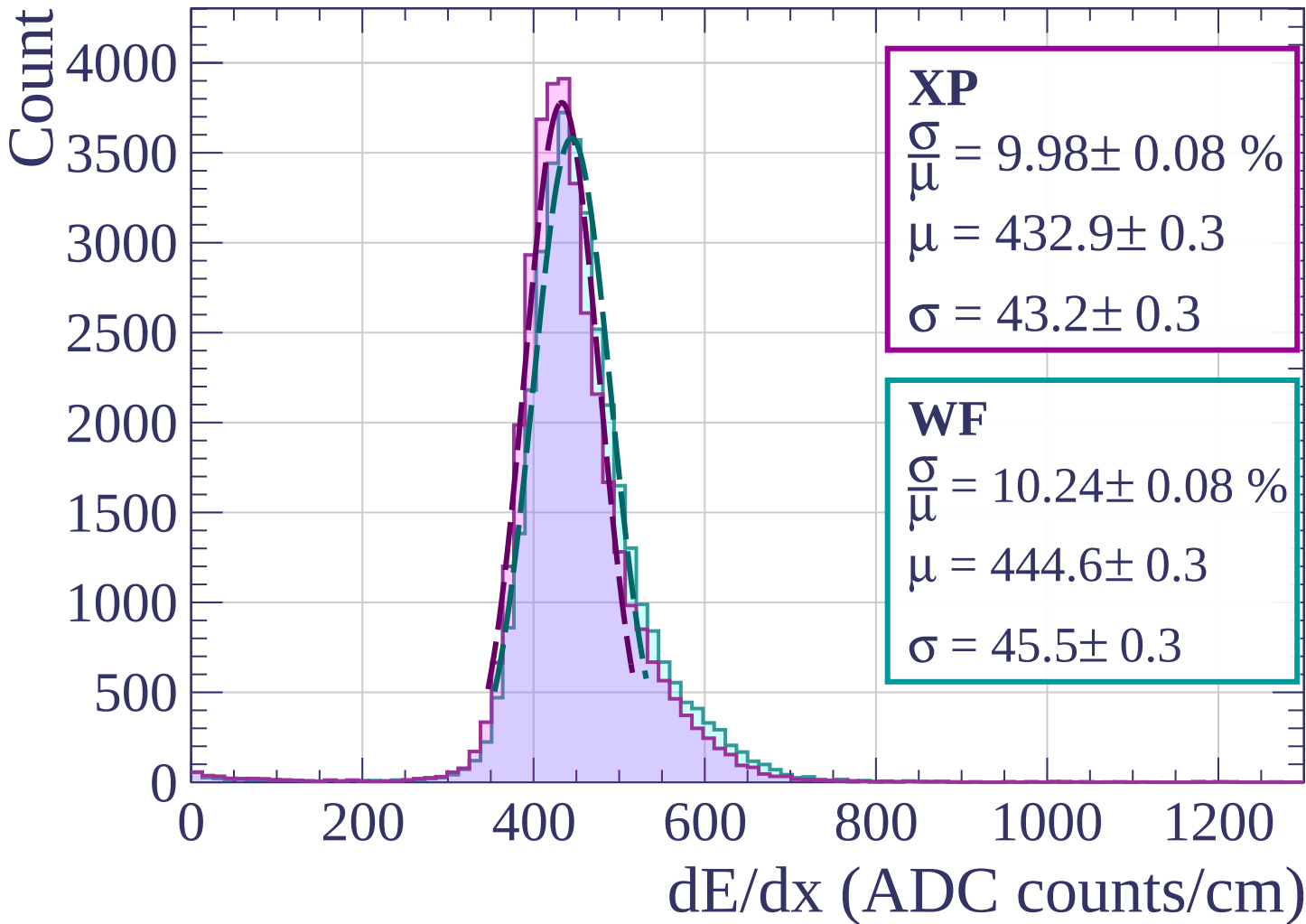
Energy loss | $800 < p < 880$



Energy loss | $880 < p < 960$



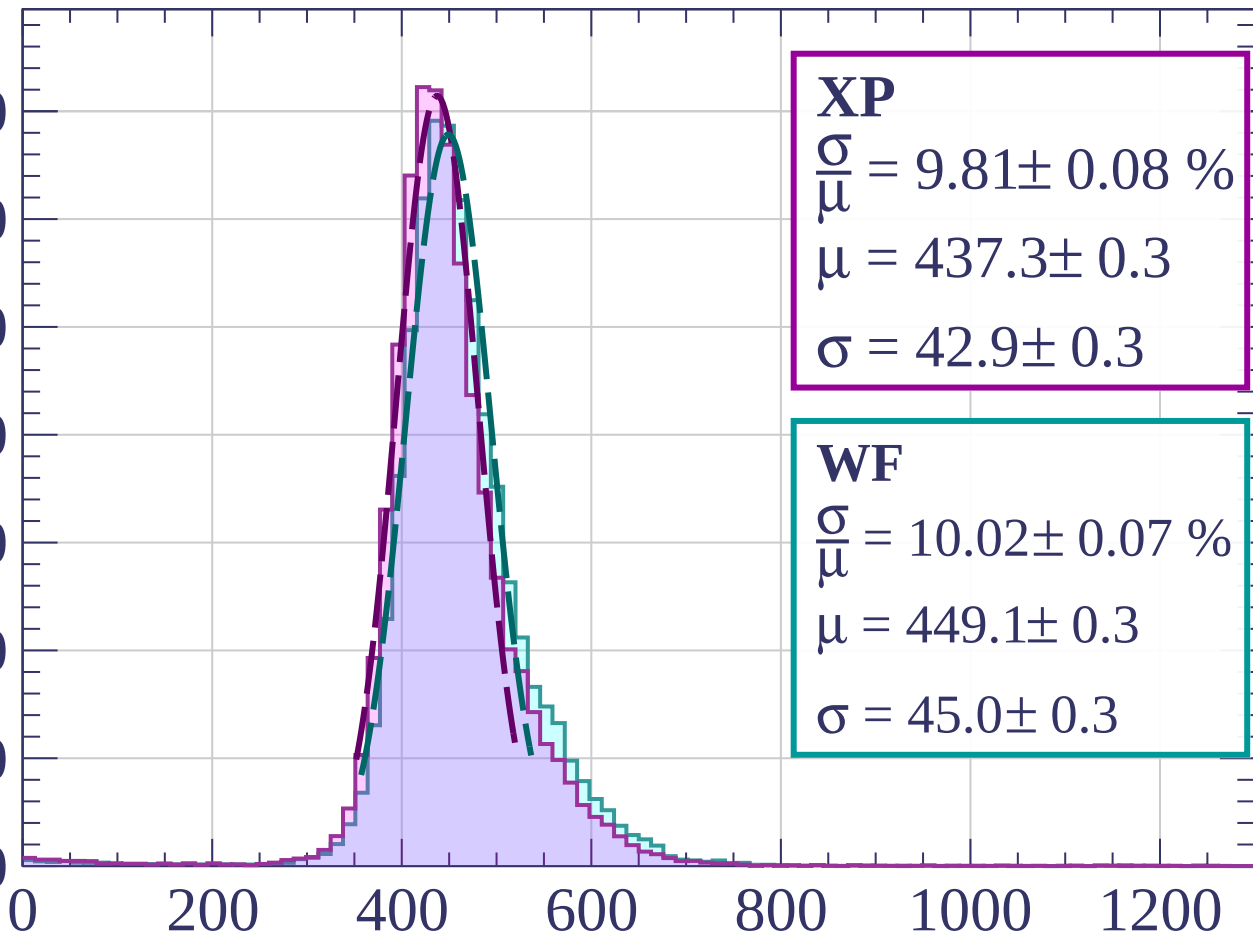
Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count

3500
3000
2500
2000
1500
1000
500
0



XP

$$\frac{\sigma}{\mu} = 9.81 \pm 0.08 \%$$

$$\mu = 437.3 \pm 0.3$$

$$\sigma = 42.9 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 10.02 \pm 0.07 \%$$

$$\mu = 449.1 \pm 0.3$$

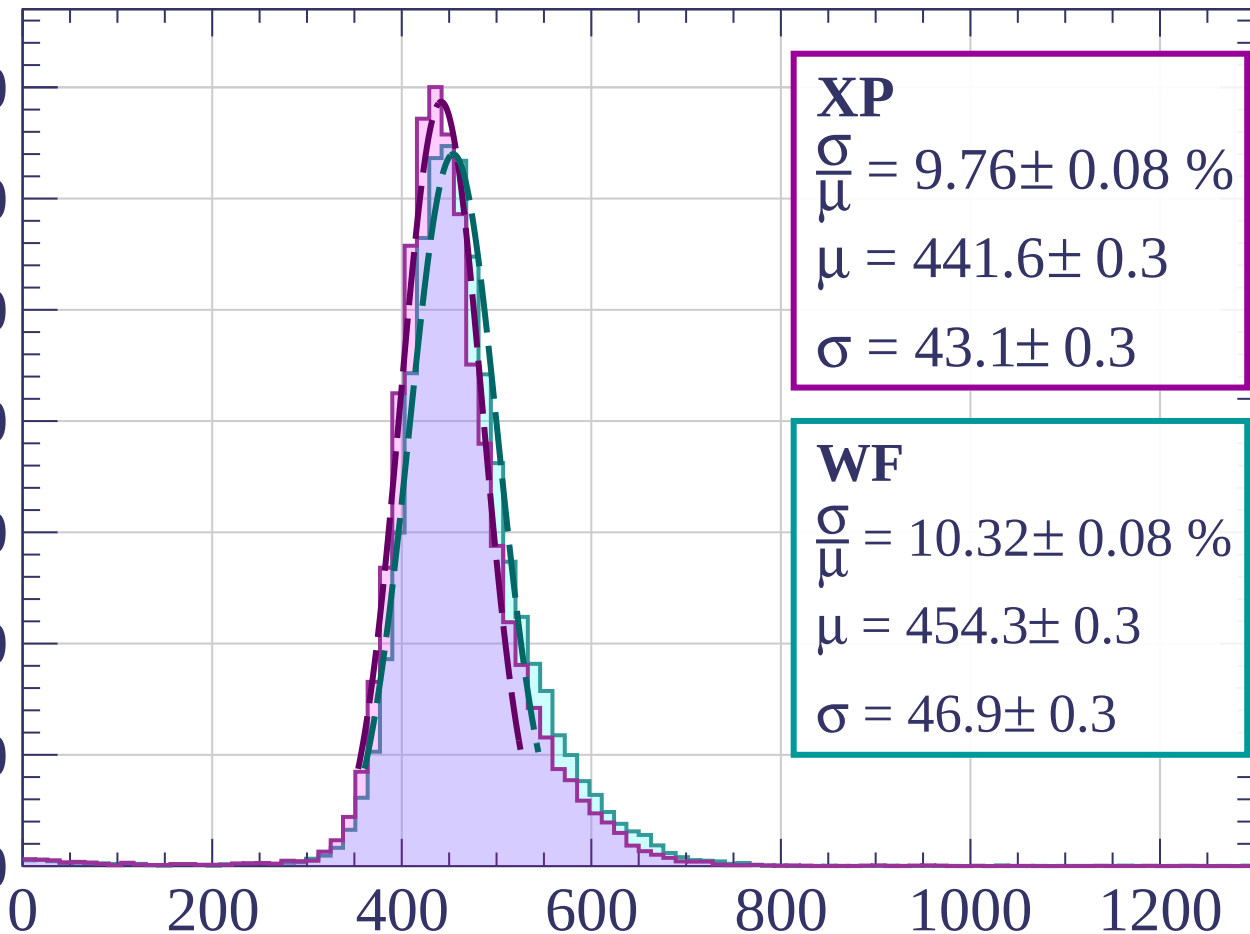
$$\sigma = 45.0 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count

3500
3000
2500
2000
1500
1000
500
0



XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.08 \%$$

$$\mu = 441.6 \pm 0.3$$

$$\sigma = 43.1 \pm 0.3$$

WF

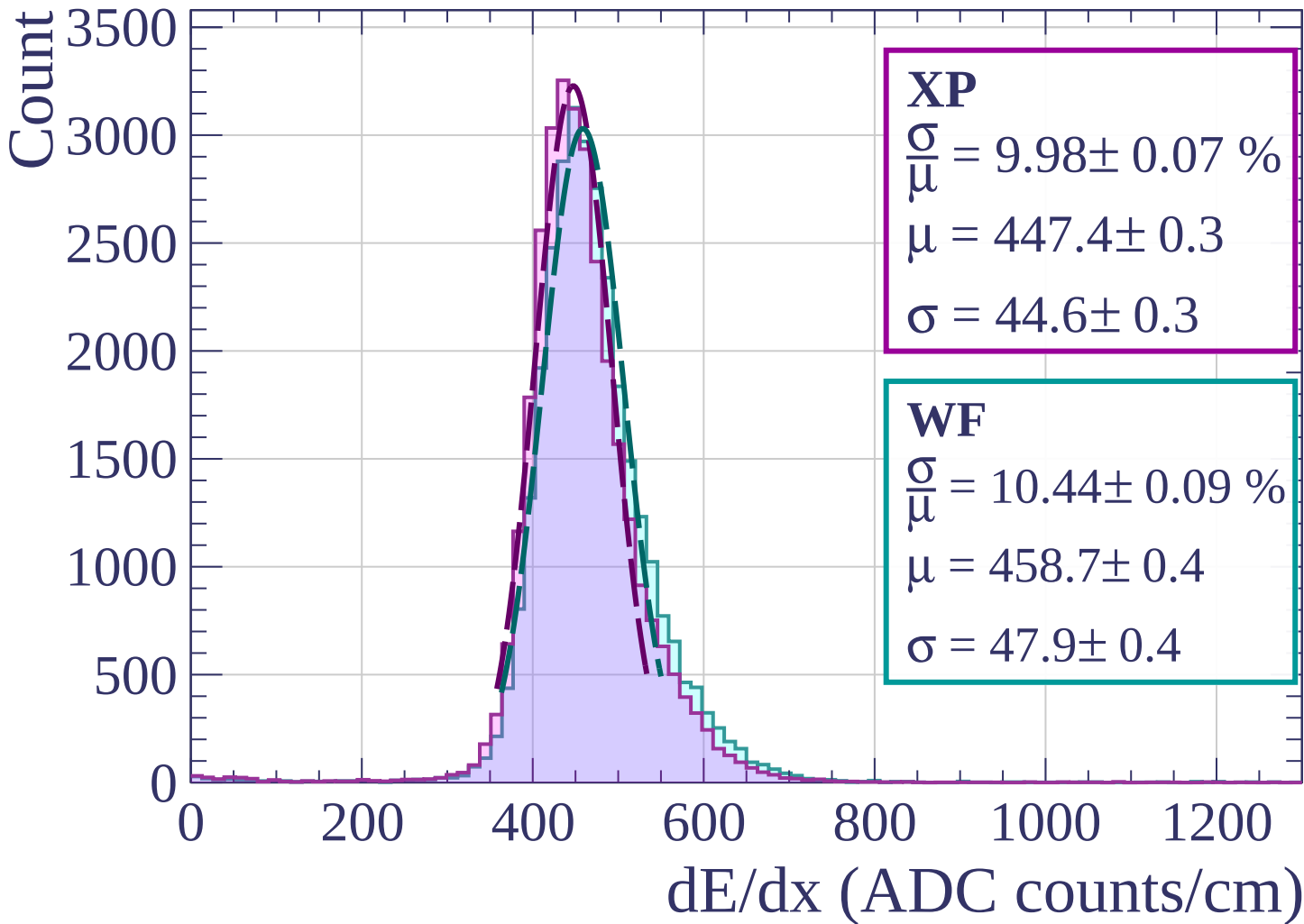
$$\frac{\sigma}{\mu} = 10.32 \pm 0.08 \%$$

$$\mu = 454.3 \pm 0.3$$

$$\sigma = 46.9 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$



Energy loss | $1280 < p < 1360$

Count

3000

2500

2000

1500

1000

500

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 10.06 \pm 0.08 \%$$

$$\mu = 450.4 \pm 0.3$$

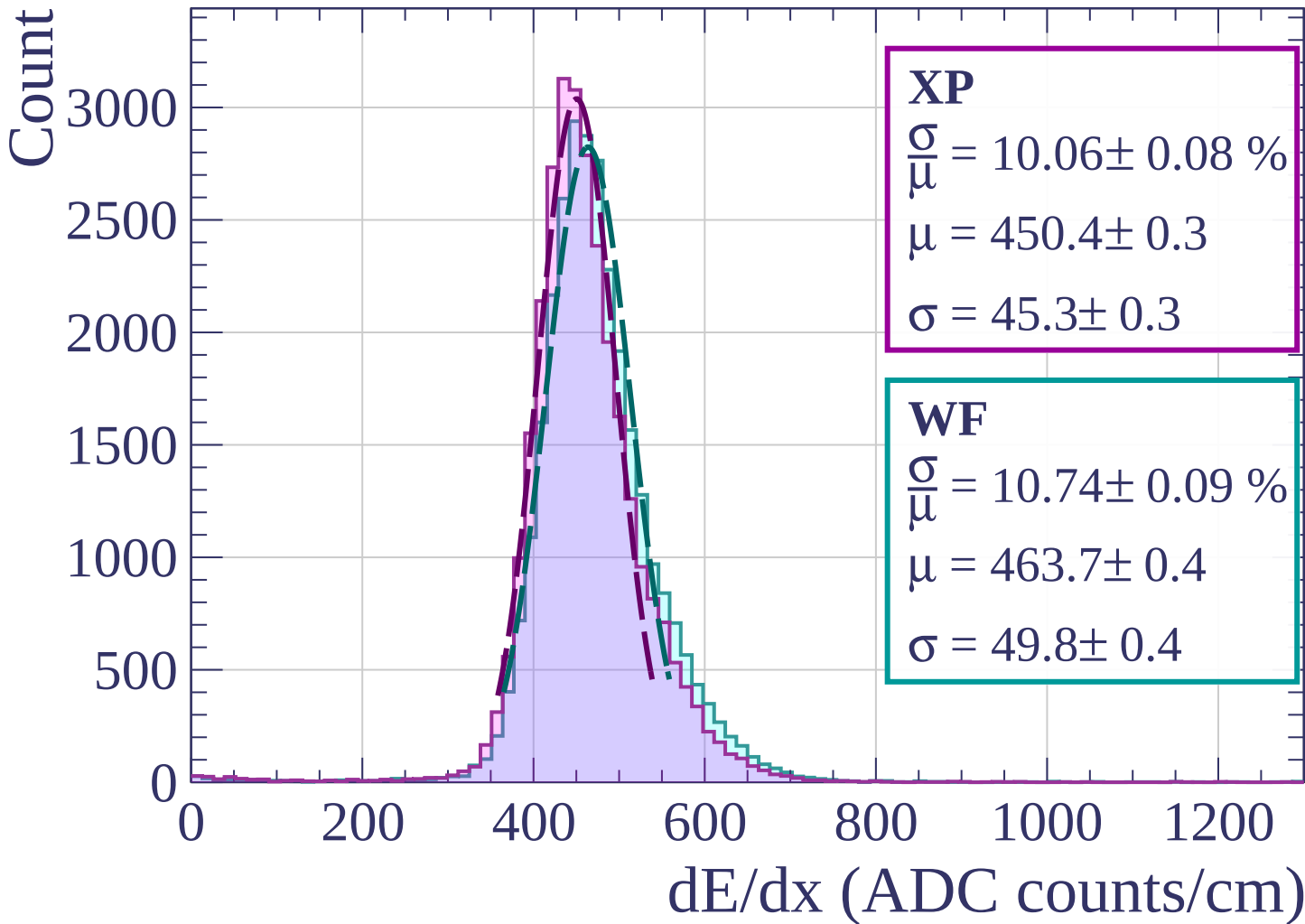
$$\sigma = 45.3 \pm 0.3$$

WF

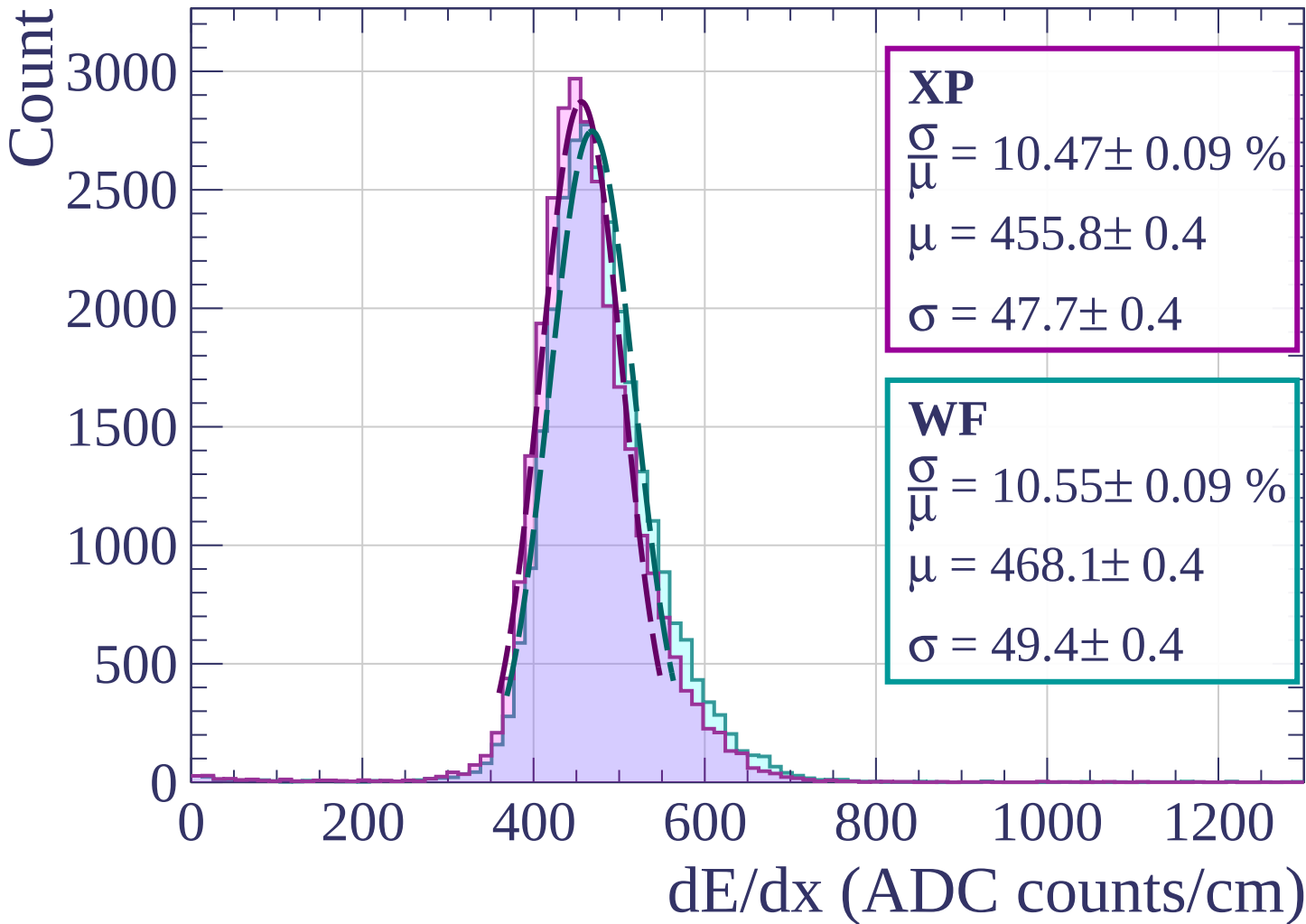
$$\frac{\sigma}{\mu} = 10.74 \pm 0.09 \%$$

$$\mu = 463.7 \pm 0.4$$

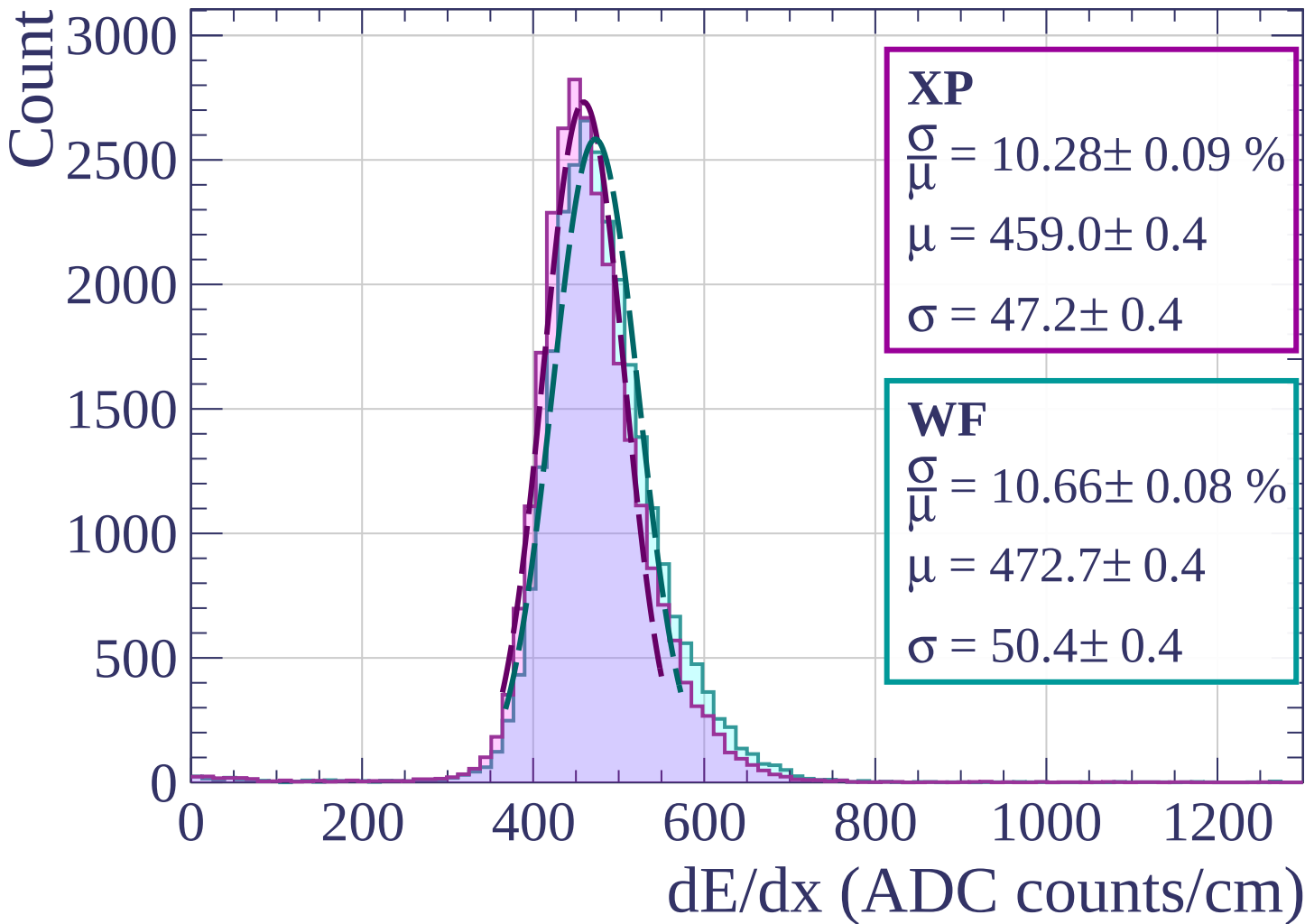
$$\sigma = 49.8 \pm 0.4$$



Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$



Energy loss | $1520 < p < 1600$

Count

2500

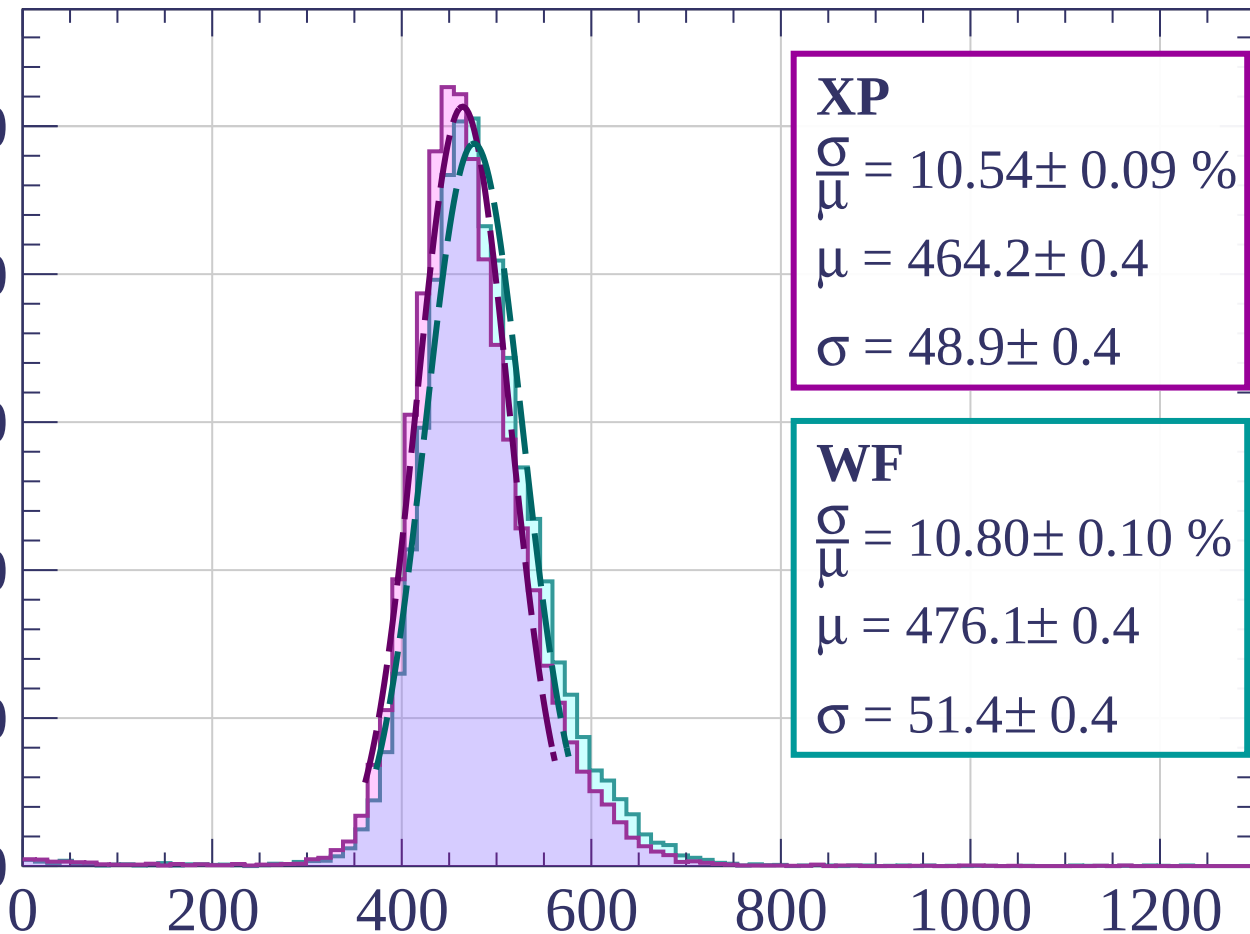
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 10.54 \pm 0.09 \%$$

$$\mu = 464.2 \pm 0.4$$

$$\sigma = 48.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 10.80 \pm 0.10 \%$$

$$\mu = 476.1 \pm 0.4$$

$$\sigma = 51.4 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count

2500

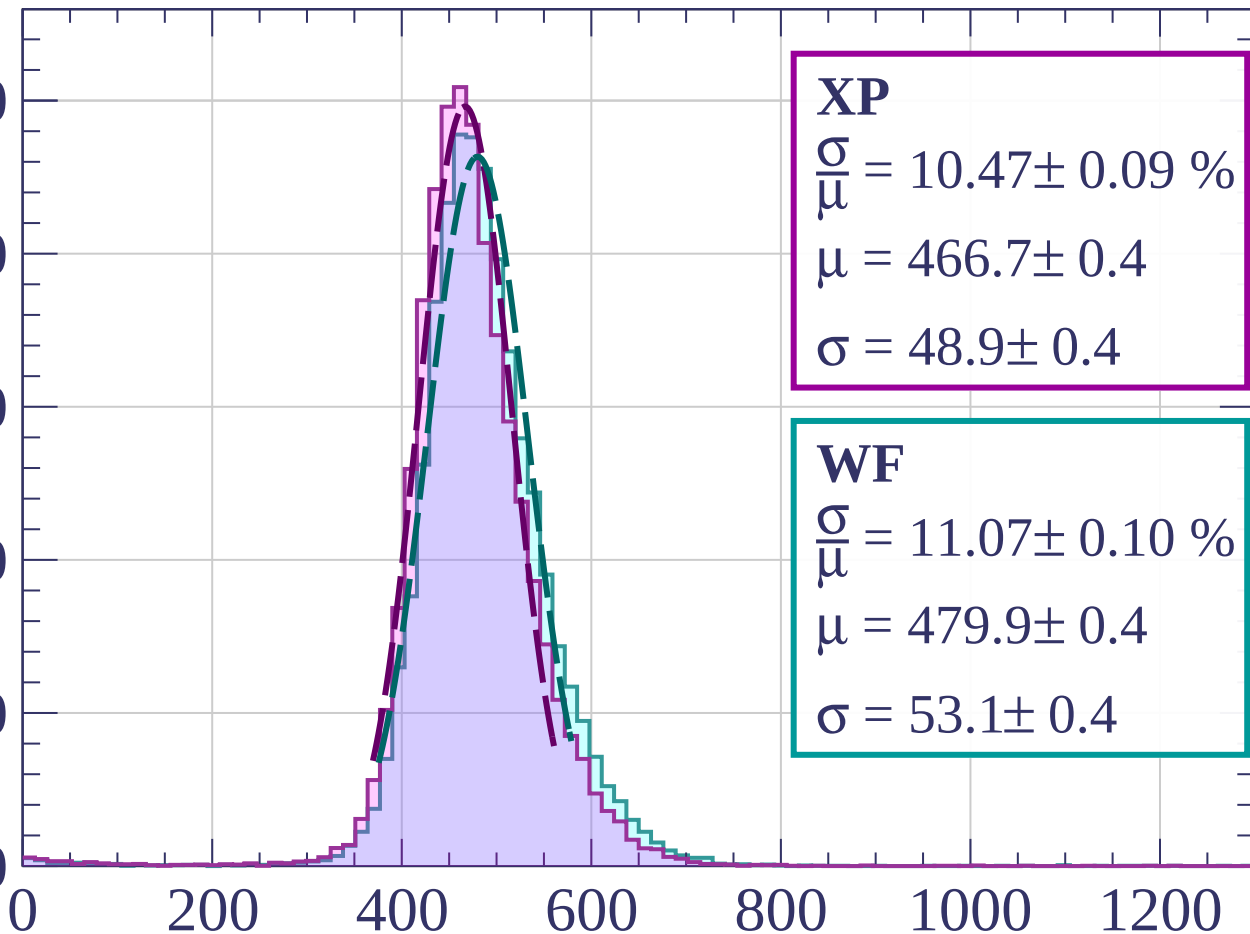
2000

1500

1000

500

0



XP

$\frac{\sigma}{\mu} = 10.47 \pm 0.09 \%$

$\mu = 466.7 \pm 0.4$

$\sigma = 48.9 \pm 0.4$

WF

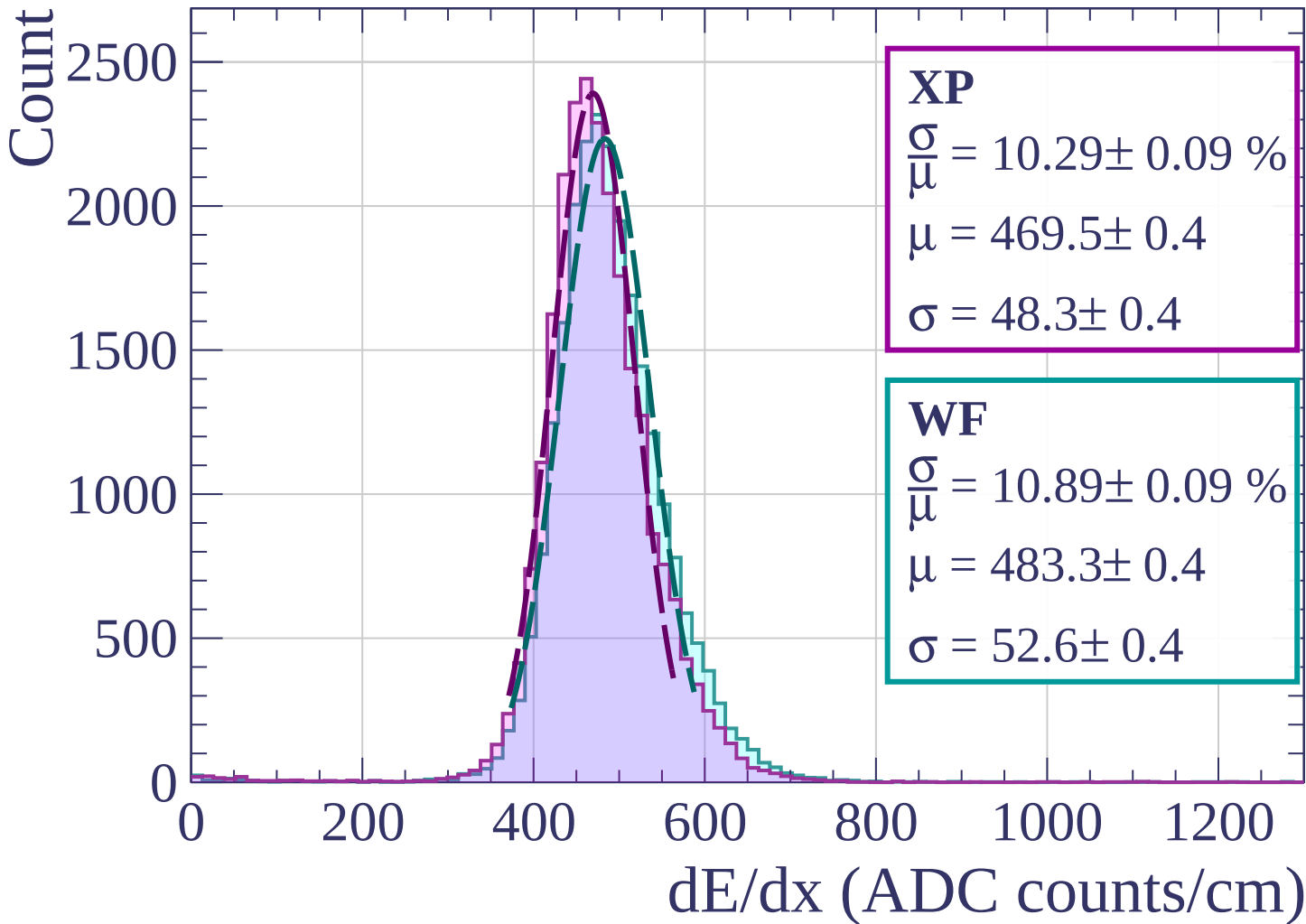
$\frac{\sigma}{\mu} = 11.07 \pm 0.10 \%$

$\mu = 479.9 \pm 0.4$

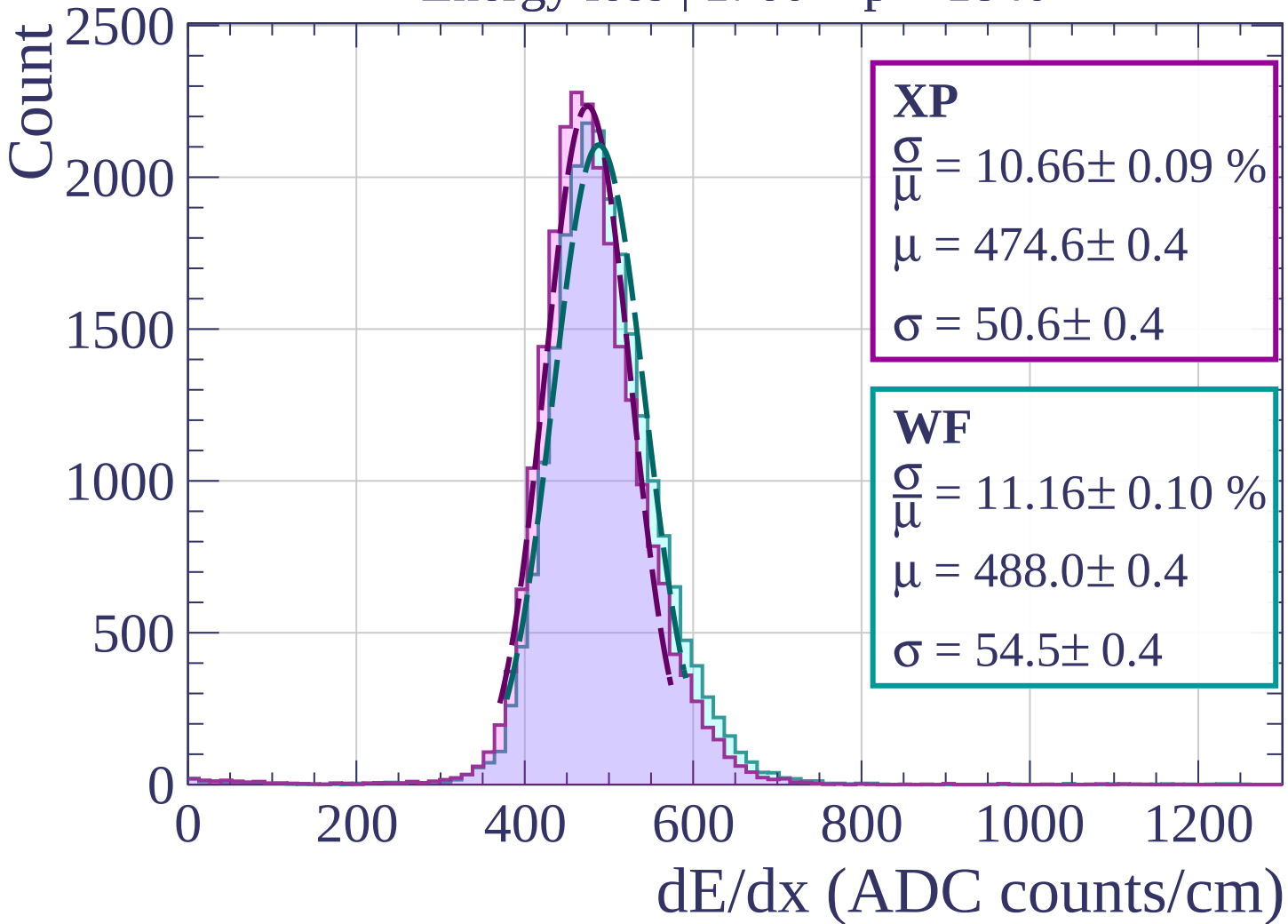
$\sigma = 53.1 \pm 0.4$

dE/dx (ADC counts/cm)

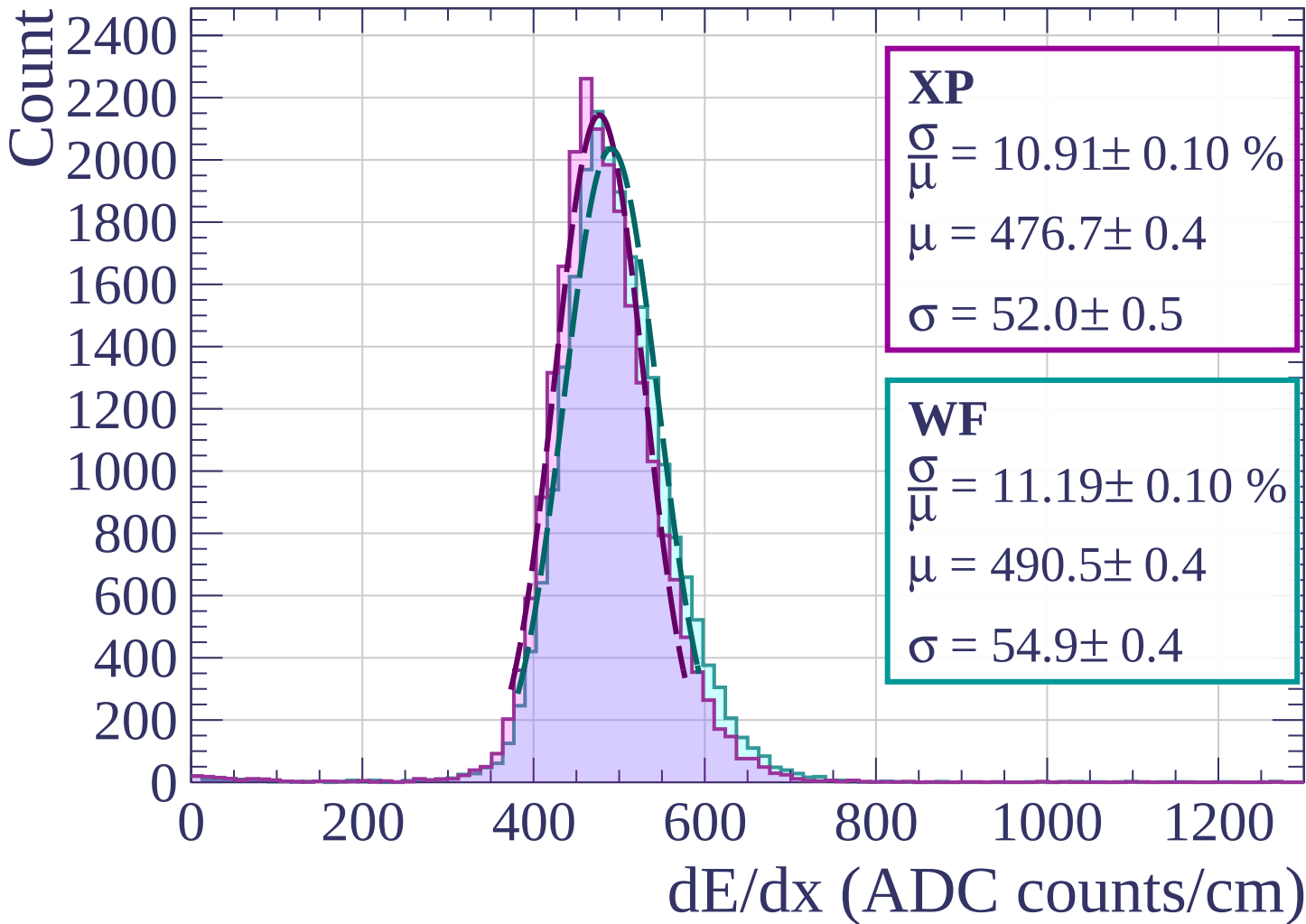
Energy loss | $1680 < p < 1760$



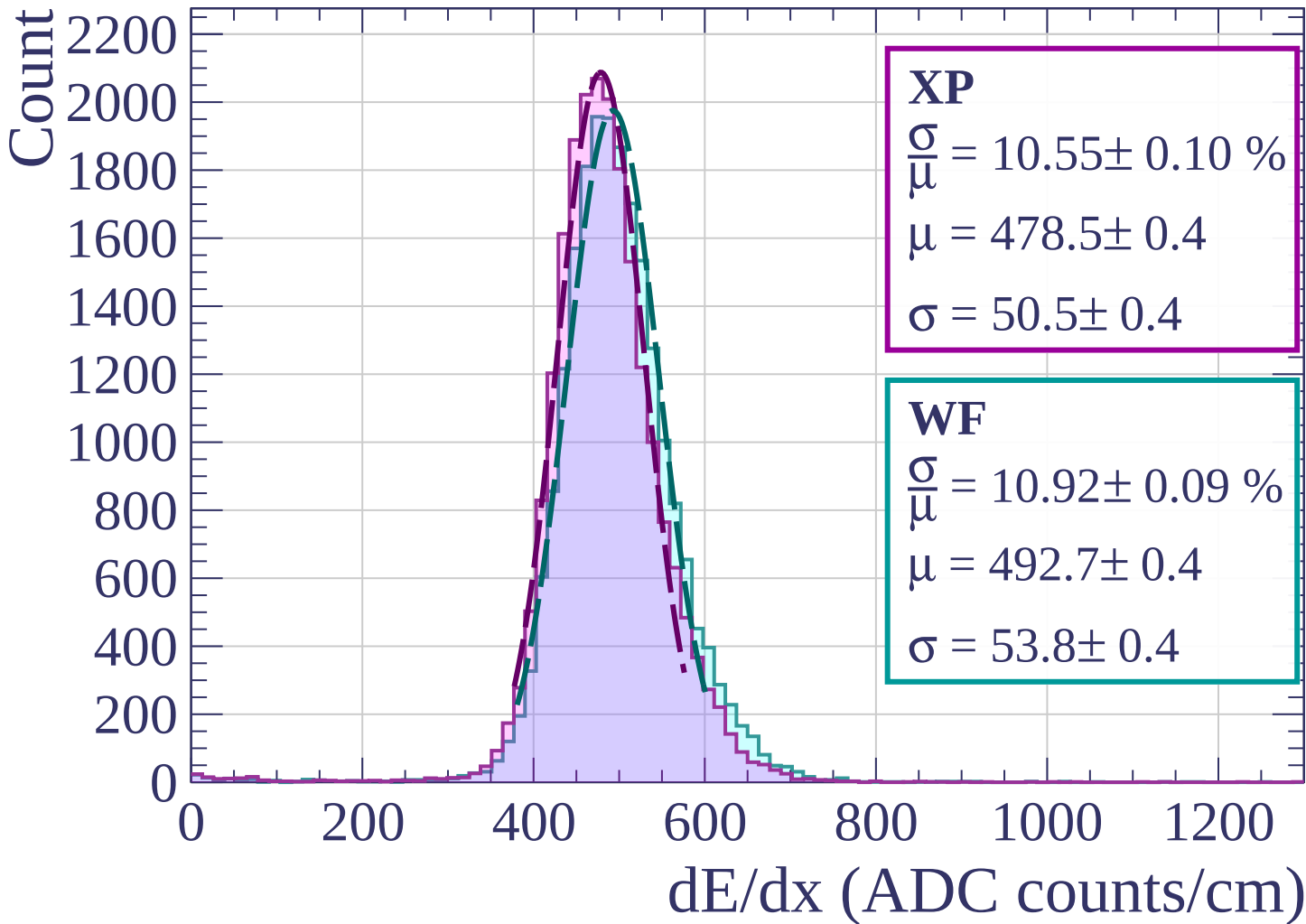
Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$

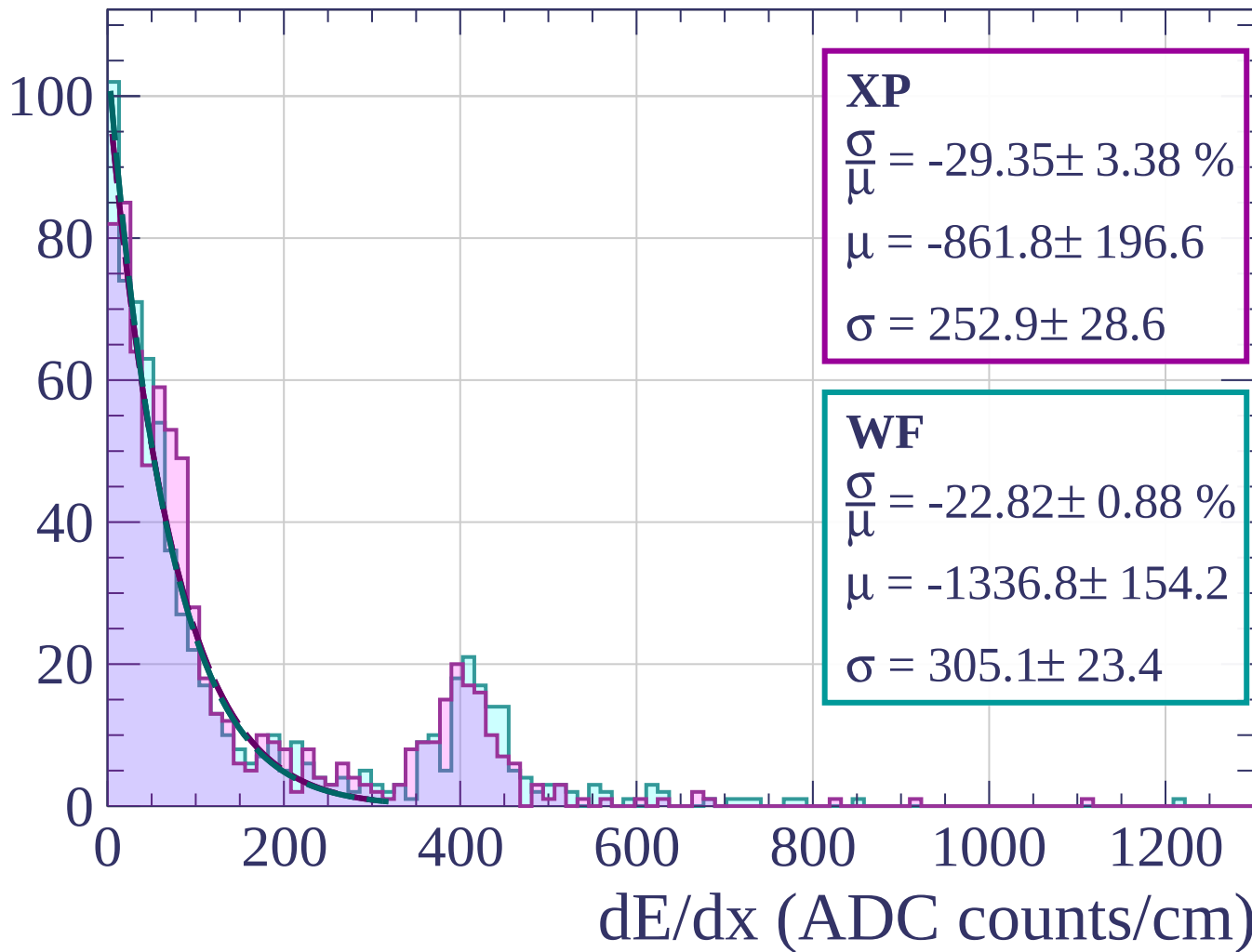


Energy loss | $1920 < p < 2000$



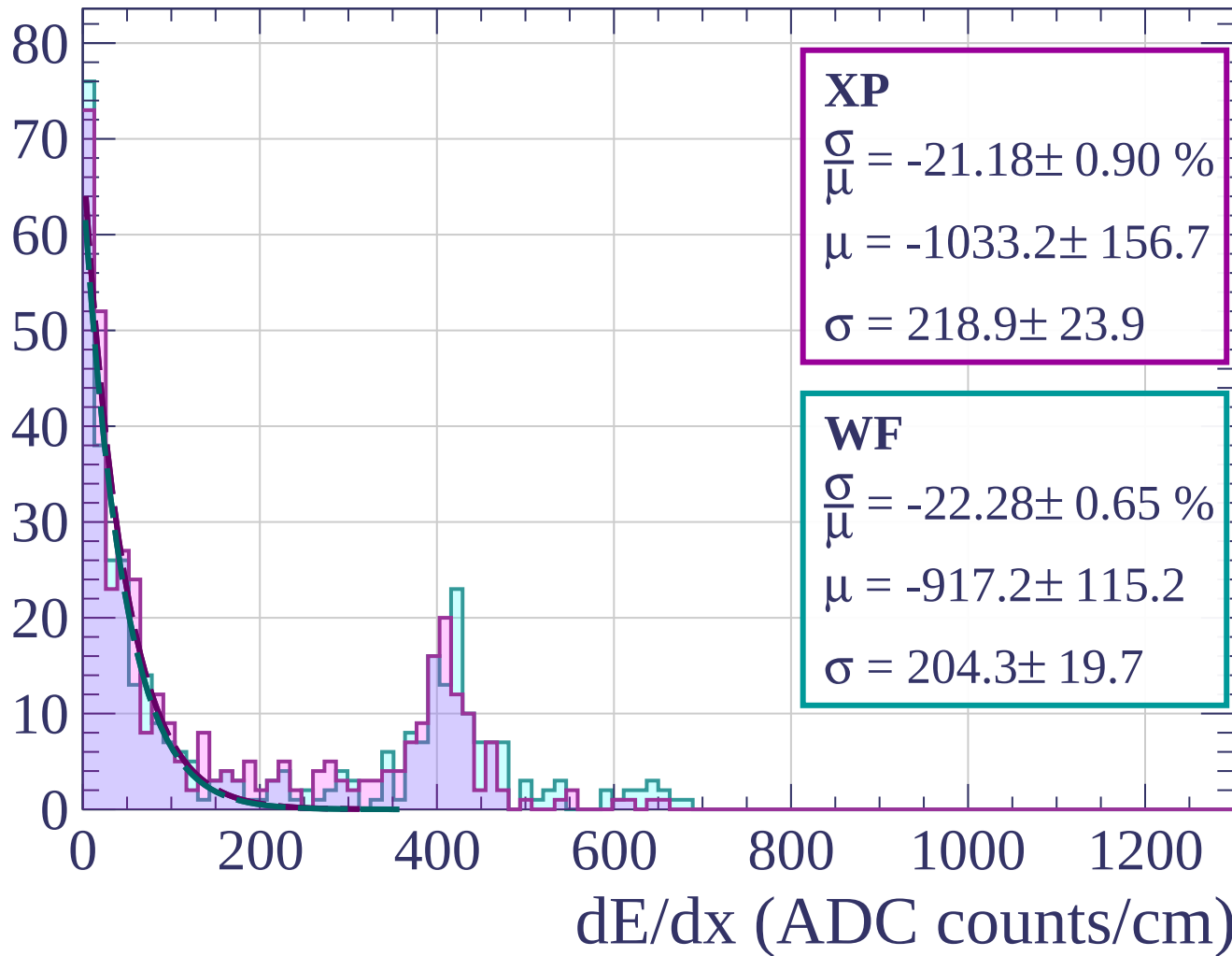
Energy loss | $0 < \phi < 3$

Count



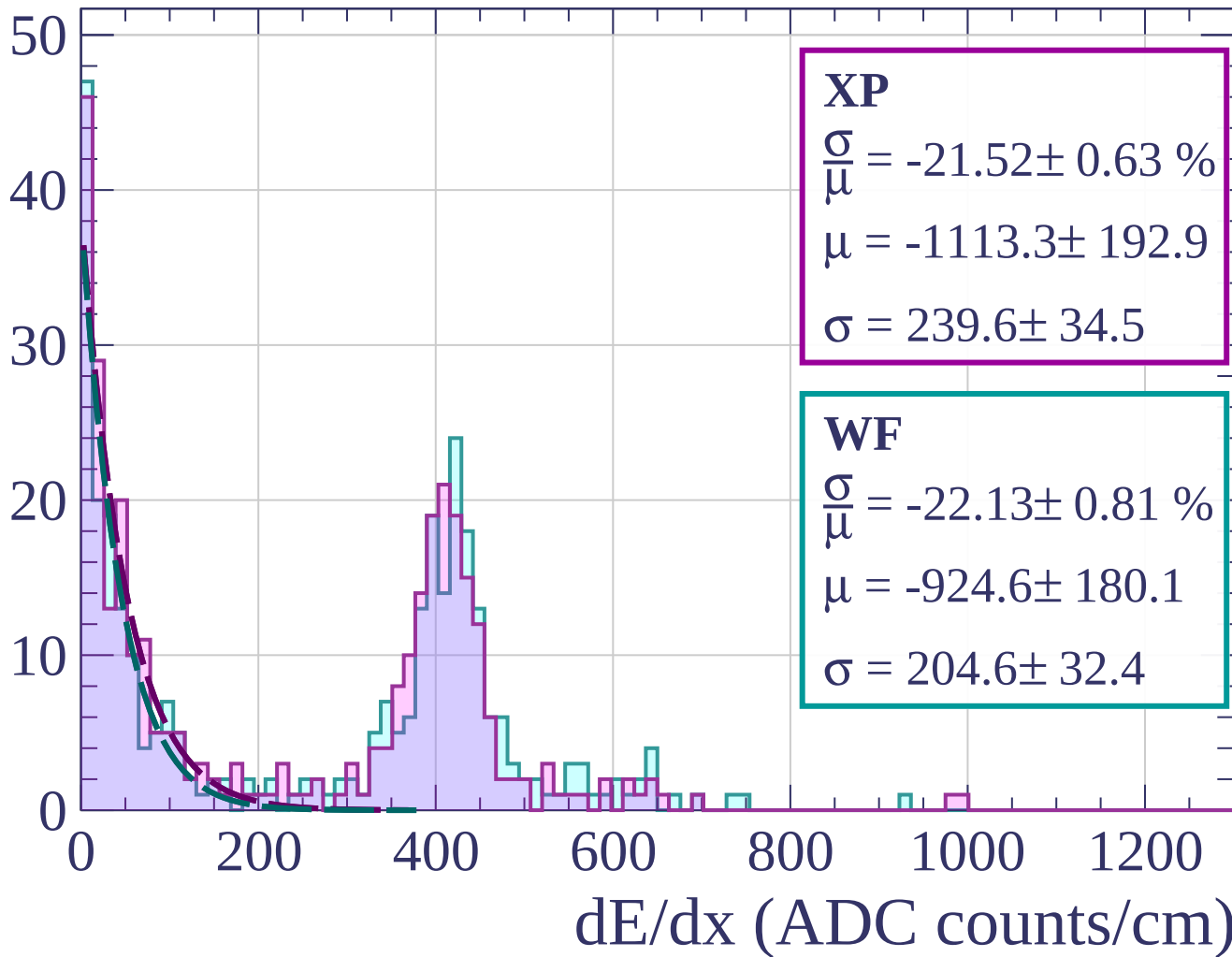
Energy loss | $3 < \phi < 6$

Count



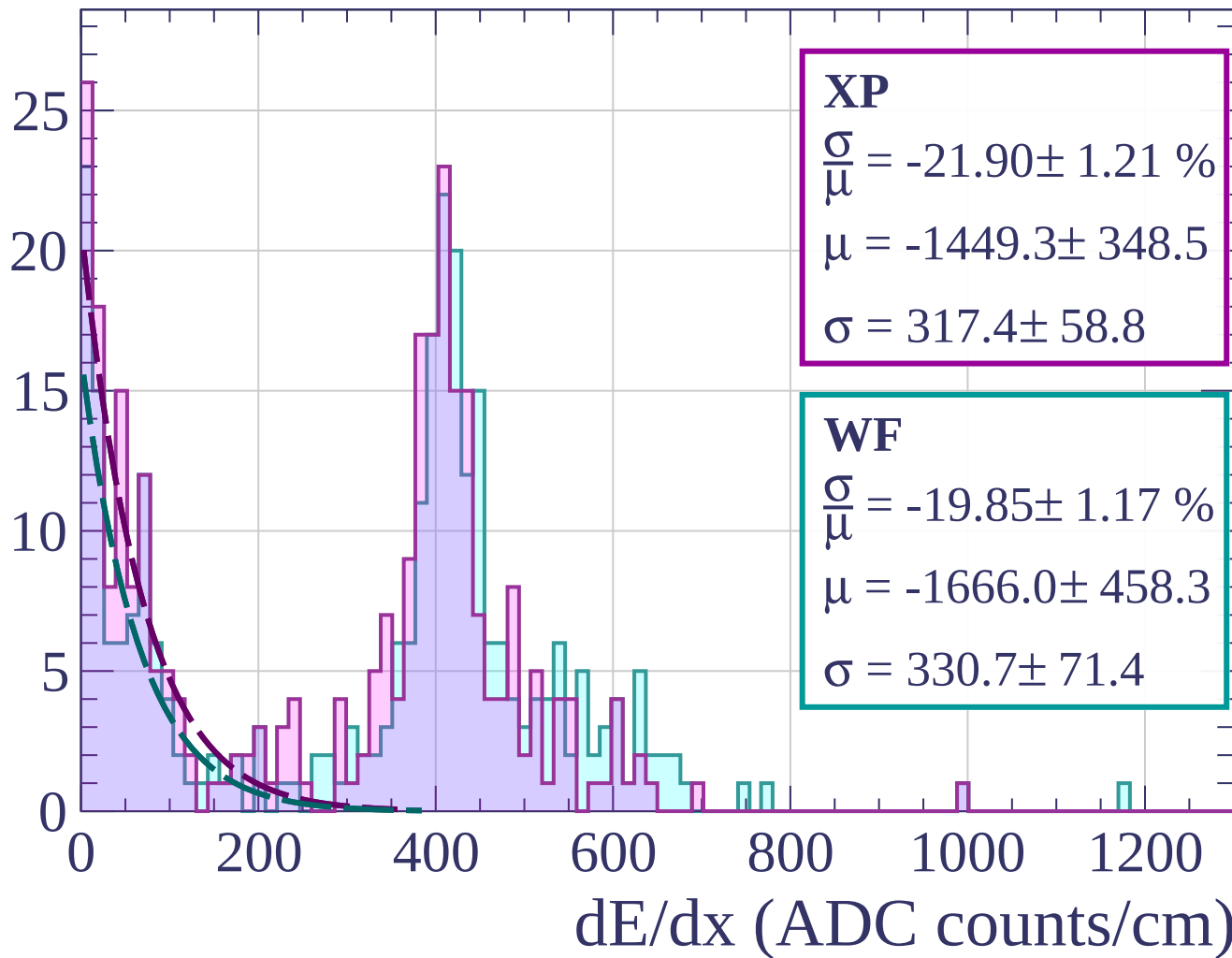
Energy loss | $6 < \phi < 9$

Count



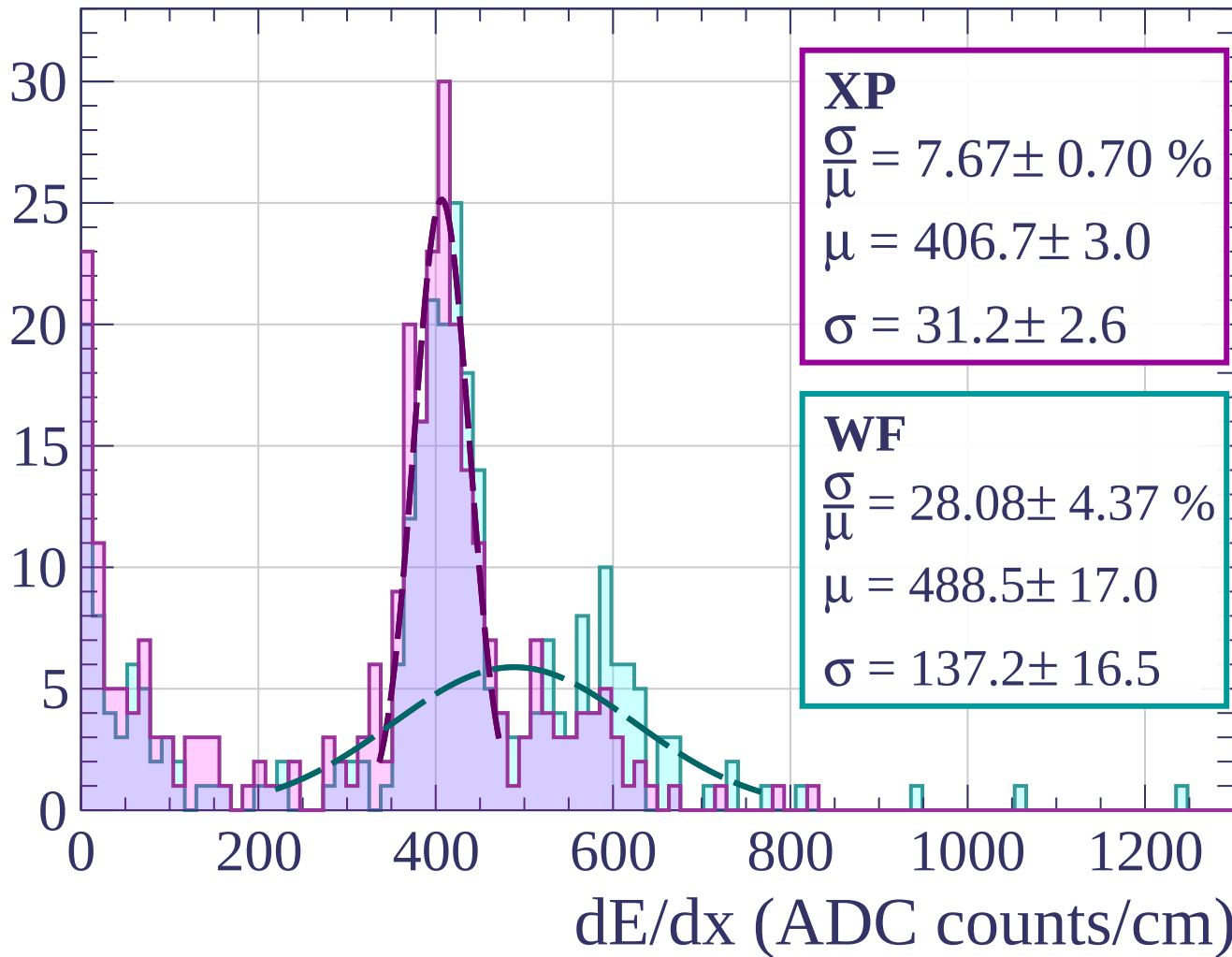
Energy loss | $9 < \phi < 12$

Count



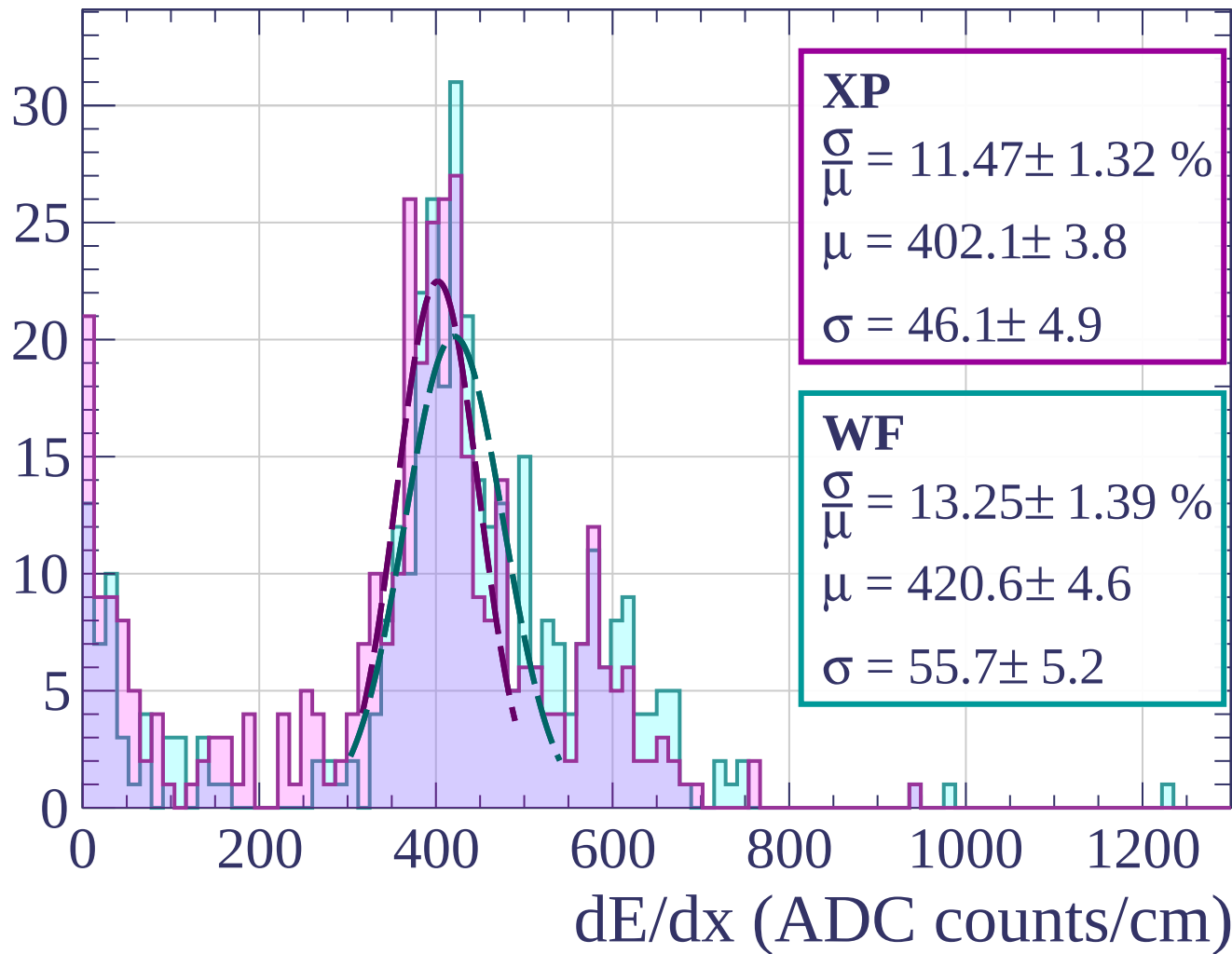
Energy loss | $12 < \phi < 15$

Count



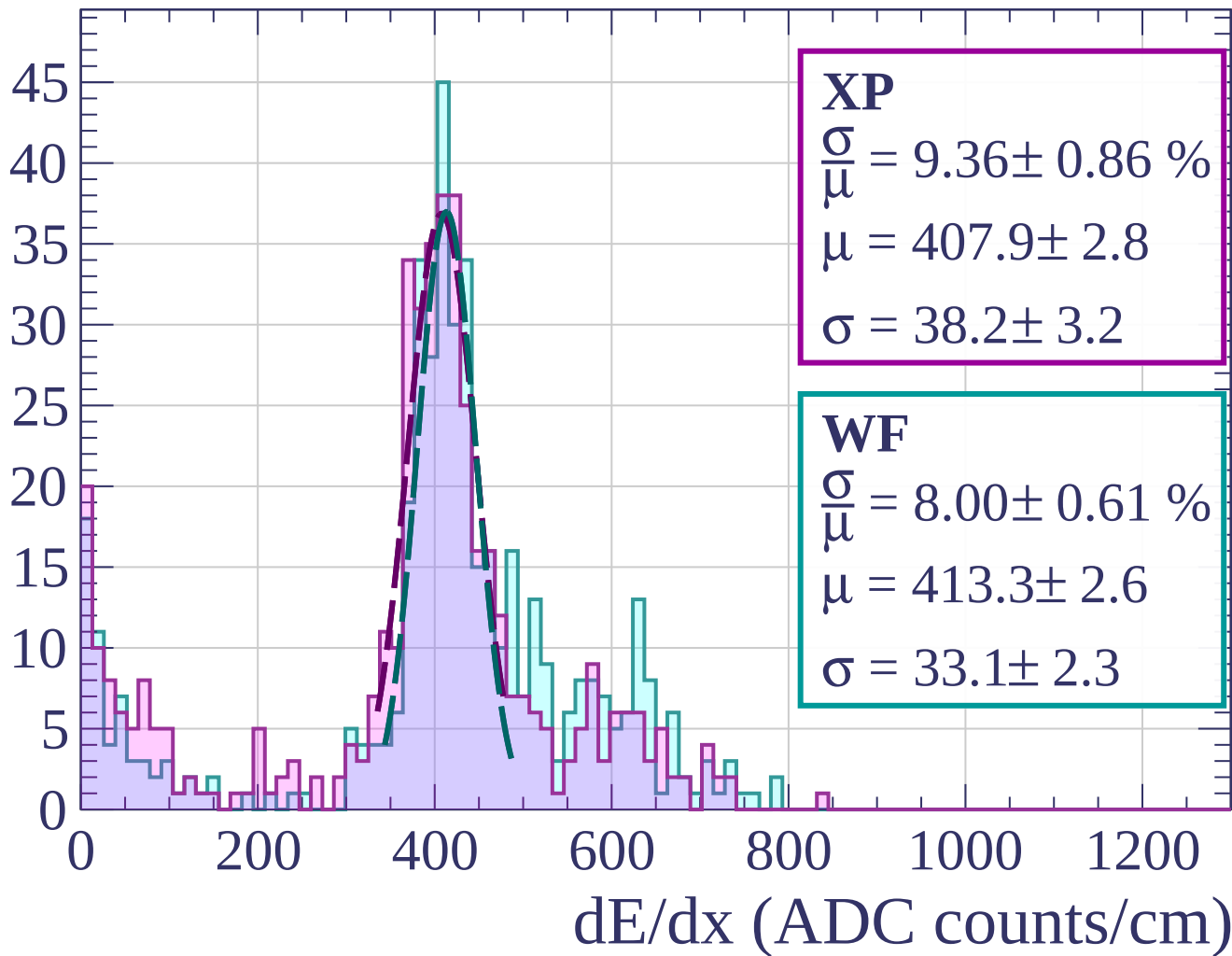
Energy loss | $15 < \phi < 18$

Count



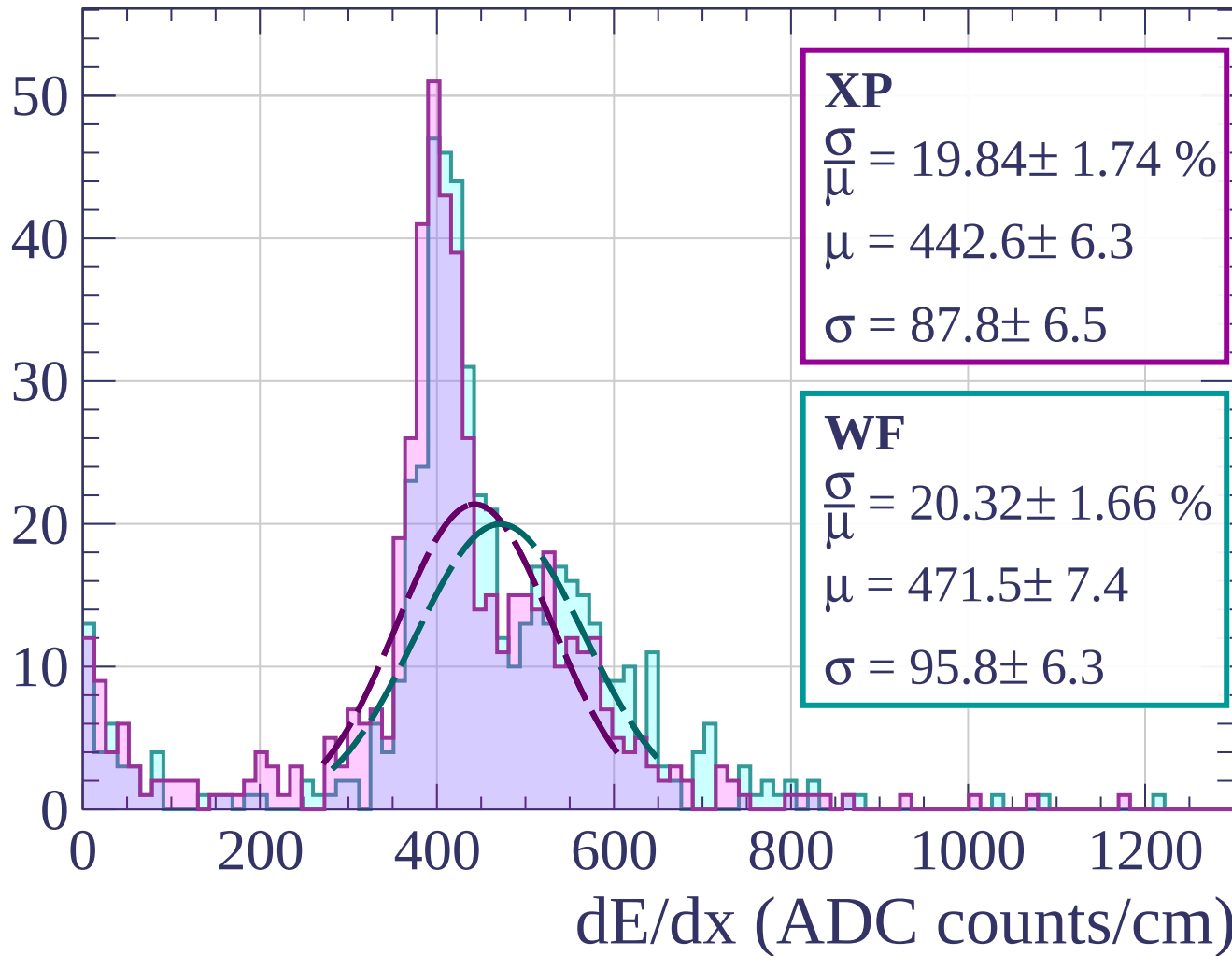
Energy loss | $18 < \phi < 21$

Count



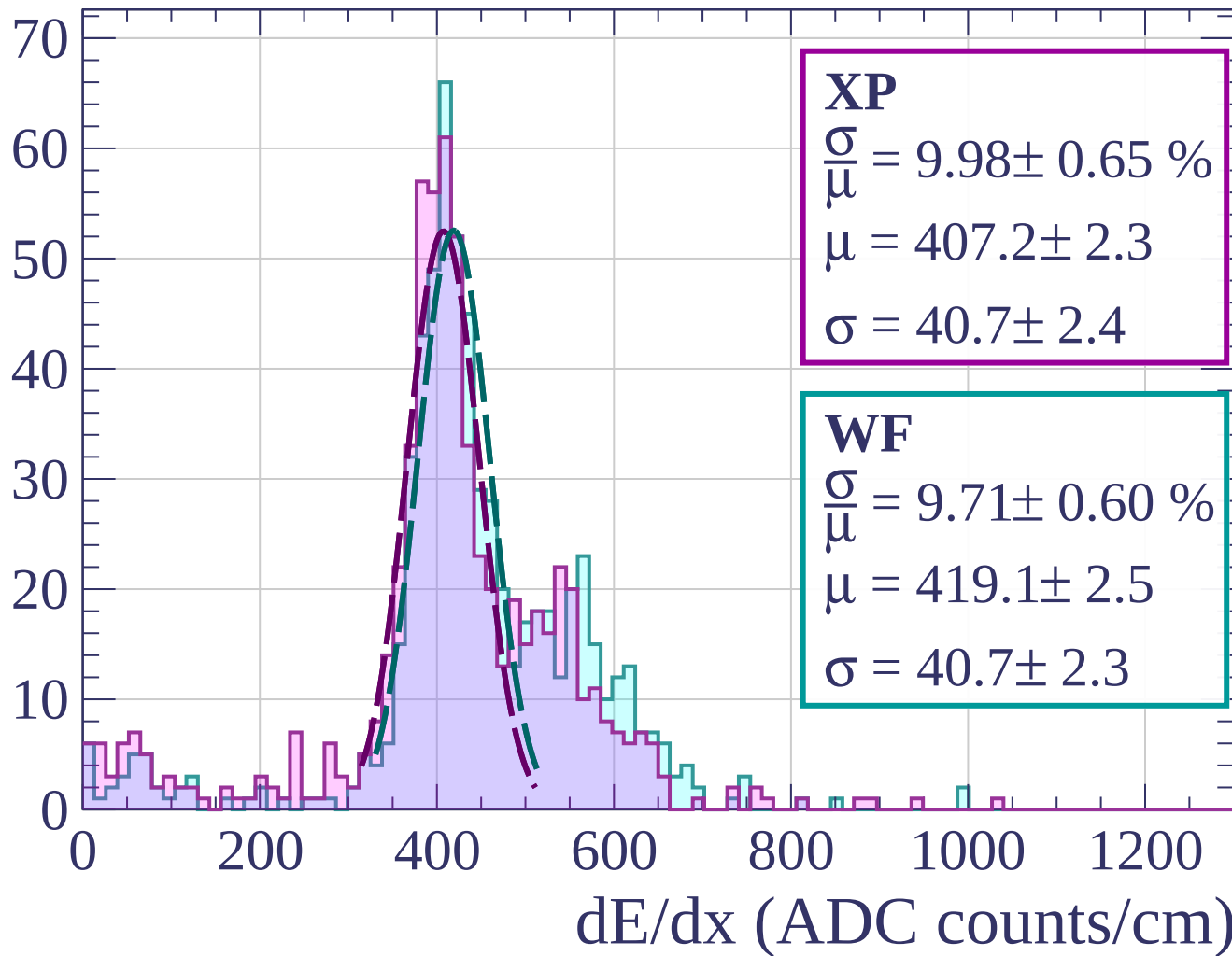
Energy loss | $21 < \phi < 24$

Count



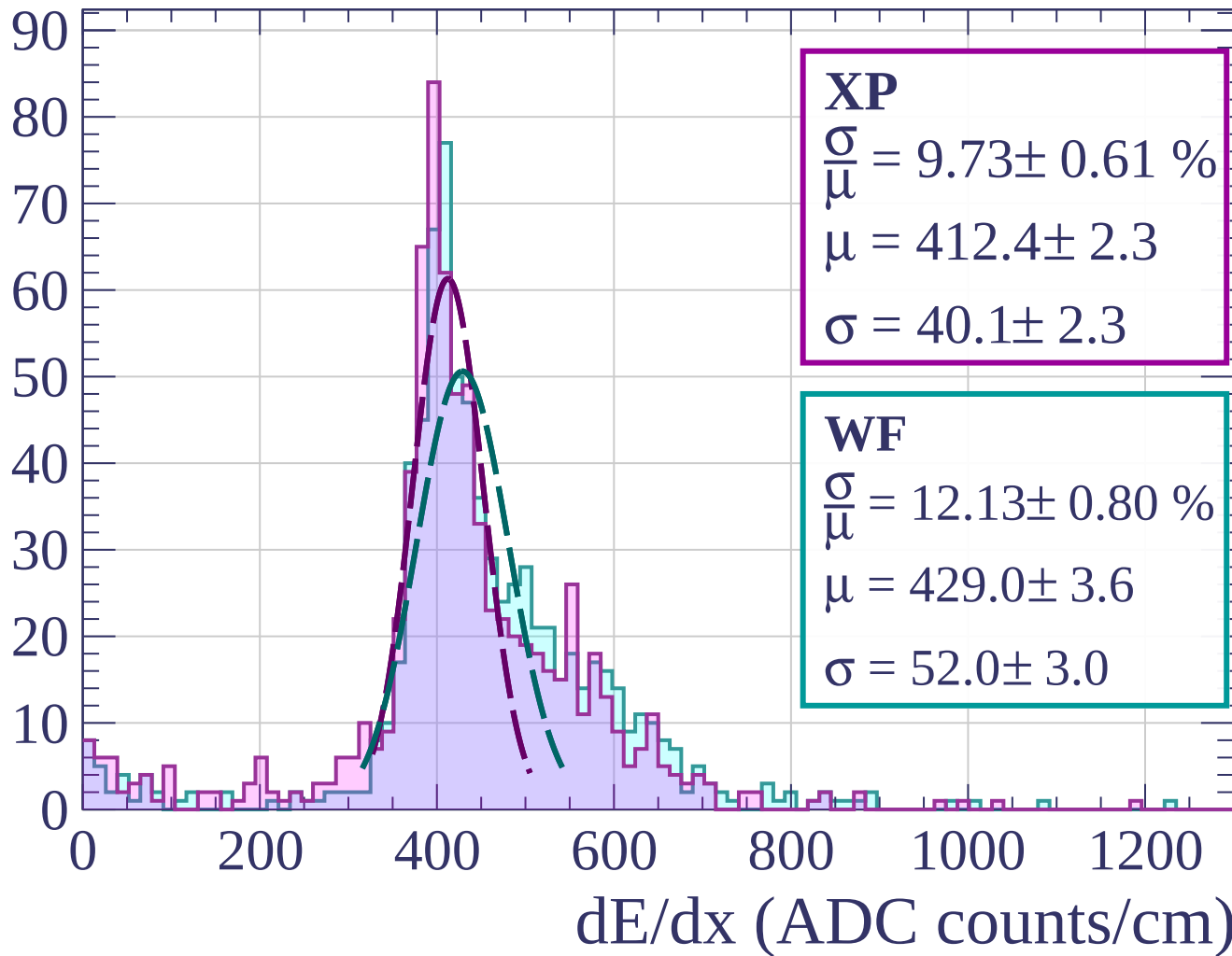
Energy loss | $24 < \phi < 27$

Count



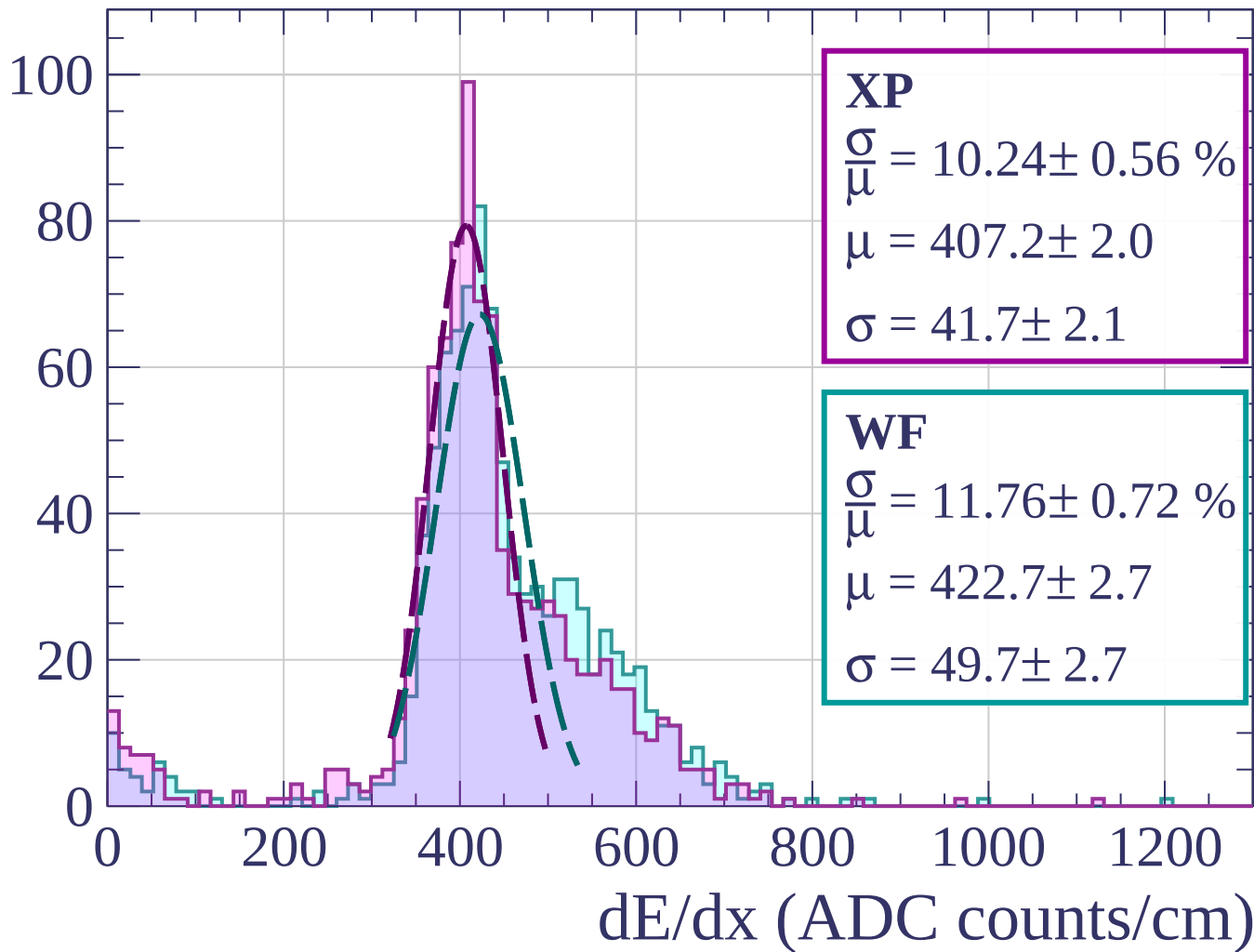
Energy loss | $27 < \phi < 30$

Count



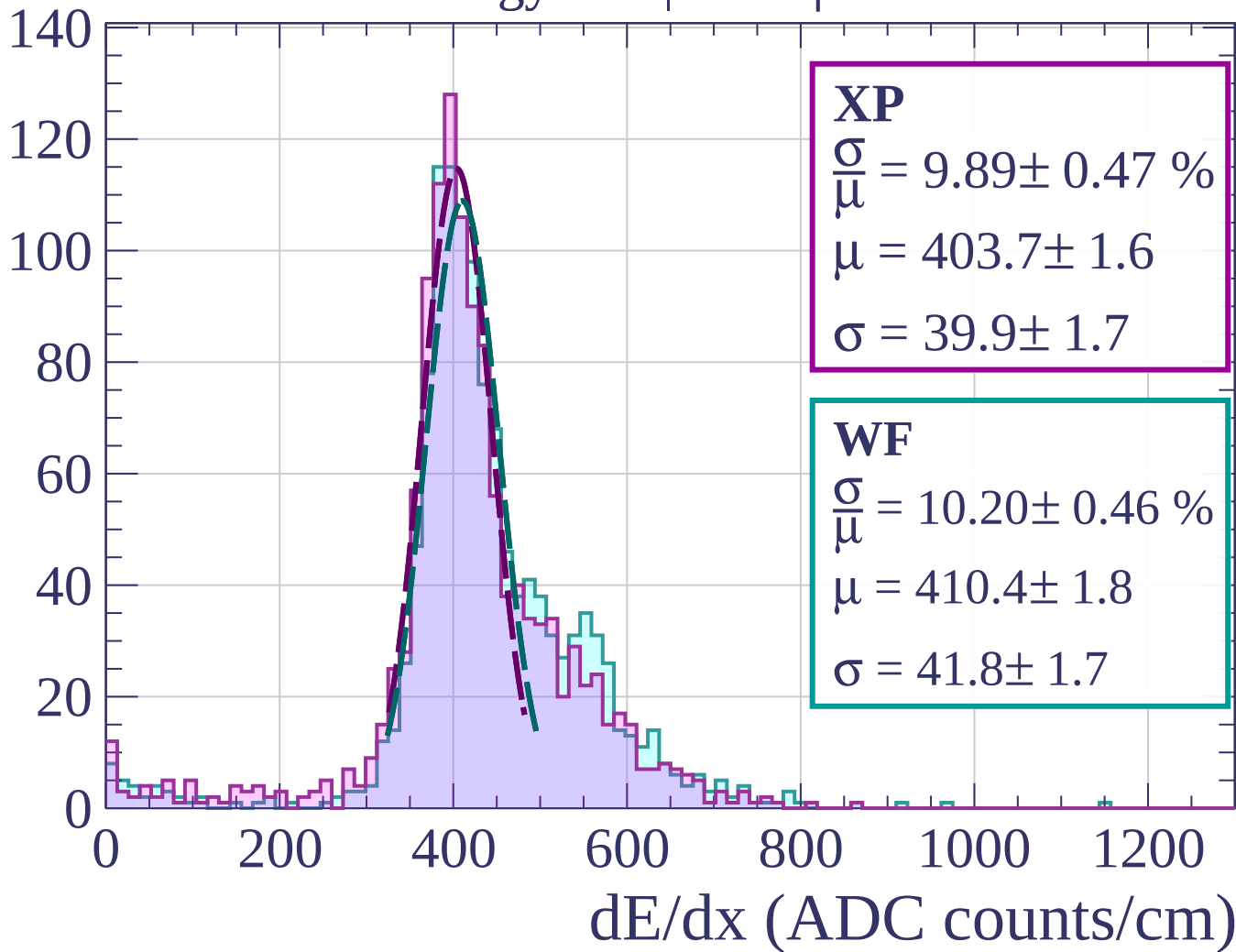
Energy loss | $30 < \phi < 33$

Count



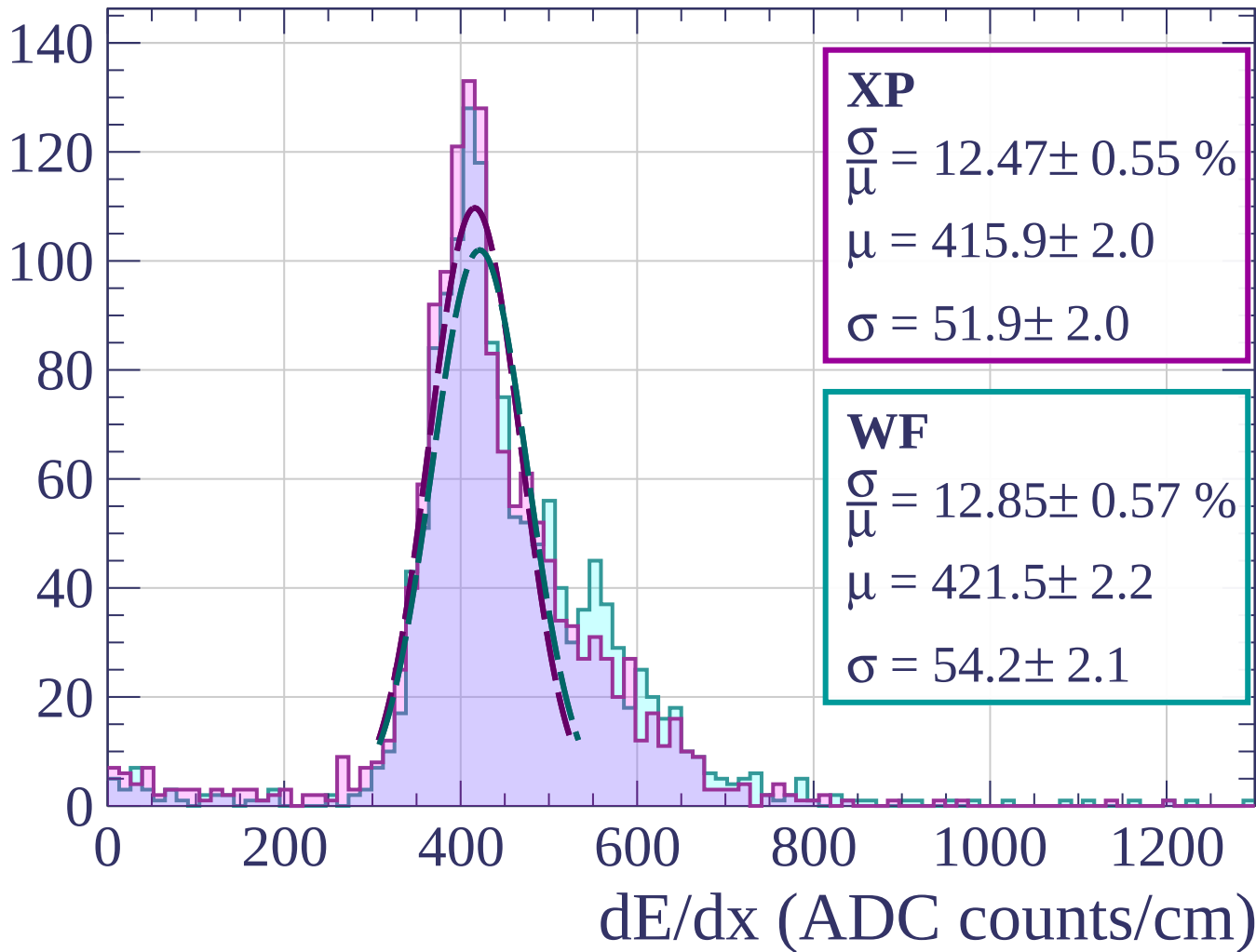
Energy loss | $33 < \phi < 36$

Count



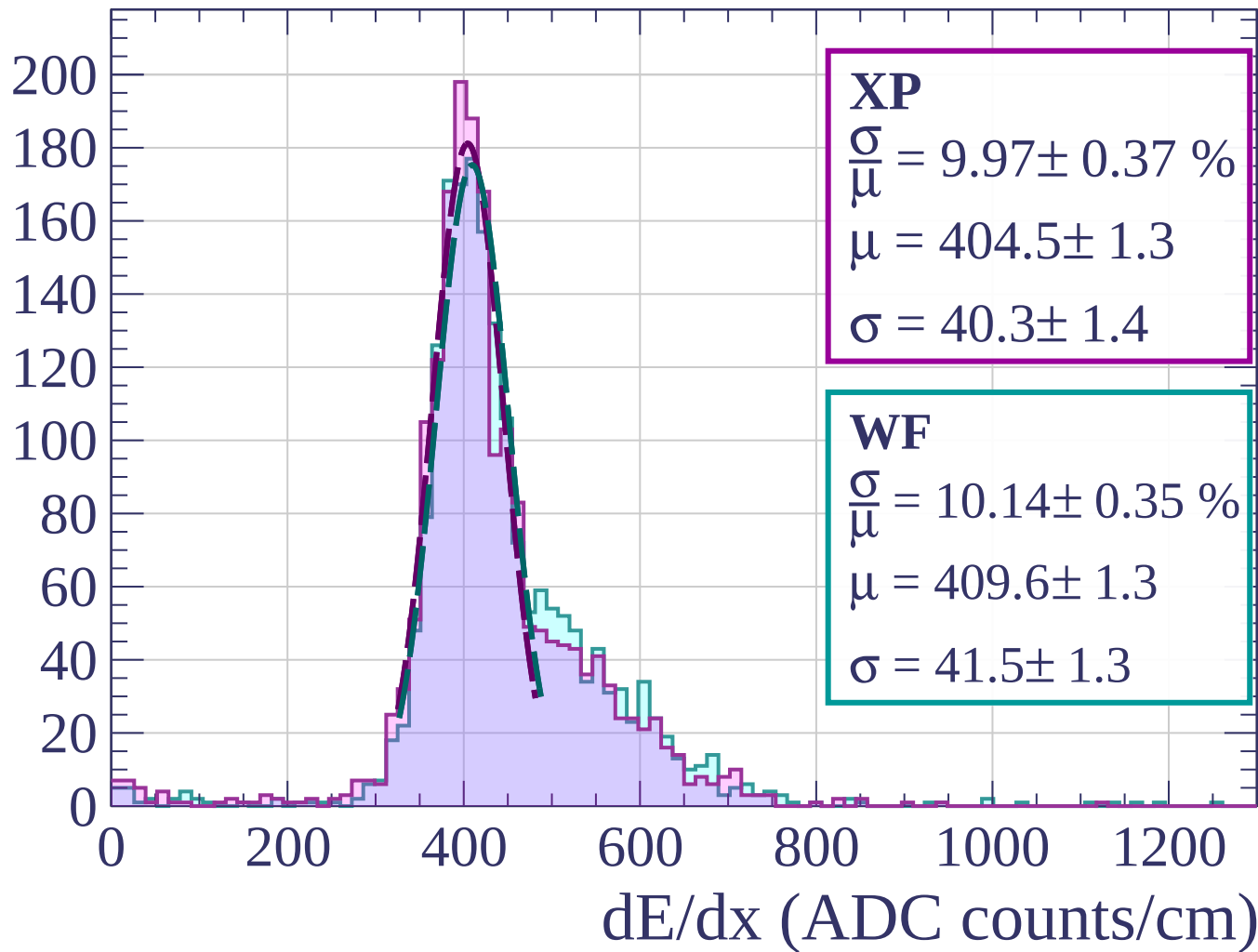
Energy loss | $36 < \phi < 39$

Count



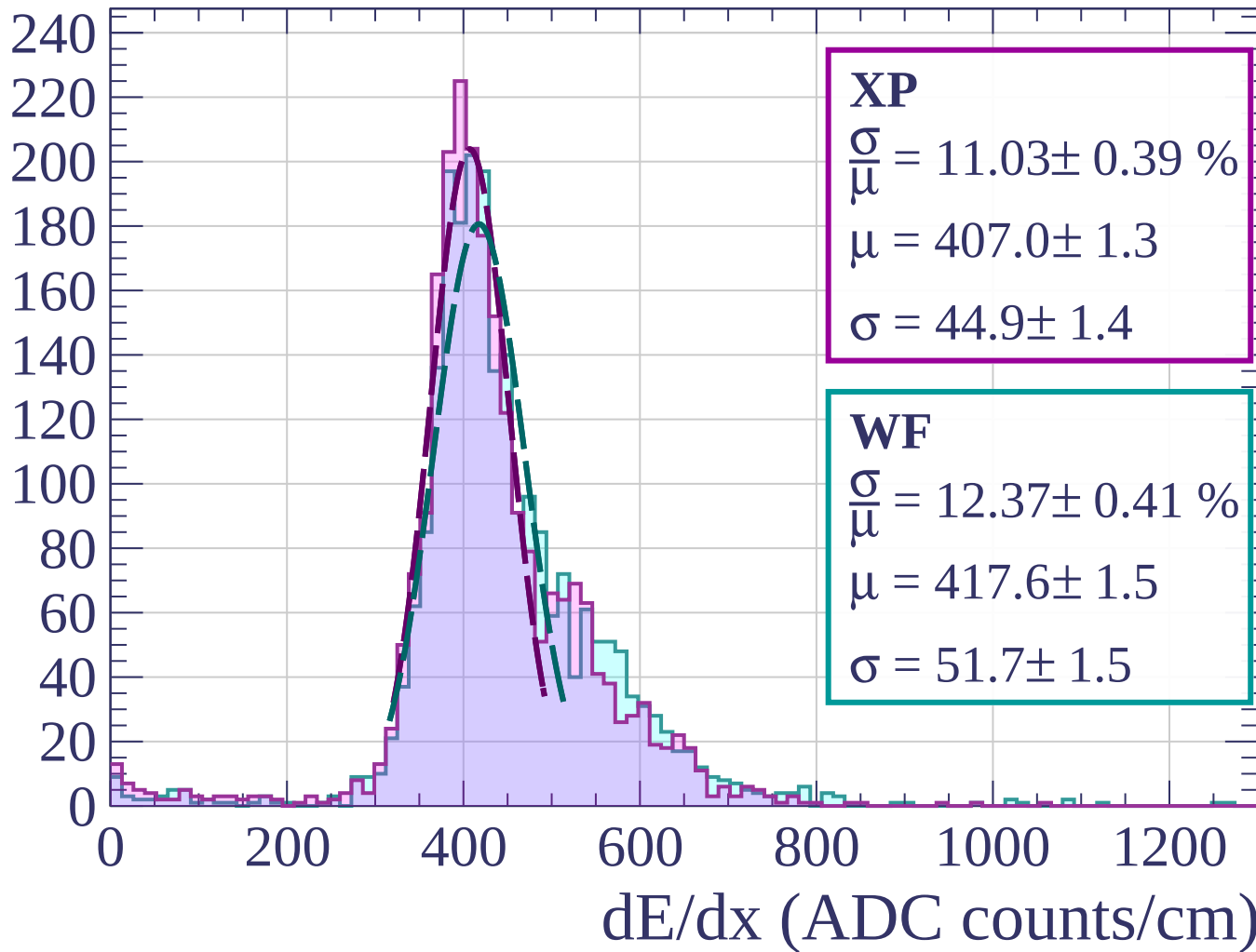
Energy loss | $39 < \phi < 42$

Count



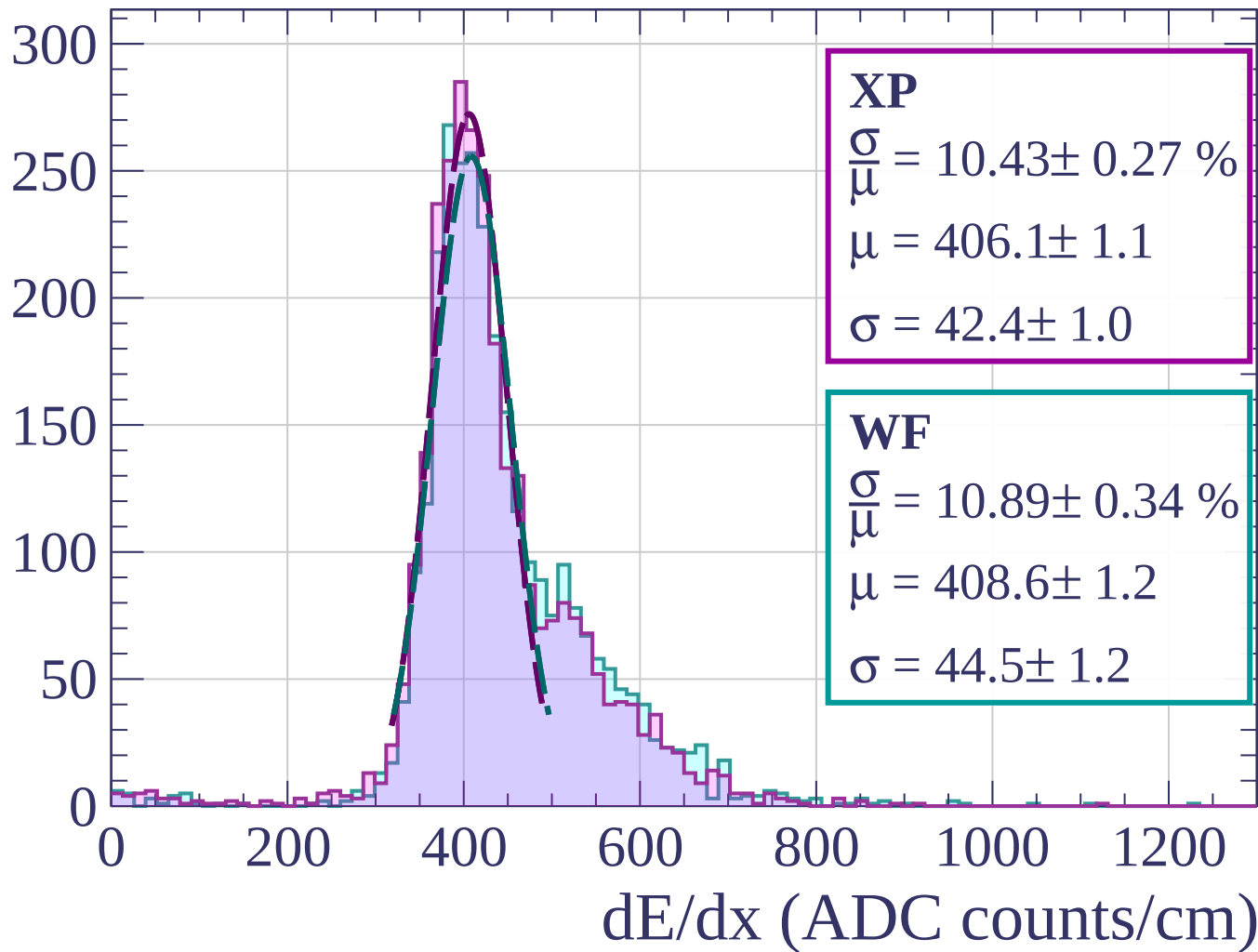
Energy loss | $42 < \phi < 45$

Count



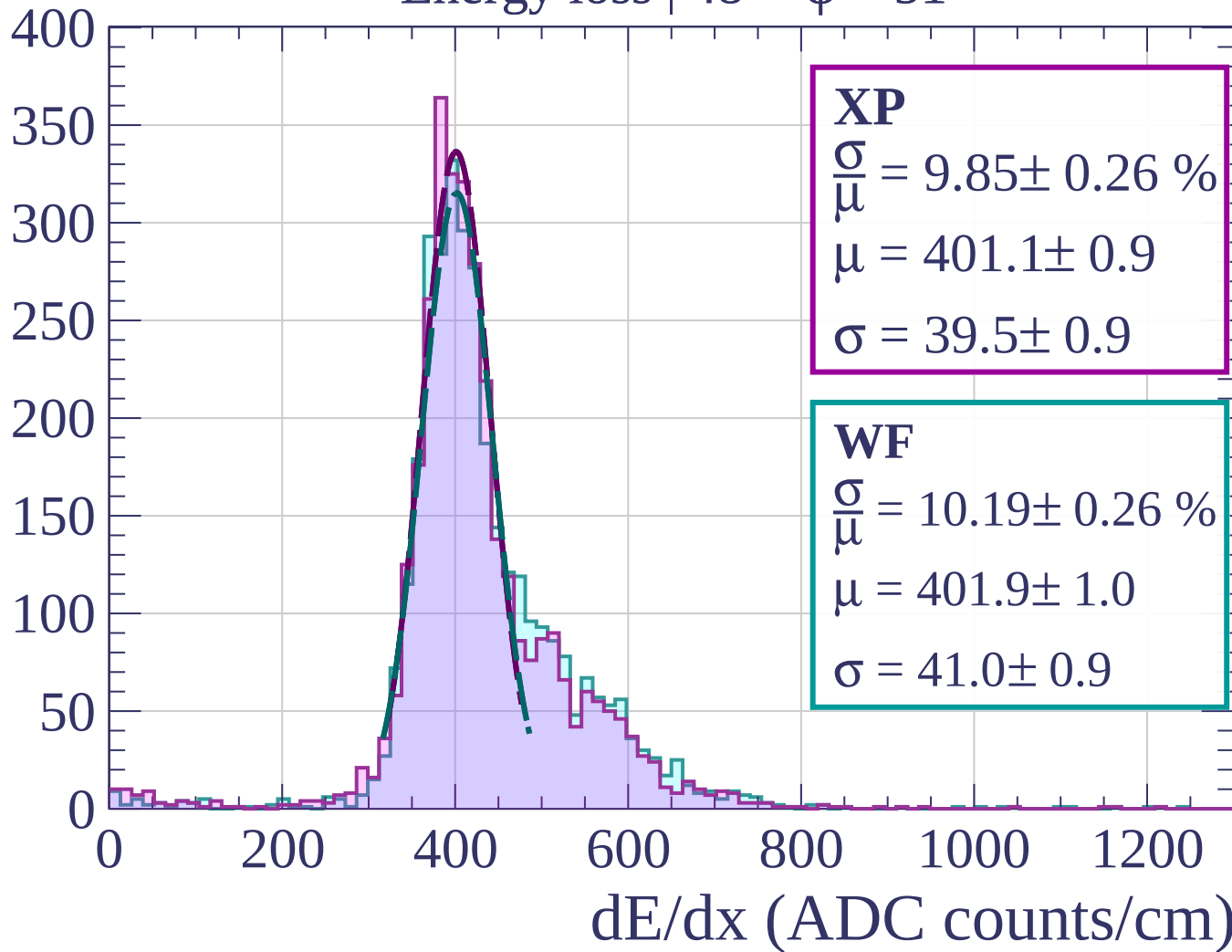
Energy loss | $45 < \phi < 48$

Count



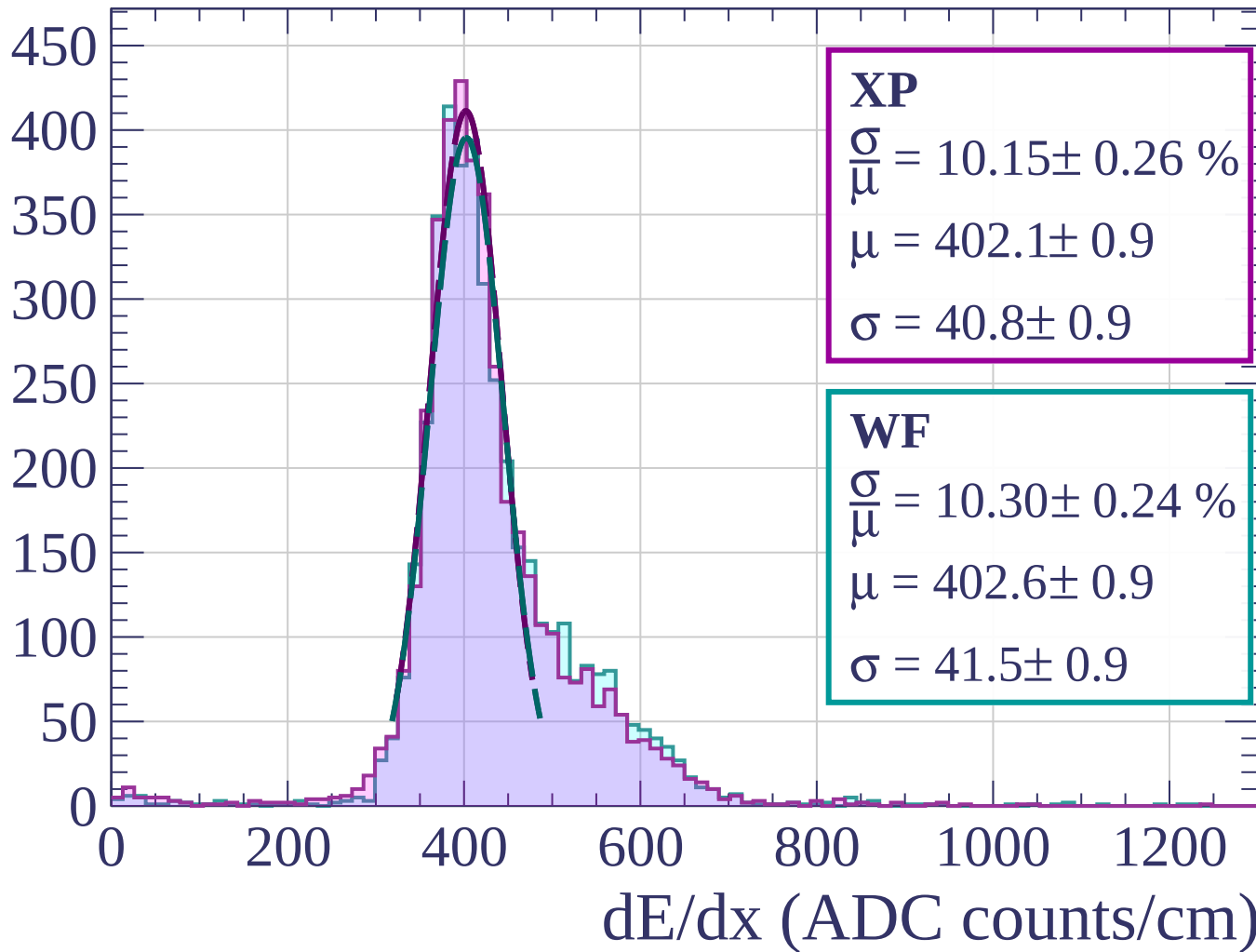
Energy loss | $48 < \phi < 51$

Count



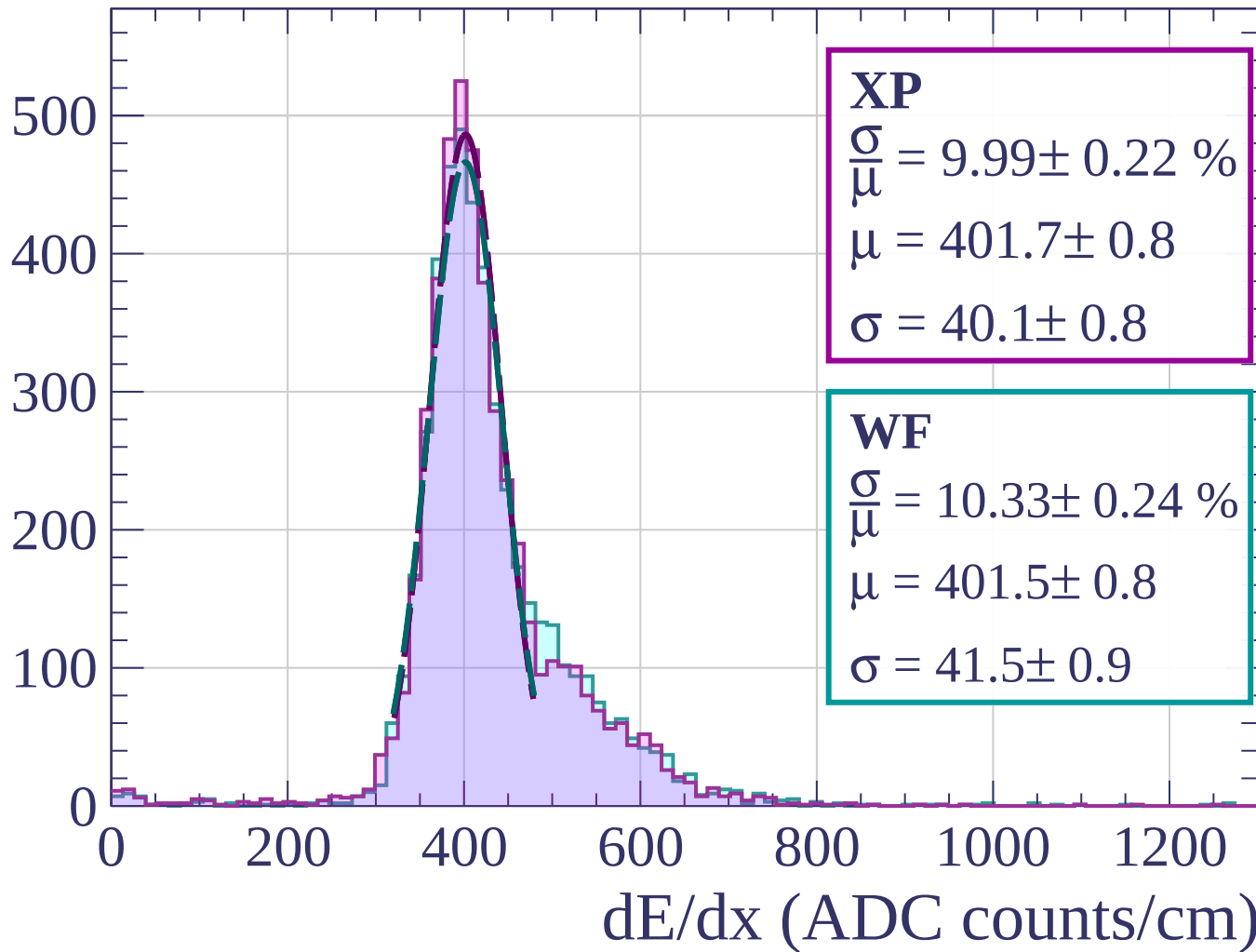
Energy loss | $51 < \phi < 54$

Count



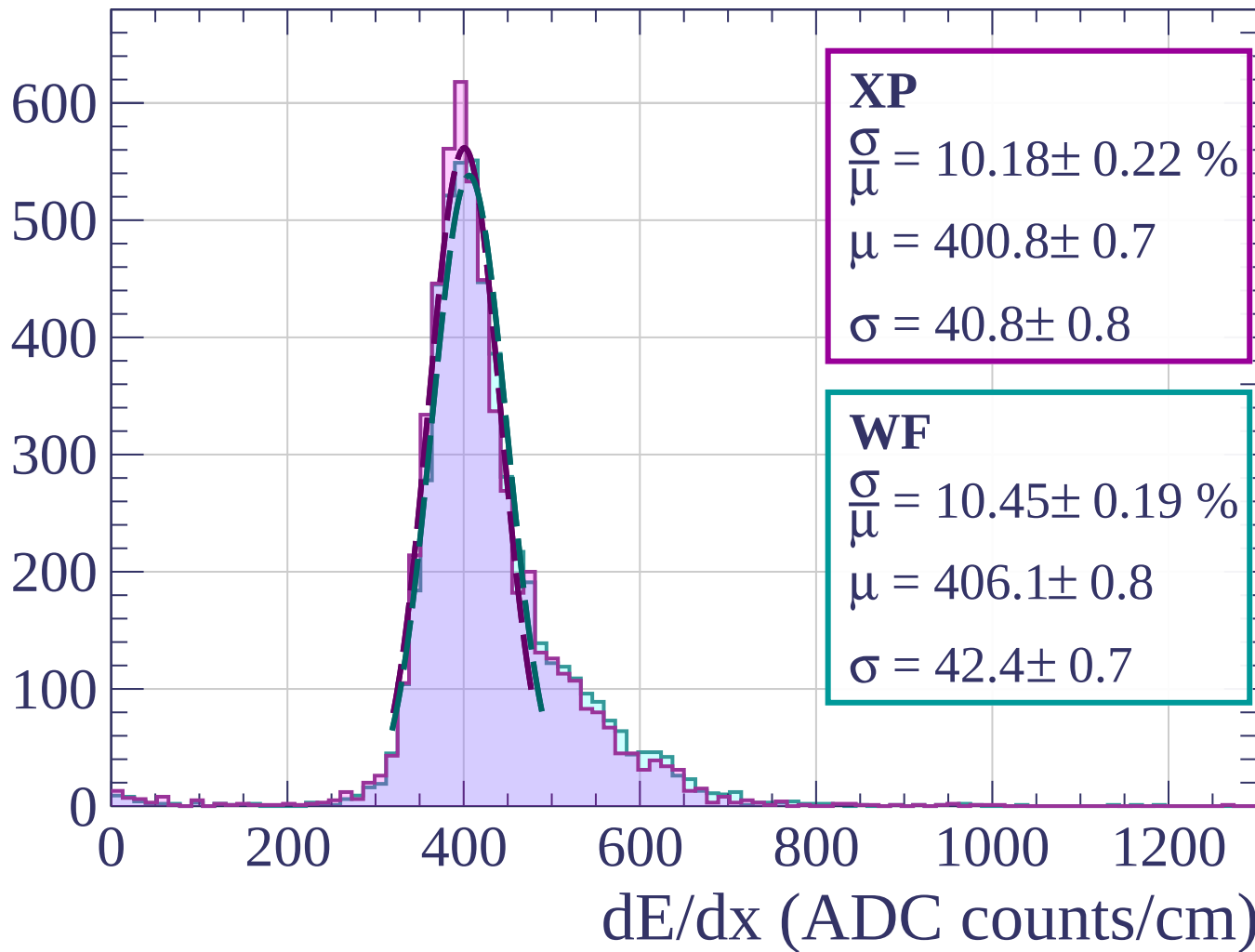
Energy loss | $54 < \phi < 57$

Count



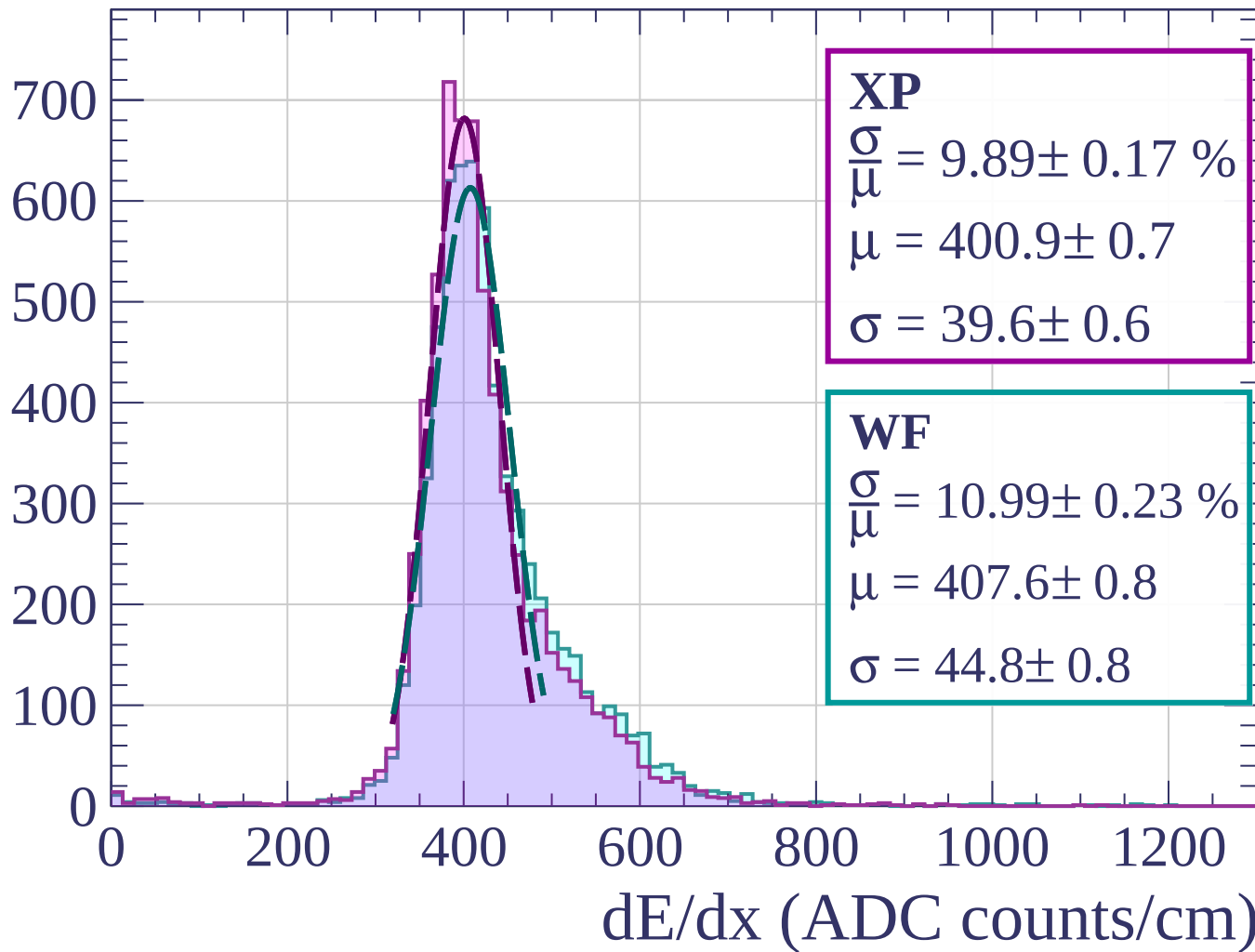
Energy loss | $57 < \phi < 60$

Count



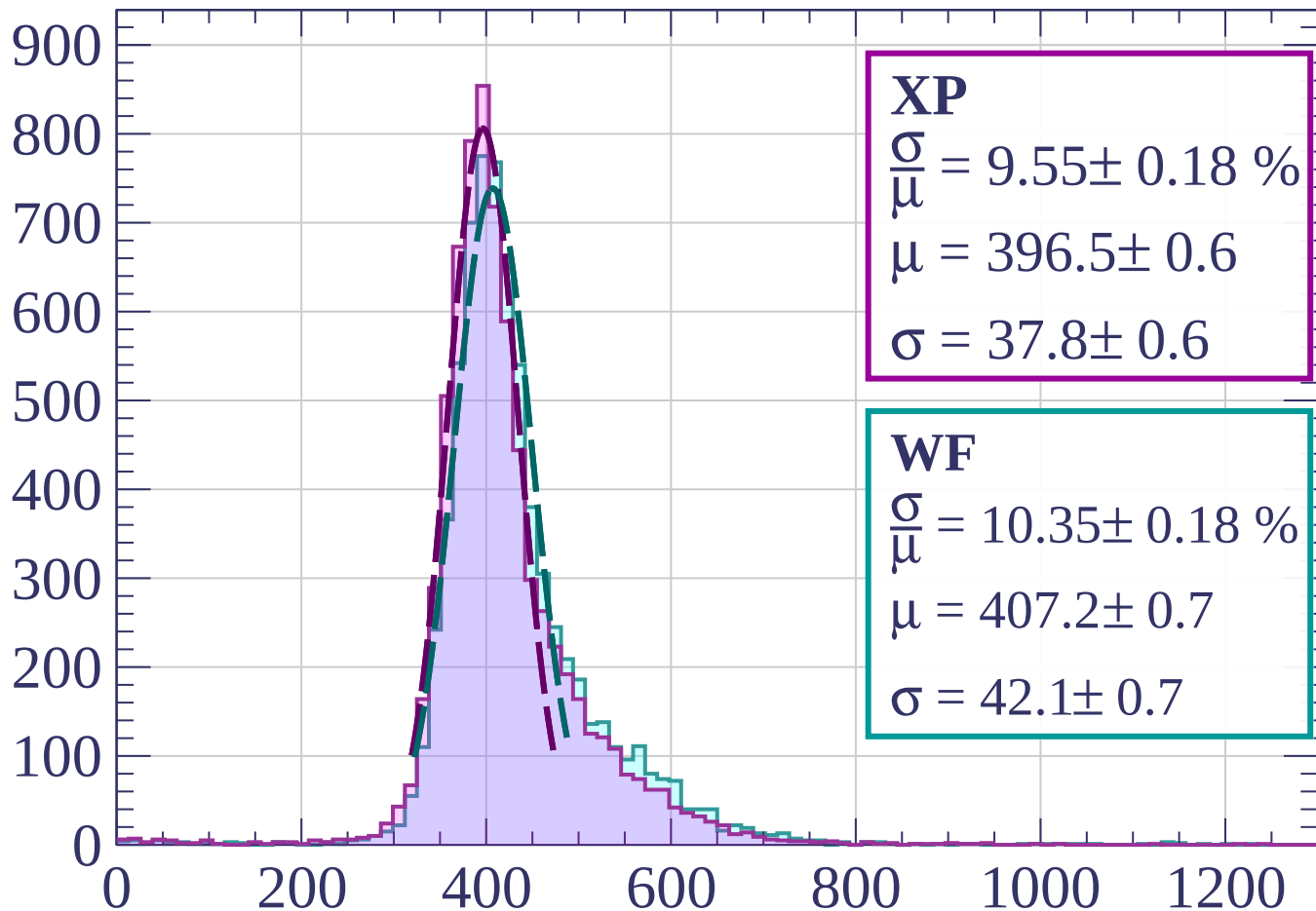
Energy loss | $60 < \phi < 63$

Count



Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 9.55 \pm 0.18 \%$$

$$\mu = 396.5 \pm 0.6$$

$$\sigma = 37.8 \pm 0.6$$

WF

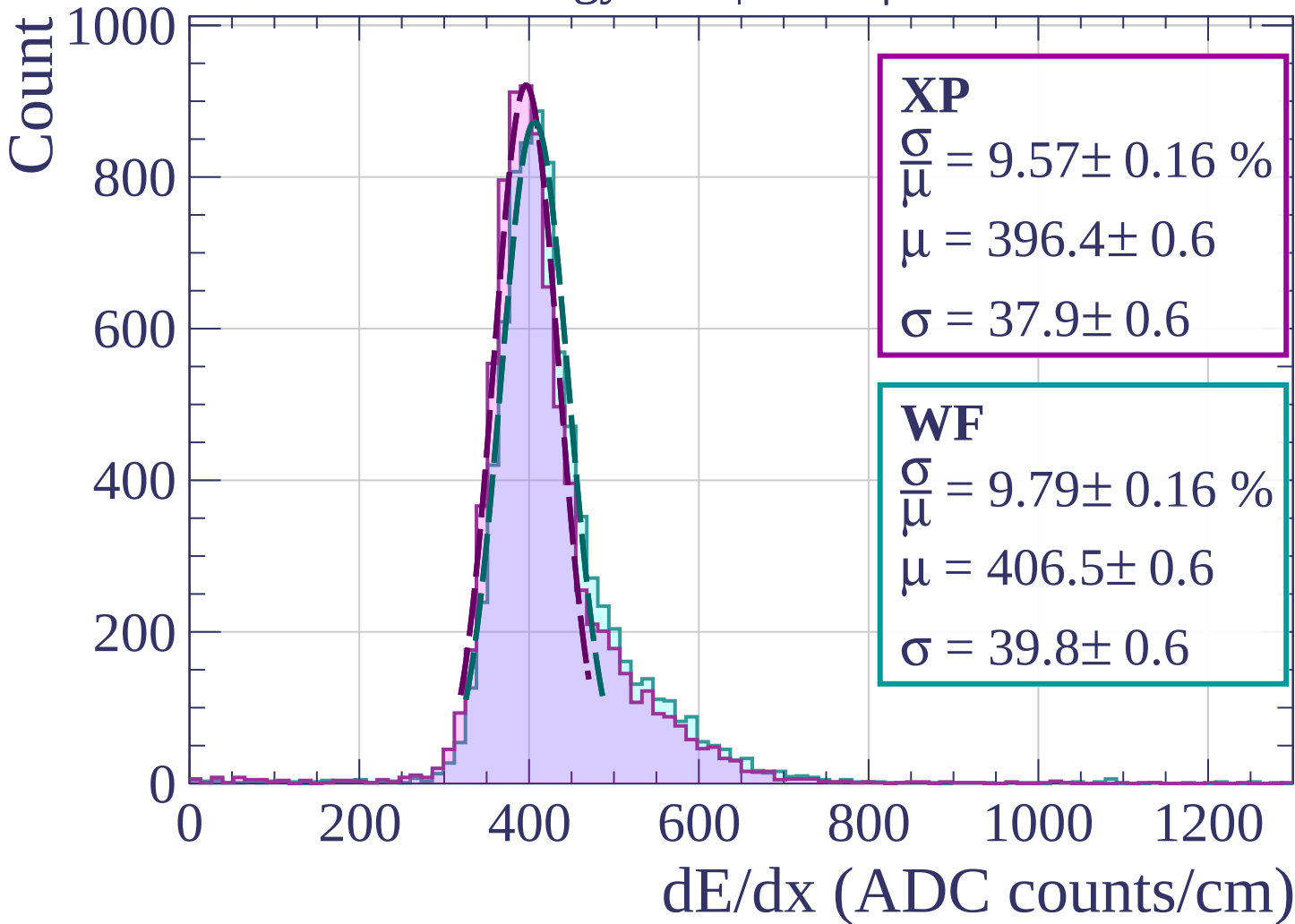
$$\frac{\sigma}{\mu} = 10.35 \pm 0.18 \%$$

$$\mu = 407.2 \pm 0.7$$

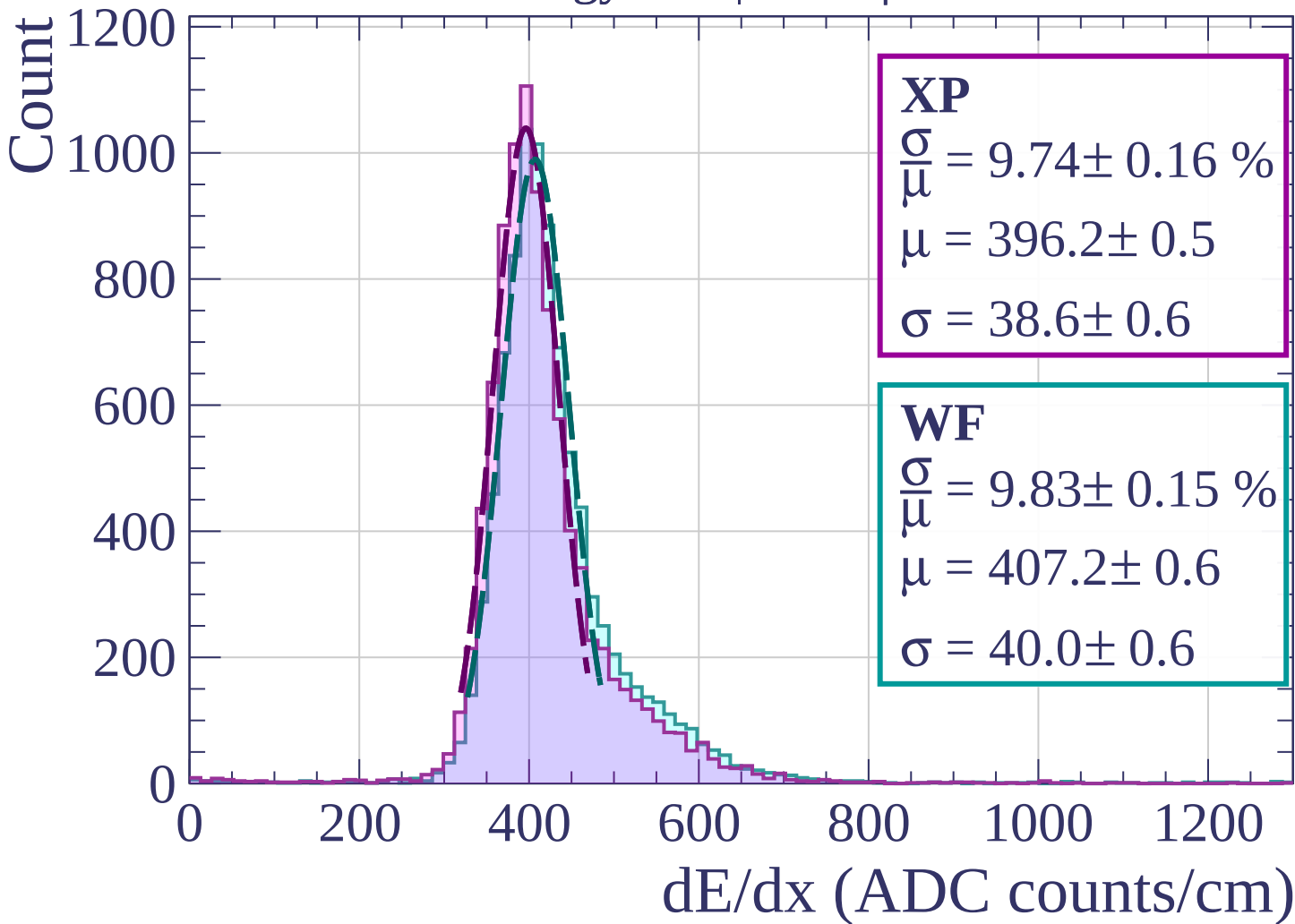
$$\sigma = 42.1 \pm 0.7$$

dE/dx (ADC counts/cm)

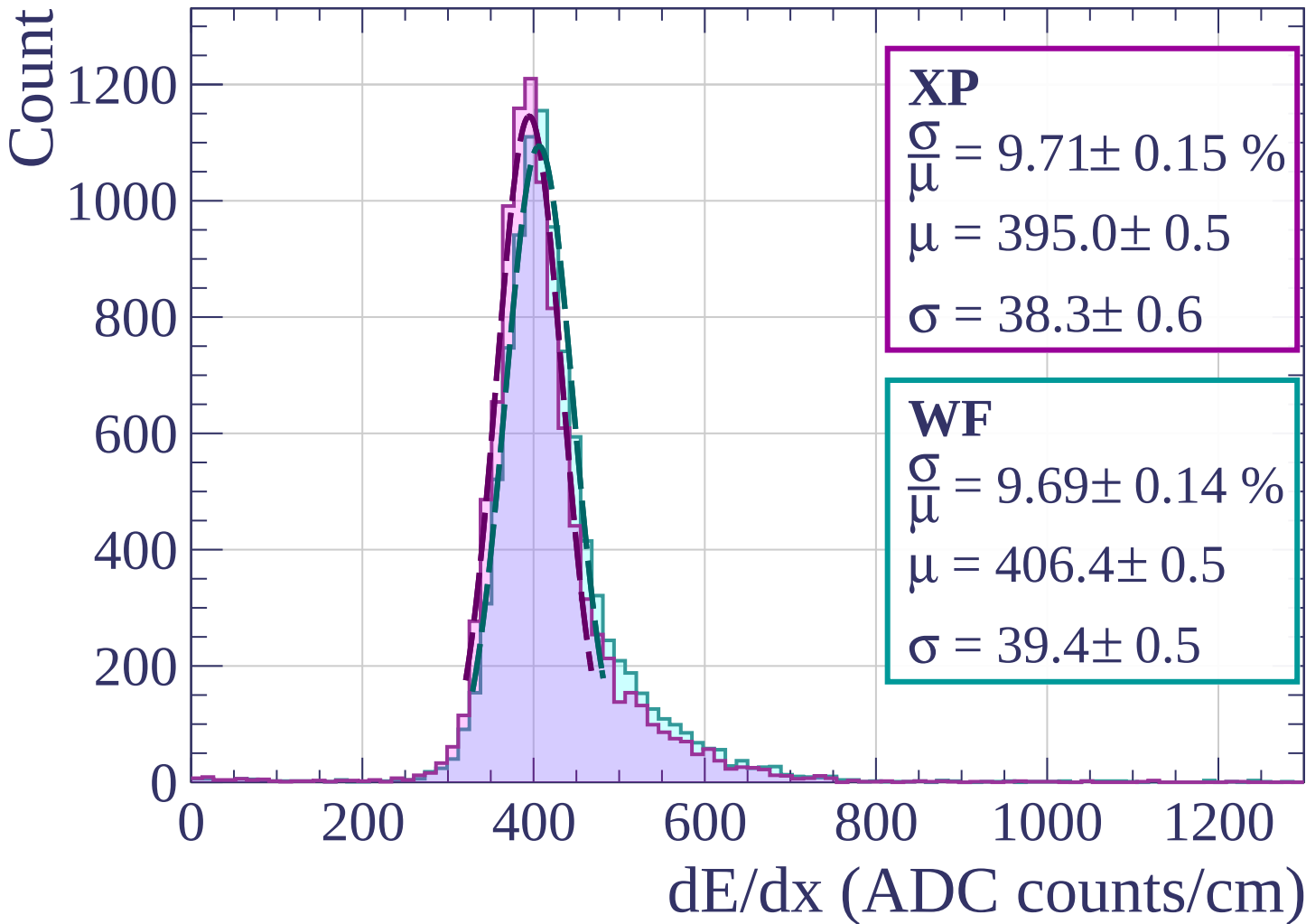
Energy loss | $66 < \phi < 69$



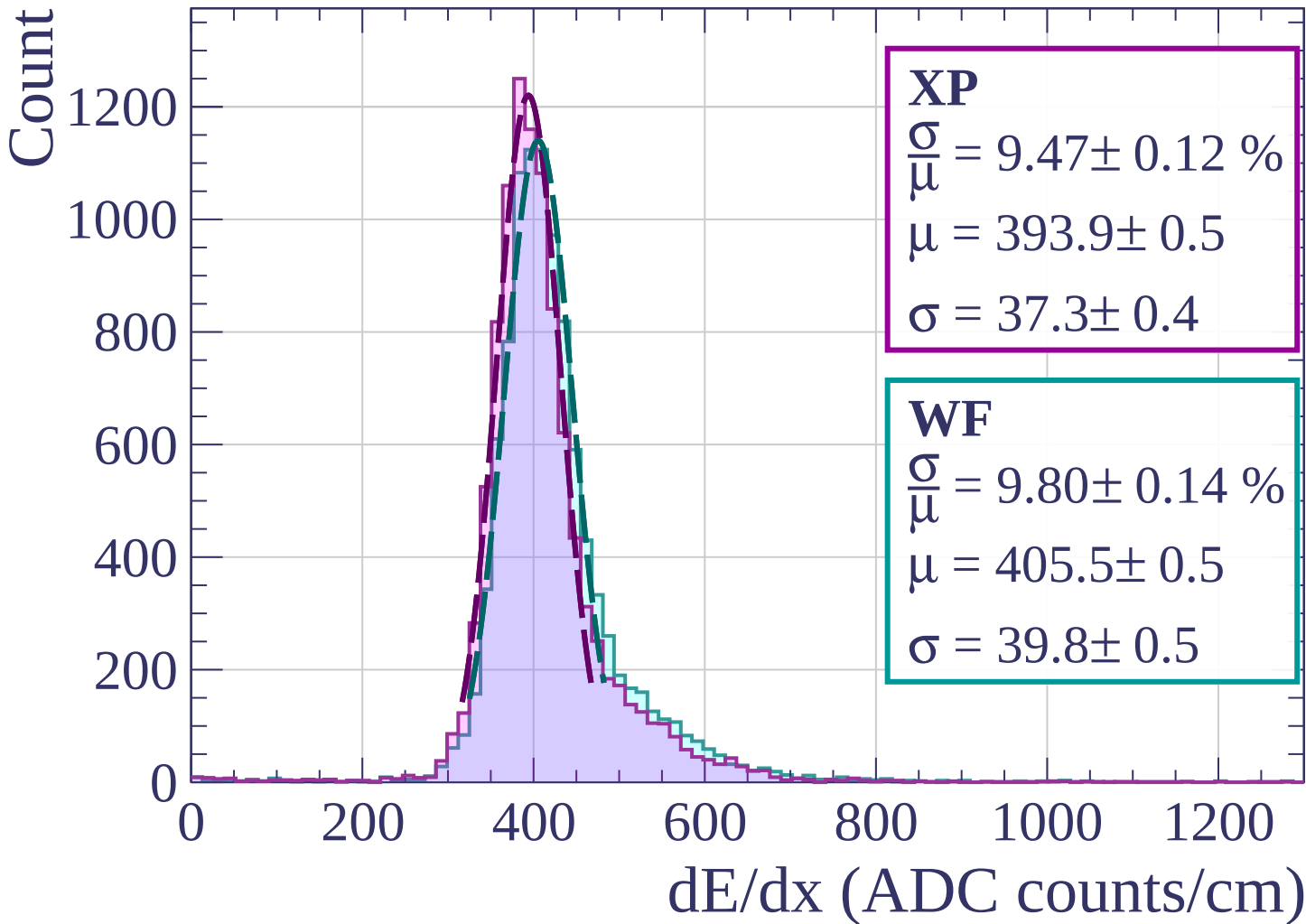
Energy loss | $69 < \phi < 72$



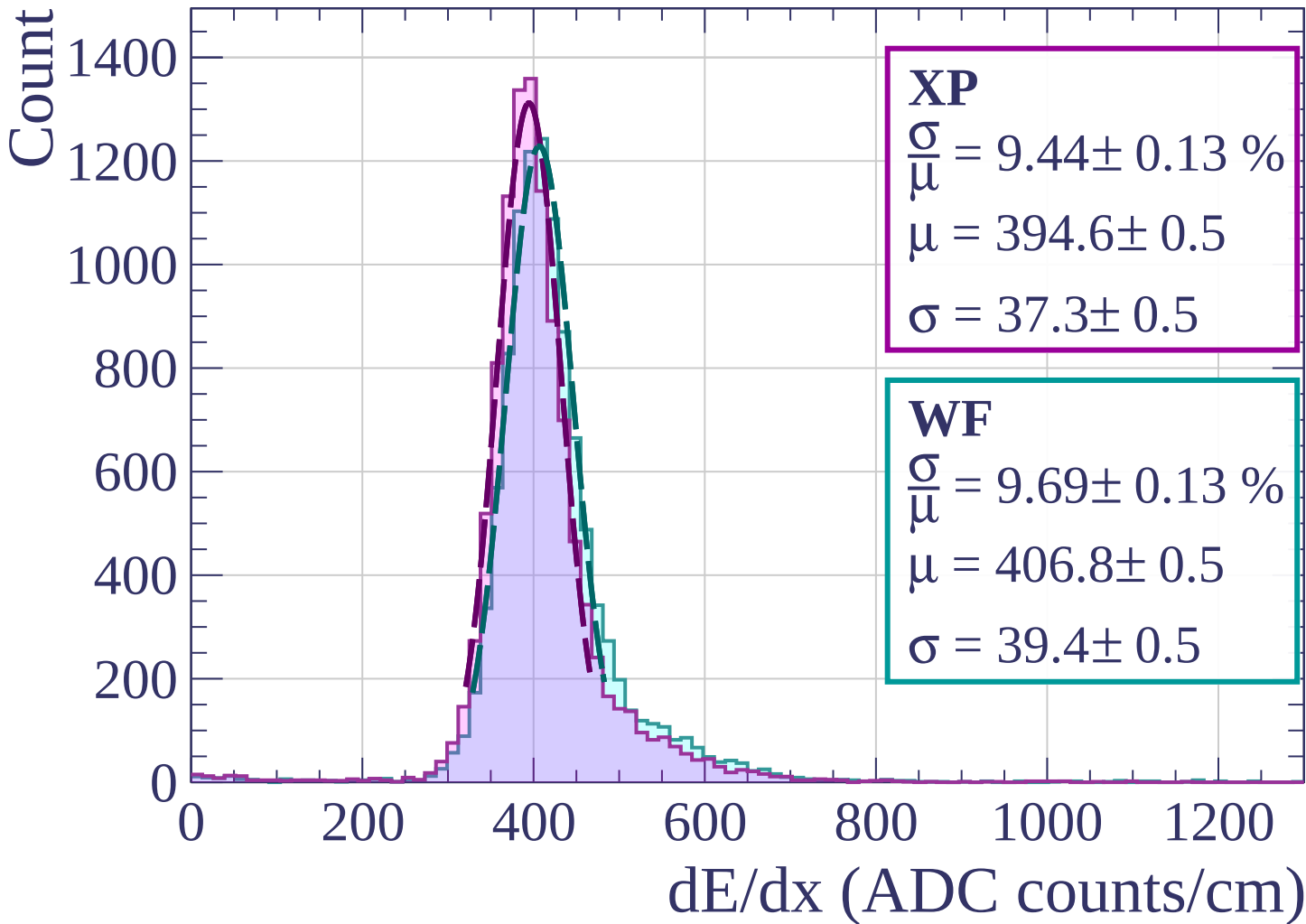
Energy loss | $72 < \phi < 75$



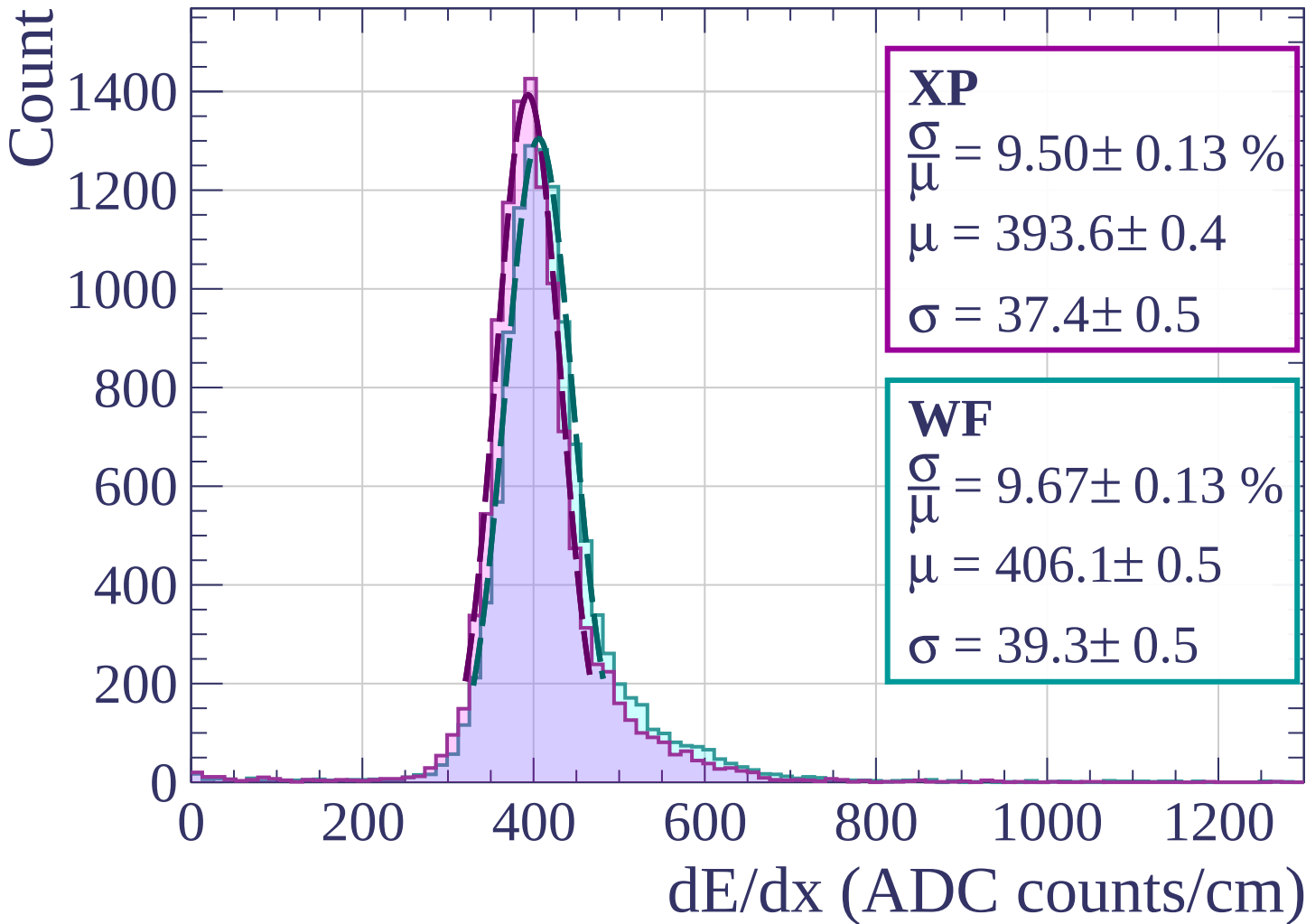
Energy loss | $75 < \phi < 78$



Energy loss | $78 < \phi < 81$

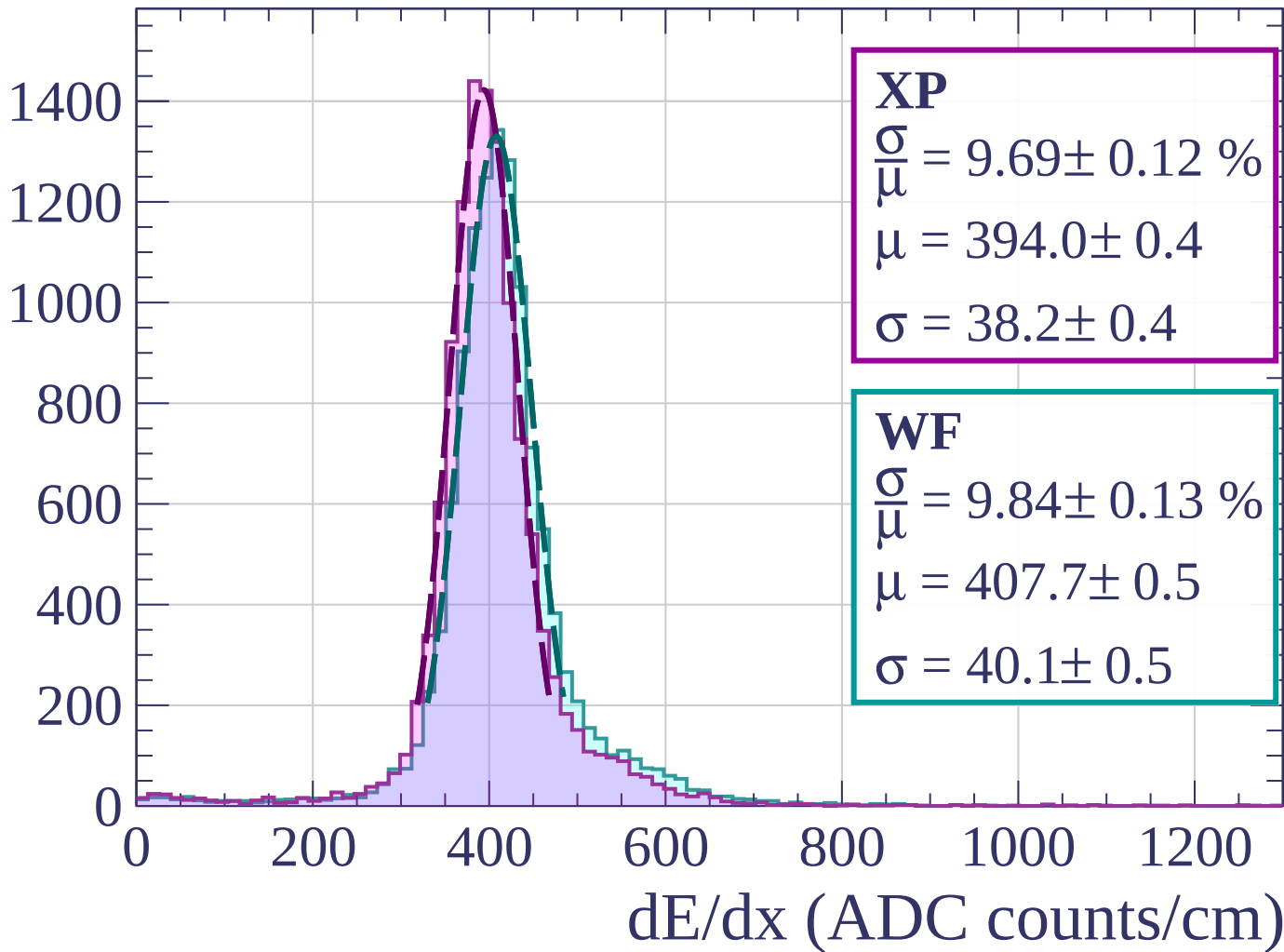


Energy loss | $81 < \phi < 84$

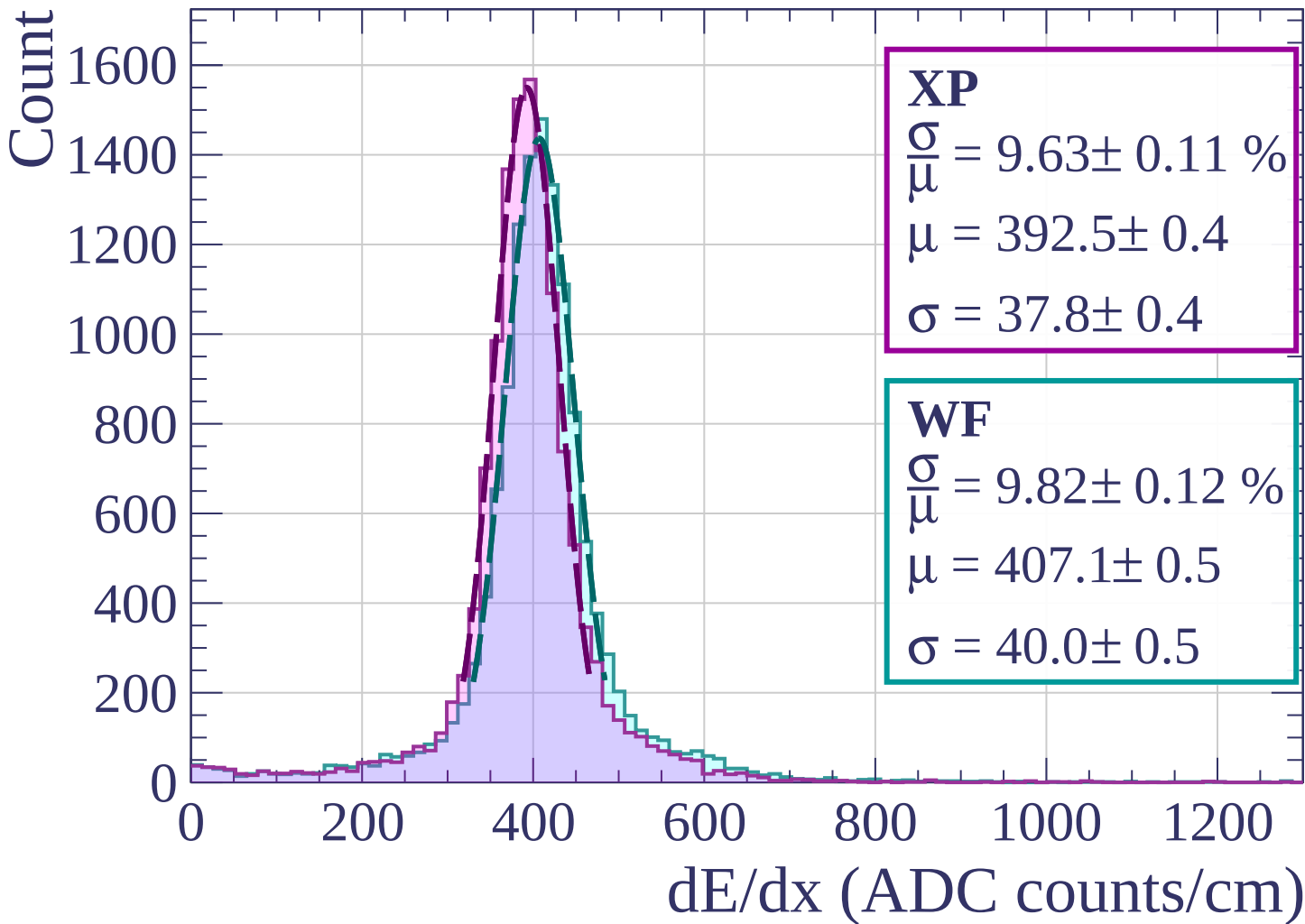


Energy loss | $84 < \phi < 87$

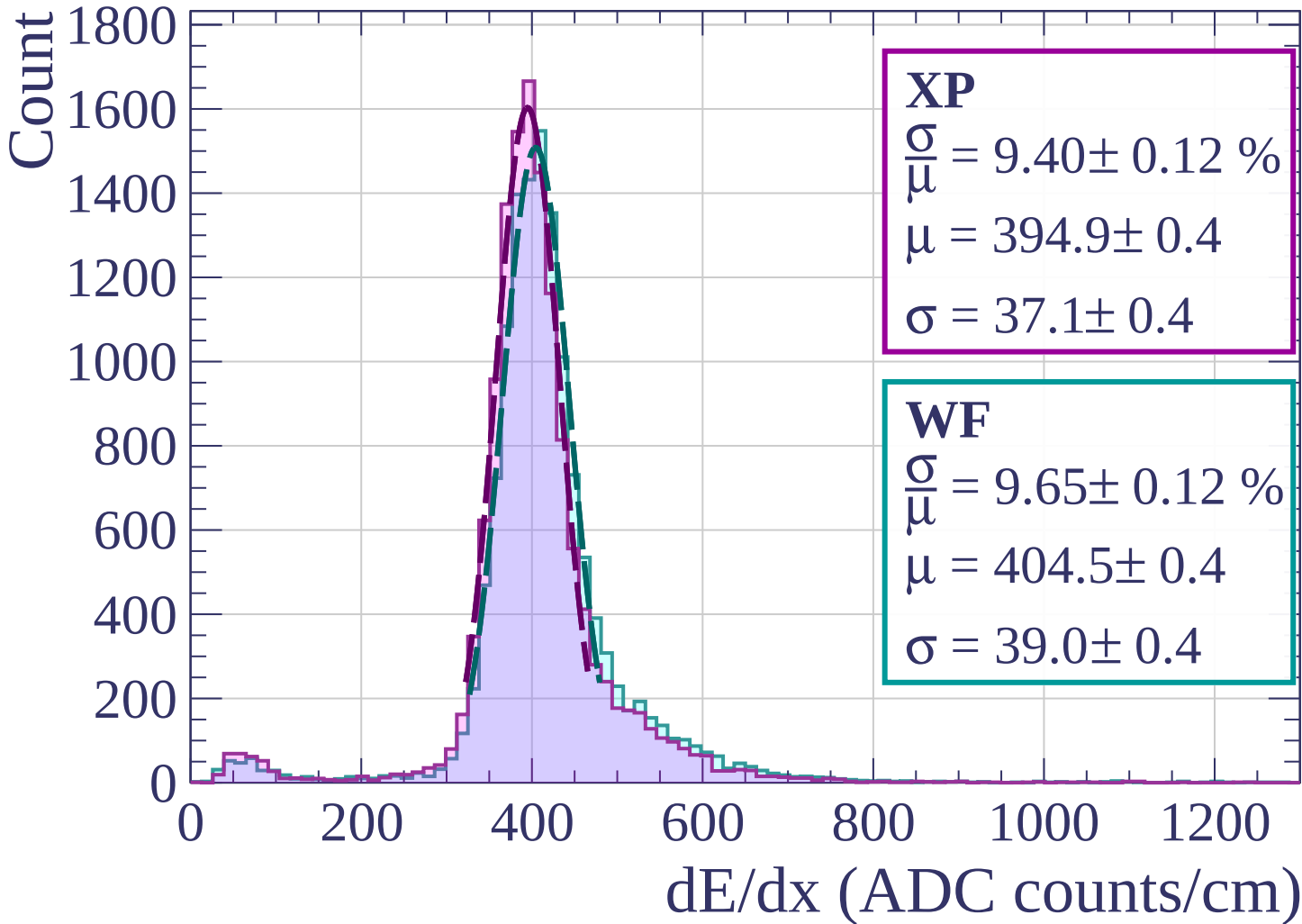
Count



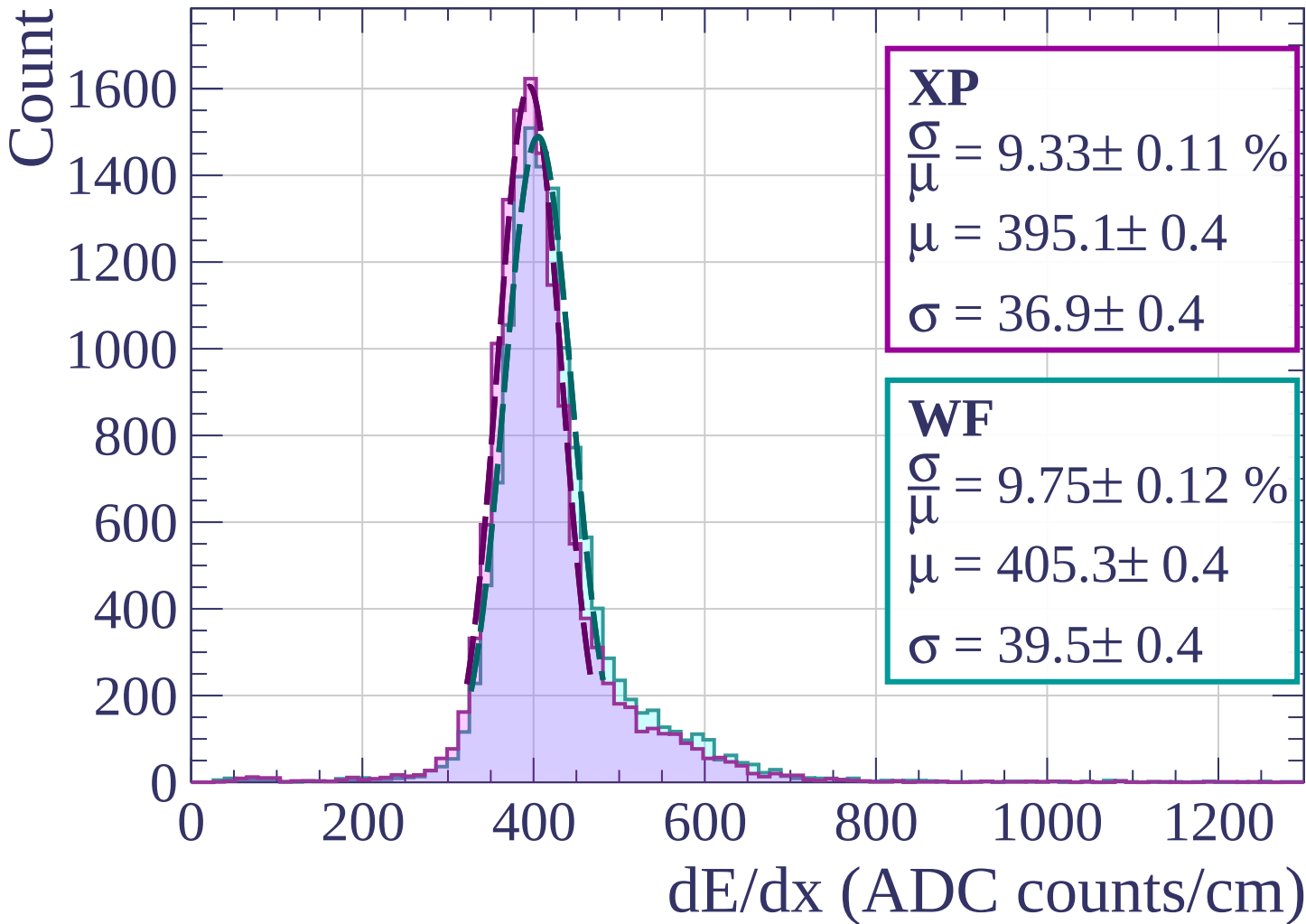
Energy loss | $87 < \phi < 90$



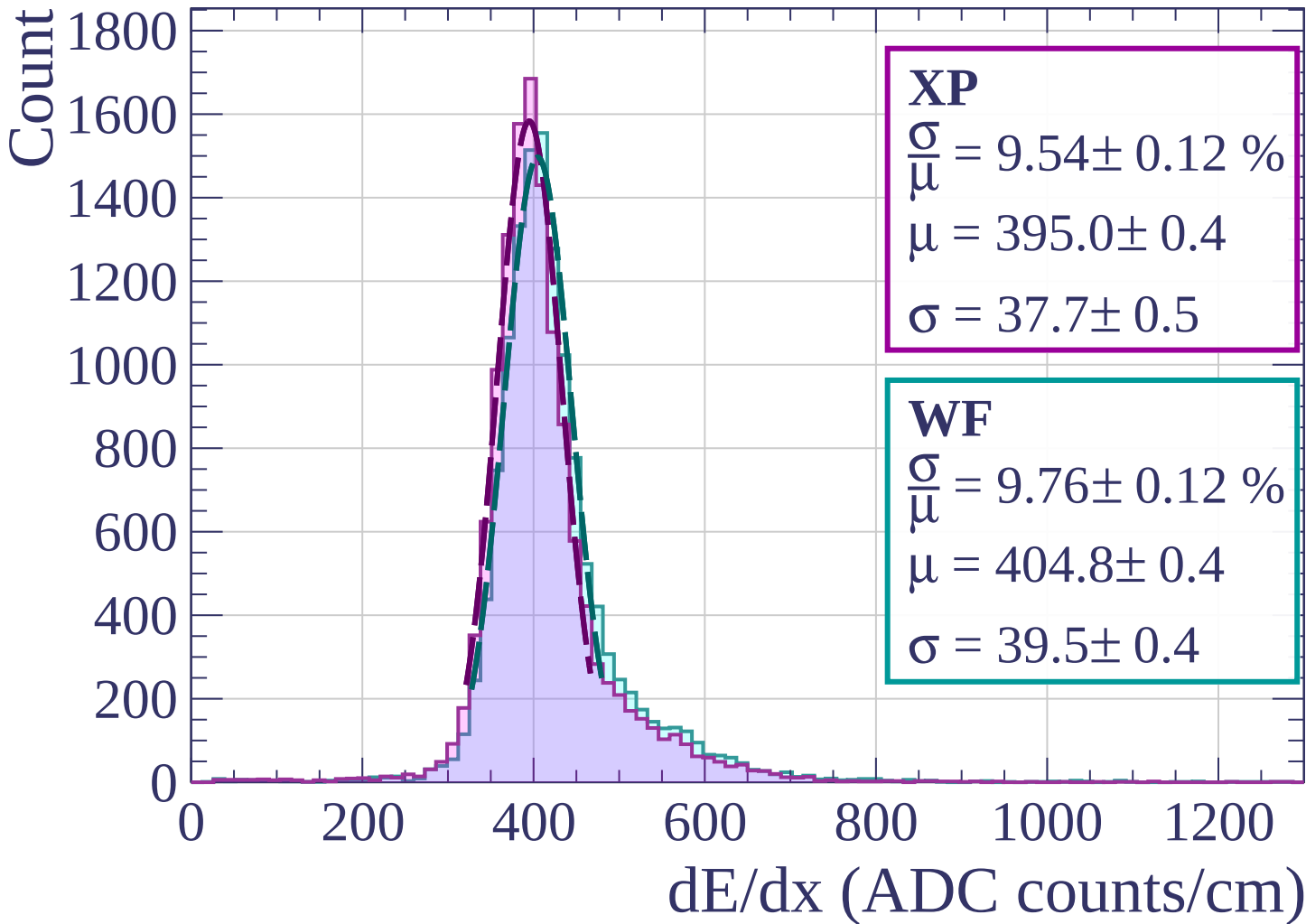
Energy loss | $0 < \theta < 3$



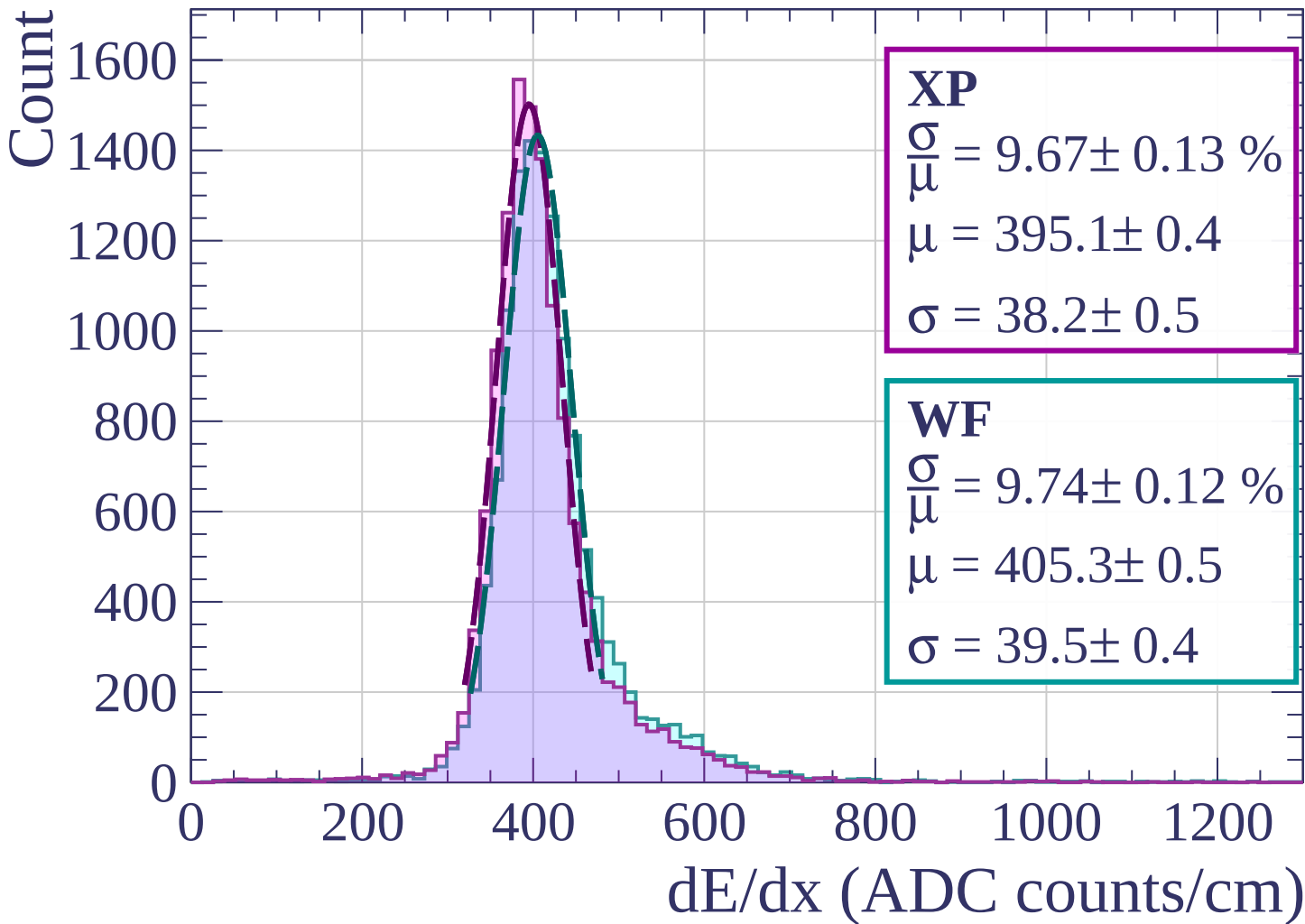
Energy loss | $3 < \theta < 6$



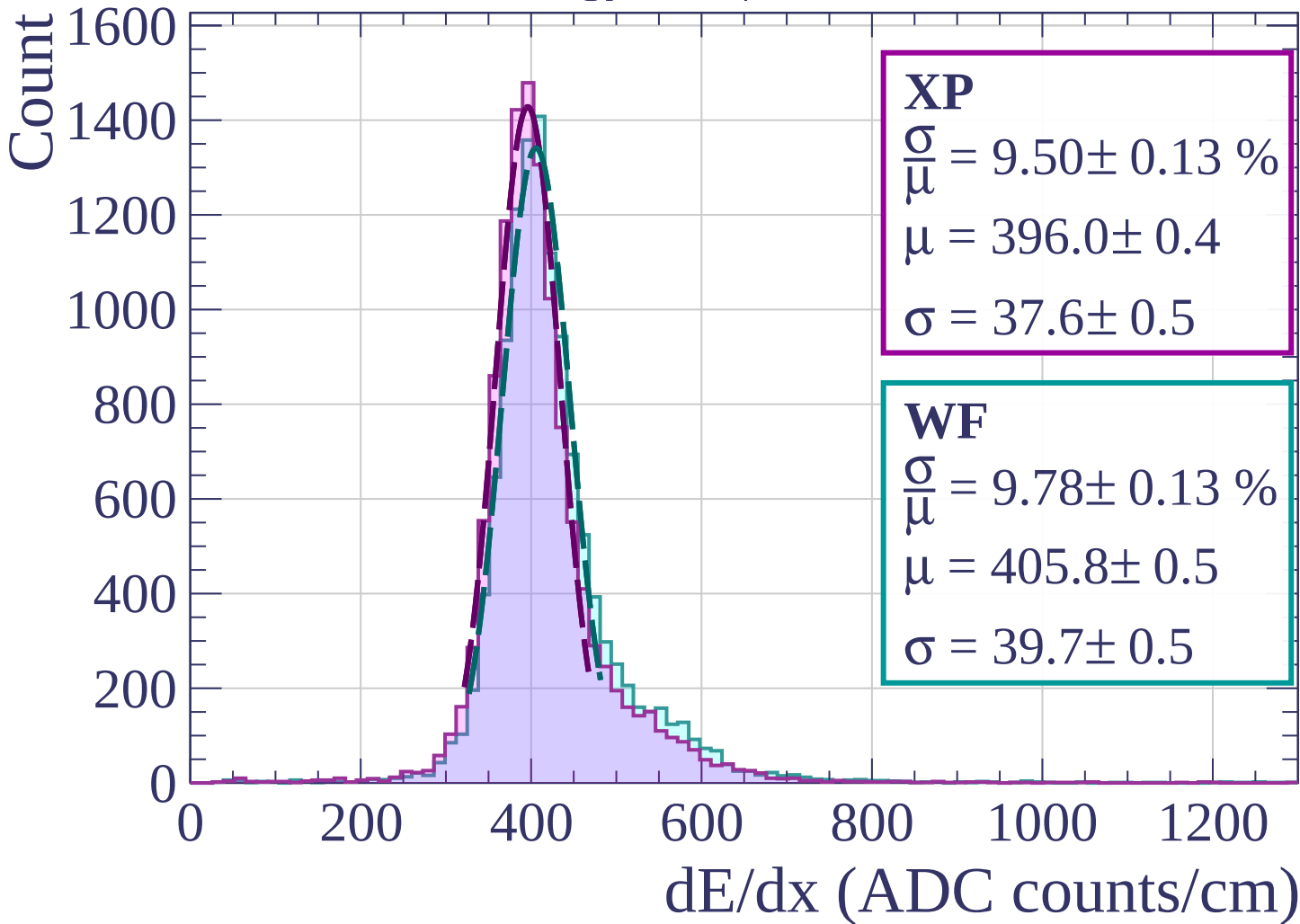
Energy loss | $6 < \theta < 9$



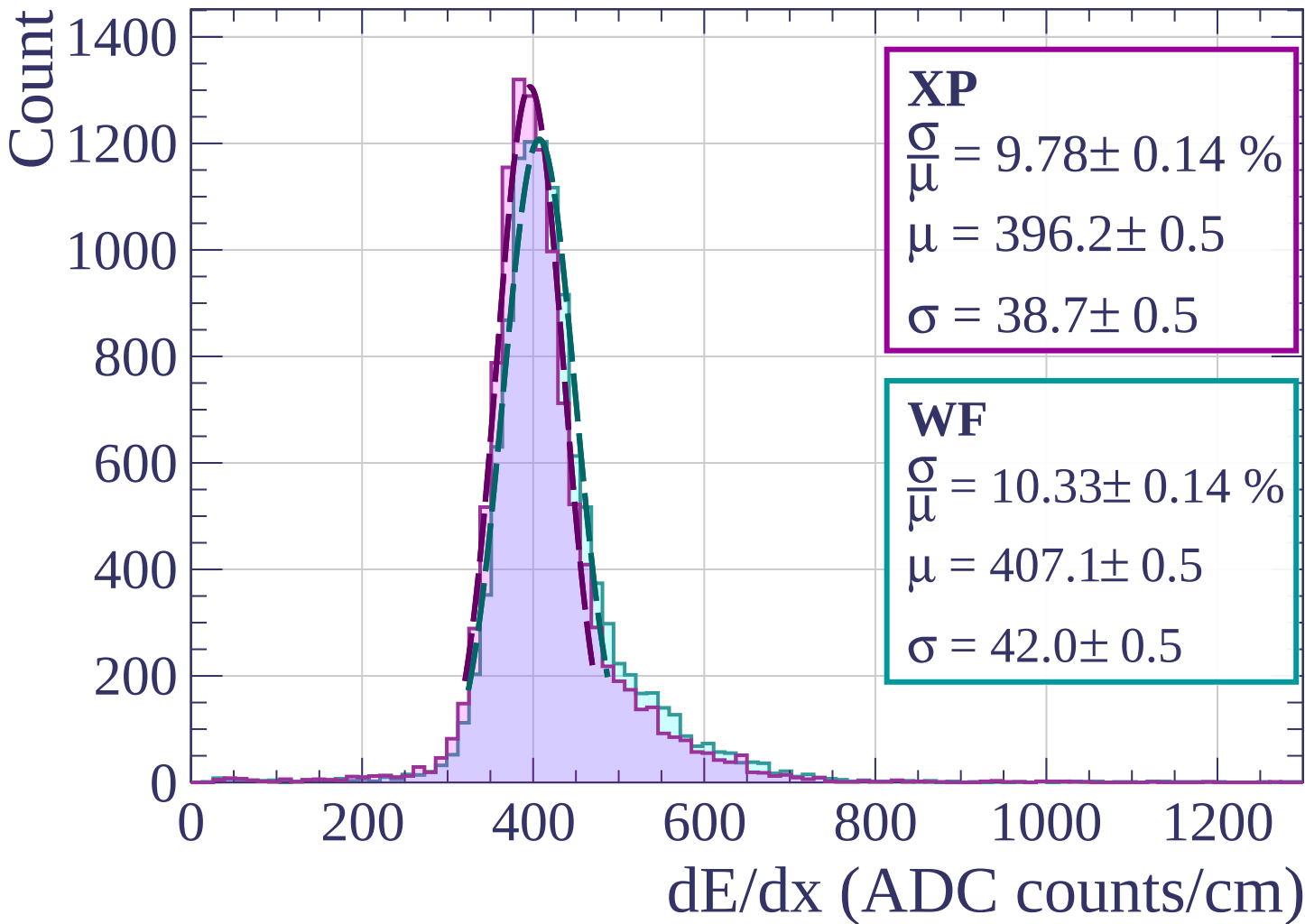
Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

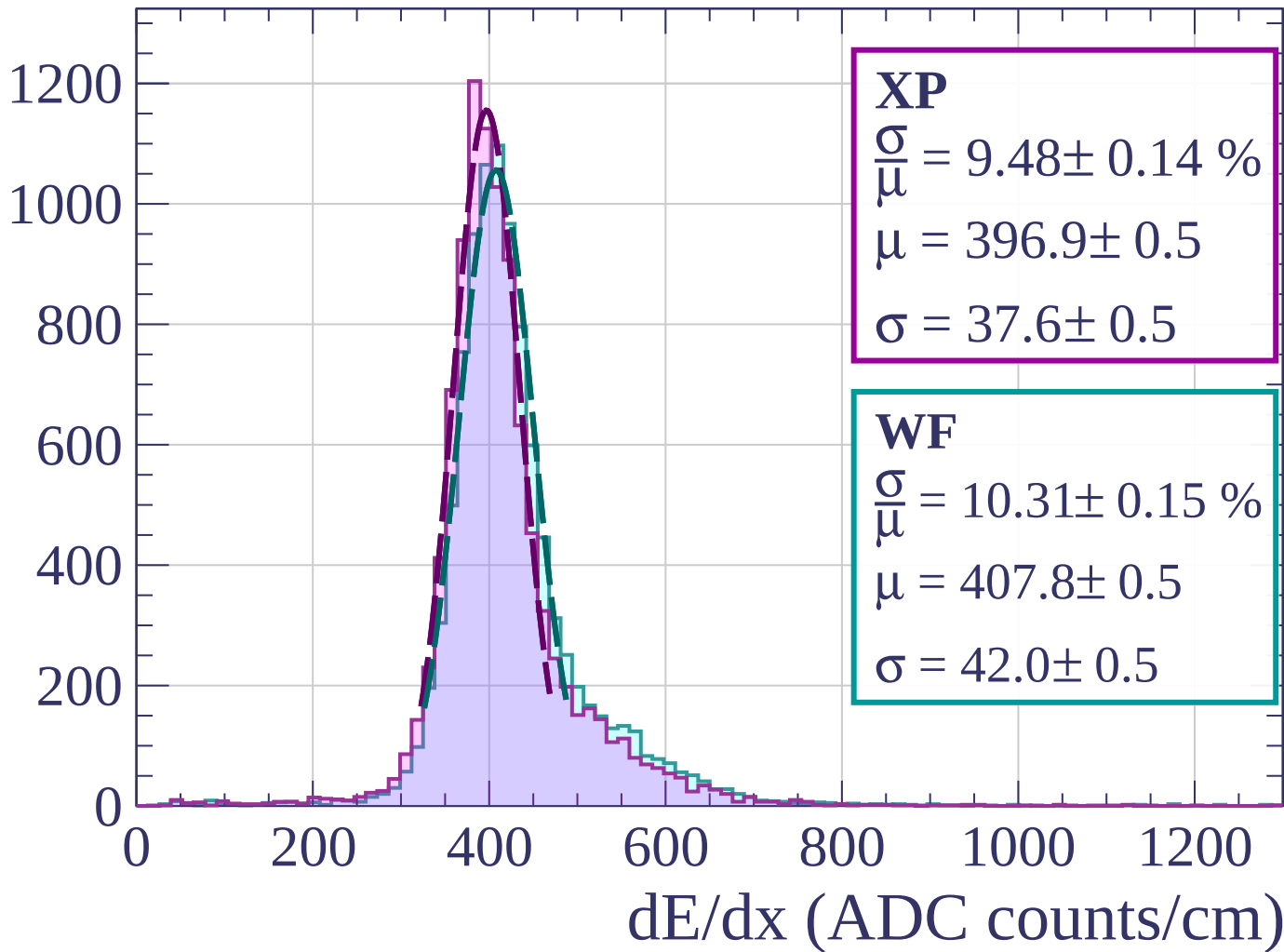


Energy loss | $15 < \theta < 18$



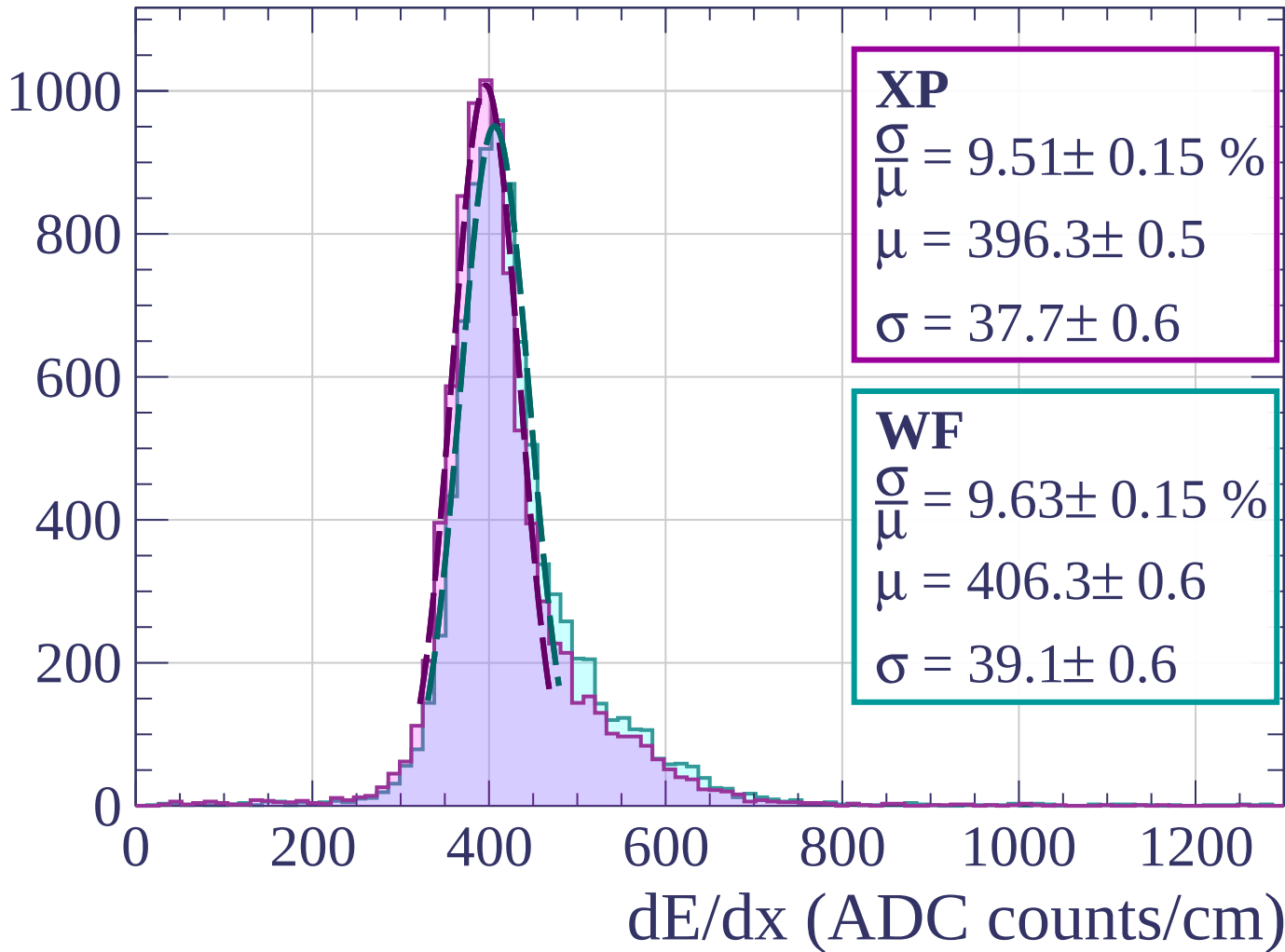
Energy loss | $18 < \theta < 21$

Count



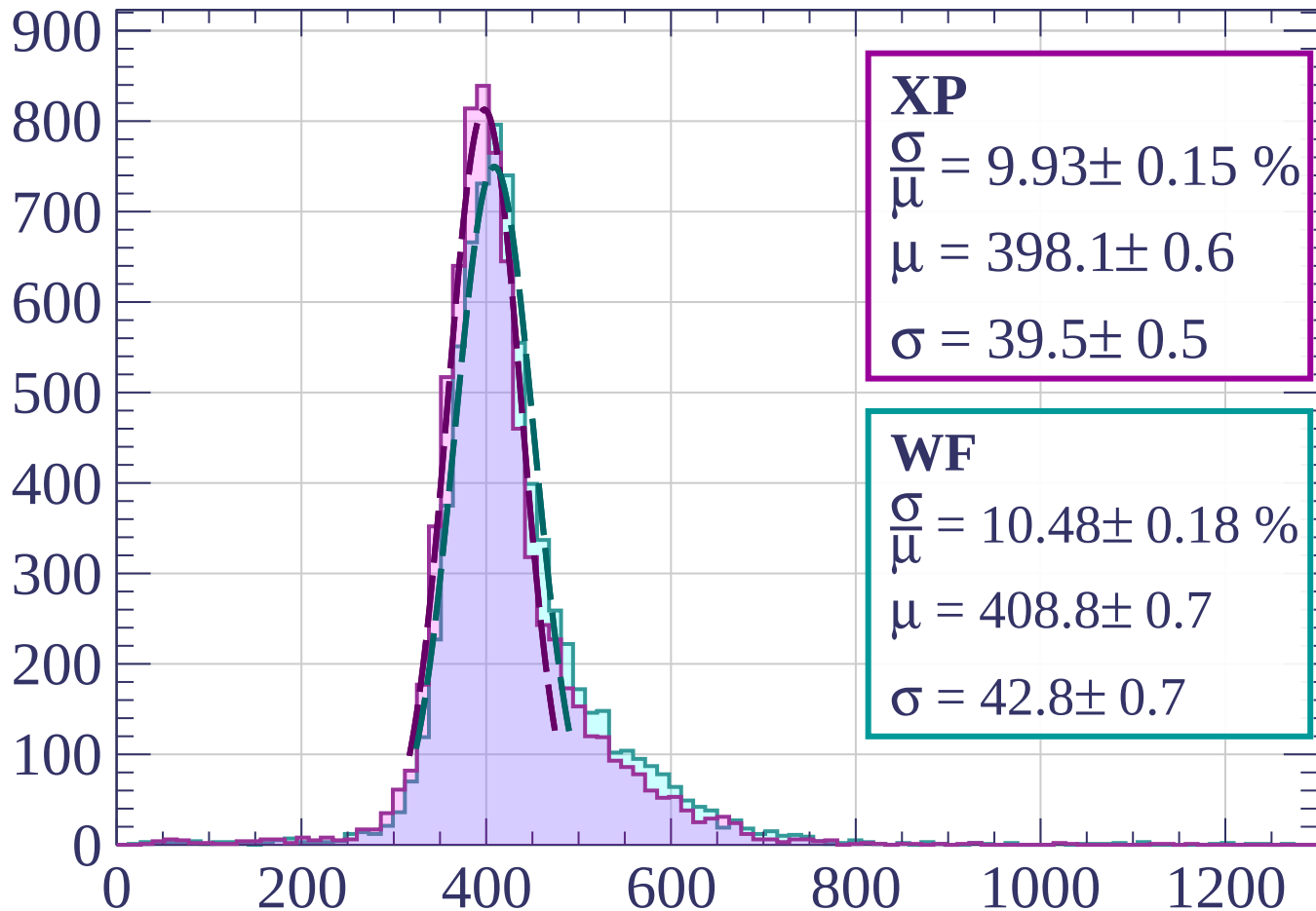
Energy loss | $21 < \theta < 24$

Count



Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 9.93 \pm 0.15 \%$$

$$\mu = 398.1 \pm 0.6$$

$$\sigma = 39.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.18 \%$$

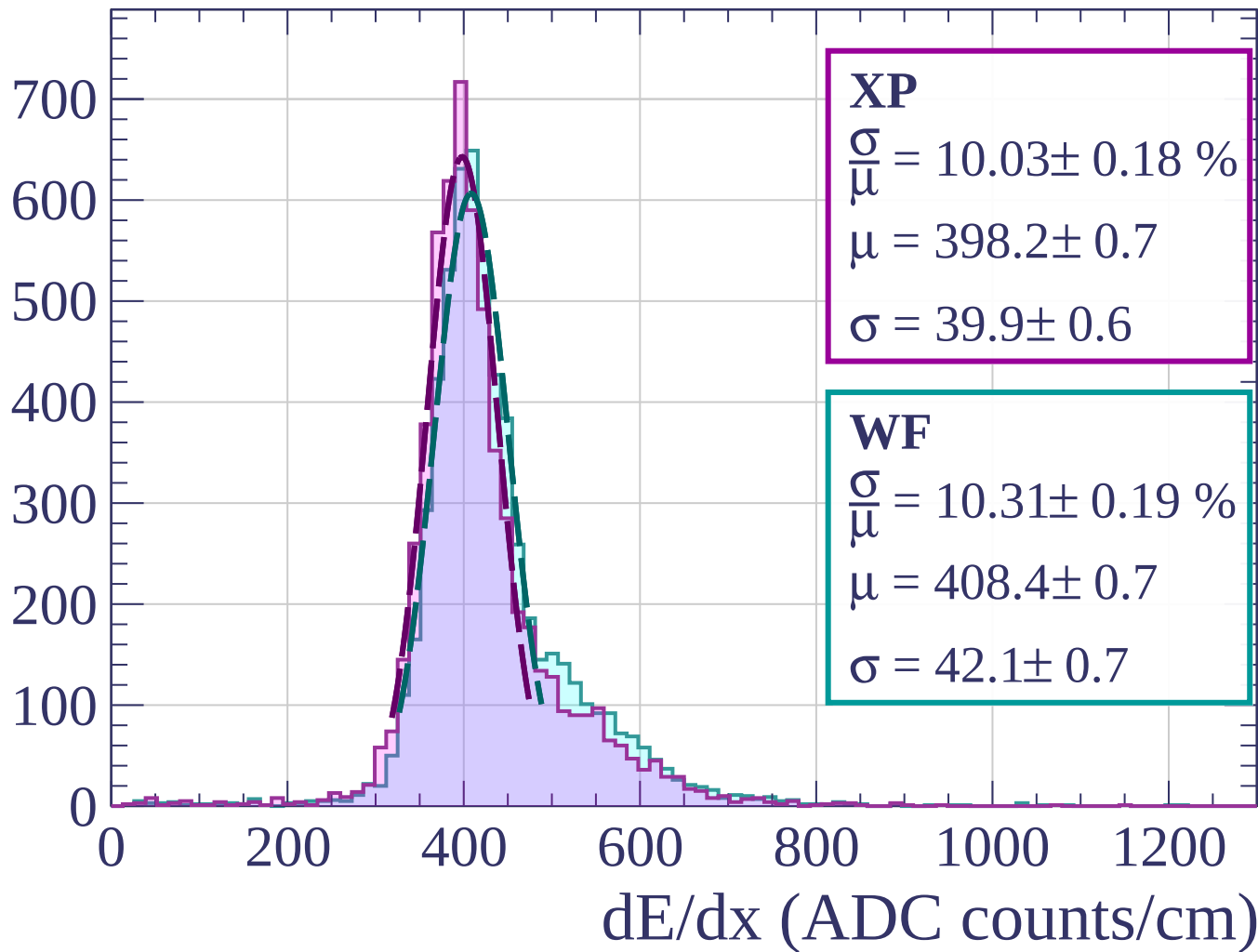
$$\mu = 408.8 \pm 0.7$$

$$\sigma = 42.8 \pm 0.7$$

dE/dx (ADC counts/cm)

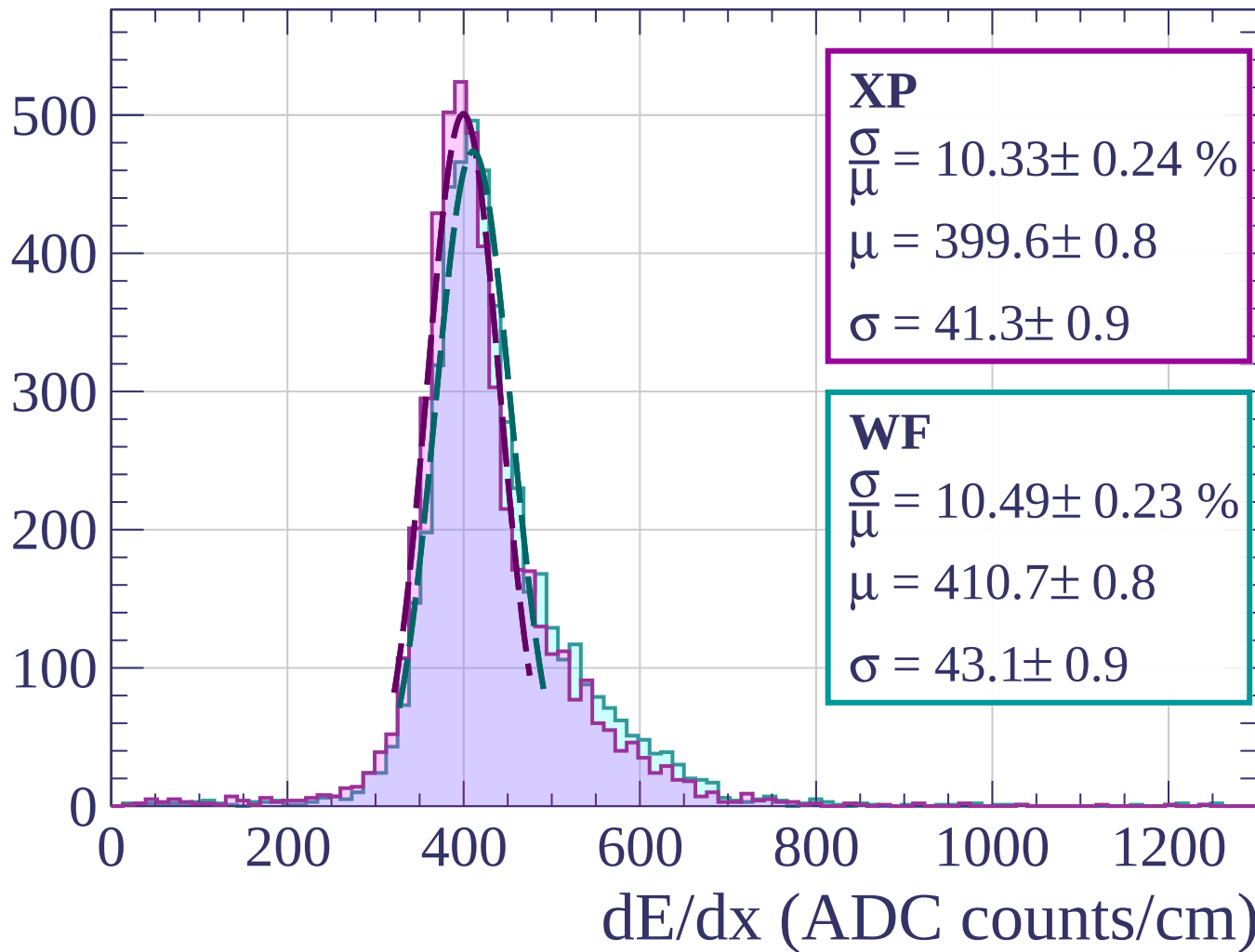
Energy loss | $27 < \theta < 30$

Count



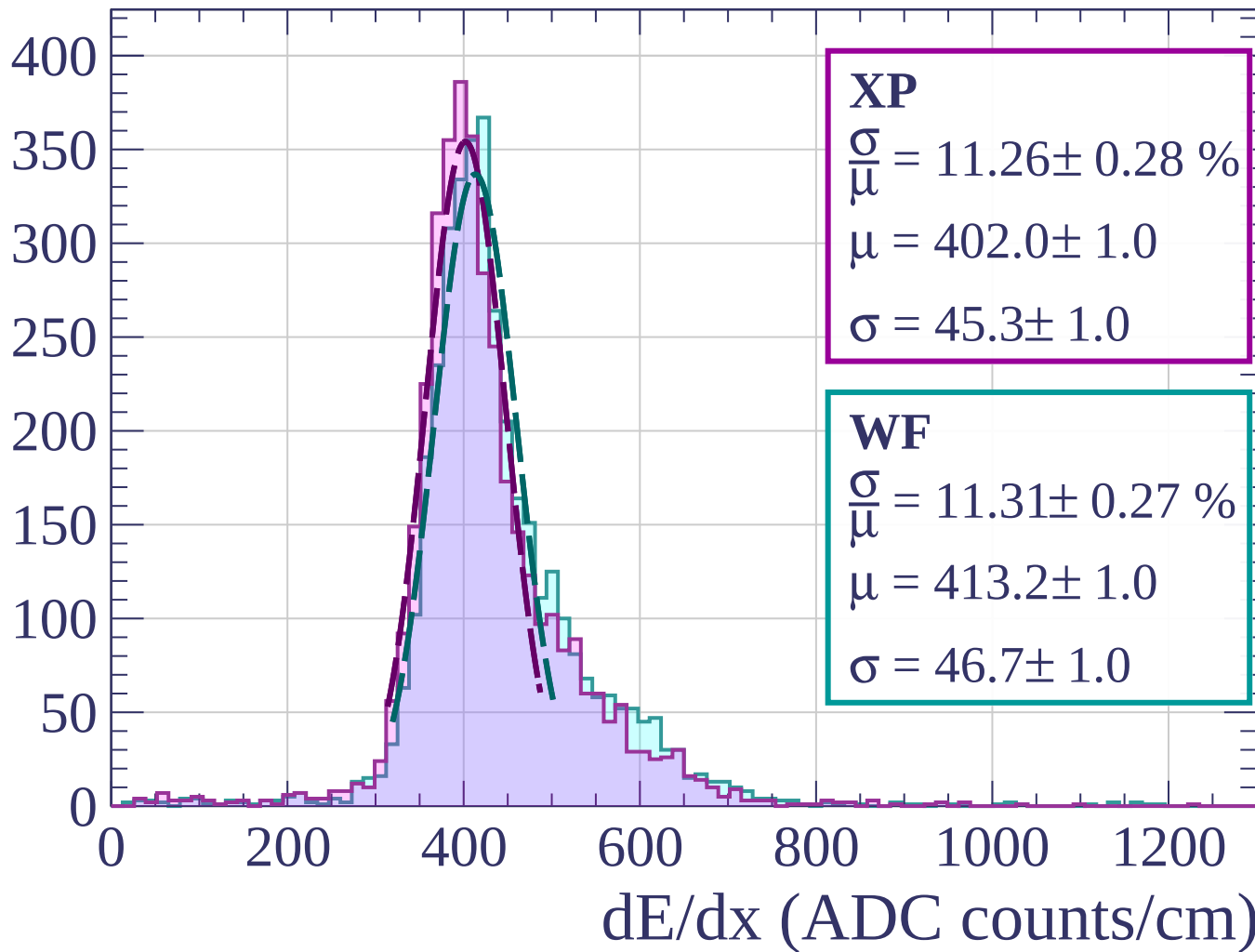
Energy loss | $30 < \theta < 33$

Count



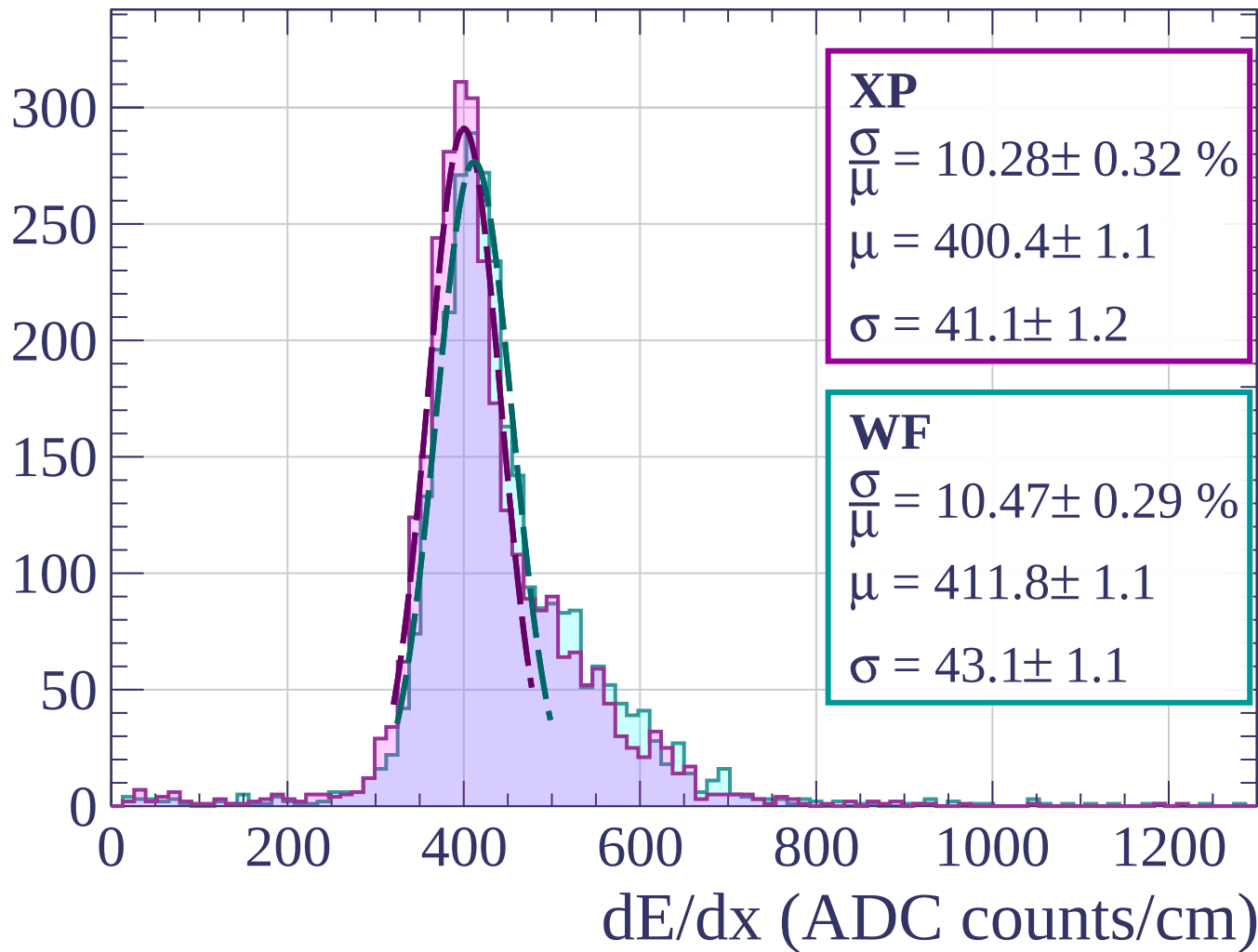
Energy loss | $33 < \theta < 36$

Count



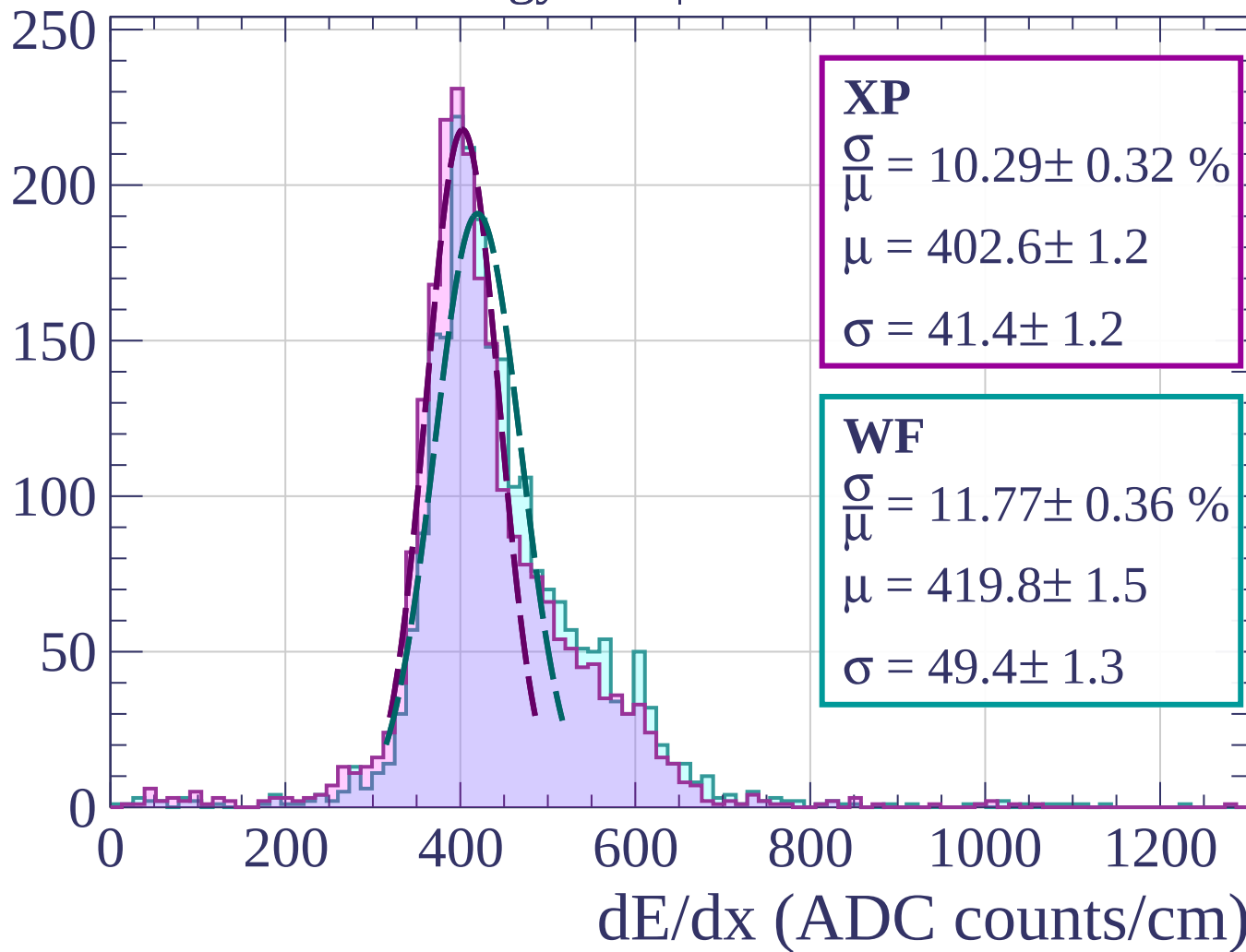
Energy loss | $36 < \theta < 39$

Count



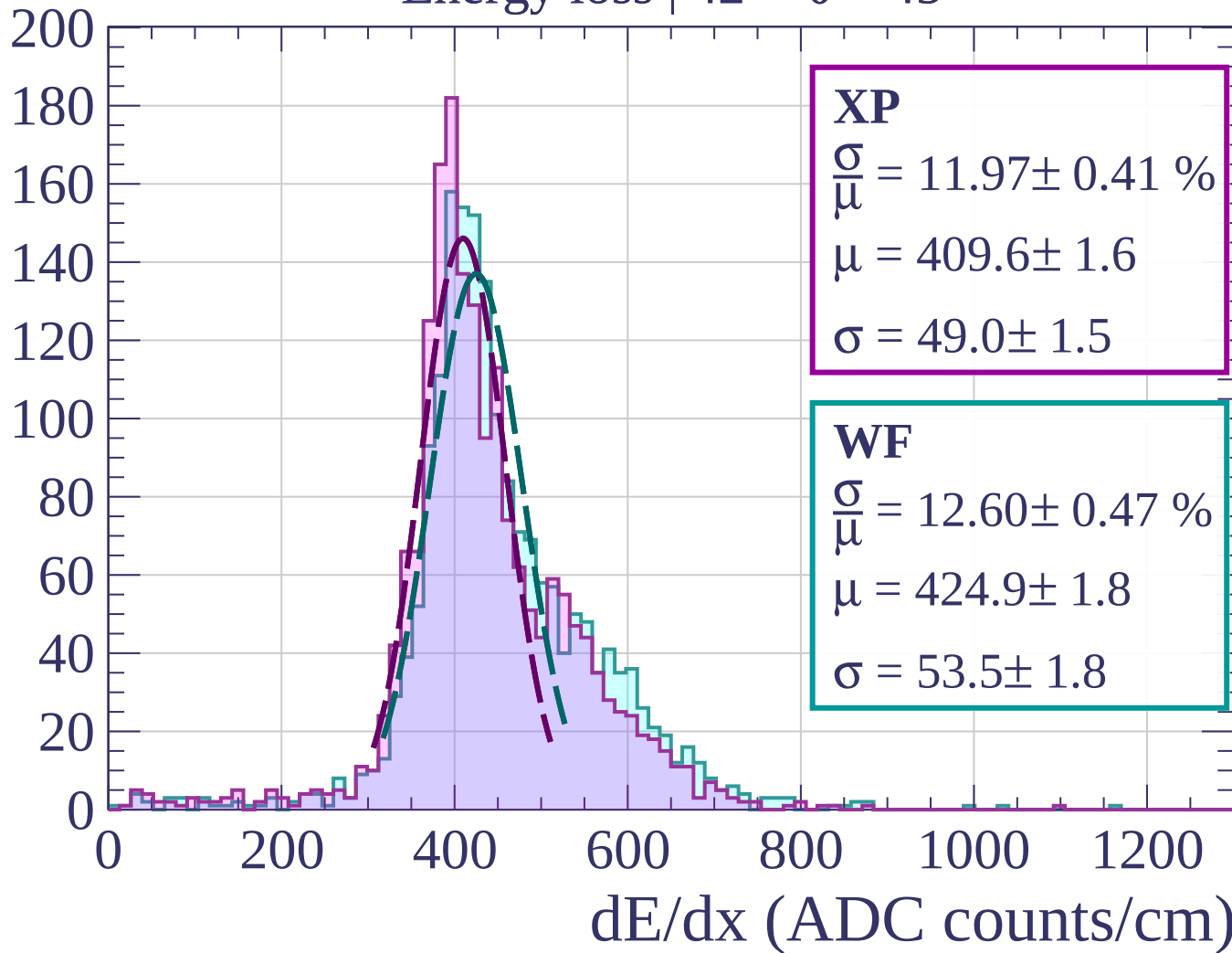
Energy loss | $39 < \theta < 42$

Count



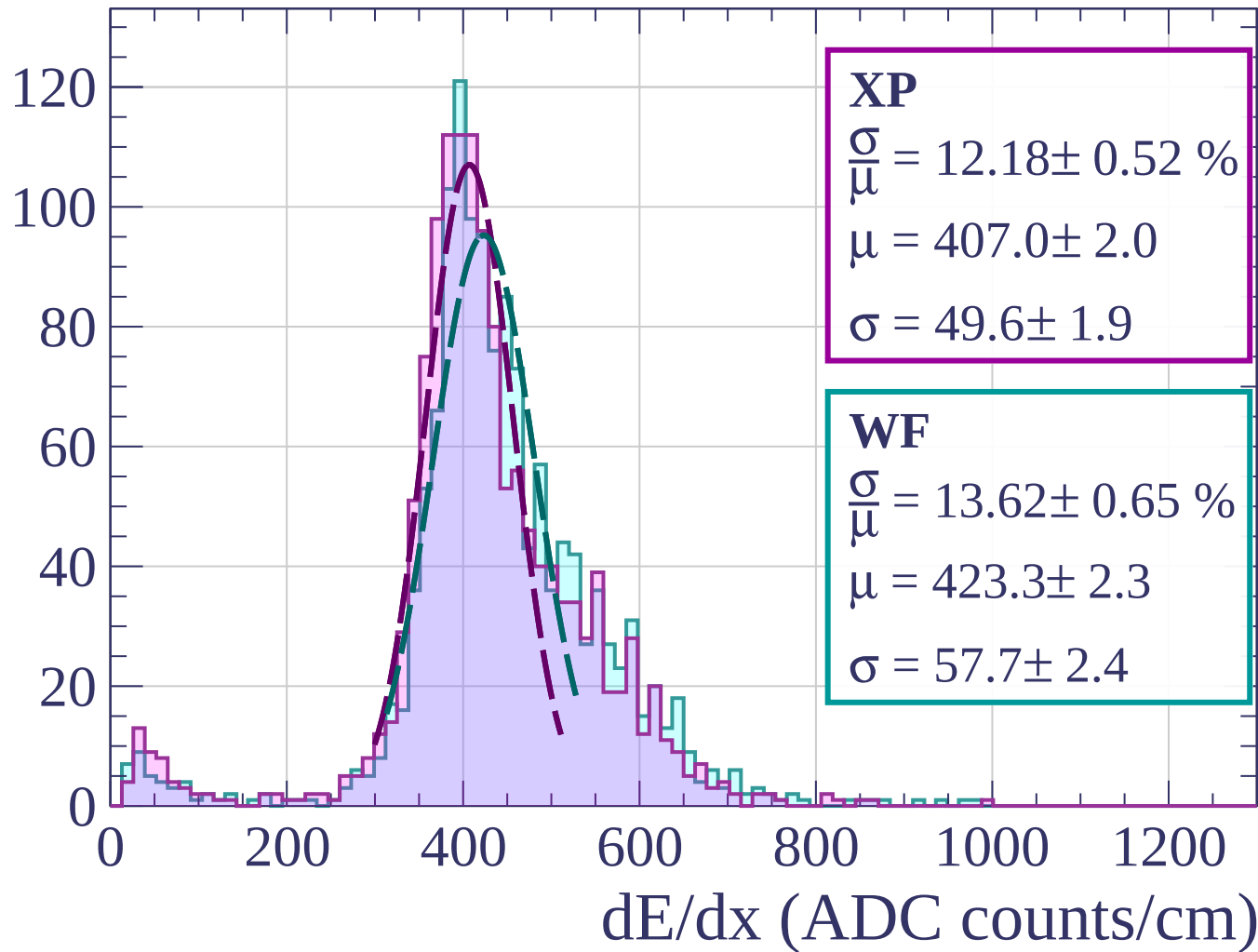
Energy loss | $42 < \theta < 45$

Count



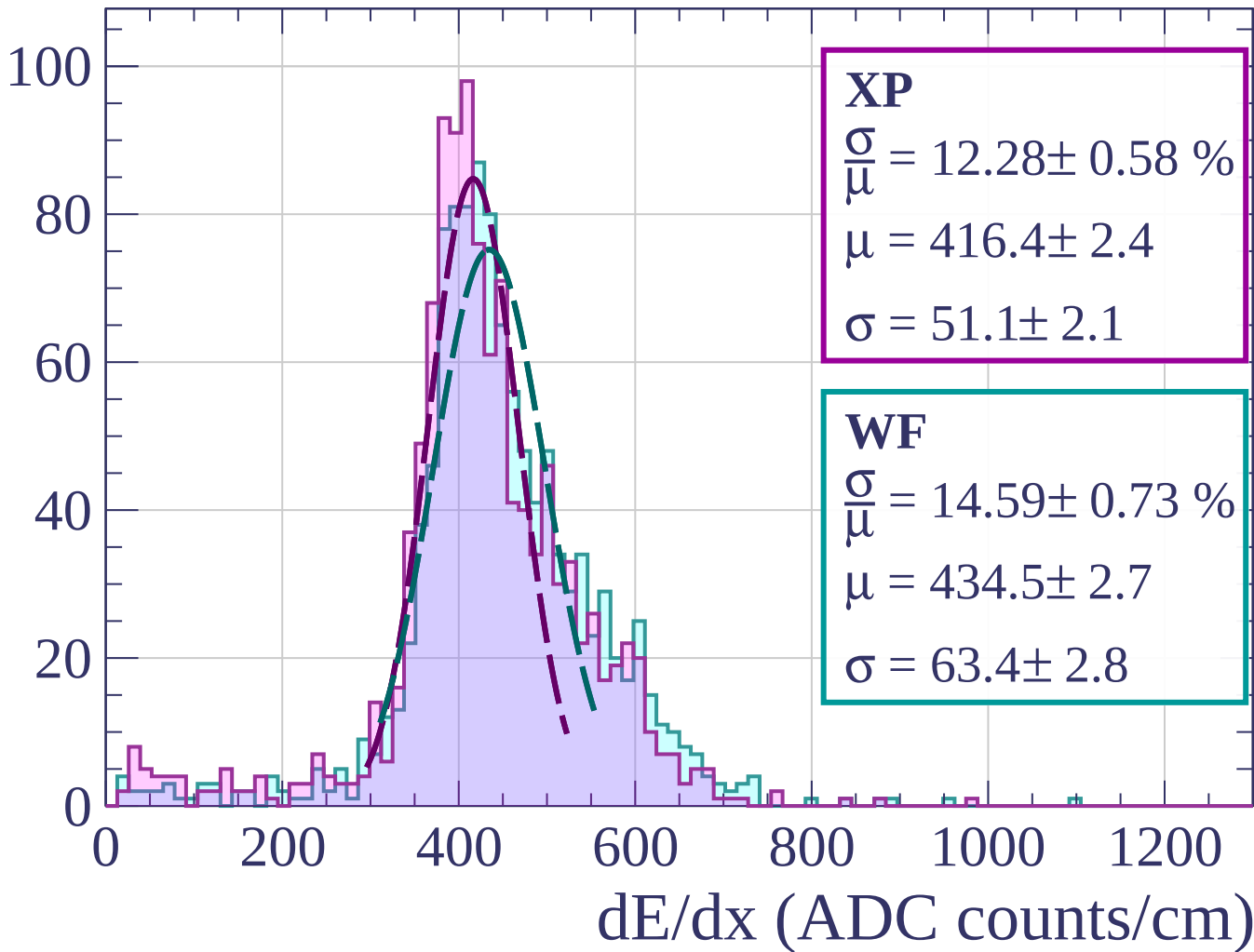
Energy loss | $45 < \theta < 48$

Count



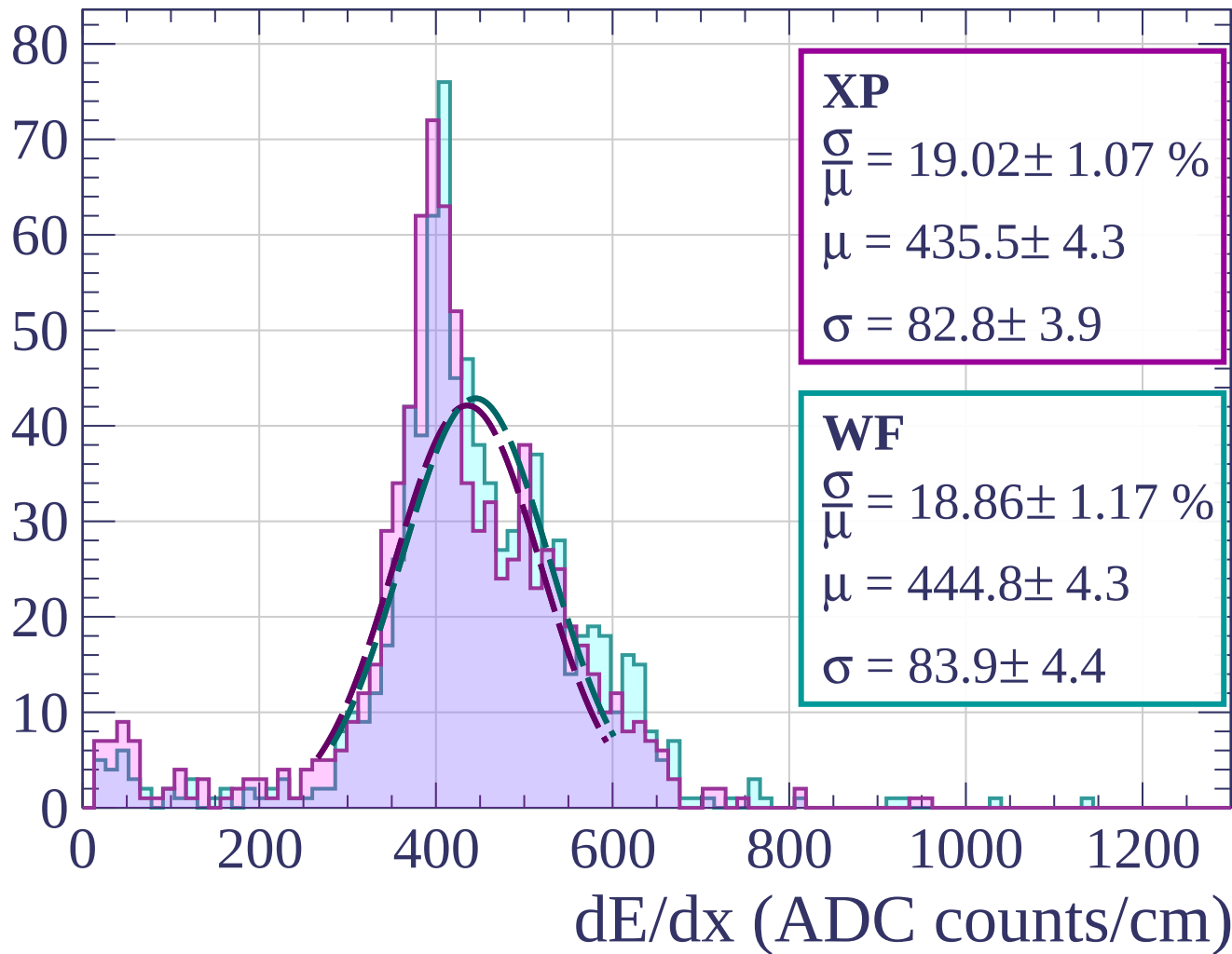
Energy loss | $48 < \theta < 51$

Count



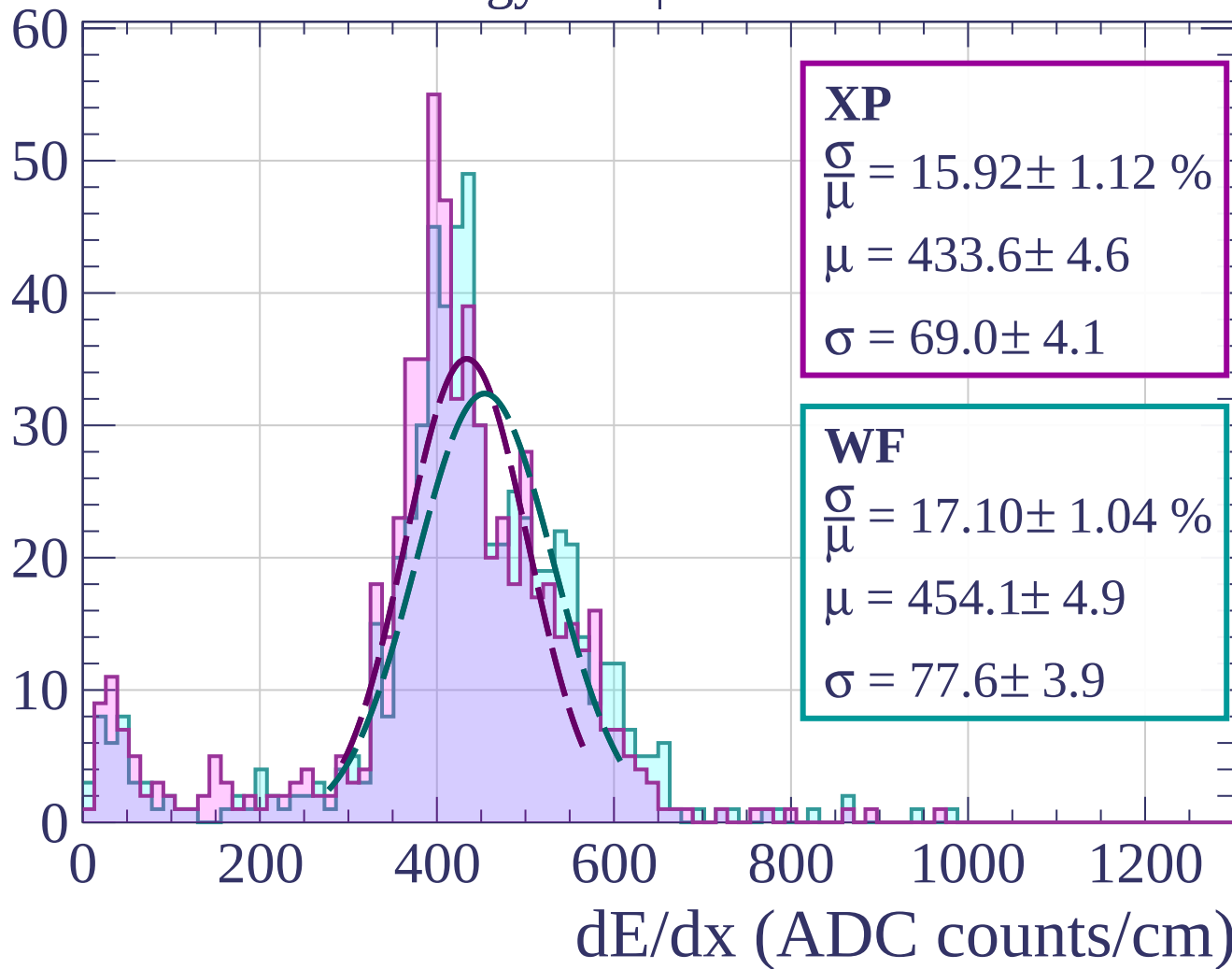
Energy loss | $51 < \theta < 54$

Count



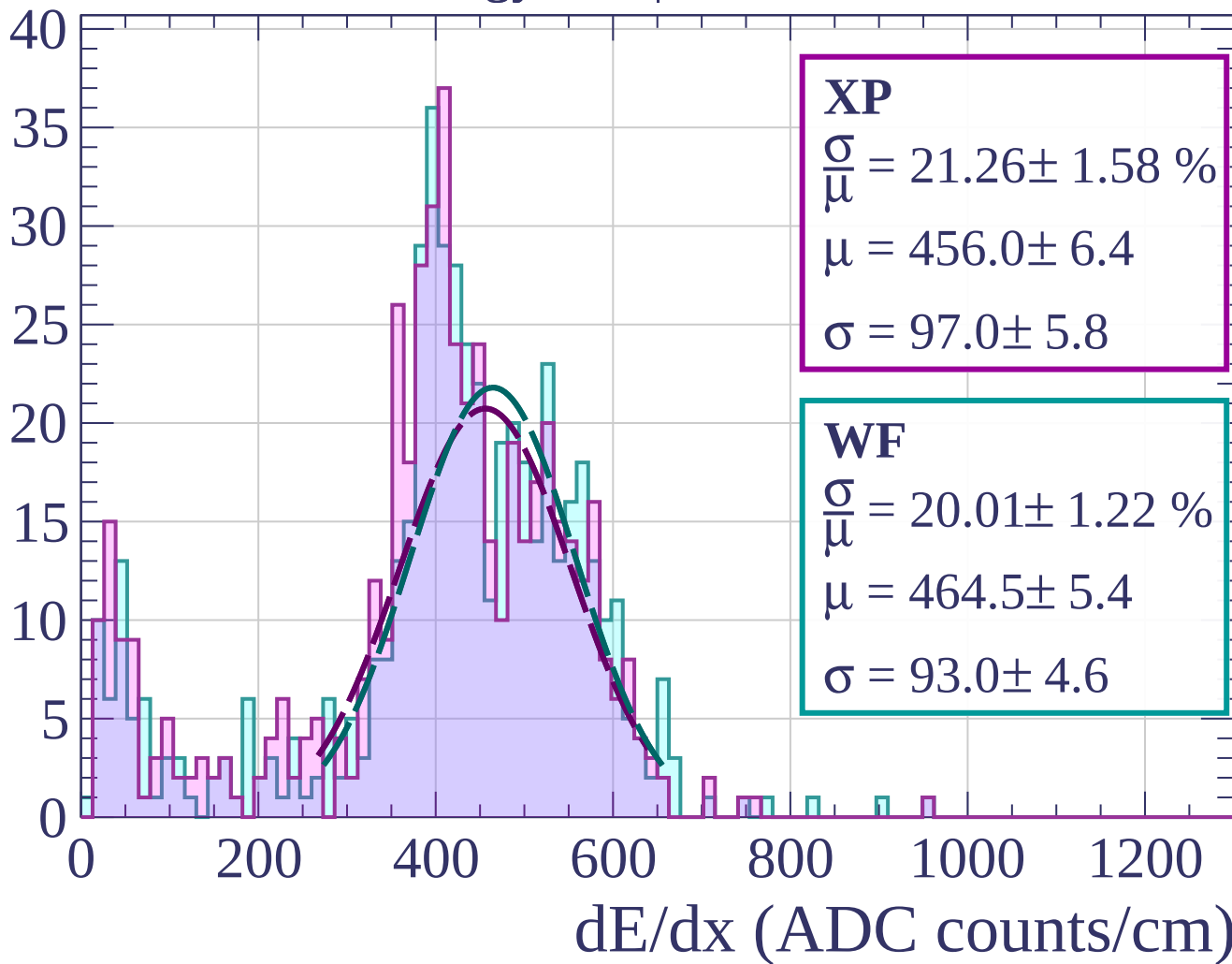
Energy loss | $54 < \theta < 57$

Count



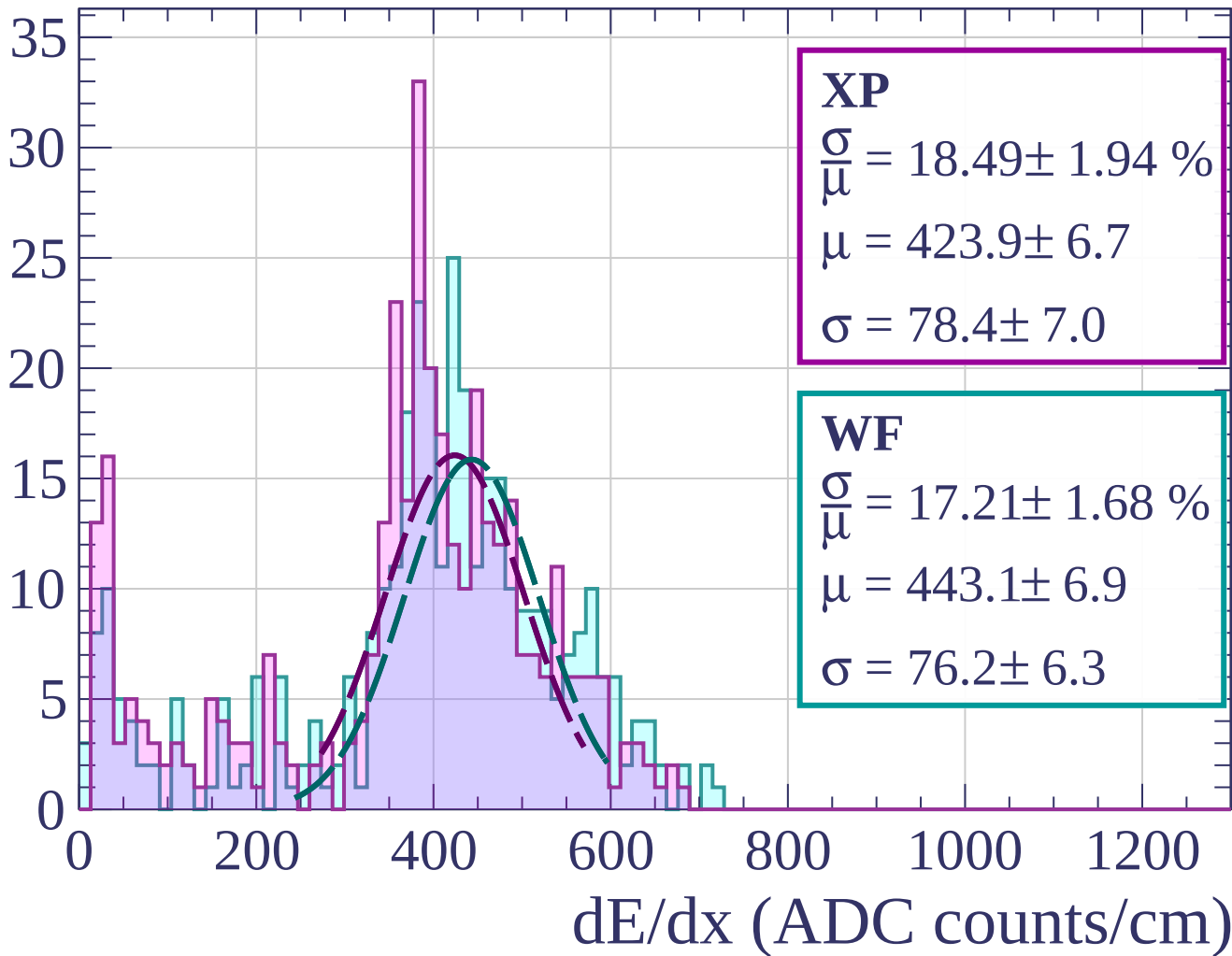
Energy loss | $57 < \theta < 60$

Count



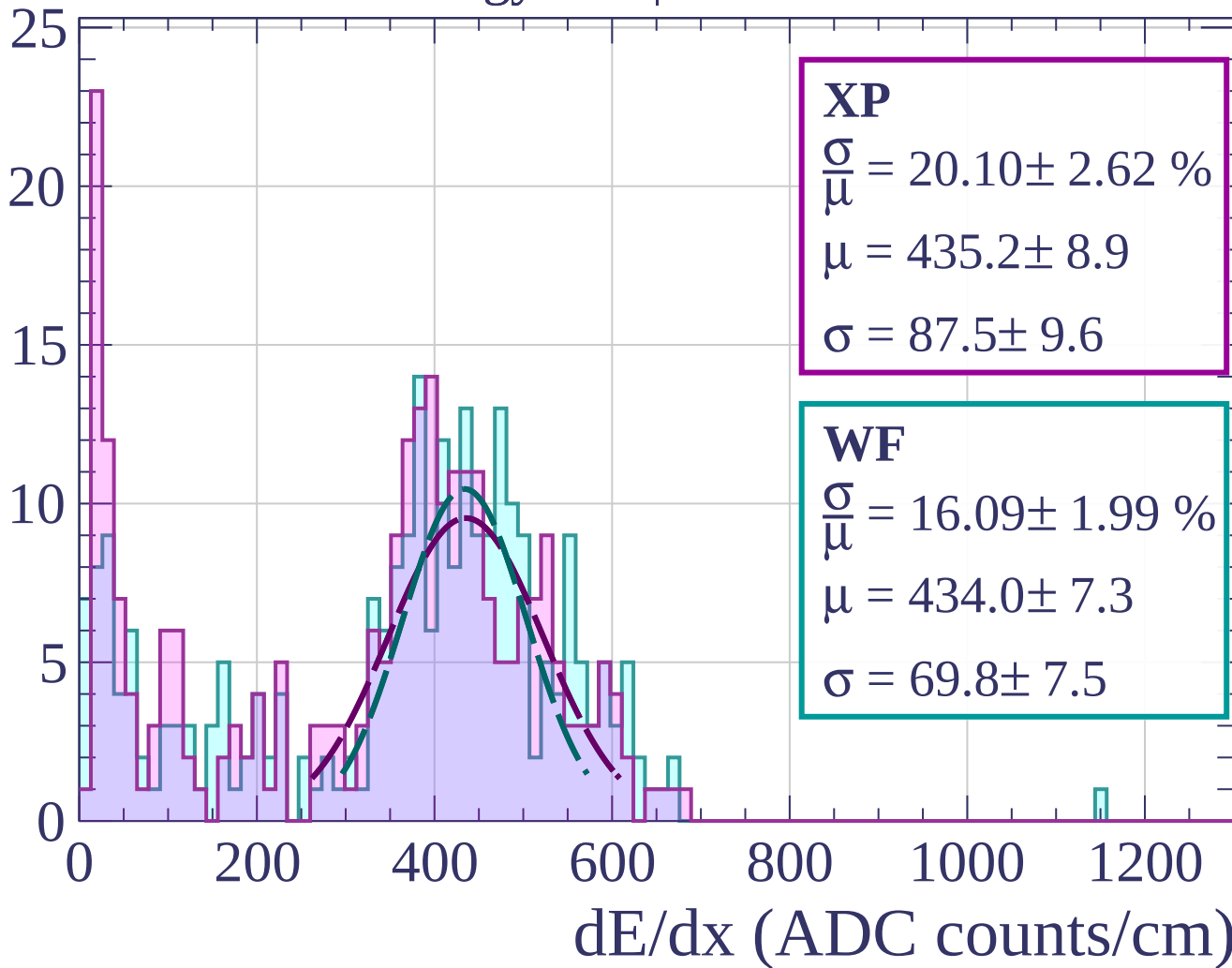
Energy loss | $60 < \theta < 63$

Count



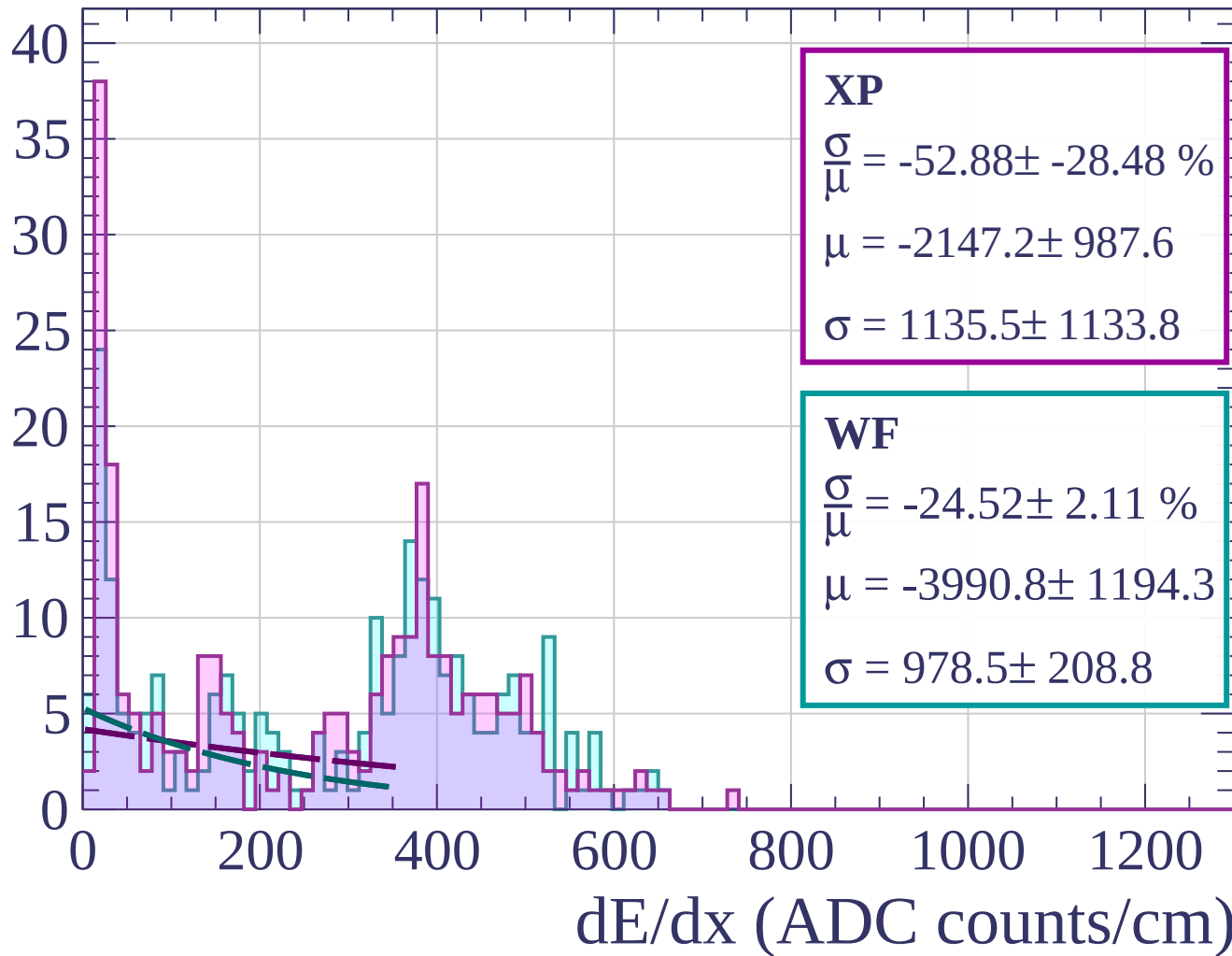
Energy loss | $63 < \theta < 66$

Count



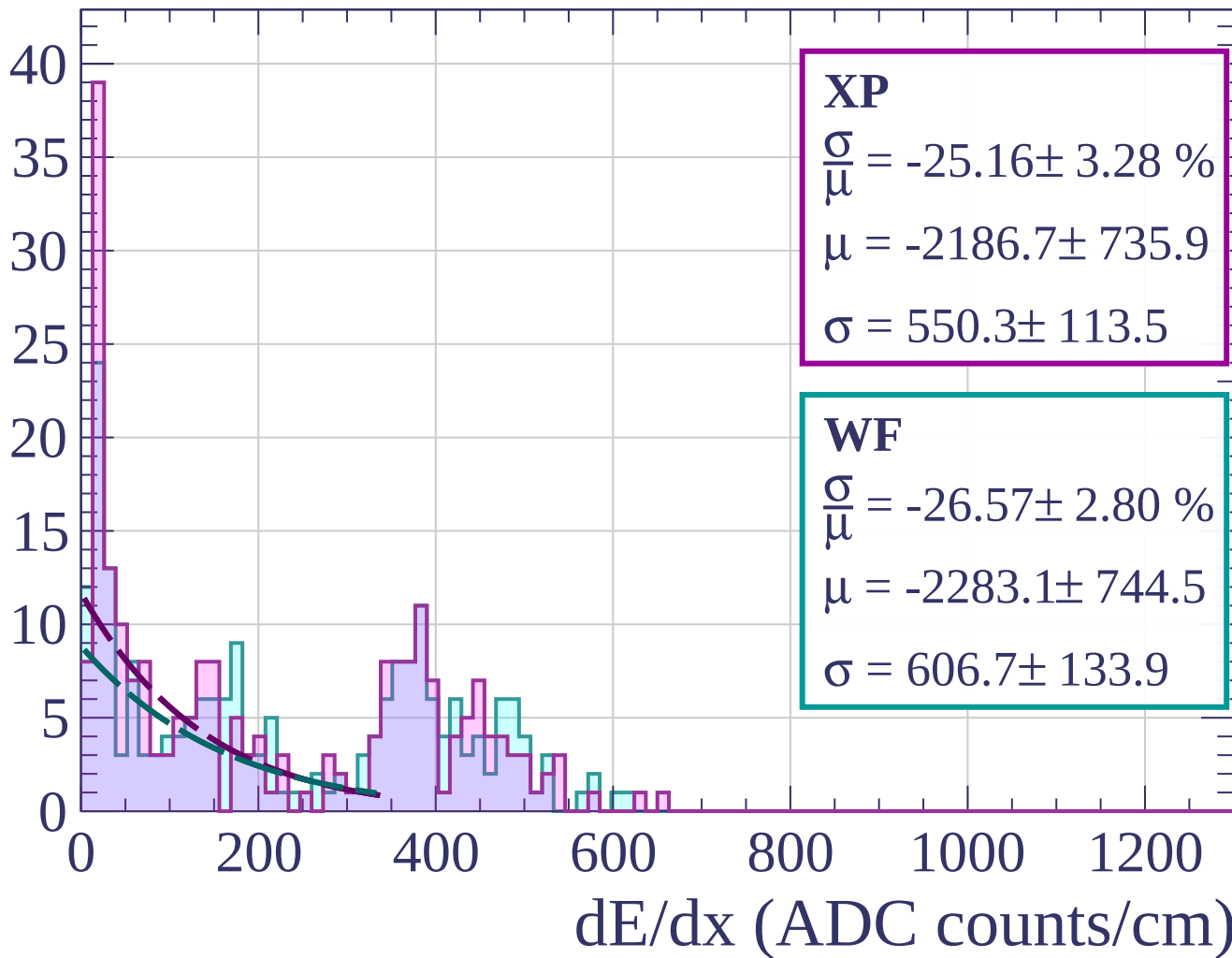
Energy loss | $66 < \theta < 69$

Count



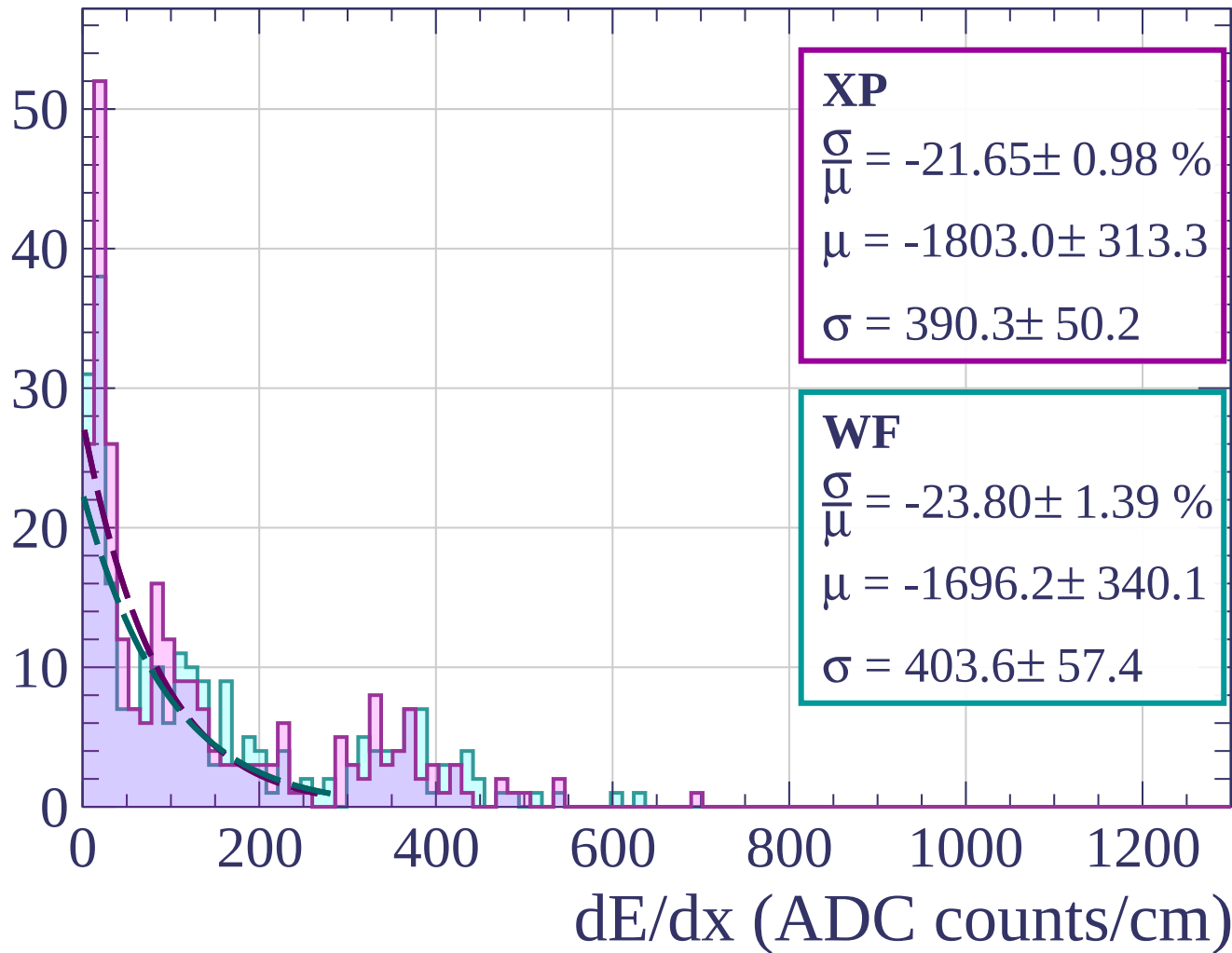
Energy loss | $69 < \theta < 72$

Count



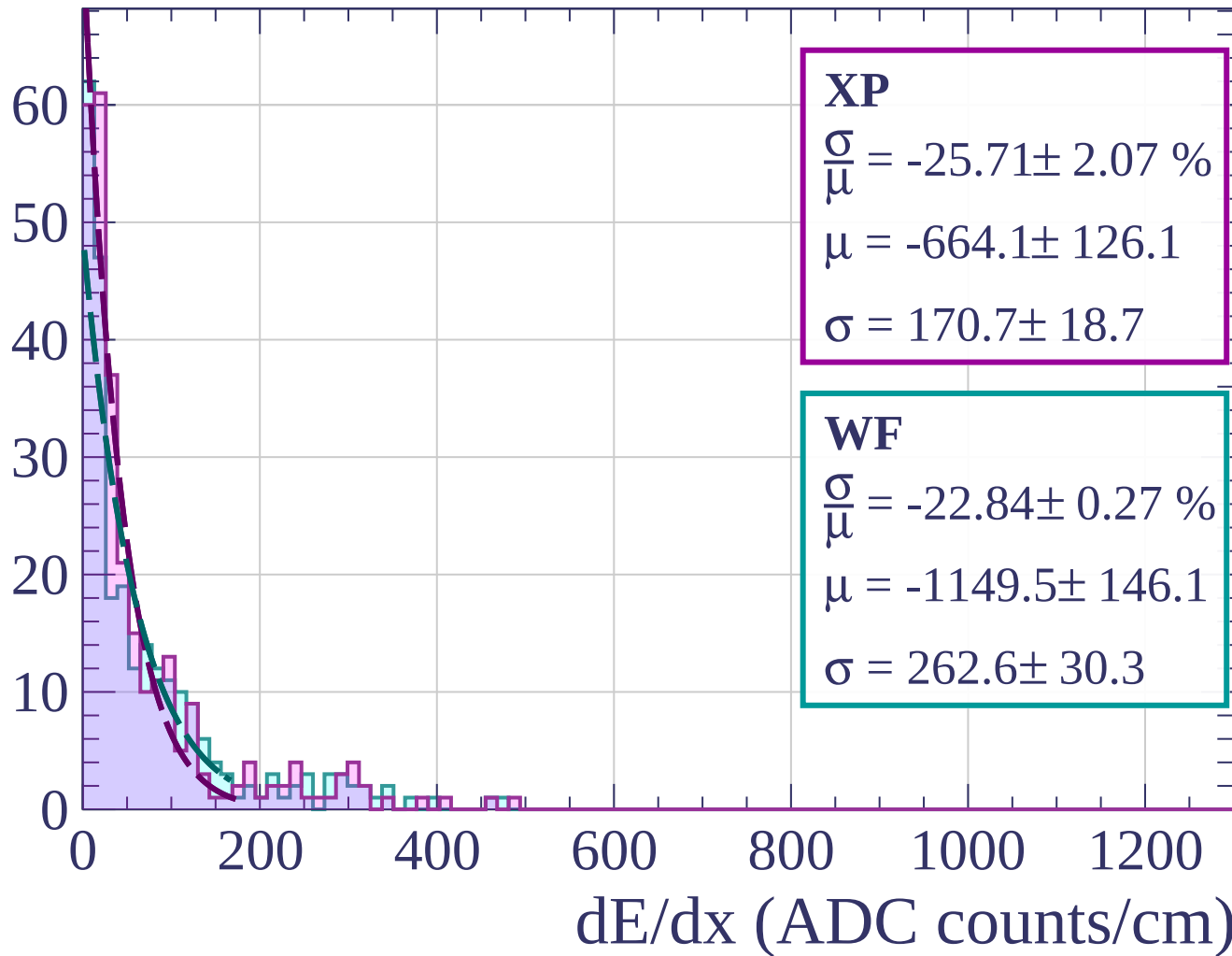
Energy loss | $72 < \theta < 75$

Count



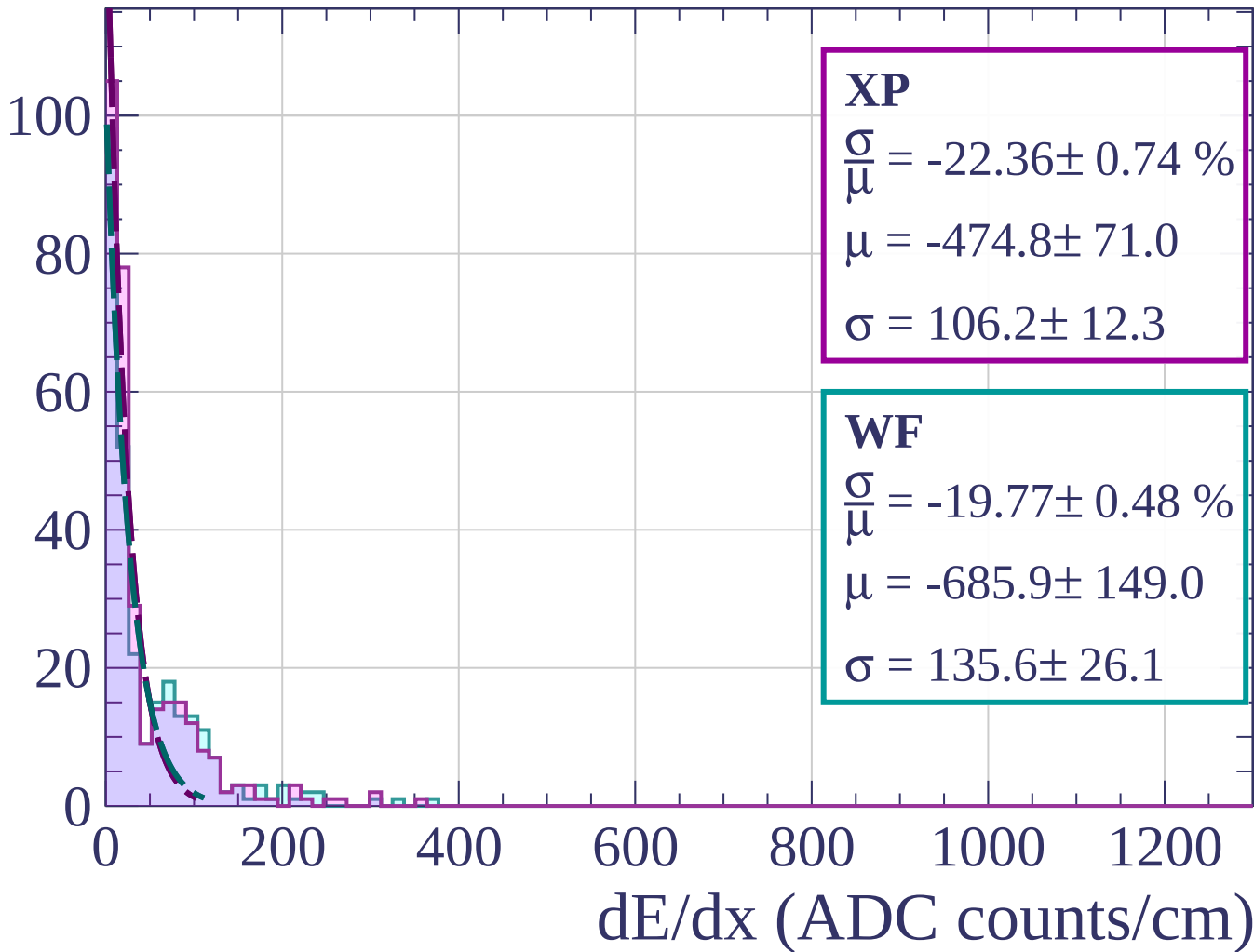
Energy loss | $75 < \theta < 78$

Count



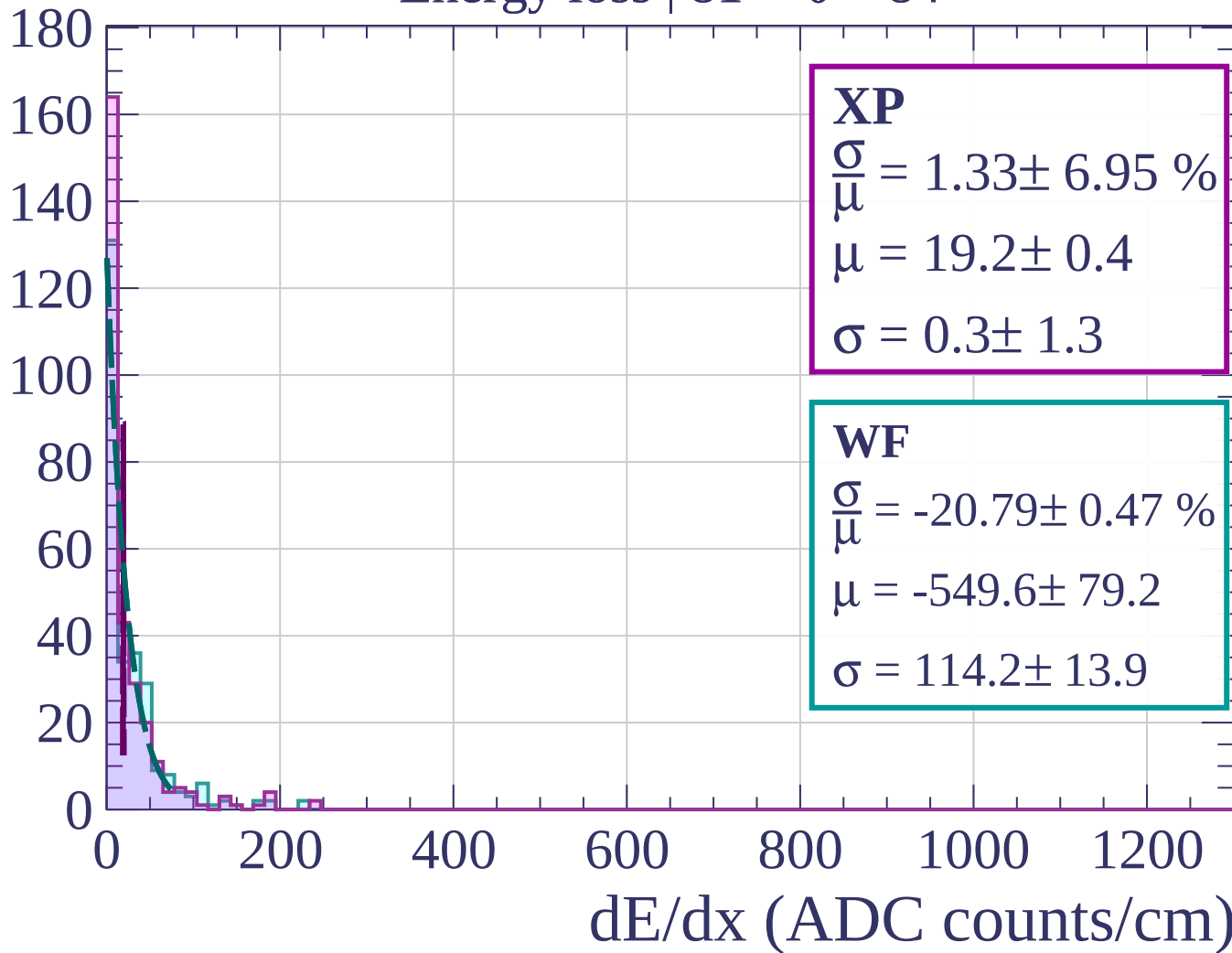
Energy loss | $78 < \theta < 81$

Count



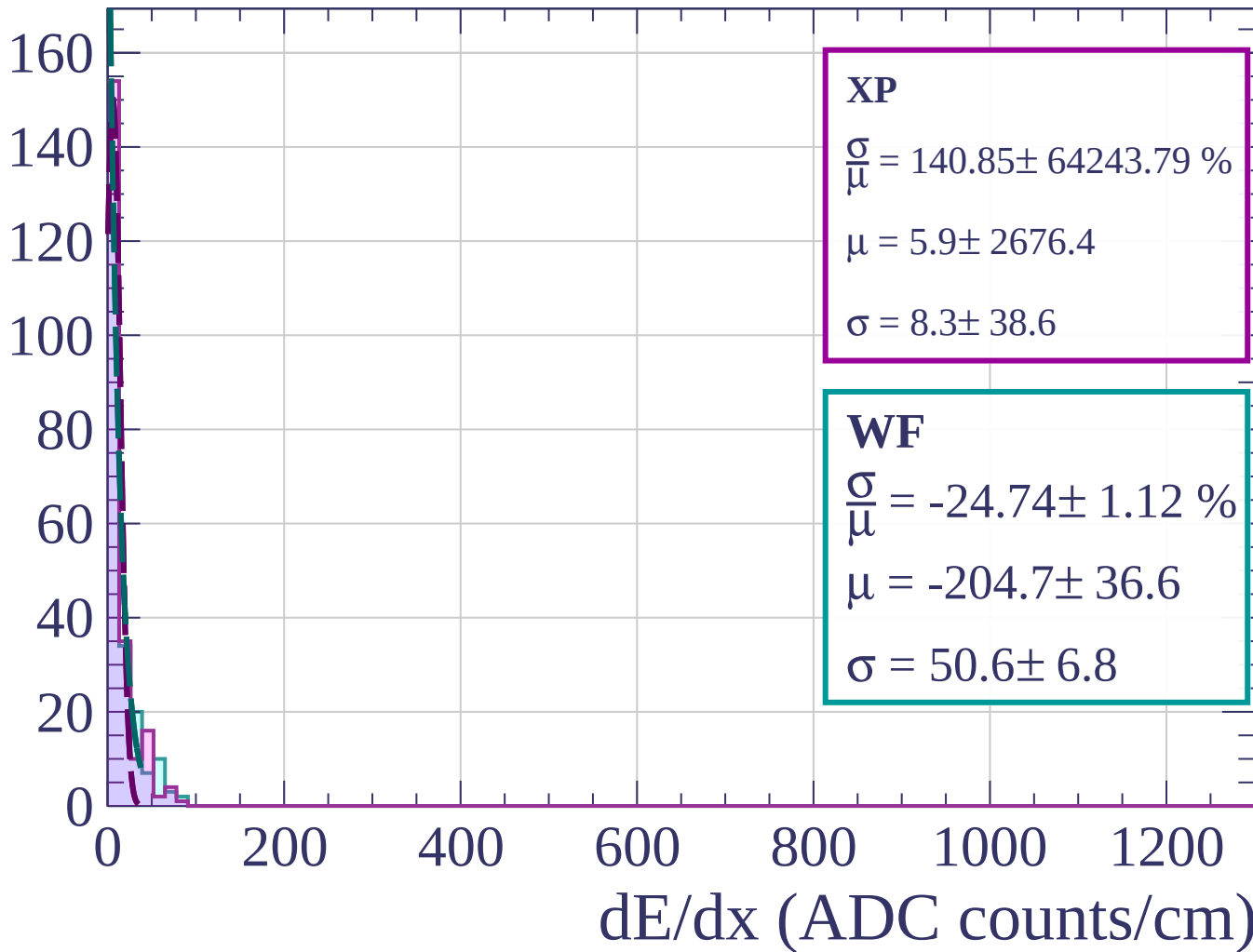
Energy loss | $81 < \theta < 84$

Count



Energy loss | $84 < \theta < 87$

Count



Energy loss | $87 < \theta < 90$

Count

